Stochastic Processes
Lecture Notes

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CHAPTER 1

Probability Space, Random Variables, Limit Theorems

1.1. Probability measure and probability space

We begin with idealizing our situation. A *probability space* is defined to be a triple $(\Omega, \mathscr{F}, \mathbb{P})$, where

$$\Omega = \text{Sample space}, \quad \mathscr{F} = \text{Events}, \quad \mathbb{P} = \text{Probability measure}. \tag{1}$$

This is our idealized world where we can precisely measure uncertainty of all possible events. First, Ω is a set called the *sample space*. This is the collection of all possible outcomes that we can observe (e.g., the six sides of a die). Second, we are going to perform some experiment on Ω (e.g., rolloing a die), and the outcome $\omega \in \Omega$ could belong to any subset E of Ω, which we call an *event*. Let us denote the collection of all events $E \subseteq \Omega$ by $\mathscr{F}$. When Ω is finite or countable, we usually take $\mathscr{F} = 2^{\Omega}$, the collection of all subsets of Ω. If Ω is uncountable (e.g., $\Omega = \mathbb{R}$), then $\mathscr{F}$ must be a special collection of subsets of Ω called a σ-algebra, In this course, we will from now on simply take $\mathscr{F} = 2^{\Omega}$ and do not worry about σ-algebra.

Lastly, $\mathbb{P}$ is a *probability measure* on the (measurable space) $(\Omega, \mathscr{F})$, which gives a quantification $\mathbb{P}(E)$ of how likely it is that the event E occurs out of our experiment. Mathematically, it is a function $\mathbb{P} : \mathscr{F} \to \mathbb{R}$ such that for each event $E \in \mathscr{F}$ (recall $E \subseteq \Omega$), it assigns a real number $\mathbb{P}(E) \in \mathbb{R}$ and satisfies the following *Kolmogorov axioms*:

(i) For each event $E \in \mathscr{F}$, $\mathbb{P}(E) \in [0,1]$;
(ii) $\mathbb{P}(\Omega) = 1$;
(iii) (σ-additivity) For a countable collection of disjoint events $A_1, A_2, \cdots \in \mathscr{F}$,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \tag{2}$$

When Ω is finite, one can replace (iii) above with the following simpler condition: If two events $E_1, E_2 \subseteq \Omega$ are disjoint, then $\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_1)$.

Exercise 1.1.1. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. Show the following.
(i) Let $E = \{x_1, x_2, \cdots x_k\} \subseteq \Omega$ be an event. Then $\mathbb{P}(E) = \sum_{i=1}^{k} \mathbb{P}(\{x_i\})$.
(ii) $\sum_{x \in \Omega} \mathbb{P}(\{x\}) = 1$.
(iii) (subadditivity) Let $A_1, A_2, \cdots \in \mathscr{F}$ be arbitrary. Then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i). \tag{3}$$

Exercise 1.1.2. Let $\mathbb{P}$ be a probability measure on sample space Ω. Show the following.
(i) Let $E = \{x_1, x_2, \cdots x_k\} \subseteq \Omega$ be an event. Then $\mathbb{P}(E) = \sum_{i=1}^{k} \mathbb{P}(\{x_i\})$.
(ii) $\sum_{x \in \Omega} \mathbb{P}(\{x\}) = 1$.

Of course, there could be many (in fact, infinitely many) different probability measures on the same sample space.

Exercise 1.1.3 (coin flip)**.** Let $\Omega = \{H, T\}$ be a sample space. Fix a parameter $p \in [0,1]$, and define a function $\mathbb{P}_p : 2^{\Omega} \to [0,1]$ by $\mathbb{P}_p(\emptyset) = 0$, $\mathbb{P}_p(\{H\}) = p$, $\mathbb{P}_p(\{T\}) = 1 - p$, $\mathbb{P}_p(\{H, T\}) = 1$. Verify that $\mathbb{P}_p$ is a probability measure on Ω for each value of p.

A typical way of constructing a probability measure is to specify how likely it is to see each individual element in Ω. Namely, let $f : \Omega \to [0,1]$ be a function that sums up to 1, i.e., $\sum_{x\in\Omega} f(x) = 1$. Define a function $\mathbb{P} : \mathscr{F} \to [0,1]$ by

$$\mathbb{P}(E) = \sum_{\omega\in E} f(\omega). \tag{4}$$

Then this is a probability measure on Ω, and f is called a *probability distribution* on Ω. For instance, the probability distribution on $\{H,T\}$ we used to define $\mathbb{P}_p$ in Exercise 1.1.3 is $f(H) = p$ and $f(T) = 1-p$.

Example 1.1.4 (Uniform probability measure)**.** Let $\Omega = \{1,2,\cdots,m\}$ be a sample space and let $\mathbb{P}$ be the *uniform probability measure* on Ω, that is,

$$\mathbb{P}(\{x\}) = 1/m \qquad \forall x \in \Omega. \tag{5}$$

Then for the event $A = \{1,2,3\}$, we have

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(\{1\}\cup\{2\}\cup\{3\}) && (6)\\ &= \mathbb{P}(\{1\}) + \mathbb{P}(\{2\}) + \mathbb{P}(\{3\}) && (7)\\ &= \frac{1}{m} + \frac{1}{m} + \frac{1}{m} = \frac{3}{m} && (8) \end{aligned}$$

Likewise, if $A \subseteq \Omega$ is any event and if we let $|A|$ denote the size (number of elements) of A, then

$$\mathbb{P}(A) = \frac{|A|}{m}. \tag{9}$$

For example, let $\Omega = \{1,2,3,4,5,6\}^2$ be the sample space of a roll of two fair dice. Let A be the event that the sum of two dice is 5. Then

$$A = \{(1,4),(2,3),(3,2),(4,1)\}, \tag{10}$$

so $|A| = 4$. Hence $\mathbb{P}(A) = 4/36 = 1/9$. ▲

Exercise 1.1.5. Show that the function $\mathbb{P} : \mathscr{F} \to [0,1]$ defined in (4) is a probability measure on Ω. Conversely, show that every probability measure on a finite sample space Ω can be defined in this way.

Remark 1.1.6 (General probability space)**.** A probability space does not need to be finite, but we need a more careful definition in that case. For example, if we take Ω to be the unit interval $[0,1]$, then we have to be careful in deciding which subset $E \subseteq \Omega$ can be an 'event': not every subset of Ω can be an event. A proper definition of general probability space is out of the scope of this course.

Exercise 1.1.7. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and let $A \subseteq \Omega$ be an event. Show that $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Example 1.1.8 (Roll of two dice)**.** Suppose we roll two dice and let X and Y be the outcome of each die. Say all possible joint outcomes are equally likely. The sample space for the roll of a single die can be written as $\{1,2,3,4,5,6\}$, so the sample space for rolling two dice can be written as $\Omega = \{1,2,3,4,5,6\}^2$. In picture, think of 6 by 6 square grid and each node represents a unique outcome (x,y) of the roll. By assumption, each out come has probability $1/36$. Namely,

$$\mathbb{P}((X,Y) = (x,y)) = 1/36 \qquad \forall 1 \le x, y \le 6. \tag{11}$$

This gives the uniform probability distribution on our sample space Ω (see Example 1.1.4).

We can compute various probabilities for this experiment. For example,

$$\begin{aligned} \mathbb{P}(\text{at least one die is even}) &= 1 - \mathbb{P}(\text{both dice are odd}) && (12)\\ &= 1 - \mathbb{P}(\{1,3,5\}\times\{1,3,5\}) = 1 - \frac{9}{36} = 3/4, && (13) \end{aligned}$$

where for the first equality we have used the complementary probability in Exercise 1.1.7.

Now think about the following question: *What is the most probable value for the sum $X+Y$?* By considering diagonal lines $x+y=k$ in the 2-dimensional plane for different values of k, we find

$$\mathbb{P}(X+Y=k)=\frac{\#\text{ of intersections between the line } x+y=k \text{ and } \Omega}{36}. \tag{14}$$

From example, $\mathbb{P}(X+Y=2)=1/36$ and $\mathbb{P}(X+Y=7)=6/36=1/6$. Moreover, from Figure 1, it is clear that the number of intersections is maximized when the diagonal line $x+y=k$ passes through the extreme points $(1,6)$ and $(6,1)$. Hence 7 is the most probable value for $X+Y$ with the probability being $1/6$.

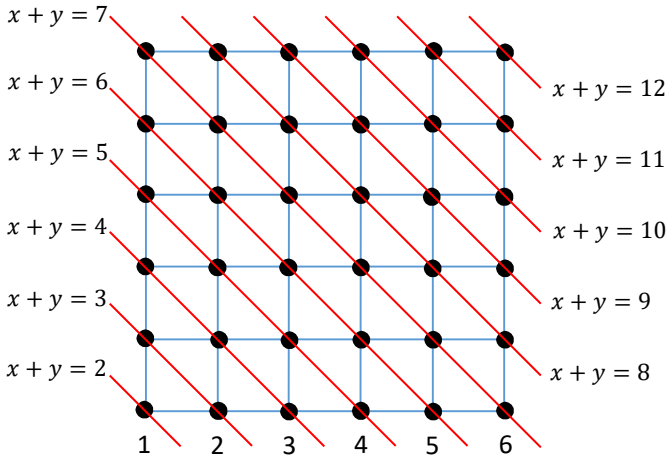


FIGURE 1. Sample space representation for roll of two independent fair dice and and events of fixed sum of two dice.

▲

1.2. Independence

When knowing something about one event does not yield any information of the other, we say the two events are independent. To make this statement a bit more precise, suppose Bob is betting \$5 for whether an event E_1 occurs or not. Suppose Alice tells him that some other event E_2 holds true. If Bob can somehow leverage on this knowledge to increase his chance of winning, then we should say the two events are not independent.

Formally, we say two events E_1 and E_2 are *independent* if

$$\mathbb{P}(E_1\cap E_2)=\mathbb{P}(E_1)\,\mathbb{P}(E_2). \tag{15}$$

We say two events or RVs are *dependent* if they are not independent.

Example 1.2.1. Flip two fair coins at the same time and let X and Y be their outcome, labeled by H and T. Clearly knowing about one coin does not give any information of the other. For instance, the first coin lands on heads with probability $1/2$. Whether the first coin lands on heads or not, the second coin will land on heads with probability $1/2$. So

$$\mathbb{P}(X=H \text{ and } Y=H)=\frac{1}{2}\cdot\frac{1}{2}=\mathbb{P}(X=H)\mathbb{P}(Y=H). \tag{16}$$

▲

Exercise 1.2.2. Suppose two events A_1 and A_2 are independent and suppose that $\mathbb{P}(A_2)>0$. Show that

$$\mathbb{P}(A_1\,|\,A_2)=\mathbb{P}(A_1). \tag{17}$$

1.3. Random variables

In this section, we introduce the notion of random variables along with a number of important examples. We also introduce expectation and variance as summary statistics for random variables, and learn how to compute them.

1.3.1. Discrete random variables. Given a finite probability space $(\Omega, \mathbb{P})$, a (discrete) *random variable* (RV) is any real-valued function $X : \Omega \to \mathbb{R}$. We can think of it as the outcome of some experiment on Ω (e.g., height of a randomly selected friend). We often forget the original probability space and specify a RV by *probability mass function* (PMF) $f_X : \mathbb{R} \to [0,1]$,

$$f_X(x) = \mathbb{P}(X = x) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) = x\}). \tag{18}$$

Namely, $\mathbb{P}(X = x)$ is the likelihood that the RV X takes value x.

Example 1.3.1. Say you win \$1 if a fair coin lands heads and lose \$1 if lands tails. We can set up our probability space $(\Omega, \mathbb{P})$ by $\Omega = \{H, T\}$ and $\mathbb{P}$ = uniform probability measure on Ω. The RV $X : \Omega \to \mathbb{R}$ for this game is $X(H) = 1$ and $X(T) = -1$. The PMF of X is given by $f_X(1) = \mathbb{P}(X = 1) = \mathbb{P}(\{H\}) = 1/2$ and likewise $f_X(-1) = 1/2$.

Exercise 1.3.2. Let $(\Omega, \mathbb{P})$ be a probability space and $X : \Omega \to \mathbb{R}$ be a RV. Let f_X be the PMF of X, that is, $f_X(x) = \mathbb{P}(X = x)$ for all x. Show that f_X adds up to 1, that is,

$$\sum_x f_X(x) = 1, \tag{19}$$

where the summation runs over all numerical values x that X can take.

There are two useful statistics of a RV to summarize its two most important properties: Its average and uncertainty. First, if one has to guess the value of a RV X, what would be the best choice? It is the *expectation* (or mean) of X, defined as below:

$$\mathbb{E}(X) = \sum_x x\,\mathbb{P}(X = x). \tag{20}$$

Example 1.3.3 (Doubling strategy)**.** Suppose we bet \$$x$ on a game where we have to predict whether a fair coin clip comes up heads. We win \$$x$ on heads and lose \$$x$ on tails. Suppose we are playing the 'doubling strategy'. Namely, we start betting \$1, and until the first time we win, we double our bet; upon the first win, we quit the game. Let X be the random variable giving the net gain of the overall game. How can we evaluate this strategy?

Let N be the random number of coin flips we have to encounter until we see the first head. For instance,

$$\mathbb{P}(N = 1) = \mathbb{P}(\{H\}) = 1/2, \tag{21}$$

$$\mathbb{P}(N = 2) = \mathbb{P}(\{(T, H)\}) = 1/2^2 \tag{22}$$

$$\mathbb{P}(N = 3) = \mathbb{P}(\{(T, T, H)\}) = 1/2^3. \tag{23}$$

In general, we have

$$\mathbb{P}(N = k) = 1/2^k. \tag{24}$$

Note that on the event that $N = k$ (i.e., we bet k times), the net gain $X|\{N = k\}$ is

$$X|\{N = k\} = (-1) + (-2) + (-2^2) + (-2^3) + \cdots + (-2^{k-1}) + 2^k \tag{25}$$

$$= -(2^k - 1) + 2^k = 1. \tag{26}$$

Since the last expression do not depend on k, we conclude that

$$\mathbb{P}(X=1) = \sum_{k=1}^{\infty} \mathbb{P}(X=1 \mid N=k)\mathbb{P}(N=k) \tag{27}$$

$$= \sum_{k=1}^{\infty} \mathbb{P}(N=k) = 1. \tag{28}$$

Hence our net gain is \$1 with probability 1. In particular, the expected net gain is also 1:

$$\mathbb{E}[X] = 1 \cdot \mathbb{P}(X=1) = 1. \tag{29}$$

What if we use tripling strategy? ▲

Exercise 1.3.4. For any RV X and real numbers $a, b \in \mathbb{R}$, show that

$$\mathbb{E}[aX+b] = a\mathbb{E}[X] + b. \tag{30}$$

Exercise 1.3.5 (Tail sum formula for expectation)**.** Let X be a RV taking values on positive integers.

(i) For any x, show that

$$\mathbb{P}(X \geq x) = \sum_{y=x}^{\infty} \mathbb{P}(X=y). \tag{31}$$

(ii) Use (i) and Fubini's theorem to show

$$\sum_{x=1}^{\infty} \mathbb{P}(X \geq x) = \sum_{y=1}^{\infty} \sum_{x=1}^{y} \mathbb{P}(X=y) \tag{32}$$

(iii) From (ii), conclude that

$$\mathbb{E}(X) = \sum_{x=1}^{\infty} \mathbb{P}(X \geq x). \tag{33}$$

On the other hand, say you play two different games where in the first game, you win or lose \$1 depending on a fair coin flip, and in the second game, you win or lose \$10. In both games, your expected winning is 0. But the two games are different in how much the outcome fluctuates around the mean. This notion if fluctuation is captured by the following quantity called *variance*:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}(X))^2]. \tag{34}$$

Namely, it is the expected squared difference between X and its expectation $\mathbb{E}(X)$.

The following exercise ties the expectation and the variance of a RV into a problem of finding a point estimator that minimizes the mean squared error.

Exercise 1.3.6 (Variance as minimum MSE)**.** Let X be a RV. Let $\hat{x} \in \mathbb{R}$ be a number, which we consider as a 'guess' (or 'estimator' in Statistics) of X. Let $\mathbb{E}[(X-\hat{x})^2]$ be the *mean squared error* (MSE) of this estimation.

(i) Show that

$$\mathbb{E}_Y[(X-\hat{x})^2] = \mathbb{E}_Y[X^2] - 2\hat{x}\mathbb{E}[X] + \hat{x}^2 \tag{35}$$

$$= (\hat{x} - \mathbb{E}[X])^2 + \mathbb{E}[X^2] - \mathbb{E}[X]^2 \tag{36}$$

$$= (\hat{x} - \mathbb{E}[X])^2 + \mathrm{Var}(X). \tag{37}$$

(ii) Conclude that the MSE is minimized when $\hat{x} = \mathbb{E}[X]$ and the global minimum is $\mathrm{Var}(X)$. In this sense, $\mathbb{E}[X]$ is the 'best guess' for X and $\mathrm{Var}(X)$ is the corresponding MSE.

Here are some of the simplest and yet most important discrete RVs.

Exercise 1.3.7 (Bernoulli RV)**.** A RV X is a *Bernoulli* variable with (success) probability $p \in [0,1]$ if it takes value 1 with probability p and 0 with probability $1-p$. In this case we write $X \sim \text{Bernoulli}(p)$. Show that $\mathbb{E}(X) = p$ and $\text{Var}(X) = p(1-p)$.

Exercise 1.3.8 (Indicator variables)**.** Let $(\Omega, \mathbb{P})$ be a probability space and let $E \subseteq \Omega$ be an event. The *indicator variable* of the event E, which is denoted by $\mathbf{1}_E$, is the RV such that $\mathbf{1}_E(\omega) = 1$ if $\omega \in E$ and $\mathbf{1}_E(\omega) = 0$ if $\omega \in E^c$. Show that $\mathbf{1}_E$ is a Bernoulli variable with success probability $p = \mathbb{P}(E)$.

Example 1.3.9 (Binomial RV)**.** Let $X_1, X_2, \cdots, X_n$ be independent and identically distributed Bernoulli p variables. Let $X = X_1 + \cdots + X_n$. One can think of flipping the same probability p coin n times. Then X is the total number of heads. Note that X has the following PMF

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^k \tag{38}$$

for k nonnegative integer, and $\mathbb{P}(X = k) = 0$ otherwise. We say X follows the Binomial distribution with parameters n and p, and write $X \sim \text{Binomial}(n, p)$.

We can compute the mean and variance of X using the above PMF directly, but it is much easier to break it up into Bernoulli variables and use linearity. Recall that $X_i \sim \text{Bernoulli}(p)$ and we have $\mathbb{E}(X_i) = p$ and $\text{Var}(X_i) = p(1-p)$ for each $1 \le i \le n$ (from Exercise 1.3.7). So by linearity of expectation (Exercise 1.3.22),

$$\mathbb{E}(X) = \mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n) = np. \tag{39}$$

On the other hand, since X_i's are independent, variance of X is the sum of variance of X_i's (Exercise 1.3.31) so

$$\text{Var}(X) = \text{Var}(X_1 + \cdots + X_n) = \text{Var}(X_1) + \cdots + \text{Var}(X_n) = np(1-p). \tag{40}$$

Example 1.3.10 (Geometric RV)**.** Suppose we flip a probability p coin until it lands heads. Let X be the total number of trials until the first time we see heads. Then in order for $X = k$, the first $k-1$ flips must land on tails and the kth flip should land on heads. Since the flips are independent with each other,

$$\mathbb{P}(X = k) = \mathbb{P}(\{T, T, \cdots, T, H\}) = (1-p)^{k-1} p. \tag{41}$$

This is valid for k positive integer, and $\mathbb{P}(X = k) = 0$ otherwise. Such a RV is called a *Geometric RV* with (success) parameter p, and we write $X \sim \text{Geom}(p)$.

The mean and variance of X can be easily computed using its moment generating function, which we will learn soon in this course. For their direct computation, note that

$$\begin{aligned}
\mathbb{E}(X) - (1-p)\mathbb{E}(X) &= (1-p)^0 p + 2(1-p)^1 p + 3(1-p)^2 p + 4(1-p)^3 p \cdots && (42)\\
&\quad - \left[(1-p)^1 p + 2(1-p)^2 p + 3(1-p)^3 p + \cdots\right] && (43)\\
&= (1-p)^0 p + (1-p)^1 p + (1-p)^2 p + (1-p)^3 p \cdots && (44)\\
&= \frac{p}{1-(1-p)} = 1, && (45)
\end{aligned}$$

where we recognized the series after the second equality as a geometric series. This gives

$$\mathbb{E}(X) = 1/p. \tag{46}$$

(In fact, one can apply Exercise 1.3.5 and quickly compute the expection of a Geometric RV.)

Exercise 1.3.11. Let $X \sim \text{Geom}(p)$. Use a similar computation as we had in Example 1.3.10 to show $\mathbb{E}(X^2) = (2-p)/p^2$. Using the fact that $\mathbb{E}(X) = 1/p$, conclude that $\text{Var}(X) = (1-p)/p^2$.

Example 1.3.12 (Poisson RV)**.** A RV X is a *Poisson RV* with rate $\lambda > 0$ if

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{47}$$

for all nonnegative integers $k \ge 0$. We write $X \sim \text{Poisson}(\lambda)$. A computation shows $\mathbb{E}(X) = \text{Var}(X) = \lambda$.

1.3.2. Continuous Random Variables. So far we have only considered discrete RVs, which takes either finitely many or countably many values. While there are many examples of discrete RVs, there are also many instances of RVs which various continuously (e.g., temperature, height, weight, price, etc.). To define a discrete RV, it was enough to specify its PMF. For a continuous RV, *probability distribution function* (PDF) plays an analogous role of PMF. We also need to replace summation $\sum$ with an integral $\int dx$.

Namely, X is a *continuous RV* if there is a function $f_X : \mathbb{R} \to [0,\infty)$ such that for any interval $[a,b]$, the probability that X takes a value from an interval $(a,b]$ is given by integrating f_X over the interval $(a,b]$:

$$\mathbb{P}(X \in (a,b]) = \int_a^b f_X(x)\,dx. \tag{48}$$

The *cumulative distribution function* (CDF) of a RV X (either discrete or continuous), denoted by F_X, is defined by

$$F_X(x) = \mathbb{P}(X \le x). \tag{49}$$

By definition of PDF, we get

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt. \tag{50}$$

Conversely, PDFs can be obtained by differentiating corresponding CDFs.

Exercise 1.3.13. Let X be a continuous RV with PDF f_X. Let a be a continuity point of f_X, that is, f_X is continuous at a. Show that $F_X(x)$ is differentiable at $x = a$ and

$$\frac{dF_X}{dx}\Big|_{x=a} = f_X(a). \tag{51}$$

The expectation of a continuous RV X with pdf f_X is defined by

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx, \tag{52}$$

and its variance $\mathrm{Var}(X)$ is defined by the same formula (34).

Exercise 1.3.14 (Tail sum formula for expectation)**.** Let X be a continuous RV with PDF f_X and suppose $f_X(x) = 0$ for all $x < 0$. Use Fubini's theorem to show that

$$\mathbb{E}(X) = \int_0^{\infty} \mathbb{P}(X \ge t)\,dt. \tag{53}$$

Example 1.3.15 (Higher moments)**.** Let X be a continuous RV with PDF f_X and suppose $f_X(x) = 0$ for all $x < 0$. We will show that for any real number $\alpha > 0$,

$$\mathbb{E}[X^\alpha] = \int_0^{\infty} x^\alpha f_X(x)\,dx. \tag{54}$$

First, Use Exercise 1.3.14 and to write

$$\mathbb{E}[X^\alpha] = \int_0^{\infty} \mathbb{P}(X^\alpha \ge x)\,dx \tag{55}$$

$$= \int_0^{\infty} \mathbb{P}(X \ge x^{1/\alpha})\,dx \tag{56}$$

$$= \int_0^{\infty} \int_{x^{1/\alpha}}^{\infty} f_X(t)\,dt\,dx. \tag{57}$$

We then use Fubini's theorem to change the order of integral. This gives

$$\mathbb{E}(X) = \int_0^{\infty} \int_{x^{1/\alpha}}^{\infty} f_X(t)\,dt\,dx = \int_0^{\infty} \int_0^{t^\alpha} f_X(t)\,dx\,dt = \int_0^{\infty} t^\alpha f_X(t)\,dt, \tag{58}$$

as desired. ▲

Exercise 1.3.16. Let X be a continuous RV with PDF f_X and suppose $f_X(x) = 0$ for all $x < 0$. Let $g : \mathbb{R} \to \mathbb{R}$ be a strictly increasing function. Use Fubini's theorem and tail sum formula for expectation to show

$$\mathbb{E}[g(X)] = \int_0^\infty g(x) f_X(x)\, dx. \tag{59}$$

Next, we introduce three important continuous RVs.

Example 1.3.17 (Uniform RV)**.** X is a *uniform RV* on the interval $[a, b]$ (denoted by $X \sim \text{Uniform}([a, b])$) if it has PDF

$$f_X(x) = \frac{1}{b-a} \mathbf{1}(a \le x \le b). \tag{60}$$

An easy computation gives its CDF:

$$\mathbb{P}(X \le x) = \begin{cases} 0 & x < a \\ (x-a)/(b-a) & a \le x \le b \\ 1 & x > b. \end{cases} \tag{61}$$

▲

Exercise 1.3.18. Let $X \sim \text{Uniform}([a, b])$. Show that

$$\mathbb{E}(X) = \frac{a+b}{2}, \qquad \text{Var}(E) = \frac{(b-a)^2}{12}. \tag{62}$$

Example 1.3.19 (Exponential RV)**.** X is an *exponential* RV with rate λ (denoted by $X \sim \text{Exp}(\lambda)$) if it has PDF

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}(x \ge 0). \tag{63}$$

Integrating the PDF gives its CDF

$$\mathbb{P}(X \le x) = (1 - e^{-\lambda x}) \mathbf{1}(x \ge 0). \tag{64}$$

Using Exercise 1.3.14, we can compute

$$\mathbb{E}(X) = \int_0^\infty e^{-\lambda t}\, dt = \left[-\frac{e^{-\lambda t}}{\lambda} \right]_0^\infty = 1/\lambda. \tag{65}$$

▲

Exercise 1.3.20. Let $X \sim \text{Exp}(\lambda)$. Show that $\mathbb{E}(X) = 1/\lambda$ directly using definition (52). Also show that $\text{Var}(X) = 1/\lambda^2$.

Example 1.3.21 (Normal RV)**.** X is a *normal* RV with mean μ and variance σ^2 (denoted by $X \sim N(\mu, \sigma^2)$) if it has PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \tag{66}$$

If $\mu = 0$ and $\sigma^2 = 1$, then X is called a standard normal RV. Note that if $X \sim N(\mu, \sigma^2)$, then $Y := X - \mu$ has PDF

$$f_Y(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}. \tag{67}$$

Since this is an even function, it follows that $\mathbb{E}(Y) = 0$. Hence $\mathbb{E}(X) = \mu$. Using the 'Gaussian integral', one can also show that $\text{Var}(X) = \sigma^2$. ▲

1.3.3. Expectation and variance of sums of RVs. In this subsection, we learn how we can compute expectation and variance for sums of RVs. For expectation, we will see that we can swap the order to summation and expectation. This is called the *linearity of expectation*, whose importance cannot be overemphasized.

Exercise 1.3.22 (Linearity of expectation)**.** In this exercise, we will show that the expectation of sum of RVs is the sum of expectation of individual RVs.

(i) Let X and Y be RVs. Show that

$$\sum_y \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x). \tag{68}$$

(ii) Verify the following steps:

$$\begin{aligned}
\mathbb{E}(X+Y) &= \sum_z z\,\mathbb{P}(X+Y=z) && (69)\\
&= \sum_z \sum_{\substack{x,y\\ x+y=z}} (x+y)\mathbb{P}(X=x,\,Y=y) && (70)\\
&= \sum_{x,y} (x+y)\,\mathbb{P}(X=x,\,Y=y) && (71)\\
&= \sum_{x,y} x\,\mathbb{P}(X=x,\,Y=y) + \sum_{x,y} y\,\mathbb{P}(X=x,\,Y=y) && (72)\\
&= \sum_x x\left(\sum_y \mathbb{P}(X=x, y=y)\right) + \sum_y y\left(\sum_x \mathbb{P}(X=x,\,Y=y)\right) && (73)\\
&= \sum_x x\mathbb{P}(X=x) + \sum_y y\mathbb{P}(Y=y) && (74)\\
&= \mathbb{E}(X) + \mathbb{E}(Y). && (75)
\end{aligned}$$

(iii) Use induction to show that for any RVs $X_1, X_2, \cdots, X_n$, we have

$$\mathbb{E}(X_1 + X_2 + \cdots + X_n) = \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_n). \tag{76}$$

Our first application of linearity of expectation is the following useful formula for variance.

Exercise 1.3.23. For any RV X, show that

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2. \tag{77}$$

Next, we will see a nice application of linearity of expectation.

Exercise 1.3.24 (Inclusion-exclusion)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. Let $A_1, A_2, \cdots, A_k \subseteq \Omega$. We will show the following inclusion-exclusion principle:

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{i=1}^k A_i\right) &= \sum_{i_1=1}^k \mathbb{P}(A_{i_1}) - \sum_{1\le i_1<i_2\le k} \mathbb{P}(A_{i_1}\cap A_{i_2}) && (78)\\
&\quad + \sum_{1\le i_1<i_2<i_3\le k} \mathbb{P}(A_{i_1}\cap A_{i_2}\cap A_{i_3}) - \cdots + (-1)^k \mathbb{P}\left(\bigcap_{i=1}^k A_i\right). && (79)
\end{aligned}$$

The method is so-called the 'indicator trick'.

For each $1 \le i \le k$, let $X_i = \mathbf{1}(A_i)$ be the indicator variable for the event A_i. Consider the following RV

$$Y = (1 - X_1)(1 - X_2)\cdots(1 - X_k). \tag{80}$$

(i) By expanding the product and using linearity of expectation, show that

$$\mathbb{E}[Y] = 1 - \sum_{i_1=1}^{k} \mathbb{E}[X_{i_1}] + \sum_{1\le i_1<i_2\le k} \mathbb{E}[X_{i_1}X_{i_2}] \tag{81}$$

$$- \sum_{1\le i_1<i_2<i_3\le k} \mathbb{E}[X_{i_1}X_{i_2}X_{i_3}] + \cdots - (-1)^k \mathbb{E}[X_1X_2\cdots X_k]. \tag{82}$$

(ii) Show that Y is the indicator variable of the event $\bigcap_{i=1}^k A_i^c$. Conclude that

$$\mathbb{E}[Y] = \mathbb{P}\left(\bigcap_{i=1}^k A_i^c\right) = 1 - \mathbb{P}\left(\bigcup_{i=1}^k A_i\right). \tag{83}$$

(iii) From (i) and (ii), deduce the inclusion-exclusion principle.

Exercise 1.3.25 (Finite second moment implies finite first moment)**.** Let X be a RV with $\mathbb{E}[X^2] < \infty$.

(i) Show that

$$\mathbb{E}[|X|] = \mathbb{E}[|X|\mathbf{1}(|X|\le 1)] + \mathbb{E}[|X|\mathbf{1}(|X|\le 1)] \le 1 + \mathbb{E}[X^2\mathbf{1}(|X|\le 1)] \le 1 + \mathbb{E}[X^2] < \infty. \tag{84}$$

Deduce that $\mathbb{E}[X] \in (-\infty, \infty)$.

(ii) Deduce that $\mathrm{Var}(X) < \infty$.

Next, we study how we can simplify the variance of sums of RVs. Unlike expectation, we cannot exchange the order of summation and variance in general.

Example 1.3.26. Flip two coins at the same time. The sample space for this experiment is $\Omega = \{H, T\}^2$ and the probability measure $\mathbb{P}$ on Ω is given by

$$\mathbb{P}(\{(H,H)\}) = \mathbb{P}(\{(T,T)\}) = 0 \tag{85}$$

$$\mathbb{P}(\{(H,T)\}) = \mathbb{P}(\{(T,H)\}) = 1/2. \tag{86}$$

Namely somehow we know for sure that the two coins will come up with the opposite sides with equal probability.

Define a RV $X : \Omega \to \{-1, 1\}$ by

$$X = \begin{cases} 1 & \text{if the first coin comes up a head} \\ -1 & \text{if the first coin comes up a tail} \end{cases} \tag{87}$$

Define another RV $Y : \Omega \to \{-1, 1\}$ similarly for the second coin. Then

$$\mathbb{E}[X] = 1 \cdot \mathbb{P}(X = 1) + (-1)\mathbb{P}(X = -1) \tag{88}$$

$$= \mathbb{P}(\{(H,H),(H,T)\}) - \mathbb{P}(\{(T,H),(T,T)\}) = 0, \tag{89}$$

$$\mathbb{E}[X^2] = 1^2 \cdot \mathbb{P}(X = 1) + (-1)^2\mathbb{P}(X = -1) = 1, \tag{90}$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 1. \tag{91}$$

A similar computation shows $\mathrm{Var}(Y) = 1$. However, note that $X + Y$ takes value 0 with probability 1. Hence $\mathrm{Var}(X + Y) = 0$. Thus $0 = \mathrm{Var}(X + Y) \neq \mathrm{Var}(X) + \mathrm{Var}(Y) = 2$. ▲

In general, when we try to write down the variance of a sum of RVs, in addition to the variance of each RV, we have extra contribution from each pairs of RVs called the covariance.

Exercise 1.3.27. In this exercise, we will see how we can express the variance of sums of RVs. For two RVs X and Y, define their *covariance* $\mathrm{Cov}(X, Y)$ by

$$\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y). \tag{92}$$

(i) Use Exercises 1.3.23 and 1.3.22 to show that

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y). \tag{93}$$

(ii) Use induction to show that for RVs $X_1, X_2 \cdots, X_n$

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2 \sum_{1 \le i,j \le n} \mathrm{Cov}(X_i, X_j) \tag{94}$$

When knowing something about one RV does not yield any information of the other, we say the two RVs are independent. Formally, we say two RVs X and Y are *independent* if for any two subsets $A_1, A_2 \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A_1 \text{ and } Y \in A_2) = \mathbb{P}(X \in A_1)\mathbb{P}(Y \in A_2). \tag{95}$$

We say two events or RVs are *dependent* if they are not independent.

Exercise 1.3.28. Suppose two RVs X and Y are independent. Then for any subsets $A_1, A_2 \subseteq \mathbb{R}$ such that $\mathbb{P}(Y \in A_2) > 0$, show that

$$\mathbb{P}(X \in A_1 \mid Y \in A_2) = \mathbb{P}(X \in A_1). \tag{96}$$

Example 1.3.29. Flip two fair coins at the same time, and let $X = 1$ if the first coin lands heads and $X = -1$ if it lands tails. Let Y be a similar RV for the second coin. Suppose each of the four outcomes in $\Omega = \{H, T\}^2$ are equality likely. Clearly knowing about one coin does not give any information of the other. For instance, the first coin lands on heads with probability $1/2$. Whether the first coin lands on heads or not, the second coin will land on heads with probability $1/2$. So

$$\mathbb{P}(X = 1 \text{ and } Y = 1) = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(X = 1)\mathbb{P}(Y = 1). \tag{97}$$

One can verify the above equation for possible outcomes. Hence X and Y are independent. ▲

Exercise 1.3.30. Let X, Y be two indepenent RVs. Show that

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]. \tag{98}$$

In the following exercise, we will see that we can swap summation and variance as long as the RVs we are adding up are independent.

Exercise 1.3.31. Recall the definition of covariance given in Exercise 1.3.27.

(i) Show that if two RVs X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$

(ii) Use Exercise 1.3.27 to conclude that if $X_1, \cdots, X_n$ are independent RVs, then

$$\mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n). \tag{99}$$

1.4. Conditional expectation

Let X, Y be discrete RVs. Recall that the expectation $\mathbb{E}(X)$ is the 'best guess' on the value of X when we do not have any prior knowledge on X. But suppose we have observed that some possibly related RV Y takes value y. What should be our best guess on X, leveraging this added information? This is called the *conditional expectation of X given $Y = y$*, which is defined by

$$\mathbb{E}[X|Y = y] = \sum_x x\,\mathbb{P}(X = x|Y = y). \tag{100}$$

This best guess on X given $Y = y$, of course, depends on y. So it is a function in y. Now if we do not know what value Y might take, then we omit y and $\mathbb{E}[X|Y]$ becomes a RV, which is called the *conditional expectation of X given Y*.

Exercise 1.4.1. Let X, Y be discrete RVs. Show that for any function $g : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}_X[Xg(Y) \mid Y] = g(Y)E_X[X \mid Y]. \tag{101}$$

Exercise 1.4.2 (Iterated expectation)**.** Let X, Y be discrete RVs. Use Fubini's theorem to show that

$$\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}_X[X|Y]]. \tag{102}$$

In case when X or Y are continuous RVs, we simply replace the sum by integral and PMF by PDF. For instance, if both X and Y are continuous with PDFs f_X, f_Y and joint PDF $f_{X,Y}$, then

$$\mathbb{E}[X|Y=y] = \int_{-\infty}^{\infty} x f_{X|Y=y}(x)\,dx, \tag{103}$$

where $f_{X|Y=y}$ is the conditional PDF of X given $Y=y$ defined by

$$f_{X|Y=y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}. \tag{104}$$

To summarize how we compute the iterated expectations when we condition on discrete and continuous RV:

$$\mathbb{E}[\mathbb{E}[X\,|\,Y]] = \begin{cases} \sum_y \mathbb{E}[X\,|\,Y=y]\mathbb{P}(Y=y) & \text{if } Y \text{ is discrete} \\ \int_{-\infty}^{\infty} \mathbb{E}[X\,|\,Y=y] f_Y(y)\,dy & \text{if } Y \text{ is continuous.} \end{cases} \tag{105}$$

An important use of iterated expectation is that we can compute probabilities using conditioning, since probability of an event is simply the expectation of the corresponding indicator variable.

Exercise 1.4.3 (Iterated expectation for probability)**.** Let X, Y be RVs.

(i) For any subset $A \subseteq \mathbb{R}$, show that $\mathbb{P}(X \in A) = \mathbb{E}[\mathbf{1}(X \in A)]$.
(ii) By using iterated expectation, show that

$$\mathbb{P}(X \le x) = \mathbb{E}_Y[\mathbb{P}(X \in A\,|\,Y)]. \tag{106}$$

Proposition 1.4.4. *Let X be a discrete random variable and $\mathbf{Y} = (Y_1, \dots, Y_k)$ a discrete random vector. Let $g : \mathbb{R}^k \to \mathbb{R}$ be a function.*

(i) *If X and $\mathbf{Y}$ are independent, then $\mathbb{E}[X\,|\,\mathbf{Y}] = \mathbb{E}[X]$.*
(ii) *Assume that $\mathbb{E}[|g(\mathbf{Y})|] < \infty$. Then $\mathbb{E}[g(\mathbf{Y})\,|\,\mathbf{Y}] = g(\mathbf{Y})$.*
(iii) *Assume that $\mathbb{E}[|Xg(\mathbf{Y})|] < \infty$. Then $\mathbb{E}[Xg(\mathbf{Y})\,|\,\mathbf{Y}] = \mathbb{E}[X\,|\,\mathbf{Y}]\,g(\mathbf{Y})$.*

PROOF. To show **(i)**, note that

$$\mathbb{E}[X\,|\,\mathbf{Y}=\mathbf{y}] = \sum_x x\,\mathbb{P}(X=x\,|\,\mathbf{Y}=\mathbf{y}) = \sum_x x\,\mathbb{P}(X=x) = \mathbb{E}[X], \tag{107}$$

where the second equality follows from the independence of X and $\mathbf{Y}$.

To show **(ii)**, note that

$$\mathbb{E}[g(\mathbf{Y})\,|\,\mathbf{Y}=\mathbf{y}] = \sum_{\mathbf{y}'} g(\mathbf{y}')\,\mathbb{P}\left(\mathbf{Y}=\mathbf{y}'\,|\,\mathbf{Y}=g(\mathbf{y})\right) = g(\mathbf{y})\,\mathbb{P}\left(\mathbf{Y}=\mathbf{y}\,|\,\mathbf{Y}=\mathbf{y}\right) = g(\mathbf{y}). \tag{108}$$

Since this holds for all $\mathbf{y} \in \mathbb{R}^k$, the assertion holds.

Lastly, to show **(iii)**, note that

$$\begin{aligned} \mathbb{E}[Xg(\mathbf{Y})\,|\,\mathbf{Y}=\mathbf{y}] &= \sum_{x,\mathbf{y}'} x g(\mathbf{y}')\,\mathbb{P}(X=x, \mathbf{Y}=\mathbf{y}'\,|\,\mathbf{Y}=\mathbf{y}) && (109)\\ &= \sum_x x g(\mathbf{y})\,\mathbb{P}(X=x, \mathbf{Y}=\mathbf{y}\,|\,\mathbf{Y}=\mathbf{y}) && (110)\\ &= \sum_x x g(\mathbf{y})\,\mathbb{P}(X=x, |\,\mathbf{Y}=\mathbf{y}) && (111)\\ &= g(\mathbf{y}) \sum_x x\,\mathbb{P}(X=x, |\,\mathbf{Y}=\mathbf{y}) && (112)\\ &= g(\mathbf{y})\mathbb{E}\left[X\,|\,\mathbf{Y}=\mathbf{y}\right]. && (113) \end{aligned}$$

Since this holds for all $\mathbf{y} \in \mathbb{R}^k$, the assertion holds. □

1.5. Conditional variance

As we have defined conditional expectation, we could define the variance of a RV X given that anther RV Y takes a particular value. Recall that the (unconditioned) variance of X is defined by

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}(X))^2]. \tag{114}$$

Note that there are two places where we take expectation. Given Y, we should improve both expectations so the *conditional variance of X given Y is defined by*

$$\mathrm{Var}(X \mid Y) = \mathbb{E}[(X - \mathbb{E}[X \mid Y])^2 \mid Y]. \tag{115}$$

Proposition 1.5.1. *Let X and Y be RVs. Then*

$$\mathrm{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - \mathbb{E}[X \mid Y]^2. \tag{116}$$

PROOF. Using linearity of conditional expectation and the fact that $\mathbb{E}[X \mid Y]$ is not random given Y,

$$\begin{aligned}
\mathrm{Var}(X \mid Y) &= \mathbb{E}[X^2 - 2X\mathbb{E}[X \mid Y] + \mathbb{E}[X \mid Y]^2 \mid Y] && (117)\\
&= \mathbb{E}[X^2 \mid Y] - \mathbb{E}[2X\mathbb{E}[X \mid Y] \mid Y] + \mathbb{E}[\mathbb{E}[X \mid Y]^2 \mid Y] && (118)\\
&= \mathbb{E}[X^2 \mid Y] - \mathbb{E}[X \mid Y]\mathbb{E}[2X \mid Y] + \mathbb{E}[X \mid Y]^2\mathbb{E}[1 \mid Y] && (119)\\
&= \mathbb{E}[X^2 \mid Y] - 2\mathbb{E}[X \mid Y]^2 + \mathbb{E}[X \mid Y]^2 && (120)\\
&= \mathbb{E}[X^2 \mid Y] - \mathbb{E}[X \mid Y]^2. && (121)
\end{aligned}$$

□

The following exercise explains in what sense the conditional expectation $\mathbb{E}[X \mid Y]$ is the best guess on X given Y, and that the minimum possible mean squared error is exactly the conditional variance $\mathrm{Var}(X \mid Y)$.

Exercise 1.5.2. Let X, Y be RVs. For any function $g : \mathbb{R} \to \mathbb{R}$, consider $g(Y)$ as an estimator of X. Let $\mathbb{E}_Y[(X - g(Y))^2 \mid Y]$ be the *mean squared error*. Recall the conditional variance $V(X|Y)$ is defined by

$$\mathrm{Var}(X \mid Y) = \mathbb{E}[(X - \mathbb{E}[X \mid Y])^2 \mid Y]. \tag{122}$$

(i) Show that

$$\begin{aligned}
\mathbb{E}_Y[(X - g(Y))^2 \mid Y] &= \mathbb{E}_Y[X^2 \mid Y] - 2g(Y)\mathbb{E}_Y[X \mid Y] + g(Y)^2 && (123)\\
&= (g(Y) - \mathbb{E}_Y(X \mid Y))^2 + \mathbb{E}_Y[X^2 \mid Y] - \mathbb{E}_Y[X \mid Y]^2 && (124)\\
&= (g(Y) - \mathbb{E}_Y(X \mid Y))^2 + \mathrm{Var}(X \mid Y). && (125)
\end{aligned}$$

(ii) Conclude that the mean squared error is minimized when $g(Y) = \mathbb{E}_Y[X \mid Y]$ and the global minimum is $\mathrm{Var}(X \mid Y)$.

(iii) Show that the law of total variance:

$$\mathrm{Var}(X) = \mathbb{E}(\mathrm{Var}(X \mid Y)) + \mathrm{Var}(\mathbb{E}[X \mid Y]). \tag{126}$$

Next, we study how we can decompose the variance of X by conditioning on Y.

Proposition 1.5.3 (Law of total variance)**.** *Let X and Y be RVs. Then*

$$\mathrm{Var}(X) = \mathbb{E}(\mathrm{Var}(X \mid Y)) + \mathrm{Var}(\mathbb{E}[X \mid Y]). \tag{127}$$

PROOF. Using previous result, iterated expectation, and linearity of expectation, we have

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 && (128)\\
&= \mathbb{E}_Y(\mathbb{E}(X^2|Y)) - (\mathbb{E}_Y(\mathbb{E}(X|Y)))^2 && (129)\\
&= \mathbb{E}_Y(\mathrm{Var}(X|Y) + (\mathbb{E}(X|Y))^2) - (\mathbb{E}_Y(\mathbb{E}(X|Y)))^2 && (130)\\
&= \mathbb{E}_Y(\mathrm{Var}(X|Y)) + \left[\mathbb{E}_Y(\mathbb{E}(X|Y))^2) - (\mathbb{E}_Y(\mathbb{E}(X|Y)))^2\right] && (131)\\
&= \mathbb{E}_Y(\mathrm{Var}(X|Y)) + \mathrm{Var}_Y(\mathbb{E}(X|Y)). && (132)
\end{aligned}$$

□

Here is a handwavy explanation on why the above is true. Given Y, we should measure the fluctuation of $X \mid Y$ from the conditional expectation $\mathbb{E}[X \mid Y]$, and this is measured as $\mathrm{Var}(X \mid Y)$. Since we don't know Y, we average over all Y, giving $\mathbb{E}(\mathrm{Var}(X \mid Y))$. But the reference point $\mathbb{E}[X \mid Y]$ itself varies with Y, so we should also measure its own fluctuation by $\mathrm{Var}(\mathbb{E}[X \mid Y])$. These fluctuations add up nicely like Pythagorian theorem because $\mathbb{E}[X \mid Y]$ is an optimal estimator so that these two fluctuations are 'orthogonal'.

Exercise 1.5.4. Let X, Y be RVs and denote $\hat{X} = \mathbb{E}[X|Y]$, meaning that $\hat{X}$ is an estimator of X given Y. Let $\tilde{X} = \hat{X} - X$ be the *estimation error*.

(i) Show that $\hat{X}$ is an *unbiased* estimator of X, that is, $\mathbb{E}(\hat{X}) = \mathbb{E}(X)$.
(ii) Show that $\mathbb{E}[\hat{X}|Y] = \hat{X}$. Hence knowing Y does not improve our current best guess $\hat{X}$.
(iii) Show that $\mathbb{E}[\tilde{X}] = 0$.
(iv) Show that $\mathrm{Cov}(\hat{X}, \tilde{X}) = 0$. Conclude that

$$\mathrm{Var}(X) = \mathrm{Var}(\hat{X}) + \mathrm{Var}(\tilde{X}). \tag{133}$$

Exercise 1.5.5. Let X, Y be RVs. Write $\bar{X} = \mathbb{E}[X \mid Y]$ and $\tilde{X} = X - \mathbb{E}[X \mid Y]$ so that $X = \bar{X} + \tilde{X}$. Here $\bar{X}$ is the estimate of X given Y and $\tilde{X}$ is the estimation error.

(i) Using Exercise 1.5.4 (iii) and iterated expectation, show that

$$\mathbb{E}[\tilde{X}^2] = \mathrm{Var}(\mathbb{E}[X \mid Y]). \tag{134}$$

(ii) Using Exercise 1.5.4 (iv), conclude that

$$\mathrm{Var}(X) = \mathbb{E}(\mathrm{Var}(X \mid Y)) + \mathrm{Var}(\mathbb{E}[X \mid Y]). \tag{135}$$

Example 1.5.6. Let $Y \sim \mathrm{Uniform}([0,1])$ and $X \sim \mathrm{Binomial}(n, Y)$. Since $X \mid Y = y \sim \mathrm{Binomial}(n, y)$, we have $\mathbb{E}[X \mid Y = y] = ny$ and $\mathrm{Var}(X \mid Y = y) = ny(1-y)$. Also, since $Y \sim \mathrm{Uniform}([0,1])$, we have

$$\mathrm{Var}(\mathbb{E}[X \mid Y]) = \mathrm{Var}(nY) = \frac{n^2}{12}. \tag{136}$$

So by iterated expectation, we get

$$\mathbb{E}(X) = \mathbb{E}_Y(\mathbb{E}[X \mid Y]) = \int_0^1 ny\, dy = \frac{n}{2}. \tag{137}$$

On the other hand, by law of total variance,

$$\begin{aligned} \mathrm{Var}(X) &= \mathbb{E}(\mathrm{Var}(X \mid Y)) + \mathrm{Var}(\mathbb{E}(X \mid Y)) && (138)\\ &= \int_0^1 ny(1-y)\, dy + \mathrm{Var}(nY) && (139)\\ &= n\left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^1 + \frac{n^2}{12} && (140)\\ &= \frac{n^2}{12} + \frac{n}{6}. && (141) \end{aligned}$$

▲

Exercise 1.5.7. Suppose we have a stick of length L. Break it into two pieces at a uniformly chosen point and let X_1 be the length of the longer piece. Break this longer piece into two pieces at a uniformly chosen point and let X_2 be the length of the longer one. Define $X_3, X_4, \cdots$ in a similar way.

(i) Let $U \sim \mathrm{Uniform}([0, L])$. Show that X_1 takes values from $[L/2, L]$, and that $X_1 = \max(U, L-U)$.

(ii) From (i), deduce that for any $L/2 \le x \le L$, we have

$$\mathbb{P}(X_1 \ge x) = \mathbb{P}(U \ge x \text{ or } L-U \ge x) = \mathbb{P}(U \ge x) + \mathbb{P}(U \le L-x) = \frac{2(L-x)}{L}. \tag{142}$$

Conclude that $X_1 \sim \text{Uniform}([L/2, L])$. What is $\mathbb{E}[X_1]$?

(iii) Show that $X_2 \sim \text{Uniform}([x_1/2, x_1])$ conditional on $X_1 = x_1$. Using iterated expectation, show that $\mathbb{E}[X_2] = (3/4)^2 L$.

(iv) In general, show that $X_{n+1} \mid X_n \sim \text{Uniform}([X_n/2, X_n])$. Conclude that $\mathbb{E}[X_n] = (3/4)^n L$.

Exercise 1.5.8 (Exercise 1.5.7 continued)**.** Let $X_1, X_2, \cdots, X_n$ be as in Exercise 1.5.7.

(i) Show that $\text{Var}(X_1) = L^2/48$.

(ii) Show that $\text{Var}(X_2) = (7/12)\,\text{Var}(X_1) + (1/48)\mathbb{E}(X_1)^2$.

(iii) Show that $\text{Var}(X_{n+1}) = (7/12)\,\text{Var}(X_n) + (1/48)\mathbb{E}(X_n)^2$ for any $n \ge 1$.

(iv) Using Exercise 1.5.7, show the following recursion on variance holds:

$$\text{Var}(X_{n+1}) = \frac{7}{12}\,\text{Var}(X_n) + \frac{1}{48}\left(\frac{9}{16}\right)^n L^2. \tag{143}$$

Furthermore, compute $\text{Var}(X_2)$ and $\text{Var}(X_3)$.

(v)* Let $A_n = \left(\frac{16}{9}\right)^n \text{Var}(X_n)$. Show that A_n's satisfy

$$A_{n+1} + L^2 = \left(\frac{28}{27}\right)(A_n + L^2). \tag{144}$$

(vi)* Show that $A_n = \left(\frac{28}{27}\right)^{n-1}(A_1 + L^2) - L^2$ for all $n \ge 1$.

(vii)* Conclude that

$$\text{Var}(X_n) = \left[\left(\frac{7}{12}\right)^n - \left(\frac{9}{16}\right)^n\right] L^2. \tag{145}$$

1.6. Elementary limit theorems

The primary subject in this section is the sequence of *independent and identically distributed* (i.i.d.) RVs and their partial sums. Namely, let $X_1, X_2, \cdots$ be an (infinite) sequence of i.i.d. RVs, and define their nth partial sum $S_n = X_1 + X_2 + \cdots + X_n$ for all $n \ge 1$. If we call X_i the ith step size or *increment*, then the sequence of RVs $(S_n)_{n\ge1}$ is called a *random walk*, where we usually set $S_0 = 0$. Think of X_i as the gain or loss after betting once in a casino. Then S_n is the net gain of fortune after betting n times. Of course there are ups and downs in the short term, but what we want to analyze using probability theory is the long-term behavior of the random walk $(S_n)_{n\ge1}$. Results of this type is called limit theorems.

Suppose each increment X_k has a finite mean μ. Then by linearity of expectation and independence of the increments, we have

$$\mathbb{E}\left(\frac{S_n}{n}\right) = \frac{\mathbb{E}[S_n]}{n} = \mu, \tag{146}$$

$$\text{Var}\left(\frac{S_n}{n}\right) = \frac{\text{Var}(S_n)}{n^2} = \frac{n\,\text{Var}(X_1)}{n^2} = \frac{\text{Var}(X_1)}{n}. \tag{147}$$

So the sample mean S_n/n has constant expectation and shrinking variance. Hence it makes sense to guess that it should behave as the constant μ, without taking the expectation. That is,

$$\lim_{n\to\infty} \frac{S_n}{n} = \mu. \tag{148}$$

But this expression is shaky, since the left hand side is a limit of RVs while the right hand side is a constant. In what sense the random sample means converge to μ? This is the content of the *laws of large numbers*, for which we will prove a weak and a strong versions.

The first limit theorem we will encounter is called the Weak Law of Large Numbers (WLLN), which is stated below:

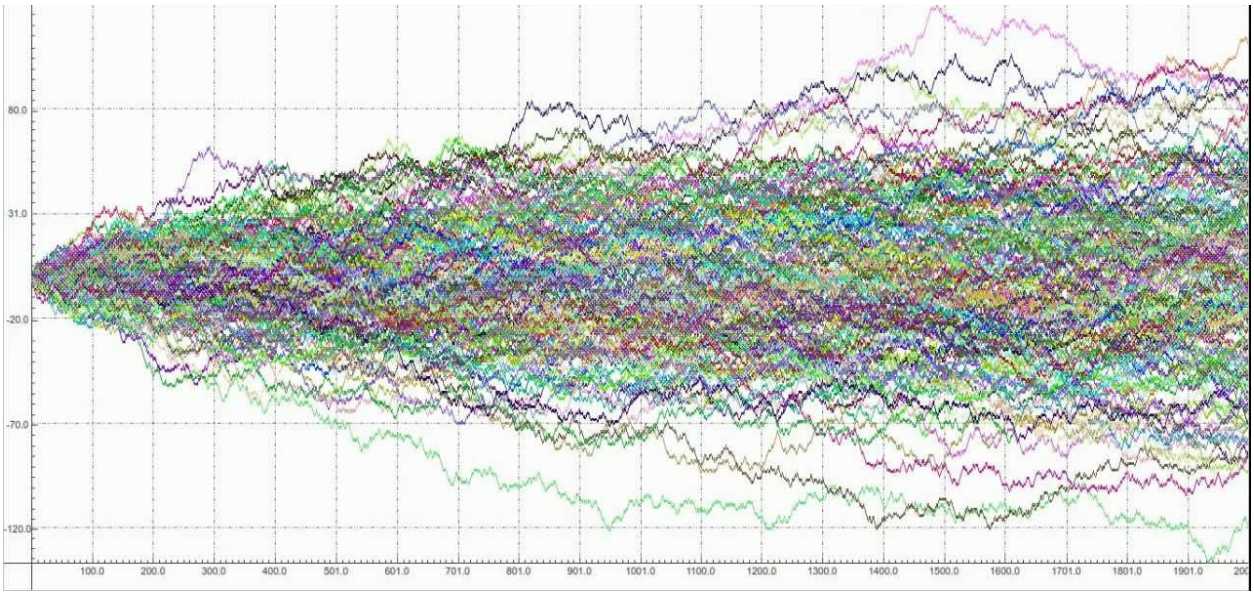


FIGURE 2. Simulation of simple random walks

Theorem 1.6.1 (WLLN). *Let $(X_k)_{k\geq 1}$ be i.i.d. RVs with mean $\mu < \infty$ and let $S_n = \sum_{k=1}^n X_i$, $n \geq 1$ be a random walk. Then for any positive constant $\varepsilon > 0$,*

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) = 0. \tag{149}$$

In words, the probability that the sample mean S_n/n is *not* within ε distance from its expectation μ decays to zero as n tends to infinity. In this case, we say the sequence of RVs $(S_n/n)_{n\geq 1}$ converges to μ *in probability*.

The second version of law of large numbers is call the *strong law of large numbers* (SLLN), which is available if the increments have finite variance.

Theorem 1.6.2 (SLLN). *Let $(X_k)_{k\geq 1}$ be i.i.d. RVs and let $S_n = \sum_{k=1}^n X_i$, $n \geq 1$ be a random walk. Suppose $\mathbb{E}[X_1] = \mu < \infty$ and $\mathbb{E}[X_1^2] < \infty$. Then*

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{S_n}{n} = \mu\right) = 1. \tag{150}$$

To make sense out of this, notice that the limit of sample mean $\lim_{n\to} S_n/n$ is itself a RV. Then SLLN says that this RV is well defined and its value is μ with probability 1. In this case, we say the sequence of RVs $(S_n/n)_{n\geq 1}$ converges to μ *with probability 1* or *almost surely*.

Perhaps one of the most celebrated theorems in probability theory is the *central limit theorem* (CLT), which tells about how the sample mean S_n/n "fluctuates" around its mean μ. From 147, if we denote $\sigma^2 = \mathrm{Var}(X_1) < \infty$, we know that $\mathrm{Var}(S_n/n) = \sigma^2/n \to 0$ as $n \to \infty$. So the fluctuation decays as we add up more increments. To see the effect of fluctuation, we first center the sample mean by subtracting its expectation and "zoom in" by dividing by the standard deviation $\sigma/\sqrt{n}$. This is where the name 'central limit' comes from: it describes the limit of centered random walks.

Theorem 1.6.3 (CLT). *Let $(X_k)_{k\geq 1}$ be i.i.d. RVs and let $S_n = \sum_{k=1}^n X_i$, $n \geq 1$ be a random walk. Suppose $\mathbb{E}[X_1] = \mu < \infty$ and $\mathbb{E}[X_1^2] = \sigma^2 < \infty$. Let $Z \sim N(0,1)$ be a standard normal RV and define*

$$Z_n = \frac{S_n - \mu n}{\sigma\sqrt{n}} = \frac{S_n/n - \mu}{\sigma/\sqrt{n}}. \tag{151}$$

Then for all $z \in \mathbb{R}$,

$$\lim_{n\to\infty} \mathbb{P}(Z_n \le z) = \mathbb{P}(Z \le z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{x^2}{2}}\, dx. \tag{152}$$

In words, the centered and rescaled RV Z_n is asymptotically distributed as a standard normal RV $Z \sim N(0,1)$. In this case, we say Z_n converges to Z as $n \to \infty$ *in distribution*. This is a remarkable result since as long as the increments X_k have finite mean and variance, it does not matter which distribution that they follow: the 'central limit' always looks like a standard normal distribution. We will not delve into proof details in this course, but we do mention that in a standard proof of CLT, one first computes the moment generating function (MGF) of S_n and Taylor-expands it up to the second order term, and use the uniqueness of MGF.

CHAPTER 2

Renewal Processes

2.1. Definition of renewal processes and Renewal SLLN

An *arrival* process is a sequence of strictly increasing RVs $0 < T_1 < T_2 < \cdots$ with the convention of setting $T_0 = 0$. For each integer $k \geq 1$, its kth *inter-arrival time* is defined by $\tau_k = T_k - T_{k-1}$. For a given arrival process $(T_k)_{k\geq 1}$, the associated *counting process* $(N(t))_{t\geq 0}$ is defined by

$$N(t) = \sum_{k=1}^{\infty} \mathbf{1}(T_k \leq t) = \#(\text{arrivals up to time } t). \tag{153}$$

Note that these three processes (arrival times, inter-arrival times, and counting) determine each other:

$$(T_k)_{k\geq 1} \Longleftrightarrow (\tau_k)_{k\geq 1} \Longleftrightarrow (N(t))_{t\geq 0}. \tag{154}$$

Exercise 2.1.1. 6.12 Let $(T_k)_{k\geq 1}$ be any arrival process and let $(N(t))_{t\geq 0}$ be its associated counting process. Show that these two processes determine each other by the following relation

$$\{T_n \leq t\} = \{N(t) \geq n\}. \tag{155}$$

In words, nth customer arrives by time t if and only if at least n customers arrive up to time t.

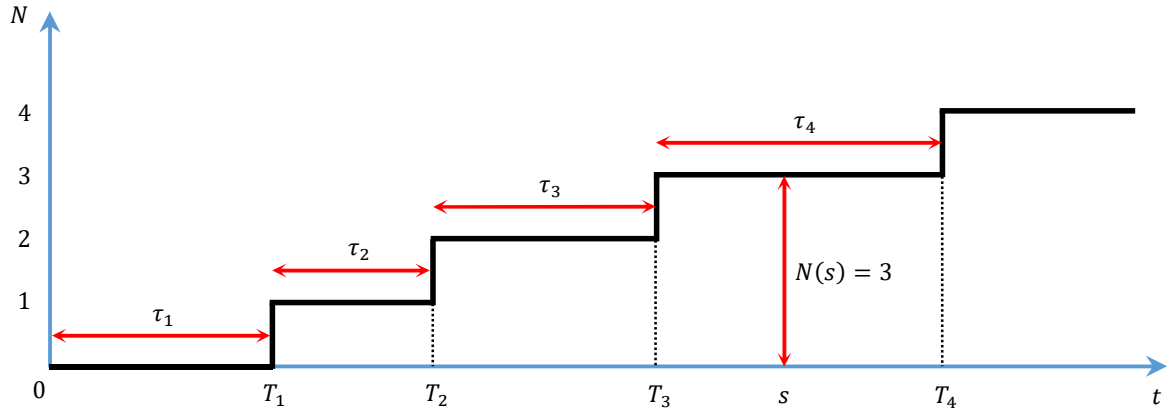


FIGURE 1. Illustration of a continuous-time arrival process $(T_k)_{k\geq 1}$ and its associated counting process $(N(t))_{t\geq 0}$. τ_k's denote inter-arrival times. $N(t) \equiv 3$ for $T_3 < t \leq T_4$.

Exercise 2.1.2 (Well-definedness of the counting process)**.** Let $(N(t))_{t\geq 0}$ denote the counting process of an arrival process with inter-arrival times $(\tau_k)_{k\geq 1}$ whose expectations are at least some positive constant $\delta > 0$ (but not necessarily independent or identically distributed). Recall that $N(t)$ and τ_k's are associated as

$$N(t) = \sum_{n=1}^{\infty} \mathbf{1}(\tau_1 + \cdots + \tau_n \leq t). \tag{156}$$

(i) Show that $T_{N(t)} \leq t$ for all $t \geq 0$. Deduce that

$$\mathbb{E}\left[\frac{N(t)}{t}\right] \leq \mathbb{E}\left[\frac{N(t)}{T_{N(t)}}\right] \overset{\text{Jensen}}{\leq} \frac{1}{\mathbb{E}\left[T_{N(t)}/N(t)\right]} \leq \frac{1}{\delta}. \tag{157}$$

(ii) From **(i)**, deduce that $\mathbb{E}[N(t)] \le t/\delta < \infty$ for all $t \ge 0$. Furthermore, deduce that $\mathbb{P}(N(t) < \infty) = 1$ for all $t \ge 0$.

Definition 2.1.3 (Renewal process)**.** A counting process $(N(t))_{t\ge 0}$ is called a *renewal process* if its inter-arrival times $\tau_1, \tau_2, \cdots$ are i.i.d. with $\mathbb{E}[\tau_1] < \infty$.

A cornerstone in the theory of renewal processes is the following strong law of large numbers for renewal processes. We first recall the strong law of large numbers.

Theorem 2.1.4 (SLLN)**.** *Let $(X_k)_{k\ge 1}$ be i.i.d. RVs and let $S_n = \sum_{k=1}^n X_i$, $n \ge 1$ be a random walk. Suppose $\mathbb{E}[|X_1|] < \infty$ and let $\mathbb{E}[X_1] = \mu$. Then*

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{S_n}{n} = \mu\right) = 1. \tag{158}$$

PROOF. See Durrett [Dur10, Thm. 2.4.1]. □

Exercise 2.1.5. Let $(X_k)_{k\ge 1}$ be i.i.d. RVs and let $S_n = \sum_{k=1}^n X_i$, $n \ge 1$ be a random walk. Suppose $\mathbb{E}[|X_1|] < \infty$ and let $\mathbb{E}[X_1] = \mu$. Let $(N(t))_{t\ge 0}$ be any counting process such that $N(t) \to \infty$ as $t \to \infty$. Show that

$$\mathbb{P}\left(\lim_{t\to\infty} \frac{S_{N(t)}}{N(t)} = \mu\right) = 1. \tag{159}$$

Theorem 2.1.6 (Renewal SLLN)**.** *Let $(T_k)_{k\ge 0}$ be a renewal process and let $(\tau_k)_{k\ge 0}$ and $(N(t))_{t\ge 0}$ be the associated inter-arrival times and counting process, respectively. Let $\mathbb{E}[\tau_k] = \mu$ be the mean inter-arrival time. If $0 < \mu < \infty$, then*

$$\mathbb{P}\left(\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{\mu}\right) = 1. \tag{160}$$

PROOF. First, write $T_k = \tau_1 + \tau_2 + \cdots + \tau_k$. Since the inter-arrival times are i.i.d. with mean $\mu < \infty$, the strong law of large numbers imply

$$\mathbb{P}\left(\lim_{k\to\infty} \frac{T_k}{k} = \mu\right) = 1. \tag{161}$$

Next, fix $t \ge 0$ and let $N(t) = n$, so that there are total n arrivals up to time t. Then the nth arrival time T_n must occur by time t, whereas the $n+1$st arrival time T_{n+1} must occur after time t. Hence $T_n \le t < T_{n+1}$. In general, we have

$$T_{N(t)} \le t < T_{N(t)+1}. \tag{162}$$

Dividing by $N(t)$, we get

$$\frac{T_{N(t)}}{N(t)} \le \frac{t}{N(t)} < \frac{T_{N(t)+1}}{N(t)+1}\frac{N(t)+1}{N(t)}. \tag{163}$$

To take the limit as $t \to \infty$, we note that $\mathbb{P}(T_k < \infty) = 1$ for all k since $\mathbb{E}[\tau_k] = \mu < \infty$. This yields $N(t) \ge k$ for some large enough t. Since k was arbitrary, this yields $N(t) \nearrow \infty$ as $t \to \infty$ with probability 1. Therefore, according to (161) and Exercise 2.1.5, we get

$$\mathbb{P}\left(\lim_{t\to\infty} \frac{T_{N(t)}}{N(t)} = \mu\right) = \mathbb{P}\left(\lim_{t\to\infty} \frac{T_{N(t)+1}}{N(t)+1} = \mu\right) = \mathbb{P}\left(\lim_{t\to\infty} \frac{N(t)+1}{N(t)} = 1\right) = 1. \tag{164}$$

Hence (163) gives

$$\mathbb{P}\left(\lim_{t\to\infty} \frac{t}{N(t)} = \mu\right) = 1. \tag{165}$$

Since $\mu > 0$, we can take the reciprocal inside the above probability. This shows the assertion. □

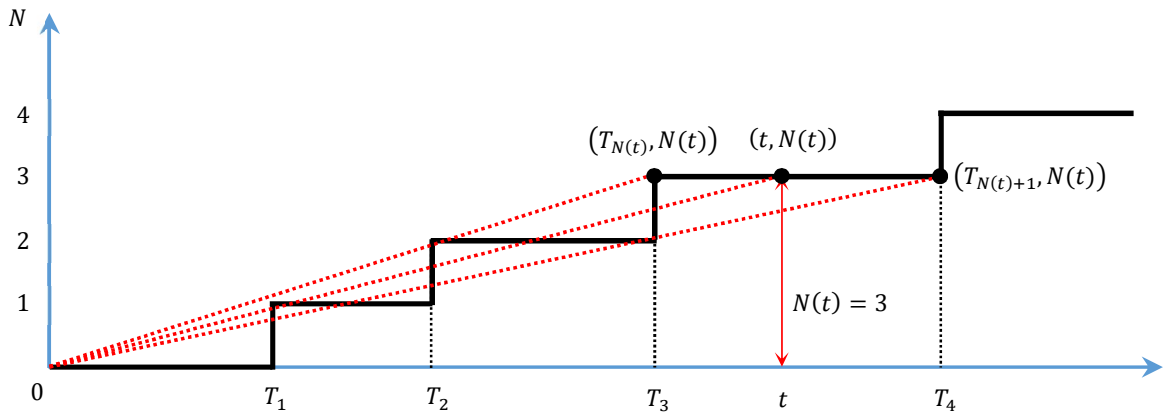


FIGURE 2. Illustration of the inequalities (163).

Example 2.1.7 (Poisson process)**.** Suppose $(N(t))_{t\geq 0}$ is a renewal process with its inter-arrival times $(\tau_k)_{k\geq 1}$ as $\mathrm{Exp}(\lambda)$ distribution with some $\lambda > 0$. In this case, we call $(N(t))_{t\geq 0}$ a "Poisson process with arrival rate λ". Note that the mean inter-arrival time is $\mathbb{E}[\tau_1] = 1/\lambda$, so the renewal SLLN yields

$$\mathbb{P}\left(\lim_{t\to\infty} \frac{N(t)}{t} = \lambda\right) = 1. \tag{166}$$

Namely, with probability 1, we tend to see about λt arrivals during $[0, t]$ as $t \to \infty$. In other words, we tend to see λ arrivals during an interval of unit length. Hence it makes sense to call the parameter λ as the 'arrival rate' of the process. ▲

2.2. Renewal reward processes and Renewal Reward SLLN

In this section, we consider a renewal process together with rewards, which are given for each inter-arrival times. This simple extension of the renewal processes will greatly improve the applicability of our theory.

Let $(T_k)_{k\geq 0}$ be a renewal process and let $(\tau_k)_{k\geq 0}$ and $(N(t))_{t\geq 0}$ be the associated inter-arrival times and counting process, respectively. Suppose we have a sequence of *rewards* $(Y_k)_{k\geq 1}$, so we regard a reward of Y_k is given at the kth arrival time T_k. We define the *reward process* $(R(t))_{t\geq 0}$ associated with the sequence $(\tau_k, Y_k)_{k\geq 1}$ as

$$R(t) = \sum_{k=1}^{N(t)} Y_k. \tag{167}$$

Namely, upon the kth arrival at time $T_k = \tau_1 + \cdots + \tau_k$, we receive a reward of Y_k. Then $R(t)$ is the total reward up to time t.

As we looked at the average number of arrivals $N(t)/t$ as $t \to \infty$, a natural quantity to look at for the reward process is the 'average reward' $R(t)/t$ as $t \to \infty$. Intuitively, since everything refreshes upon new arrivals, we should expect

$$\frac{R(t)}{t} \to \frac{\text{expected reward during one 'cycle'}}{\text{expected duration of one 'cycle'}} \tag{168}$$

as $t \to \infty$ almost surely. This is made precise by the following result.

Theorem 2.2.1 (Renewal reward SLLN)**.** *Let $(R(t))_{t\geq 0}$ be the reward process associated with an i.i.d. sequence of inter-arrival times $(\tau_k)_{k\geq 1}$ and an i.i.d. sequence of rewards $(Y_k)_{k\geq 1}$. Suppose $\mathbb{E}[Y_k] < \infty$ and $\mathbb{E}[\tau_k] \in (0, \infty)$. Then*

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{R(t)}{t} = \frac{\mathbb{E}[Y_1]}{\mathbb{E}[\tau_1]}\right) = 1. \tag{169}$$

PROOF. Let $(\tau_k)_{k\geq 0}$ denote the inter-arrival times for the renewal process $(T_k)_{k\geq 0}$. Note that

$$\frac{R(t)}{t} = \left(\frac{1}{N(t)}\sum_{k=1}^{N(t)} Y_k\right)\frac{N(t)}{t}. \tag{170}$$

Hence by Exercise 2.1.5, the 'average reward' up to time t in the bracket converges to $\mathbb{E}[Y_1]$ almost surely. Moreover, the average number of arrivals $N(t)/t$ converges to $1/\mathbb{E}[\tau_1]$ by Theorem 2.1.6. Hence the assertion follows. □

Remark 2.2.2. Theorem 2.1.6 can be obtained as a special case of the above reward version of SLLN, simply by choosing $Y_k = \tau_k$ for $k \geq 1$ so that $R(t) = N(t)$.

Example 2.2.3 (Long run car costs)**.** This example is excerpted from [Dur99]. Mr. White do not drive the same car more than t^* years, where $t^* > 0$ is some fixed number. He changes to a new car when the old one breaks down or reaches t^* years. Let X_k be the life time of the kth car that Mr. White drives, which are i.i.d. with finite expectation. Let τ_k be the duration of his kth car. According to his policy, we have

$$\tau_k = \min(X_k, t^*). \tag{171}$$

Let $T_k = \tau_1 + \cdots + \tau_k$ be the time that Mr. White is done with the kth car. Then $(T_k)_{k\geq 0}$ is a renewal process. Note that the expected running time for the kth car is

$$\mathbb{E}[\tau_k] = \mathbb{E}[\tau_k \,|\, X_k < t^*]\mathbb{P}(X_k < t^*) + \mathbb{E}[\tau_k \,|\, X_k \geq t^*]\mathbb{P}(X_k \geq t^*) \tag{172}$$

$$= \mathbb{E}[X_k \,|\, X_k < t^*]\mathbb{P}(X_k < t^*) + t^*\mathbb{P}(X_k \geq t^*). \tag{173}$$

Suppose that the car cost g during each cycle is given by

$$g(t) = \begin{cases} A+B & \text{if } t < t^* \\ A & \text{if } t \geq t^*. \end{cases} \tag{174}$$

Namely, if the car breaks down by t^* years, then Mr. White has to pay $A+B$ dolors; otherwise, the cost is only A dolors. Then the expected cost for one cycle is

$$\mathbb{E}[g(\tau_k)] = A + B\mathbb{P}(\tau_k < t^*) = A + B\mathbb{P}(X_k < t^*). \tag{175}$$

Thus by Theorem 2.2.1, the long-run car cost of Mr. White is

$$\lim_{t\to\infty}\frac{R(t)}{t} = \frac{\mathbb{E}[g(\tau_k)]}{\mathbb{E}[\tau_k]} = \frac{A + B\mathbb{P}(X_k < t^*).}{\mathbb{E}[X_k \,|\, X_k < t^*]\mathbb{P}(X_k < t^*) + t^*\mathbb{P}(X_k \geq t^*)}. \tag{176}$$

For more concrete example, let $X_k \sim \text{Uniform}([0,10])$ and let $A = 10$ and $B = 3$. Then

$$\mathbb{E}[g(\tau_k)] = 10 + 3t^*/10. \tag{177}$$

On other other hand,

$$\mathbb{E}[\tau_k] = \mathbb{E}[X_k \,|\, X_k < t^*]\mathbb{P}(X_k < t^*) + t^*\mathbb{P}(X_k \geq t^*) \tag{178}$$

$$= \frac{t^*}{2}\frac{t^*}{10} + t^*\frac{10-t^*}{10} = t^* - (t^*)^2/20. \tag{179}$$

Note that for $\mathbb{E}[X_k \,|\, X_k < t^*] = t^*/2$, observe that a uniform RV over $[0,10]$ conditioned on being $[0,t^*]$ is uniformly distributed over $[0,t^*]$. This yields

$$\frac{\mathbb{E}[g(\tau_k)]}{\mathbb{E}[\tau_k]} = \frac{10 + 0.3t^*}{t^*(1 - t^*/20)}. \tag{180}$$

Lastly, in order to minimize the above long-run cost, we differentiate it in t^* and find global minimum. A straightforward computation shows that the long-run cost is minimized at

$$t^* = \frac{-1+\sqrt{1.6}}{0.03} \approx 8.83. \tag{181}$$

Thus the optimal strategy for Mr. White in this situation is to drive each car up to 8.83 years. ▲

CHAPTER 3

Markov chains

Say we would like to model the USD price of bitcoin. We could observe the actual price at every hour and record it by a sequence of real numbers $x_1, x_2, \cdots$. However, it is more interesting to build a 'model' that could predict the price of bitcoin at time t, or at least give some meaningful insight how the actual bitcoin price behaves over time. Since there are so many factors affecting the price at every time, it might be reasonable that the price at time t should be given by a certain RV, say X_t. Then our sequence of predictions would be a sequence of RVs, $(X_t)_{t\geq 0}$. This is an example of *stochastic processes*. Here 'process' means that we are not interest in just a single RV, that their sequence as a whole: 'stochastic' means that the way the RVs evolve in time might be random.

In this note, we will be studying a very important class of stochastic processes called *Markov chains*. The importance of Markov chains lies two places: 1) They are applicable for a wide range of physical, biological, social, and economical phenomena, and 2) the theory is well-established and we can actually compute and make predictions using the models.

3.1. Definition and examples

Roughly speaking, *Markov processes* are used to model temporally changing systems where future state only depends on the current state. For instance, if the price of bitcoin tomorrow depends only on its price today, then bitcoin price can be modeled as a Markov process. (Of course, the entire history of price often affects decisions of buyers/sellers so it may not be a realistic assumption.)

Even through Markov processes can be defined in vast generality, we concentrate on the simplest setting where the state and time are both discrete. Let $\Omega = \{1, 2, \cdots, m\}$ be a finite set, which we call the *state space*. Consider a sequence $(X_t)_{t\geq 0}$ of Ω-valued RVs, which we call a *chain*. We call the value of X_t the *state* of the chain at time t. In order to narrow down the way the chain $(X_t)_{t\geq 0}$ behaves, we introduce the following properties:

(i) (Markov property) The distribution of X_{t+1} given the history $X_0, X_1, \cdots, X_t$ depends only on X_t. That is, for any values of $j_0, j_1, \cdots, j_t, k \in \Omega$,

$$\mathbb{P}(X_{t+1} = k \,|\, X_t = j_t, X_{t-1} = j_{t-1}, \cdots, X_1 = j_1, X_0 = j_0) = \mathbb{P}(X_{t+1} = k \,|\, X_t = j_t). \tag{182}$$

(ii) (Time-homogeneity) The transition probabilities

$$p_{ij} = \mathbb{P}(X_{t+1} = j \,|\, X_t = i) \qquad i, j \in \Omega \tag{183}$$

do not depend on t.

When the chain $(X_t)_{t\geq 0}$ satisfies the above two properties, we say it is a (discrete-time and time-homogeneous) *Markov chain*. We define the *transition matrix* P to be the $m \times m$ matrix of transition probabilities:

$$P = (p_{ij})_{1\leq i,j\leq m} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix}. \tag{184}$$

Finally, since the state X_t of the chain is a RV, we represent its probability mass function (PMF) via a row vector

$$\mathbf{r}_t = [\mathbb{P}(X_t = 1), \mathbb{P}(X_t = 2), \cdots, \mathbb{P}(X_t = m)]. \tag{185}$$

Example 3.1.1. Let $\Omega = \{1,2\}$ and let $(X_t)_{t\geq 0}$ be a Markov chain on Ω with the following transition matrix

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}. \tag{186}$$

We can also represent this Markov chain pictorially as in Figure 4, which is called the 'state space diagram' of the chain $(X_t)_{t\geq 0}$.

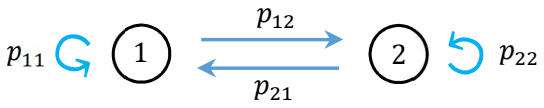


FIGURE 1. State space diagram of a 2-state Markov chain

For some concrete example, suppose

$$p_{11} = 0.2, \quad p_{12} = 0.8, \quad p_{21} = 0.6, \quad p_{22} = 0.4. \tag{187}$$

If the initial state of the chain X_0 is 1, then

$$\mathbb{P}(X_1 = 1) = \mathbb{P}(X_1 = 1 \,|\, X_0 = 1)\mathbb{P}(X_0 = 1) + \mathbb{P}(X_1 = 1 \,|\, X_0 = 2)\mathbb{P}(X_0 = 2) \tag{188}$$

$$= \mathbb{P}(X_1 = 1 \,|\, X_0 = 1) = p_{11} = 0.2 \tag{189}$$

and similarly,

$$\mathbb{P}(X_1 = 2) = \mathbb{P}(X_1 = 2 \,|\, X_0 = 1)\mathbb{P}(X_0 = 1) + \mathbb{P}(X_1 = 2 \,|\, X_0 = 2)\mathbb{P}(X_0 = 2) \tag{190}$$

$$= \mathbb{P}(X_1 = 2 \,|\, X_0 = 1) = p_{12} = 0.8. \tag{191}$$

Also we can compute the distribution of X_2. For example,

$$\mathbb{P}(X_2 = 1) = \mathbb{P}(X_2 = 1 \,|\, X_1 = 1)\mathbb{P}(X_1 = 1) + \mathbb{P}(X_2 = 1 \,|\, X_1 = 2)\mathbb{P}(X_1 = 2) \tag{192}$$

$$= p_{11}\mathbb{P}(X_1 = 1) + p_{21}\mathbb{P}(X_1 = 2) \tag{193}$$

$$= 0.2 \cdot 0.2 + 0.6 \cdot 0.8 = 0.04 + 0.48 = 0.52. \tag{194}$$

In general, the distribution of X_{t+1} can be computed from that of X_t via a simple linear algebra. Note that for $i = 1,2$,

$$\mathbb{P}(X_{t+1} = i) = \mathbb{P}(X_{t+1} = i \,|\, X_t = 1)\mathbb{P}(X_t = 1) + \mathbb{P}(X_{t+1} = i \,|\, X_t = 2)\mathbb{P}(X_t = 2) \tag{195}$$

$$= p_{1i}\mathbb{P}(X_t = 1) + p_{2i}\mathbb{P}(X_t = 2). \tag{196}$$

This can be written as

$$[\mathbb{P}(X_{t+1} = 2), \mathbb{P}(X_{t+1} = 2)] = [\mathbb{P}(X_{t+1} = 2), \mathbb{P}(X_{t+1} = 2)] \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}. \tag{197}$$

That is, if we represent the distribution of X_t as a row vector, then the distribution of X_{t+1} is given by multiplying the transition matrix P to the left. ▲

Example 3.1.2 (Gambler's ruin)**.** Suppose a gambler has fortune of k dolors initially and starts gambling. At each time he wins or loses 1 dolor independently with probability p and $1-p$, respectively. The game ends when his fortune reaches either 0 or N dolors. What is the probability that he wins N dolors and goes home happy?

We use Markov chains to model his fortune after betting t times. Namely, let $\Omega = \{0,1,2,\cdots,N\}$ be the state space. Let $(X_t)_{t\geq 0}$ be a sequence of RVs where X_t is the gambler's fortune after betting t times. We first draw the state space diagram for $N = 4$ below: Next, we can write down its transition probabilities as

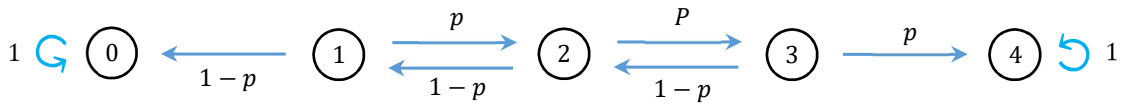


FIGURE 2. State space diagram of a 5-state gambler's chain

$$\begin{cases} \mathbb{P}(X_{t+1} = k+1 \,|\, X_t = k) = p & \forall 1 \le k < N \\ \mathbb{P}(X_{t+1} = k \,|\, X_t = k+1) = 1-p & \forall 1 \le k < N \\ \mathbb{P}(X_{t+1} = 0 \,|\, X_t = 0) = 1 \\ \mathbb{P}(X_{t+1} = N \,|\, X_t = N) = 1. \end{cases} \tag{198}$$

For example, the transition matrix P for $N = 5$ is given by

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1-p & 0 & p & 0 & 0 & 0 \\ 0 & 1-p & 0 & p & 0 & 0 \\ 0 & 0 & 1-p & 0 & p & 0 \\ 0 & 0 & 0 & 1-p & 0 & p \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{199}$$

We call the resulting Markov chain $(X_t)_{t\ge 0}$ the *gambler*'s chain. ▲

Example 3.1.3 (Ehrenfest Chain)**.** This chain is originated from the physics literature as a model for two cubical volumes of air connected by a thin tunnel. Suppose there are total N indistinguishable balls split into two "urns" A and B. At each step, we pick up one of the N balls uniformly at random, and move it to the other urn. Let X_t denote the number of balls in urn A after t steps. This is a Markov chain called the *Ehrenfest chain*. (See the state space diagram in Figure 3.)

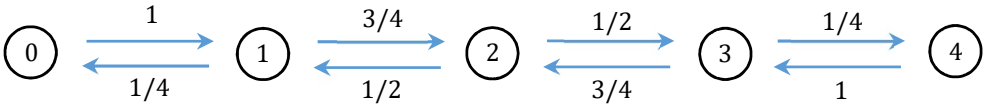


FIGURE 3. State space diagram of the Ehrenfest chain with 4 balls

It is easy to figure out the transition probabilities by considering different cases. If $X_t = k$, then urn B has $N-k$ balls at time t. If $0 < k < N$, then with probability k/N we move one ball from A to B and with probability $(N-k)/N$ we move one from B to A. If $k = 0$, then we must pick up a ball from urn B so $X_{t+1} = 1$ with probability 1. If $k = N$, then we must move one from A to B and $X_{t+1} = N-1$ with probability 1. Hence, the transition kernel is given by

$$\begin{cases} \mathbb{P}(X_{t+1} = k+1 \,|\, X_t = k) = (N-k)/N & \forall 0 \le k < N \\ \mathbb{P}(X_{t+1} = k-1 \,|\, X_t = k) = k/N & \forall 0 < k \le N \\ \mathbb{P}(X_{t+1} = 1 \,|\, X_t = 0) = 1 \\ \mathbb{P}(X_{t+1} = N-1 \,|\, X_t = N) = 1. \end{cases} \tag{200}$$

For example, the transition matrix P for $N = 5$ is given by

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1/5 & 0 & 4/5 & 0 & 0 & 0 \\ 0 & 2/5 & 0 & 3/5 & 0 & 0 \\ 0 & 0 & 3/5 & 0 & 2/5 & 0 \\ 0 & 0 & 0 & 4/5 & 0 & 1/5 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}. \tag{201}$$

▲

Exercise 3.1.4. Repeat rolling two four sided dices with numbers 1,2,3, and 4 on them. Let Y_k be the some of the two dice at the kth roll. Let $S_n = Y_1 + Y_2 + \cdots + Y_n$ be the total of the first n rolls, and define $X_t = S_t \,(\text{mod} 6)$. Show that $(X_t)_{t\geq 0}$ is a Markov chain on the state space $\Omega = \{0,1,2,3,4,5\}$. Furthermore, identify its transition matrix.

We generalize our observation in Example 3.1.1 in the following exercise.

Exercise 3.1.5. Let $(X_t)_{t\geq 0}$ be a Markov chain on state space $\Omega = \{1,2,\cdots,m\}$ with transition matrix $P = (p_{ij})_{1\leq i,j\leq m}$. Let $\mathbf{r}_t = [\mathbb{P}(X_t = 1),\cdots,\mathbb{P}(X_t = m)]$ denote the row vector of the distribution of X_t.

(i) Show that for each $i \in \Omega$,

$$\mathbb{P}(X_{t+1} = i) = \sum_{j=1}^{m} p_{ji}\mathbb{P}(X_t = j). \tag{202}$$

(ii) Show that for each $t \geq 0$,

$$\mathbf{r}_{t+1} = \mathbf{r}_t P. \tag{203}$$

(iii) Show by induction that for each $t \geq 0$,

$$\mathbf{r}_t = \mathbf{r}_0 P^t. \tag{204}$$

Exercise 3.1.6. Let $(X_t)_{t\geq 0}$ be a Markov chain on state space $\Omega = \{1,2,\cdots,m\}$ with transition matrix $P = (p_{ij})_{1\leq i,j\leq m}$.

(i) (Multi-step transition prob.) Show that for each $x, y \in \Omega$ and $a, b \geq 1$,

$$\mathbb{P}\left(X_{a+b} = y \,|\, X_a = x\right) = P^b(x,y). \tag{205}$$

Hint: Use Exercise 3.1.5.

(ii) (The Chapman-Kolmogorov eq.) Show that for each $x, y \in \Omega$ and $n, m \geq 1$,

$$P^{n+m}(x,y) = \sum_{z\in\Omega} P^n(x,z)P^m(z,y). \tag{206}$$

While right-multiplication of P advances a given row vector of distribution one step forward in time, left-multiplication of P on a column vector computes the expectation of a given function with respect to the future distribution. This point is clarified in the following exercise.

Exercise 3.1.7. Let $(X_t)_{t\geq 0}$ be a Markov chain on a state space $\Omega = \{1,2,\cdots,m\}$ with transition matrix P. Let $f : \Omega \to \mathbb{R}$ be a function. Suppose that if the chain X_t has state x at time t, then we get a 'reward' of $f(x)$. Let $\mathbf{r}_t = [\mathbb{P}(X_t = 1),\cdots,\mathbb{P}(X_t = m)]$ be the distribution of X_t. Let $\mathbf{v} = [f(1), f(2),\cdots, f(m)]^T$ be the column vector representing the reward function f.

(i) Show that the expected reward at time t is given by

$$\mathbb{E}[f(X_t)] = \sum_{i=1}^{m} f(i)\mathbb{P}(X_t = i) = \mathbf{r}_t\mathbf{v}. \tag{207}$$

(ii) Use part (i) and Exercise 3.1.5 to show that

$$\mathbb{E}[f(X_t)] = \mathbf{r}_0 P^t \mathbf{v}. \tag{208}$$

(iii) The total reward up to time t is a RV given by $R_t = \sum_{k=0}^{t} f(X_t)$. Show that

$$\mathbb{E}[R_t] = \mathbf{r}_0(I + P + P^2 + \cdots + P^t)\mathbf{v}. \tag{209}$$

Exercise 3.1.8. Suppose that the probability it rains today is 0.4 if neither of the last two days was rainy, but 0.5 if at least one of the last two days was rainy. Let $\Omega = \{S, R\}$, where S=sunny and R=rainy. Let W_t be the weather of day t.

(i) Show that $(W_t)_{t\geq 0}$ is *not* a Markov chain.

(ii) Expand the state space into the set of pairs $\Sigma := \Omega^2$. For each $t \geq 0$, define $X_t = (W_{t-1}, W_t) \in \Sigma$. Show that $(X_t)_{t\geq 0}$ is a Markov chain on Σ. Identify its transition matrix.
(iii) What is the two-step transition matrix?
(iv) What is the probability that it will rain on Wednesday if it didn't rain on Sunday and Monday?

3.2. Stationary distribution and examples

3.2.1. Stationary distribution. Let $(X_t)_{t\geq 0}$ be a Markov chain on state space $\Omega = \{1, 2, \cdots, m\}$ with transition matrix P. If π is a distribution on Ω such that

$$\pi = \pi P, \tag{210}$$

then we say π is a *stationary distribution* of the Markov chain $(X_t)_{t\geq 0}$.

In Exercise 3.1.5, we observed that we can simply multiply the transition matrix P to a given row vector $\mathbf{r}_t$ of distribution on the state space Ω in order to get the next distribution $\mathbf{r}_{t+1}$. Hence if the initial distribution of the chain is π, then its distribution is invariant in time.

Example 3.2.1. Consider the 2-state Markov chain $(X_t)_{t\geq 0}$ with transition matrix (as in Exercise 3.1.1)

$$P = \begin{bmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{bmatrix}. \tag{211}$$

Then $\pi = [3/7, 4/7]$ is a stationary distribution of X_t. Indeed,

$$[3/7, 4/7] = [3/7, 4/7] \begin{bmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{bmatrix}. \tag{212}$$

Furthermore, this is the unique stationary distribution. To see this, let $\pi = [\pi_1, \pi_2]$ be a stationary distribution of X_t. Then $\pi = \pi P$ gives

$$0.2\pi_1 + 0.6\pi_2 = \pi_1 \tag{213}$$
$$0.8\pi_1 + 0.4\pi_2 = \pi_2. \tag{214}$$

These equations lead to

$$4\pi_1 = 3\pi_2. \tag{215}$$

Since π is a probability distribution, $\pi_1 + \pi_2 = 1$ so $\pi = [3/7, 4/7]$ is the only solution. This shows the uniqueness of the stationary distribution for X_t. ▲

A Markov chain may have multiple stationary distributions, as the following example illustrates.

Example 3.2.2. Let $(X_t)_{t\geq 0}$ be a 2-state Markov chain with transition matrix

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{216}$$

Then any distribution $\pi = [p, 1-p]$ is a stationary distribution for the chain $(X_t)_{t\geq 0}$.

Stationary distributions are closely related with eigenvectors and eigenvalues of the transition matrix P. Namely, by taking transpose,

$$\pi^T = P^T \pi^T. \tag{217}$$

Hence, the transpose of any stationary distribution is an eigenvector of P^T associated with eigenvalue 1. More properties of stationary distribution in this line of thought are given by the following exercise.

Exercise 3.2.3. Given a matrix A, a row vector v, and a real number λ, we say v is a *left eigenvector* of A associated with *left eigenvalue* λ if

$$\mathbf{v}A = \lambda \mathbf{v}. \tag{218}$$

If $\mathbf{v}$ is a column vector and if $A\mathbf{v} = \lambda\mathbf{v}$, then we say v is a *(right) eigenvector* associated with *(right) eigenvalue* λ. Let $(X_t)_{t\geq 0}$ be a Markov chain on state space $\Omega = \{1, 2, \cdots, m\}$ with transition matrix $P = (p_{ij})_{1\leq i,j\leq m}$.

(i) Show that a distribution π on Ω is a stationary distribution for the chain $(X_t)_{t\geq 0}$ if and only if it is a left eigenvector of P associated with left eigenvalue 1.

(ii) Show that 1 is a right eigenvalue of P with right eigenvector $[1, 1, \cdots, 1]^T$.

(iii) Recall that a square matrix and its transpose have the same (right) eigenvalues and corresponding (right) eigenspaces have the same dimension. Show that the Markov chain $(X_t)_{t\geq 0}$ has a unique stationary distribution if any only if $[1, 1, \cdots, 1]^T$ spans the (right) eigenspace of P associated with (right) eigenvalue 1.

In the following exercise, we compute the stationary distribution of the so-called birth-death chain.

Exercise 3.2.4 (Birth-Death chain)**.** Let $\Omega = \{0, 1, 2, \cdots, N\}$ be the state space. Let $(X_t)_{t\geq 0}$ be a Markov chain on Ω with transition probabilities

$$\begin{cases} \mathbb{P}(X_{t+1} = k+1 \,|\, X_t = k) = p & \forall 0 \leq k < N \\ \mathbb{P}(X_{t+1} = k-1 \,|\, X_t = k) = 1-p & \forall 1 \leq k \leq N \\ \mathbb{P}(X_{t+1} = 0 \,|\, X_t = 0) = 1-p \\ \mathbb{P}(X_{t+1} = N \,|\, X_t = N) = p. \end{cases} \tag{219}$$

This is called a Birth-Death chain. Its state space diagram is as below.

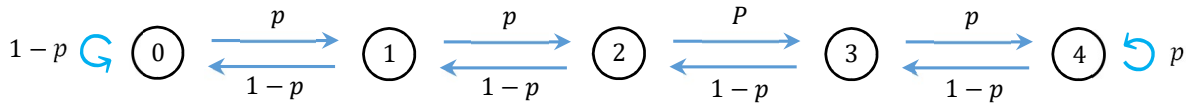


FIGURE 4. State space diagram of a 5-state Birth-Death chain

(i) Let $\pi = [\pi_0, \pi_1, \cdots, \pi_N]$ be a distribution on Ω. Show that π is a stationary distribution of the Birth-Death chain if and only if it satisfy the following 'balance equation'

$$p\pi_k = (1-p)\pi_{k+1} \qquad 0 \leq k < N. \tag{220}$$

(ii) Let $\rho = p/(1-p)$. From (ii), deduce that $\pi_k = \rho^k \pi_0$ for all $0 \leq k < N$.

(iii) Using the normalization condition $\pi_0 + \pi_1 + \cdots + \pi_N = 1$, show that $\pi_0 = 1/(1 + \rho + \rho^2 + \cdots + \rho^N)$. Conclude that

$$\pi_k = \frac{\rho^k}{1 + \rho + \rho^2 + \cdots + \rho^N} \qquad 0 \leq k \leq N. \tag{221}$$

Conclude that the Birth-Death chain has a unique stationary distribution given by (221), which becomes the uniform distribution on Ω when $p = 1/2$.

Exercise 3.2.5 (Success run chain)**.** Consider a Markov chain $(X_n)_{n\geq 0}$ on the state space $\Omega = \mathbb{Z}_{\geq 0}$ of non-negative integers with the following transition matrix P:

$$P(x, x+1) = p, \quad P(x, 0) = 1-p \qquad \text{for all } x \in \mathbb{Z}_{\geq 0}, \tag{222}$$

where $p \in [0, 1]$ is a fixed parameter. In words, one flips independent coins with success probability p; every time it comes up heads with probability p X_n increases by 1, and when it comes up tails, X_n resets back to 0.

(i) Let $\pi : \mathbb{Z}_{\geq 0} \to [0,\infty)$ be a function that satisfies $\pi P = \pi$. Show that the following recursion holds:

$$\pi(0) = (1-p)\sum_{k=0}^{\infty} \pi(k) \tag{223}$$

$$\pi(1) = p\pi(0) \tag{224}$$

$$\pi(2) = p\pi(1) \tag{225}$$

$$\vdots \tag{226}$$

Deduce that $\pi(k) = p^k \pi(0)$ for all $k \geq 1$.

(ii) Show that π is an stationary distribution on $\mathbb{Z}_{\geq 0}$ if and only if $\pi(k) = (1-p)p^k$ for $k \geq 0$. (Such π is called the Geometric(p) distribution that puts probability $p^k(1-p)$ on each $k \in \mathbb{Z}_{\geq 0}$. The number of success run of probability-p coin flips has this distribution.)

Next, we take a look at an example of an important class of Markov chains, which is called the random walk on graphs. This is the basis of many algorithms involving machine learning on networks (e.g., Google's PageRank).

Example 3.2.6 (Random walk on graphs)**.** We first introduce some notions in graph theory. A *graph* G consists of a pair (V,E) of sets of nodes V and edges $E \subseteq V^2$. A graph G can be concisely represented as a $|V| \times |V|$ matrix A_G, which is called the *adjacency matrix* of G. Namely, the (i,j) entry of A_G is defined by

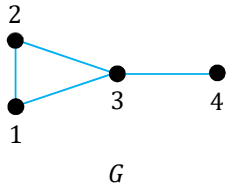
$$A_G(i,j) = \mathbf{1}(\text{nodes } i \text{ and } j \text{ are adjacent in } G). \tag{227}$$

We say G is *simple* if $(i,j) \in E$ implies $(j,i) \in E$ and $(i,i) \notin E$ for all $i \in V$. For a simple graph $G = (V,E)$, we say a node j is *adjacent* to i if $(i,j) \in E$. We denote $\deg_G(i)$ the number of neighbors of i in G, which we call the *degree* of i.

Consider we hop around the nodes of a given simple graph $G = (V,E)$: at each time, we jump from one node to one of the neighbors with equal probability. For instance, if we are currently at node 2 and if 2 is adjacent to 3, 5, and 6, then we jump to one of the three neighbors with probability 1/3. The location of this jump process at time t can be described as a Markov chain. Namely, a Markov chain $(X_t)_{t\geq 0}$ on the node set V is called a *random walk on* G if

$$\mathbb{P}(X_{t+1} = j \,|\, X_t = i) = \frac{A_G(i,j)}{\deg_G(i)}. \tag{228}$$

Note that its transition matrix P is obtained by normalizing each row of the adjacency matrix A_G by the corresponding degree.



G

$$A_G = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad P = \begin{bmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

FIGURE 5. A 4-node simple graph G, its adjacency matrix A_G, and associated random walk transition matrix P

What is the stationary distribution of random walk on G? There could be many (see Exercise 3.7.1), but here is a typical one that always works. Define a probability distribution π on V by

$$\pi(i) = \frac{\deg_G(i)}{\sum_{j\in V} \deg_G(i)} = \frac{\deg_G(i)}{2|E|}. \tag{229}$$

Namely, the probability that $X_t = i$ is proportional to the degree of node i. Then π is a stationary distribution for P. Indeed,

$$\sum_{i\in V} \pi(j)P(j,i) = \sum_{i\in V} \frac{\deg_G(i)}{2|E|}\frac{A_G(i,j)}{\deg_G(i)} \tag{230}$$

$$= \frac{1}{2|E|}\sum_{i\in V} A_G(i,j) = \frac{\deg_G(j)}{2|E|} = \pi(j). \tag{231}$$

Hence if we perform a random walk on facebook network, then we are about ten times more likely to visit a person of 100 friends than to visit a person of 10 friends. ▲

Next, we will discuss two classes of Markov chains for which it is very easy to compute stationary distributions.

3.2.2. Reversible chains. A probability distribution μ on Ω is *reversible* for transition matrix $P : \Omega^2 \to [0,1]$ if the following *detailed balance equation* is satisfied:

$$\mu(x)\,P(x,y) = \mu(y)\,P(y,x) \qquad \text{for all } x, y \in \Omega. \tag{232}$$

Below we observe that reversiblity implies stationarity.

Proposition 3.2.7. *Let P be a transition matrix on state space Ω. If a probability distribution μ on Ω is reversible for P, then it is also stationary for P.*

PROOF. Note that

$$\left[\mu P\right](x) = \sum_{y\in\Omega} \mu(y)P(y,x) \overset{\text{reversibility}}{=} \sum_{y\in\Omega} \mu(x)P(x,y) = \mu(x)\sum_{y\in\Omega} P(x,y) = \mu(x). \tag{233}$$

□

Example 3.2.8 (Mass transport)**.** In order to give some concrete intuition, consider the following thought experiment. Imagine Ω is the note set of some graph, and $\mu(x)$ represents the mass of sand at node x for each $x \in \Omega$. Overnight, a truck runs around the graph and transport $P(x,y)$ proportion of sand $\mu(x)$ at node x to node y. Hence

$$(\text{Amount of mass transported from } x \text{ to } y) = \mu(x)P(x,y). \tag{234}$$

Let μ' denote the new distribution of sands on Ω. Observe that

$$(\text{Total amount of mass transported from } x) = \sum_{y\in\Omega} \mu(x)P(x,y) \tag{235}$$

$$= \mu(x)\sum_{y\in\Omega} P(x,y) = \mu(x). \tag{236}$$

Hence the new sand distribution μ' is given by

$$\mu'(x) = (\text{Total amount of mass transported into } x) = \sum_{y\in\Omega} \mu(y)P(y,x). \tag{237}$$

In other words, $\mu' = \mu P$. From this, we see that

$$\mu = \mu' \iff \mu = \mu P \iff \mu \text{ is stationary for } \mathbb{P}. \tag{238}$$

On the other hand, the detailed balance equation (232) can be interpreted as

$$(\text{mass sent from } x \text{ to } y) = (\text{mass sent from } y \text{ to } x). \tag{239}$$

From this, it is obvious why a reversible distribution is also stationary. ▲

A *reversible Markov chain* is a Markov chain whose initial distribution is reversible. Namely, let $(X_n)_{n\geq 0}$ denote a Markov chain on state space Ω with transition matrix P. Let the initial state X_0 be distributed as a reversible distribution μ. By Proposition 3.2.7, X_n is a stationary process, meaning that the distribution of X_n does not change in n. However, we can say more than just stationarity. Namely, one cannot distinguish if X_n is run *backward in time*. More precisely,

$$\mathbb{P}_\mu(X_0 = x_0, X_1 = x_1, \ldots, X_n = x_n) = \mathbb{P}_\mu(X_n = x_0, X_{n-1} = x_1, \ldots, X_0 = x_n) \tag{240}$$

for all $n \geq 1$ and states $x_0, \ldots, x_n$.

In practice, reversibility is easier to check than stationarity. So showing reveribility could be handy to stationarity.

Example 3.2.9 (SSRW on graphs is reversible)**.** Let $(X_n)_{n\geq 0}$ be a simple symmetric random walk $G = (V, E)$. Let the initial state X_0 be distribued as the distribution $\mu(x) = \frac{\deg(x)}{2|E|}$. From Example 3.2.6, we already know that μ is a stationary distribution. However, the detailed balance equation is also easily verified:

$$\mu(x)P(x,y) = \frac{\deg(x)}{2|E|}\frac{A_G(x,y)}{\deg(x)} = \frac{\deg(y)}{2|E|}\frac{A_G(y,x)}{\deg(y)} = \mu(y)P(y,x). \tag{241}$$

Hence X_n is not just stationary, but also reversible. Imagine X_n follows its trajectory backwards in time. The above detailed balance equation makes it so that the backward random walk still follows the same rule: Choose a neighbor uniformly at random and move. ▲

3.2.3. Doubly stochastic chains. Let P be the transition matrix of a Markov chain $(X_n)_{n\geq 0}$ on a state space Ω. Recall that the row sums of P are always 1, since each row gives the transition probabilities from a fixed state to all other possible states. We say the Markov chain $(X_n)_{n\geq 0}$ (as well as P) is *doubly stochastic* if the column sums of P are also always 1.

Definition 3.2.10. A matrix $P : \Omega \times \Omega \to [0, \infty)$ is *doubly stochastic* if its row and column sums are all one, that is, for all $x, y \in \Omega$,

$$\sum_{z\in\Omega} P(x,z) = \sum_{z\in\Omega} P(z,y) = 1. \tag{242}$$

For doubly stochastic Markov chains, the uniform distribution on Ω (which makes sense if Ω is finite) is always a stationary distribution.

Proposition 3.2.11. *Let $(X_n)_{n\geq 0}$ be a doubly stochastic Markov chain on a finite state space Ω. Then $\pi = \text{Uniform}(\Omega)$ is a stationary distribution for the chain.*

PROOF. Let P denote the transition matrix of the chain $(X_n)_{n\geq 0}$. Note that $\pi(x) = 1/|\Omega|$ for all $x \in \Omega$. Then, viewing π as a row vector, for each $x \in \Omega$,

$$(\pi P)(x) = \sum_{z\in\Omega} \pi(z)P(z,x) = \frac{1}{|\Omega|}\sum_{z\in\Omega} P(z,x) = \frac{1}{|\Omega|} = \pi(x). \tag{243}$$

Hence π is stationary for P. □

Example 3.2.12 (Random Walk on a ring)**.** Let $(X_n)_{n\geq 0}$ denote a random walk on the integer ring $\mathbb{Z}/m\mathbb{Z}$, which consists of integers $0, \ldots, m-1$ and we view $m \equiv 0$ modulo m. The chain moves by 1 clockwise with probability p and 1 counterclockwise with probability $1-p$. Then it is easy to verify the corresponding transition matrix is doubly stochastic. Hence by Proposition 3.2.11, the uniform distribution is a stationary distribution. This is quite resonable since the process is invariant under 'rotation'. ▲

Exercise 3.2.13. Suppose a Markov chain has the uniform distribution as a stationary distribution. Does it have to be a doubly stochastic chain?

3.3. Markov property to the infinite future

The Markov property (182) states that, knowing the state X_n at time n, the distribution of the immediate future X_{n+1} does not depend on the past $X_0,\dots,X_{n-1}$. How about one step ahead, X_{n+2}? As one can guess, the distribution of X_{n+2} also does not depend on the past $X_0,\dots,X_{n-1}$ given the present. As usual, the argument is by conditioning on the 'missing step' X_{n+1} and applying the markov property (182):

$$\begin{aligned}
&\mathbb{P}(X_{n+2}=j_{n+2}\,|\,X_n=j_n,\dots,X_0=j_0) && (244)\\
&\quad=\sum_{j_{n+1}}\mathbb{P}(X_{n+2}=j_{n+2}\,|\,X_{n+1}=j_{n+1},X_n=j_n,\dots,X_0=j_0)\mathbb{P}(X_{n+1}=j_{n+1}\,|\,X_n=j_n,\dots,X_0=j_0) && (245)\\
&\quad=\sum_{j_{n+1}}\mathbb{P}(X_{n+2}=j_{n+2}\,|\,X_{n+1}=j_{n+1})\mathbb{P}(X_{n+1}=j_{n+1}\,|\,X_n=j_n) && (246)\\
&\quad=\sum_{j_{n+1}}\mathbb{P}(X_{n+2}=j_{n+2},\,X_{n+1}=j_{n+1}\,|\,X_n=j_n) && (247)\\
&\quad=\mathbb{P}(X_{n+2}=j_{n+2}\,|\,X_n=j_n). && (248)
\end{aligned}$$

Note that the second equality applies the Markov property to each conditional probabilites to forget the past information. Using a similar argument, one can show that any distant future, X_{n+k} for $k\geq 1$, does not depend on the past $X_0,\dots,X_{n-1}$ given X_n. In fact, this is ture for 'infinite future' as well, meaning the entire process $(X_{n+k})_{k\geq 1}$. Here we give a formal statement.

Proposition 3.3.1. *Let $(X_n)_{n\geq 0}$ be a Markov chain on a finite state space Ω with transition matrix P. Then*

$$\mathbb{P}\left((X_{n+1},X_{n+2},\dots)\in U\,|\,X_n=x,\,(X_0,\dots,X_{n-1})\in B\right)=\mathbb{P}\left((X_{n+1},X_{n+2},\dots)\in U\,|\,X_n=x\right). \tag{249}$$

PROOF. Omitted. □

Example 3.3.2 (Absorption in Gambler's ruin)**.** Recall the Gambler's ruin Markov chain X_n, which is on the state space $\Omega=\{0,1,\dots,N\}$ and trasition matrix P with $P(i,i+1)=p$ for $i\in\{0,\dots,N-1\}$, $P(i-1,i)=1-p$ for $i\in\{1,\dots,N\}$, $P(0,0)=1P(N,N)=1$. We will assume the initial fortune is $X_0=x$. We will compute the probability of winning, that is, the probability that the Gambler's fortune X_n absorbs into N rather than to 0. As a function of x, denote

$$\begin{aligned}
h(x)&=\mathbb{P}_x(\text{state } N \text{ is reached before } 0) && (250)\\
&:=\mathbb{P}(\text{state } N \text{ is reached before } 0\,|\,X_0=x). && (251)
\end{aligned}$$

The goal is to solve for $h(x)$. Note that it has the following bounary values

$$h(0)=0,\qquad h(N)=1. \tag{252}$$

If we start at $x\in\{1,\dots,N-1\}$, then the Markov chain goes to $x+1$ with probability p and renews as a Gambler's ruin chain with initial state $x+1$; with probability $q:=1-p$, it goes to $x-1$ and renews there. Hence it should be evident that

$$h(x)=ph(x+1)+qh(x)\qquad \text{for } 0<x<N. \tag{253}$$

Noting that $p+q=1$, (254) can be re-written as

$$ph(x)+qh(x)=ph(x+1)+qh(x)\qquad \text{for } 0<x<N. \tag{254}$$

This gives

$$h(x+1)-h(x)=\frac{q}{p}(h(x)-h(x-1))\qquad \text{for } 0<x<N, \tag{255}$$

which is a multiplicative recursion for the differences $h(x+1)-h(x)$. It follows that

$$h(x+1)-h(x)=(q/p)^x h(1)\qquad \text{for } 0<x<N. \tag{256}$$

Summing over x and using one of the boundary condition $h(0) = 0$, we get

$$h(x+1) = h(1)\sum_{k=0}^{x}(q/p)^x \qquad \text{for } 0 < x < N \tag{257}$$

$$= \begin{cases} h(1)\frac{1-(q/p)^{x+1}}{1-(q/p)} & \text{if } p \neq q \\ h(1)(x+1) & \text{if } p = q = 1/2. \end{cases} \tag{258}$$

So it only remains to figure out $h(1)$. For this, we use the remaining boundary condition $h(N) = 1$ by setting $x = N-1$:

$$1 = h(N) = \begin{cases} h(1)\frac{1-(q/p)^{N}}{1-(q/p)} & \text{if } p \neq q \\ h(1)N & \text{if } p = q = 1/2. \end{cases} \tag{259}$$

It follows that $h(1) = \frac{1-(q/p)}{1-(q/p)^N}$ for $p \neq q$ and $h(1) = 1/N$ for $p = q = 1/2$. Thus, we conclude that

$$h(x) = \begin{cases} \frac{1-(q/p)^{x}}{1-(q/p)^N} & \text{if } p \neq q \\ x/N & \text{if } p = q = 1/2. \end{cases} \tag{260}$$

For instance, say we start the gamble with $x = 50$ dollor and win the whole game once we accumulate $N = 100$ dollors before hitting 0 dollor. If each game is fair ($p = q = 1/2$), the probabilty of winning the game is $h(50) = 50/100 = 0.5$. If it is a slightly unfavorable game with $p = 49/100$ and $q = 51/100$, then the probability of winning is $h(1) = \frac{1-(51/49)^{50}}{1-(51/49)^{100}} \approx 0.11917$. So, just make each game slightly unfavorable and your casino is on business! ▲

3.4. Strong Markov Property

In this section, we will discuss an extension of the Markov property stopped at random times. Namely, we would like to restart a Markov chain $(X_n)_{n\geq 1}$ at a (possibly random) time T and would like to say the future process $(X_n)_{n>T}$ is independent of the past $(X_n)_{0\leq n<T}$ given the present state X_T. Can we do this for any random time T? The short answer is no. For an illustration, let $(X_n)_{n\geq 0}$ be the gambler's ruin chain in Example 3.1.2 with initial fortune $X_0 = 1$ and cap $N = 100$. Consider three (possibly) random times:

1. $T = T_1 = 50$;
2. $T = T_2 =$ first time to reach \$50;
3. $T = T_3 =$ first time to reach \$50 that i immediately followed by \$51.

We will consider the probability of jumping from 50 to 51 from time T to $T+1$. If we can really restart the Markov chain at time T, this transition probability should all be p, the probability of winning one round of gamble. This is indeed the case for $T = T_1 = 50$ by using time-homogenity. On the other hand, it fails to be true for $T = T_3$, since $X_{T_3+1} = 51$ with probability 1 by the definition of T_3. In other words, the Markov chain is 'forced' to transition to 51 at time T_3, not following its transition probabilities. Roughly speaking, this change of behavior happends becuase *knowning the value of T_3 contains some information about the chain at future times*. Finally, how about $T = T_2$? At first glance, knowing the value of T_2 should not exert any information on how it would behave in the following steps, since the value of T_2 is completely determined by the history of the chain at prior times $n \leq T_2$. This leads us to the definition of 'stopping times', which we define for a general stochastic process and not necessarily only for Markov chains.

Definition 3.4.1 (Stopping time)**.** Let $(X_n)_{n\geq 0}$ be a stochastic process defined on a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let T be a random variable taking valus from $\mathbb{Z}_{\geq 0} \cup \{\infty\}$. Then T is a *stopping time* (for the

process $(X_n)_{n\geq 0}$) if for each nonnegative integer n, there is a subset $C_n \subseteq \Omega^{n+1}$ such that the following equality of events holds:

$$\{T = n\} = \{(X_0, X_1, \ldots, X_n) \in C_n\}. \tag{261}$$

In words, T is a stopping time for $(X_n)_{n\geq 0}$ if one can determine if $T = n$ by checking a certain condition for the history $X_0, X_1, \ldots, X_n$, without using any information about the future states $X_{n+1}, X_{n+2}, \ldots$. In this sense, the random time T_3 defined above is not a stopping time, but T_2 is, since we can write

$$\{T_2 = n\} = \{X_0 \neq 50,\, X_1 \neq 50,\, \ldots, X_{n-1} \neq 50,\, X_n = 50\}. \tag{262}$$

Example 3.4.2 (First hitting time is a stopping time)**.** Let $(X_n)_{n\geq 0}$ be a stochastic process defined on a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $A \subseteq \Omega$ be a subset of the sample space. We will define τ_A to be the 'first hitting time' of A, which is the first time that the process X_n enters into states in A:

$$\tau_A := \inf\{n \geq 0 \,|\, X_n \in A\}. \tag{263}$$

Here 'inf' stands for infimum, which agrees with the minimum if the set is nonempty, and conventionally taken to be ∞ if the set is empty. For example, $\tau_A = \infty$ means the process X_n never enters the states in A. In order to see that τ_A is indeed a stopping time, we simply mimic what we did in (262): For each $n \geq 0$,

$$\{\tau_A = n\} = \{X_0 \neq A,\, X_1 \neq A,\, \ldots, X_{n-1} \neq A,\, X_n \in A\}. \tag{264}$$

Similarly, define T_A to be the first time after time zero to hit some state in A:

$$T_A := \inf\{n \geq 1 \,|\, X_n \in A\}. \tag{265}$$

As before, T_A is also a stopping time since, for each $n \geq 0$,

$$\{T_A = n\} = \{X_1 \neq A,\, X_1 \neq A,\, \ldots, X_{n-1} \neq A,\, X_n \in A\}. \tag{266}$$

Hence, in general, first hitting times are stopping times. ▲.

Exercise 3.4.3. Let $(X_n)_{n\geq 0}$ be a stochastic process defined on a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $A \subseteq \Omega$ be a subset of the sample space. For each $k \geq 1$, let $T_A^{(k)}$ denote the kth time that the process X_n visits some state in A. That is,

$$T_A^{(m)} = \begin{cases} \inf\{n > T_A^{(m-1)} : X_n \in A\} & \text{if } T_A^{(m-1)} < \infty \\ \infty & \text{otherwise.} \end{cases} \tag{267}$$

Show that $T_A^{(k)}$ is a stopping time for all $k \geq 1$.

Now that we have introduced stopping times, we can state the strong Markov property of Markov chains, which states that one can restart Markov chains at stopping times.

Proposition 3.4.4 (Strong Markov property)**.** *Let $(X_n)_{n\geq 0}$ be a Markov chain on a finite state space Ω and let T denote a stopping time. Let A be any event determined by $(X_0, X_1, \ldots, X_T)$. Then for any states $x, u_1, \ldots, u_n \in \Omega$,*

$$\mathbb{P}(X_{T+1} = u_1, \ldots, X_{T+m} = u_m \,|\, T < \infty,\, A,\, X_T = x) = \mathbb{P}(X_1 = u_1, \ldots, X_m = u_m \,|\, X_0 = x). \tag{268}$$

PROOF. We will condition on the value of T and apply the usual Markov property. For instance, suppose $T = n$. Then

$$\begin{aligned} \mathbb{P}(X_{T+1} = u_1, \ldots, X_{T+m} = u_m \,|\, T = n,\, A,\, X_T = x) &= \mathbb{P}(X_{n+1} = u_1, \ldots, X_{n+m} = u_m \,|\, A,\, X_n = x) && (269)\\ &= \mathbb{P}(X_{n+1} = u_1, \ldots, X_{n+m} = u_m \,|\, X_n = x) && (270)\\ &= \mathbb{P}(X_1 = u_1, \ldots, X_m = u_m \,|\, X_0 = x), && (271) \end{aligned}$$

where the last equality uses time-homogeneity of the Markov chain and the second equality uses the Marko property. (Note that, A is determined by the history $X_0, \ldots, X_n$. Hence we can decompose the event $A \cap \{X_n = x\}$ by $A' \cap \{X_n = x\}$, where A' is an event determined by $X_0, \ldots, X_{n-1}$. Then we can forget A' in the conditioning by the Markov property.)

Now, in generall, we do not know the value of T. Hence we will use iterated expectation for probability (Exercise 1.4.3) to condition on the value of T and then average over all values of T:

$$\begin{aligned}
&\mathbb{P}(X_{T+1} = u_1, \dots, X_{T+m} = u_m \,|\, T < \infty,\, A,\, X_T = x) && (272)\\
&= \mathbb{E}_T\left[\mathbb{P}(X_{T+1} = u_1, \dots, X_{T+m} = u_m \,|\, T,\, A,\, X_T = x)\right] && (273)\\
&= \mathbb{E}_T\left[\mathbb{P}(X_1 = u_1, \dots, X_m = u_m \,|\, X_0 = x)\right] && (274)\\
&= \mathbb{P}(X_1 = u_1, \dots, X_m = u_m \,|\, X_0 = x). && (275)
\end{aligned}$$

Here, the third equailty follows from the observation we made in the first paragraph: If we know the value of $T < \infty$, the assertion holds. □

Example 3.4.5. Let $(X_n)_{n\geq 0}$ be the gambler's ruin chain in Example 3.1.2 with initial fortune $X_0 = 1$ and cap $N = 100$. Let $T = T_2$ denote the first hitting time of \$50. We have seen that it is a stopping time. So by the strong Markov property,

$$\mathbb{P}(X_{T+1} = 51 \,|\, X_T = 50) = \mathbb{P}(X_1 = 51 \,|\, X_0 = 50) = p. \tag{276}$$

▲.

3.5. Recurrence and Transience

We now have the sufficient tools to study the behavior of Markov chains. Throughout this section, we fix a Markov chain $(X_n)_{n\geq 0}$ on finite state space Ω with transition matrix P. Here are some quick highlights.

Elementary behavior of Markov chains:

(i) For each state $x \in \Omega$, the Markov chain X_n either visits x infinitely often with probability one (we say x is 'recurrent') or only finitely many times with probability one (we say x is 'transient').

(ii) Starting from any state, the Markov chain X_n must enter the set of recurrent states eventually. Once entering a recurrent state, it will never go back to any transient states.

(iii) The set of recurrent states is further decomposed into 'communicating classes'. The Markov chain eventually enters exactly one communicating class and stays there forever, visiting every states in that class infinitley often.

3.5.1. Definition of recurrence/transitions and equivalent conditions. There are multiple ways to define recurrence/transience of states under a Markov chain: using 1) return probabilities; 2) infinitely many returns; 3) expected number of returns. Our first approach is via return probabilities. Namely, for each state $x \in \Omega$, define

$$T_x = T_x^{(1)} := \inf\{n \geq 1 : X_n = x\}, \tag{277}$$

which is the first time $n \geq 1$ at which the chain X_n hits the state x. In Example 3.4.2, we have seen that T_x is a stopping time for each $x \in \Omega$. For states $x, y \in \Omega$, let

$$\rho_{xy} = \mathbb{P}(T_y < \infty \,|\, X_0 = x), \tag{278}$$

which is the probability that the chain X_n eventually visits y starting from x.

Definition 3.5.1. Let $(X_n)_{n\geq 0}$ be a Markov chain on finite state space Ω. Each state $x \in \Omega$ is said to be *recurrent* if $\rho_{xx} = 1$ and *transient* if $\rho_{xx} < 1$.

Exercise 3.5.2. Show the following implications:

$$\rho_{xx} > 0 \qquad \Longleftrightarrow \qquad P^m(x,x) > 0 \text{ for some } m \geq 1. \tag{279}$$

Hint: Use

$$\sup_{n\geq 1} \mathbb{P}\left(X_n = x \,\middle|\, X_0 = x\right) \leq \mathbb{P}\left(\bigcup_{n=1}^{\infty} \{X_n = x\} \,\middle|\, X_0 = x\right) \leq \sum_{n=1}^{\infty} \mathbb{P}\left(X_n = x \,\middle|\, X_0 = x\right). \tag{280}$$

Proposition 3.5.3. *Fix states $x, y \in \Omega$ and let $T_y^{(m)}$ denote the time of mth visit to y, as defined in Exercise 3.4.3. Then for each $m \geq 1$,*

$$\mathbb{P}(T_y^{(m)} < \infty \,|\, X_0 = x) = \rho_{xy}\rho_{yy}^{m-1}. \tag{281}$$

PROOF. We first note that the assertion for $m = 1$ follows from the definition of ρ_{xy}:

$$\mathbb{P}(T_y^{(1)} < \infty \,|\, X_0 = x) = \rho_{xy}. \tag{282}$$

Next, suppose the assertion holds for $m = k$. We deduce the assertino for $m = k+1$ by using the strong Markov property. Namely, note that since $T_y^{(k+1)} > T_y^{(k)}$, $T_y^{(k+1)} < \infty$ implies $T_y^{(k)} < \infty$. Also, note that $X_{T_y^{(k)}} = y$ by definition. Hence,

$$\begin{aligned}
\mathbb{P}(T_y^{(k+1)} < \infty \,|\, X_0 = x) &= \mathbb{P}(T_y^{(m+1)} < \infty, T_y^{(k)} < \infty \,|\, X_0 = x) && (283)\\
&= \mathbb{P}(T_y^{(m+1)} < \infty \,|\, T_y^{(k)} < \infty, X_0 = x)\,\mathbb{P}(T_y^{(k)} < \infty \,|\, X_0 = x) && (284)\\
&= \mathbb{P}(T_y^{(m+1)} < \infty \,|\, X_{T_y^{(k)}} = y, T_y^{(k)} < \infty, X_0 = x)\,\mathbb{P}(T_y^{(k)} < \infty \,|\, X_0 = x) && (285)\\
&= \mathbb{P}(T_y^{(1)} < \infty \,|\, X_0 = y)\,\mathbb{P}(T_y^{(k)} < \infty \,|\, X_0 = x) && (286)\\
&= \rho_{yy}(\rho_{xy}\rho_{yy}^{k-1}) = \rho_{xy}\rho_{yy}^{k}, && (287)
\end{aligned}$$

where the last two equalities use the strong Markov property and the induction hypothesis, respectively. □

Theorem 3.5.4. *The following hold:*

(i) *$x \in \Omega$ is recurrent if and only if $\mathbb{P}(T_x^{(m)} < \infty$ for all $m \geq 1 \,|\, X_0 = x) = 1$.*
(ii) *$x \in \Omega$ is transient if and only if $\mathbb{P}(T_x^{(m)} < \infty$ for all $m \geq 1 \,|\, X_0 = x) = 0$.*

PROOF. Denote the probability in the assertion as p^*. Note that for each $m \geq 1$, by monotonicity of events,

$$p^* = \mathbb{P}(T_x^{(m)} < \infty \text{ for all } m \geq 1 \,|\, X_0 = x) \leq \mathbb{P}(T_x^{(m)} < \infty \,|\, X_0 = x) = \rho_{xx}^m. \tag{288}$$

Hence if $\rho_{xx} < 1$, $\rho_{xx}^m \to 0$ as $m \to \infty$ so the $p^* = 0$. Hence $p^* = 0$ if x is transient.

On the other hand, suppose x is recurrent. Then by subadditivity (3),

$$\mathbb{P}\left(\bigcup_{m=1}^{\infty} \{T_x^{(m)} = \infty\} \,\middle|\, X_0 = x\right) \leq \sum_{m=1}^{\infty} \mathbb{P}\left(T_x^{(m)} = \infty \,|\, X_0 = x\right) = 0. \tag{289}$$

Hence the probability on the left hand side equals zero. By taking the complement,

$$1 = 1 - \mathbb{P}\left(\bigcup_{m=1}^{\infty} \{T_x^{(m)} = \infty\} \,\middle|\, X_0 = x\right) = \mathbb{P}\left(\bigcap_{m=1}^{\infty} \{T_x^{(m)} < \infty\} \,\middle|\, X_0 = x\right) = p^*. \tag{290}$$

This shows $p^* = 1$ if x is recurrent.

Finally, the converse implications are automatic, since each state $x \in \Omega$ is either recurrent or transient, mutually exclusive, by definition. (Note that this also implies p^* is either 0 or 1. In other words, with probability one, either x is visited infinitely often or only finitely many times.) □

Next, consider the following counting variable

$$N_y := \sum_{n=1}^{\infty} \mathbf{1}(X_n = y) = \sum_{m=1}^{\infty} \mathbf{1}(T_y^{(m)} < \infty), \tag{291}$$

which equals the total number of visits to state $y \in \Omega$ by the Markov chain. By using Fubini's theorem, one can write

$$\mathbb{E}[N_y \,|\, X_0 = x] = \mathbb{E}\left[\sum_{m=1}^{\infty} \mathbf{1}(T_y^{(m)} < \infty) \,\middle|\, X_0 = x\right] = \sum_{m=1}^{\infty} \mathbb{E}\left[\mathbf{1}(T_y^{(m)} < \infty) \,\middle|\, X_0 = x\right] \tag{292}$$

$$= \sum_{m=1}^{\infty} \mathbb{P}\left(T_y^{(m)} < \infty \,|\, X_0 = x\right) \tag{293}$$

$$= \sum_{m=1}^{\infty} \rho_{xy}\rho_{yy}^{m-1}. \tag{294}$$

Note that the last expression equals zero if $\rho_{xy} = 0$, infinity if $\rho_{xy} > 0$ and $\rho_{yy} = 1$, and a converging geometric series if $\rho_{xy} > 0$ and $\rho_{yy} < 1$. Hence we obtain the following result:

Proposition 3.5.5. *The following hold:*

$$\mathbb{E}[N_y \,|\, X_0 = x] = \begin{cases} 0 & \text{if } \rho_{xy} = 0 \\ \infty & \text{if } \rho_{xy} > 0 \text{ and } y \text{ is recurrent} \\ \frac{\rho_{xy}}{1-\rho_{yy}} & \text{if } \rho_{xy} > 0 \text{ and } y \text{ is transient.} \end{cases} \tag{295}$$

Now we can give an alternative characterization of the recurrence of a state in terms of the sum of multi-step transition probabilities.

Theorem 3.5.6. *For each $x, y \in \Omega$, it holds that*

$$\mathbb{E}[N_y \,|\, X_0 = x] = \sum_{n=1}^{\infty} P^n(x, y). \tag{296}$$

Furthermore, a state $x \in \Omega$ is recurrent if and only if $\mathbb{E}[N_x \,|\, X_0 = x] = \sum_{m=1}^{\infty} P^m(x, x) = \infty$.

PROOF. Recall that $N_y = \sum_{n=1}^{\infty} \mathbf{1}(X_n = y)$. Hence

$$\mathbb{E}[N_y \,|\, X_0 = x] = \mathbb{E}\left[\sum_{n=1}^{\infty} \mathbf{1}(X_n = y) \,\middle|\, X_0 = x\right] = \sum_{n=1}^{\infty} \mathbb{E}\left[\mathbf{1}(X_n = y) \,\middle|\, X_0 = x\right] \tag{297}$$

$$= \sum_{n=1}^{\infty} \mathbb{P}\left(X_n = y \,\middle|\, X_0 = x\right) = \sum_{n=1}^{\infty} P^n(x, y), \tag{298}$$

verifying the first assertion. But Proposition 3.5.5 implies that x is recurrent if and only if $\mathbb{E}[N_x \,|\, X_0 = x] = \infty$. Hence the assertion follows. □

3.5.2. Recurrence and transience of random walks on $\mathbb{Z}^d$. In this subsection, we study recurrence and transience of simple random walks on the integer lattice $\mathbb{Z}^d$ in d-dimensions. As a set, $\mathbb{Z}^d$ consists of the d tuple of integer coordinates: $\mathbb{Z}^d = \{(a_1, \ldots, a_d) \,|\, a_1, \ldots, a_d \in \mathbb{Z}\}$. We will view it as a graph by giving 'nearest neighbor edges'. Namely, if we have two points $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^d$, then we connect $\mathbf{x}$ and $\mathbf{y}$ by an edge if and only if they differ by 1 at exactly a single coordiante. For example, in $\mathbb{Z}^2$, $(1,2)$ is adjacent to $(2,1)$ but not adjacent to $(2,3)$. As a graph, $\mathbb{Z}^2$ looks like a square grid in $\mathbb{R}^2$. Similarly, $\mathbb{Z}^3$ would be the 3-dimensional square lattice consisting of the grid points (a, b, c) for $a, b, c \in \mathbb{Z}$.

A random ralk on any graph is simple if it only moves along edges and do not hop multiple edges, and *symmetric* if at any node, there is an equal chance to pick any neighbor to move to[1]. Hence a simple symmetric random walk (SSRW) on $\mathbb{Z}$ moves either up or down with equal probability; SSRW on $\mathbb{Z}^2$ chooses one of the 4 directions with equal probability and moves along one edge; SRW on $\mathbb{Z}^d$ has $2d$ directions to choose from at each iteration with equal probability.

[1] In this lecture note, we assume all random walks on graphs are simple.

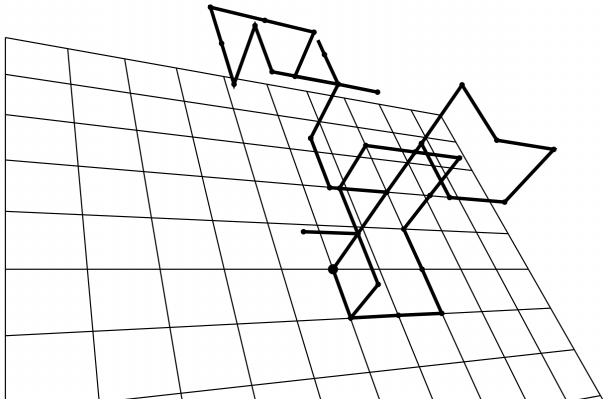


FIGURE 6. A sample path of RW on $\mathbb{Z}^3$. Figure excerpted from http://www.cmat.edu.uy/~lessa/resource/randomwalknotes.pdf

The goal of the following serious of examples is to show the following Polya's random walk theorem[2]:

$$\textit{SSRW on } \mathbb{Z}^d \textit{ is recurrent if } d \le 2 \textit{ and transient for } d \ge 3. \tag{299}$$

In Example 3.5.7, we have seen that SRW on $\mathbb{Z}$ is recurrent. For higher dimensions, we will compute the sum of return probabilities as we did before.

Example 3.5.7 (Recurrence and transience of simple random walk on $\mathbb{Z}$)**.** Let $(X_n)_{n\ge 0}$ denote a simple random walk on the 1-dimensional integer lattice $\mathbb{Z}$. Namely, it is a Markov chain with state space $\mathbb{Z}$ and transition matrix P such that $P(x, x+1) = p$ and $P(x, x-1) = 1-p$ for all $x \in \mathbb{Z}$. Here $p \in [0,1]$ is a parameter that governs the bias in one-step transition of the walker. Simply put, the walker moves up by 1 with probability p and down by 1 with probability $1-p$. Suppose $X_0 = 0$, meaning that the walkter is originally at the origin.

The question we ask is as follows: Is the origin transient or recurrent? The answer depends on p. If $p > 1/2$, then the walker has a positive bias, so it would be natural to guess the origin is transient – it will drift away toward the infinity and will never get back to the origin after some time. Likewise, if $p < 1/2$, it will drift away toward negative infinity and the origin should be transient. Lastly, if the walker is symmetric, $p = 1/2$, maybe the origin is recurrent. We will justify these guesses using Theorem 3.5.6.

The key observation here is the following combinatorial fact: In order to return to the origin, the walker must take an even number of steps with equal number of up-steps and down-steps. It follows that

$$\mathbb{P}(X_{2n} = 0 \,|\, X_0 = 0) = \binom{2n}{n} p^n (1-p)^n = \frac{(2n)!}{n!n!} (p(1-p))^n. \tag{300}$$

We will need to understand how $n!$ grows as n grows. We will apply Stirling's approximation, which states that

$$n! \sim (n/e)^n \sqrt{2\pi n}, \tag{301}$$

where $a_n \sim b_n$ means $a_n / b_n \to 1$ as $n \to \infty$. Using this, one can show that

$$\mathbb{P}(X_{2n} = 0 \,|\, X_0 = 0) \sim \frac{(2n/e)^{2n}\sqrt{4\pi n}}{(n/e)^{2n}(2\pi n)} (p(1-p))^n = \frac{1}{\sqrt{\pi n}} (4p(1-p))^n. \tag{302}$$

Now consider the symmetric case $p = 1/2$. Then $\mathbb{P}(X_{2n} = 0) \sim 1/\sqrt{\pi n}$. This implies that for some constant $c > 0$, $\mathbb{P}(X_{2n} = 0) \ge c/\sqrt{\pi n}$ for all $n \ge 1$. Now in order to see the recurrence of the origin, we

[2] Kakutani at UCLA colloquium said: "A drunk man will eventually find his way home, but a drunk bird may get lost forever"

compute the sum of all return probabilities:

$$\sum_{n=0}^{\infty} \mathbb{P}(X_n = 0 \,|\, X_0 = 0) = \sum_{n=0}^{\infty} \mathbb{P}(X_{2n} = 0 \,|\, X_0 = 0) \geq \sum_{n=0}^{\infty} \frac{c}{\sqrt{\pi n}} = \infty, \tag{303}$$

where the first equality uses the fact that the walker cannot return to the origin in odd number of steps, and the last equality uses the fact that $\sum_{n=0}^{\infty} 1/\sqrt{n} = \infty$. Hence by Theorem 3.5.6, we deduce that the origin is recurrent if $p = 1/2$. By Theorem 3.5.4 this means that the walker returns to the origin infinitely many times with probability one. By Proposition 3.5.5, this also means that the expected number of visits to the origin is infinite.

On the other hand, suppose $p \neq 1/2$. Recall that the function $p \mapsto p(1-p)$ attains its global maximum of $1/4$ at $p = 1/2$. Hence $0 \leq 4p(1-p) < 1$ if $p \in [0,1]$ and $p \neq 1/2$. In this case, for some constant $c > 0$, $\mathbb{P}(X_{2n} = 0 \,|\, X_0 = 0) \leq c\rho^n$ for all $n \geq 1$, where $\rho = 4p(1-p) \in (0,1)$. Hence

$$\sum_{n=0}^{\infty} \mathbb{P}(X_n = 0 \,|\, X_0 = 0) = \sum_{n=0}^{\infty} \mathbb{P}(X_{2n} = 0 \,|\, X_0 = 0) \leq \sum_{n=0}^{\infty} c\rho^n < \infty. \tag{304}$$

So by Theorem 3.5.6, we deduce that the origin is transient if $p \neq 1/2$. By Theorem 3.5.4 this means that the walker returns to the origin only finitely many times with probability one. By Proposition 3.5.5, this also means that the expected number of visits to the origin is finite. ▲

Example 3.5.8 (SSRW on $\mathbb{Z}^2$ is recurrent ("A drunk man eventually finds way home"))**.** Let $\mathbf{X}_n$ denote a SSRW on $\mathbb{Z}^2$, which chooses one of the 4 steps $[1,0]$, $[-1,0]$, $[0,1]$, $[0,-1]$ independently with equal probability at each move. It would be easy to analyze the return probability of $\mathbf{X}_n$ if we can treat its two coordinates as independent random walks on $\mathbb{Z}$, since we have already analyze one-dimensional random walks. However, the two coordinates of $\mathbf{X}_n$ are *not* independent. For example, if its first coordinate increased or decreased, then we know that its second coordinate did not change.

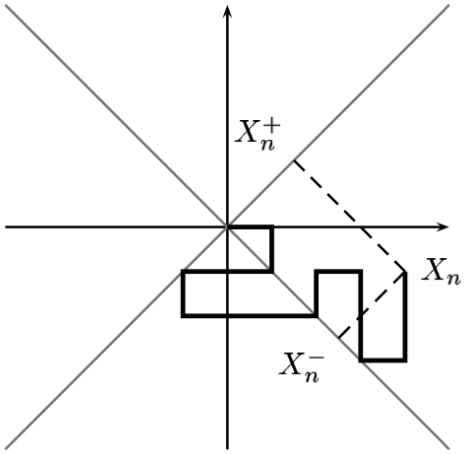


FIGURE 7. A sample path of RW on $\mathbb{Z}^2$. Projecting it on $y = \pm x$ shows that it has the same distribution as a pair of independent random walks on $\mathbb{Z}$. Figure excerpted from http://www.cmat.edu.uy/~lessa/resource/randomwalknotes.pdf

In order to handle this issue, we are going to compare $\mathbf{X}_n$ with the following auxiliary process $\mathbf{Y}_n = [Y_n^{(1)}, Y_n^{(2)}]$, where $Y_n^{(1)}$ and $Y_n^{(2)}$ are independent SSRW on $\mathbb{Z}$. So each $Y_n^{(i)}$ changes by ± 1 with equal probability, independently for each $i = 1,2$:

$$\mathbb{P}(Y_{n+1}^{(i)} - Y_n^{(i)} = 1) = \mathbb{P}(Y_{n+1}^{(i)} - Y_n^{(i)} = -1) = 1/2 \quad \text{for } i = 1,2. \tag{305}$$

Now, since $Y_n^{(1)}$ and $Y_n^{(2)}$ change by ± 1 at the same time, the vector $\mathbf{Y}_n = [Y_n^{(1)}, Y_n^{(2)}]$ on $\mathbb{Z}^2$ changes by $[1,1]$, $[-1,1]$, $[1,-1]$, $[-1,-1]$ with equal probability. This is exactly the original SSRW $\mathbf{X}_n$ rotated by 45 degrees

and scaled by $\sqrt{2}$, so that it lives on the 'checkerboard lattice' $\{(a,b) \mid a,b \in \mathbb{Z}, a+b \text{ is even}\} \subseteq \mathbb{Z}^2$. From this observation, it follows that the probability of $\mathbf{Y}_n$ returning to the origin is the same as the probability that $\mathbf{X}_n$ returning to the origin. This gives

$$\begin{aligned}
\mathbb{P}(\mathbf{X}_n = [0,0] \,|\, \mathbf{X}_0 = [0,0]) &= \mathbb{P}(\mathbf{Y}_n = [0,0] \,|\, \mathbf{Y}_0 = [0,0]) && (306)\\
&= \mathbb{P}\left(Y_n^{(1)} = 0,\, Y_n^{(2)} = 0 \,|\, Y_0^{(1)} = 0,\, Y_0^{(2)} = 0\right) && (307)\\
&= \mathbb{P}\left(Y_n^{(1)} = 0 \,|\, Y_0^{(1)} = 0\right) \mathbb{P}\left(Y_n^{(2)} = 0 \,|\, Y_0^{(2)} = 0\right) && (308)\\
&= \mathbb{P}\left(Y_n^{(1)} = 0 \,|\, Y_0^{(1)} = 0\right)^2. && (309)
\end{aligned}$$

where the last equality is due to the independence between $Y_n^{(1)}$ and $Y_n^{(2)}$. By using the analysis in Example 3.5.7, we get

$$\mathbb{P}(\mathbf{X}_{2n} = [0,0] \,|\, \mathbf{X}_0 = [0,0]) \sim \frac{1}{\pi n}. \tag{310}$$

To conclude, note that $\mathbf{X}_n$ can only return to the origin at even number of steps, and the above asymptotic yields that for some constant $c > 0$, $\mathbb{P}(\mathbf{X}_{2n} = [0,0] \,|\, \mathbf{X}_0 = [0,0]) \geq c/(\pi n)$ for all $n \geq 1$. Hence

$$\sum_{n=0}^{\infty} \mathbb{P}(\mathbf{X}_n = 0 \,|\, \mathbf{X}_0 = 0) = \sum_{n=0}^{\infty} \mathbb{P}(\mathbf{X}_{2n} = 0 \,|\, \mathbf{X}_0 = 0) \geq \sum_{n=0}^{\infty} \frac{c}{\pi n} = \infty. \tag{311}$$

Therefore $\mathbf{X}_n$ is recurrent by Theorem 3.5.6. ▲

Example 3.5.9 (SSRW on $\mathbb{Z}^d$ for $d \geq 3$ is transient ("A drunk bird never finds way home"))**.** Let $\mathbf{X}_n$ denote a SSRW on $\mathbb{Z}^d$ with $d \geq 3$, which chooses one of the $2d$ steps $[\pm 1, 0, \ldots, 0]$, $[0, \pm 1, 0, \ldots, 0]$, $\ldots$ $[0, \ldots, 0, \pm 1]$ independently with equal probability at each move. We will first apply a similar argument as in Example 3.5.8 and try to show that $\mathbf{X}_n$ is transient. This simple argument almost works, but we will see why it fails.

As we did in the two dimensional case, consider a process $\mathbf{Y}_n = [Y_n^{(1)}, \ldots, Y_n^{(d)}]$ on $\mathbb{Z}^d$ where $Y_n^{(i)}$'s are independent SSRWs on $\mathbb{Z}$. So each $Y_n^{(i)}$ changes by ± 1 with equal probability, independently for each $i = 1, \ldots, d$. As before, note that

$$\begin{aligned}
\mathbb{P}(\mathbf{X}_n = [0, \ldots, 0] \,|\, \mathbf{X}_0 = [0, \ldots, 0])\text{“ = ”}\mathbb{P}(\mathbf{Y}_n = [0, \ldots, 0] \,|\, \mathbf{Y}_0 = [0, \ldots, 0]) && (312)\\
= \mathbb{P}\left(Y_n^{(1)} = 0, \ldots, Y_n^{(d)} = 0 \,|\, Y_0^{(1)} = 0, \ldots\, Y_n^{(d)} = 0\right) && (313)\\
= \mathbb{P}\left(Y_n^{(1)} = 0 \,|\, Y_0^{(1)} = 0\right)^d. && (314)
\end{aligned}$$

Then by using the analysis in Example 3.5.7,

$$\mathbb{P}(\mathbf{X}_{2n} = \mathbf{0} \,|\, \mathbf{X}_0 = \mathbf{0}) \sim \left(\frac{1}{\sqrt{\pi n}}\right)^d. \tag{315}$$

The above probabilities are not summable if $d \geq 3$, as $\sum_{n=1}^{\infty} n^{-d/2} = \infty$ for $d \geq 3$. Hence by Theorem 3.5.6, conclude that $\mathbf{X}_n$ is transient.

The reason that the above argument fails is that the quoted equality in (313) is not true. For example, set $d = 3$. Then there are 6 steps that $\mathbf{X}_n$ takes: $[\pm 1, 0, 0,]$, $[0, \pm 1, 0]$, and $[0, 0, \pm 1]$. However, $\mathbf{Y}_n$ can take $2^3 = 8$ distinct steps in the integer lattice $\mathbb{Z}^3$ (see Figure 8). For a correct argument, we rely on a combinatorial argument we employed in Example 3.5.7, which we will demonstrate in the case of $d = 3$.

Let $d = 3$. Note that the walk can only return to the origin in even number of steps. Let a, b, c denote the number of $[1,0,0]$, $[0,1,0]$, $[0,0,1]$-steps that $\mathbf{X}_n$ takes in the first $2n$ steps, starting from the origin. In order for $\mathbf{X}_{2n} = \mathbf{0}$, we need to have $a+b+c = n$ and the number of $[-1,0,0]$, $[0,-1,0]$, $[0,0,-1]$-steps must

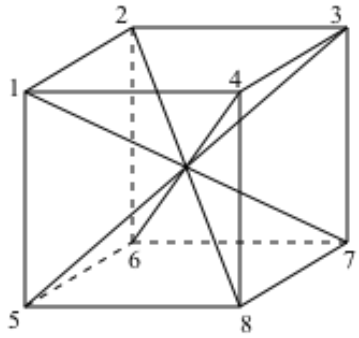


FIGURE 8. A triple of independent SSRWs on $\mathbb{Z}$ has 8 possible moves $\{-1,1\}^3$

be a,b,c, respectively. Hence

$$\mathbb{P}(\mathbf{X}_n = \mathbf{0} \,|\, \mathbf{X}_0 = \mathbf{0}) = \sum_{\substack{0\le a,b,c \\ a+b+c=n}} \frac{(2n)!}{(a!b!c!)^2} \left(\frac{1}{6}\right)^{2n} \tag{316}$$

$$= \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} \sum_{\substack{0\le a,b,c \\ a+b+c=n}} \left(\frac{n!}{(a!b!c!)}\right)^2 \left(\frac{1}{3}\right)^{2n}, \tag{317}$$

where we have used the identity $\frac{(2n)!}{(a!b!c!)^2} = \binom{2n}{n}\left(\frac{n!}{(a!b!c!)}\right)^2$. Now note that

$$1 = \sum_{\substack{0\le a,b,c \\ a+b+c=n}} \frac{n!}{(a!b!c!)} \left(\frac{1}{3}\right)^{n}, \tag{318}$$

since the right hand side equals the total probability of putting n balls into three bins uniformly at random. Also, observe that

$$\frac{n!}{(a!b!c!)} \le \frac{n!}{\lfloor n/3\rfloor!\lfloor n/3\rfloor!\lfloor n/3\rfloor!} \tag{319}$$

for all $a,b,c \ge 0$ with $a+b+c=n$ and $\lfloor n/3\rfloor$ is the largest smaller integer than $n/3$. In words, one has the most amount of arrangements of n balls into three bins if the number of balls in each bins are as equal as possible. Using this,

$$\mathbb{P}(\mathbf{X}_{2n} = \mathbf{0}) = \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} \sum_{\substack{0\le a,b,c \\ a+b+c=n}} \left(\frac{n!}{(a!b!c!)}\right)^2 \left(\frac{1}{3}\right)^{2n} \tag{320}$$

$$\le \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} \frac{n!}{\lfloor n/3\rfloor!\lfloor n/3\rfloor!\lfloor n/3\rfloor!} \left(\frac{1}{3}\right)^{n} \tag{321}$$

$$\sim \frac{1}{\sqrt{\pi n}} \frac{(n/e)^n\sqrt{2\pi n}}{(n/3e)^n\sqrt{2\pi n/3}^3} \left(\frac{1}{3}\right)^{n} = O(n^{-3/2}). \tag{322}$$

From this, one can deduce that $\sum_{n=1}^{\infty} \mathbb{P}(\mathbf{X}_n = \mathbf{0} \,|\, \mathbf{X}_0 = \mathbf{0}) \le \sum_{n=1}^{\infty} cn^{-3/2} < \infty$. Thus, by Theorem 3.5.6, conclude that $\mathbf{X}_n$ is transient. ▲

3.5.3. The set of recurrent states is closed. In this subsection, we will see that a Markov chain will eventually enter the set of recurrent states and will never leave it. This is proved in the following statement.

Definition 3.5.10 (Closed sets)**.** Fix a subset of states $A \subseteq \Omega$. We say A is *closed* if $\rho_{xy} = 0$ for all $x \in A$ and $y \in \Omega \setminus A$.

Proposition 3.5.11. *For distinct $x, y \in \Omega$, the following hold:*

(i) *Suppose $\rho_{xy} > 0$ and $\rho_{yx} < 1$. Then x is transient.*

(ii) *Suppose $\rho_{xy} > 0$ and $\rho_{xx} = 1$. Then $\rho_{yx} = 1$ and y is recurrent.*

(iii) *Suppose x is recurrent and y is transient. Then $\rho_{xy} = 0$. In particular, the set of recurrent states is closed.*

PROOF. To show **(i)**, suppose $\rho_{xy} > 0$ and $\rho_{yx} < 1$. Heuristically, this means that the chain can go from x to y with some positive probability, but there is a positive chance $1 - \rho_{yx} > 0$ that it will never return to x. Then it does not visit x any more, so x should be transient. To make this argument rigorous, we note that

$$\begin{aligned} 1 - \rho_{xx} &= \mathbb{P}(X_n \text{ never visits } x \,|\, X_0 = x) && (323)\\ &\geq \mathbb{P}(X_n \text{ visits } y \text{ and never returns to } x \,|\, X_0 = x) && (324)\\ &= \mathbb{P}(X_n \text{ never returns to } x \text{ after time } T_y \,|\, X_0 = x)\, \mathbb{P}(T_y < \infty \,|\, X_0 = x) && (325)\\ &= \mathbb{P}(X_n \text{ never returns to } x \,|\, X_0 = y)\, \rho_{xy}. && (326)\\ &= (1 - \rho_{yx})\rho_{xy} > 0, && (327) \end{aligned}$$

where we have used the strong Markov property to get the third equality. This shows $\rho_{xx} < 1$, so x is transient.

Next, in order to show **(ii)**, suppose $\rho_{xy} > 0$ and $\rho_{xx} = 1$. In this case, we first observe that $\rho_{yx} = 1$, since otherwise by part **(i)** x must be transient, which contradicts the assumption that $\rho_{xx} = 1$. Hence, heuristically, assuming $X_0 = y$, the chain almost surely visits x and there is a positive chance $\rho_{yx} > 0$ to return to y; if it fails, then it will eventually return to y and by the strong Markov property we can start the game anew. Among this infinitely many independent trials, it should succeed at least once since the success probability is $\rho_{xy}\rho_{xy} = \rho_{xy} > 0$.

More rigorously, since $\rho_{xy}, \rho_{yx} > 0$, there are integers $k, \ell \geq 1$ such that $P^{(k)}(x, y) > 0$ and $P^{\ell}(y, x) > 0$ by Exercise 3.5.2. Hence

$$\begin{aligned} \sum_{n=1}^{\infty} P^n(y, y) \geq \sum_{n=1}^{\infty} P^{k+n+\ell}(y, y) &\geq \sum_{n=1}^{\infty} P^k(y, x) P^n(x, x) P^{\ell}(y, x) && (328)\\ &= P^k(y, x) \left(\sum_{n=1}^{\infty} P^n(x, x) \right) P^{\ell}(y, x) = \infty. && (329) \end{aligned}$$

The last equality follows since $P^{(k)}(x, y), P^{\ell}(y, x) > 0$ and $\sum_{n=1}^{\infty} P^n(x, x) = \infty$ as x is recurrent (see Theorem 3.5.4). Then by Theorem 3.5.4, we deduce that y is also recurrent.

Lastly, **(iii)** is a direct consequence of **(ii)** (proof by contradiction). Heuristically, since x is visited infinitely often, any positive chance to access y every time x will make y also recurrent. This implies that the set of all recurrent states is closed. □

3.5.4. Cannonical decomposition of the state space. In this subsection, we discuss a generic structure about the state space Ω of a Markov chain.

Definition 3.5.12 (Accessibility)**.** For two states $x, y \in \Omega$, we say y is *accessible* from x, and denote as $x \longrightarrow y$, if $P^n(x, x) > 0$ for some $n \geq 0$. If x and y are mutually accessible from each other, we say x and y *communicate* and dnoete as $x \longleftrightarrow y$.

From Exercise 3.5.2, It is easy to observe that $x \longrightarrow y$ if and only if $\rho_{xy} > 0$ or $x = y$. The following exercise shows that the communication relation $\longleftrightarrow$ is an equivalence relation.

Exercise 3.5.13. Show that the binary relation '$\longleftrightarrow$' between the states defined in Definition 3.5.14 is an equivalence relation. Namely, it satisfies the following properties:

(1) (reflexivity) For each $x \in \Omega$, $x \longleftrightarrow x$.
(2) (symmetry) For each $x, y \in \Omega$, $x \longleftrightarrow y$ if and only if $y \longleftrightarrow x$.
(3) (translativity) For each $x, y, z \in \Omega$, $x \longleftrightarrow y$ and $y \longleftrightarrow z$ implies $x \longleftrightarrow z$.
(*Hint:* Using Exercise 3.1.6, show that $x \longrightarrow y$ and $y \longrightarrow z$ implies $x \longrightarrow z$.)

Next, we define irreducible and closed subsets. In words, a set of states is called 'irreducible' if all states there communicate, and 'closed' if it never leaves there once it gets there.

Definition 3.5.14 (Irreducible sets)**.** Fix a subset of states $A \subseteq \Omega$. We say A is *irreducible* if $x \longleftrightarrow y$ for all $x, y \in A$ and *closed* if $\rho_{xy} = 0$ for all $x \in A$ and $y \in \Omega \setminus A$.

Now for each state $x \in \Omega$, define its *communicating class* $\mathscr{R}_x$ by

$$\mathscr{R}_x = \{y \in \Omega \mid x \longleftrightarrow y\}. \tag{330}$$

A direct consequence of communication $\longleftrightarrow$ being an equivalence relation between states is that the state space Ω is partitioned into pairwise disjoint communicating classes. While transient states may form multiple communicating classes, it is customary to put them in one subset as stated in Theorem 3.5.15.

Theorem 3.5.15 (Cannonical decomposition of the state space)**.** *The state space Ω of a Markov chain can be decomposed as*

$$\Omega = \mathscr{T} \sqcup \mathscr{R} = \mathscr{T} \sqcup \left(\bigcup_i \mathscr{R}_i \right), \tag{331}$$

where $\mathscr{T}$ is the set of all transient states, $\mathscr{R}$ is the set of all recurrent states, and $\{\mathscr{R}_i\}$ is at most a countable collection of pairwise disjoint, closed, and irreducible sets of recurrent states.

PROOF. The first decomposition $\Omega = \mathscr{T} \sqcup \mathscr{R}$ is trivial by definition of transience and recurrence. We will further decompose $\mathscr{R}$ into the distjoint union of communicating classes $\bigcup_i \mathscr{R}_i$. Indeed, note that each recurrent state x must belong to some communicating class $\mathscr{R}_i$, as certainly $x \in \mathscr{R}_x$. Also, two distinct recurrent states $x, y \in \Omega$ should either be in the same communicating class or belongs to disjoint communicating classes by the transitivity of communication (see Exercise 3.5.13).

It remains to show that each communicating class $\mathscr{R}_i$ is closed and irreducible. First, the irreducibility follows from the symmetry of communication (Exercise 3.5.13). Lastly, it remains to show that $\rho_{xy} = 0$ if $x \in \mathscr{R}_i$ and $y \in \Omega \setminus \mathscr{R}_i$ for each i. This is essentially a consequence of Proposition 3.5.11. Namely, if $y \in \mathscr{T}$, then $\rho_{xy} = 0$ by Proposition 3.5.11 **(iii)**. If $y \in \mathscr{R}_j$ for some $j \neq i$, then $\rho_{xy} = 0$ since otherwise $\rho_{yx} = 1$ by Proposition 3.5.11 **(ii)**, which means x and y communicate so they must belong to the same communicating class, a contradiction. □

Consequently, the transition matrix P can be written as the following block form (after permutting the rows and columns):

$$P = \begin{array}{c} \\ \mathscr{R} \\ \mathscr{T} \end{array} \begin{array}{c} \begin{array}{cc} \mathscr{R} & \mathscr{T} \end{array} \\ \begin{bmatrix} P_{\mathscr{R}} & O \\ S & P_{\mathscr{T}} \end{bmatrix} \end{array} = \begin{array}{c} \\ \mathscr{R}_1 \\ \mathscr{R}_2 \\ \vdots \\ \mathscr{R}_m \\ \mathscr{T} \end{array} \begin{array}{c} \begin{array}{ccccc} \mathscr{R}_1 & \mathscr{R}_2 & \cdots & \mathscr{R}_m & \mathscr{T} \end{array} \\ \begin{bmatrix} P_{\mathscr{R}_1} & O & & & O \\ O & P_{\mathscr{R}_2} & O & & O \\ \vdots & & \ddots & O & O \\ O & O & & P_{\mathscr{R}} & O \\ S_1 & S_2 & \cdots & S_m & P_{\mathscr{T}} \end{bmatrix} \end{array}, \tag{332}$$

where the last expression assumes that there are only finitely many communicating recurrent classes $\mathscr{R}_1, \ldots, \mathscr{R}_m$.

Example 3.5.16 (Gambler's ruin in cannonical form)**.** Consider gambler's ruin with state space $\Omega = \{0,1,2,3,4,5\}$ and transition matrix P given as (199). The set recurrent states is $\mathscr{R} = \{0,5\}$ and $\mathscr{T} = \{1,2,3,4\}$ is the set of transient states (why?). Then a canonical form for the transition matrix is

$$P = \begin{array}{c} \\ 0 \\ 5 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccccc} 0 & 5 & 1 & 2 & 3 & 4 \end{array} \\ \left[\begin{array}{cc|cccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \hline 1-p & 0 & 0 & p & 0 & 0 \\ 0 & 0 & 1-p & 0 & p & 0 \\ 0 & 0 & 0 & 1-p & 0 & p \\ 0 & p & 0 & 0 & 1-p & 0 \end{array}\right] \end{array}. \tag{333}$$

▲

Definition 3.5.17. Fix a Markov chain $(X_n)_{n\geq 1}$ on state space Ω with transition matrix P. We say the chain (as well as P) is said to be *irreducible* if Ω is irreducible, and *recurrent* if it is rreducible and all states are recurrent.

Proposition 3.5.18. *The following hold:*

(i) *A finite closed set $A \subseteq \Omega$ contains at least one recurrent state. In particular, if Ω is finite, then it contains at least one recurrent state.*

(ii) *If $A \subseteq \Omega$ is finite, closed, and irreducible, then all states in A are recurrent.*

(iii) *A finite state Markov chain is recurrent if it is irreducible.*

PROOF. Let N_y denote the total number of visits to y during $[0,\infty)$. Note that $\sum_{y\in A} N_y$ counts the total number of visits to any state in A, and it is infinite since A is closed. But then

$$\infty = \mathbb{E}\left[\sum_{y\in A} N_y\right] = \sum_{y\in A} \mathbb{E}\left[N_y\right]. \tag{334}$$

Since the last expression is a finite sum, it implies that $\mathbb{E}[N_y] = \infty$ for some $y \in A$. Then such $y \in A$ is recurrent by Theorem 3.5.6. For the second part, note that Ω is a closed subset of Ω, so it must contain a recurrent state by what we have just shown. This shows **(i)**.

To show **(ii)**, suppose $A \subseteq \Omega$ is finite, closed, and irreducible. According to **(i)**, A contains at least one recurrent state. But then by the irreducibility and Proposition 3.5.11, all states in A must be recurrent. This shows **(ii)**.

Lastly, **(iii)** follows immediately from **(ii)**. □

3.5.5. Genearlized Gambler's ruin and expected time until absorption. In this subsection, we generalize the Gambler's ruin example (Example 3.1.2) and see how we can analyze various statistics related to the first recurrent state ever reached.

Namely, let $(X_n)_{n\geq 0}$ be a Markov chain on a finite state space Ω with transition matrix P. We may decompose $\Omega = \mathscr{T} \sqcup \mathscr{R}$, where $\mathscr{T}$ is the set of all transient states, $\mathscr{R}$ is the set of all recurrent states. In Proposition 3.5.18, we have seen that any finite-state Markov chain contains at least one recurrent state, so $\mathscr{R}$ must be nonempty. We also assume that $\mathscr{T}$ is nonempty (just like gambler's ruin). Define

$$T_{\mathscr{R}} = \inf\{n \geq 0 \,|\, X_n \in \mathscr{R}\} = \text{first time to be in } \mathscr{R}. \tag{335}$$

Since eventually X_n must enter $\mathscr{R}$ (why?), $T_{\mathscr{R}}$ is finite almost surely. Then

$$X_{T_{\mathscr{R}}} = \text{first recurrent state ever reached.} \tag{336}$$

We are interested in solving the following matrix $U : \mathscr{T} \times \mathscr{R} \to [0,1]$, where

$$U(x,y) = \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_0 = x). \tag{337}$$

Namely, $U(x,y)$ is the probability that, starting from a transient state x, the first ever recurrent state reached is y.

In order for our analysis, we may use the elementary decomposition of P in (332):

$$P = \begin{array}{c} \\ \mathscr{R} \\ \mathscr{T} \end{array} \begin{array}{c} \begin{array}{cc} \mathscr{R} & \mathscr{T} \end{array} \\ \begin{bmatrix} P_{\mathscr{R}} & O \\ S & P_{\mathscr{T}} \end{bmatrix} \end{array}. \tag{338}$$

Proposition 3.5.19. *$U = S + P_{\mathscr{T}} U$. In particular, $U = (I - P_{\mathscr{T}})^{-1} S$.*

PROOF. As we did in the analysis of Gambler's ruin in Example 3.3.2, we will use a 'first step analysis', which is simply to condition on the state X_1 after the first move. Fix $x \in \mathscr{T}$ and $y \in \mathscr{R}$. By conditioning on X_1, we first write

$$\begin{aligned} U(x,y) &= \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_0 = x) && (339)\\ &= \sum_{z \in \Omega} \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_1 = z, X_0 = x) P(x,z) && (340)\\ &= P(x,y) + \sum_{z \in \mathscr{T}} \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_0 = z) P(x,z) && (341)\\ &= S(x,y) + \sum_{z \in \mathscr{T}} P_{\mathscr{T}}(x,z) U(z,y). && (342) \end{aligned}$$

We will justify (341) in more detail below. The last inequality above follows easily by using definitions of S and $P_{\mathscr{R}}$. Before delving into details, we note that the above result can be translated into the matrix equation $U = S + P_{\mathscr{R}} U$. This gives

$$(I - P_{\mathscr{R}}) U = S. \tag{343}$$

Using Exercise 3.5.20, we can derive $U = (I - P_{\mathscr{T}})^{-1} S$, as desired.

Now we justify (341). For the first term, we note that

$$\sum_{z \in \mathscr{R}} \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_1 = z, X_0 = x) P(x,z) = \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_1 = y, X_0 = x) P(x,y) = P(x,y). \tag{344}$$

Namely, if the chain jumps to a recurrent state $z \in \mathscr{R}$ at time 1, then the only contribution to $U(x,y)$ is when $z = y$. Second, we note that, by conditioning on the values of the stopping time $T_{\mathscr{R}}$ and using strong Markov property,

$$\begin{aligned} \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_1 = z, X_0 = x) &= \sum_{n=2}^{\infty} \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_1 = z, X_0 = x) && (345)\\ &= \sum_{n=2}^{\infty} \mathbb{P}(X_2 \in \mathscr{T}, \ldots, X_{n-1} \in \mathscr{T}, X_n = y \,|\, T_{\mathscr{R}} = n, X_1 = z, X_0 = x) && (346)\\ &= \sum_{n=2}^{\infty} \mathbb{P}(X_1 \in \mathscr{T}, \ldots, X_{n-2} \in \mathscr{T}, X_{n-1} = y \,|\, T_{\mathscr{R}} = n-1, X_0 = z) && (347)\\ &= \mathbb{P}(X_{T_{\mathscr{R}}} = y \,|\, X_0 = z). && (348) \end{aligned}$$

□

Exercise 3.5.20. Show that $I - P_{\mathscr{R}}$ is invertible and $(I - P_{\mathscr{R}})^{-1} = \sum_{n=0}^{\infty} (P_{\mathscr{R}})^n$.

Next, we turn our attention to the expected time until first enterring the set of recurrent states. Namely, we are interested in solving the following column vector $W : \mathscr{T} \to [0,1]$, where

$$W(x) = \mathbb{E}[T_{\mathscr{R}} \,|\, X_0 = x]. \tag{349}$$

So $W(x)$ is the expect time to first hit any recurrent state starting from a transient state x.

Proposition 3.5.21. *$W = \mathbf{1}_{\mathscr{T}} + P_{\mathscr{T}} W$. In particular, $W = (I - P_{\mathscr{T}})^{-1} \mathbf{1}_{\mathscr{T}}$. Here $\mathbf{1}_{\mathscr{T}}$ denotes the $|\mathscr{T}|$-dimensional column vector of all ones.*

PROOF. As before, we use the 'first step analysis'. Fix $x \in \mathscr{T}$ and $y \in \mathscr{R}$. By conditioning on X_1, we first write

$$W(x) = \mathbb{E}[T_{\mathscr{R}} \mid X_0 = x] \tag{350}$$
$$= \sum_{z\in\Omega} \mathbb{E}[T_{\mathscr{R}} \mid X_1 = z,\, X_0 = x] P(x,z) \tag{351}$$
$$= 1 + \sum_{z\in\Omega} \mathbb{E}[T_{\mathscr{R}} \mid X_0 = z] P(x,z) \tag{352}$$
$$= 1 + \sum_{z\in\mathscr{T}} \mathbb{E}[T_{\mathscr{R}} \mid X_0 = z] P(x,z) \tag{353}$$
$$= \mathbf{1}_{\mathscr{T}}(x) + \sum_{z\in\mathscr{T}} P_{\mathscr{T}}(x,z) W(z). \tag{354}$$

The justification (353) is similar as that of (341). This gives

$$(I - P_{\mathscr{R}})W = \mathbf{1}_{\mathscr{T}}. \tag{355}$$

Using Exercise 3.5.20, we can derive $W = (I - P_{\mathscr{T}})^{-1}\mathbf{1}_{\mathscr{T}}$, as desired. □

Exercise 3.5.22. Consider a Markov chain $(X_n)_{n\geq 0}$ on state space $\Omega = \{1,2,3,4,5,6,7,8,9\}$ whose transition probabilities are given by the following state-space diagram:

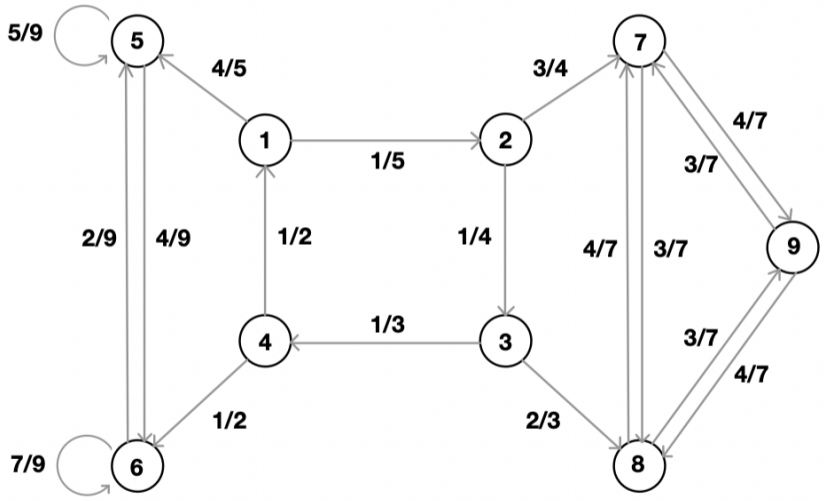


FIGURE 9. A triple of independent SSRWs on $\mathbb{Z}$ has 8 possible moves $\{-1,1\}^3$

(i) Show that $\mathscr{T} = \{1,2,3,4\}$ is the set of transient states and $\mathscr{R} = \{5,6,7,8,9\}$ is the set of recurrent states.

(ii) Let U denote the matrix $\mathscr{T} \times \mathscr{R} \to [0,1]$ where $U(x,y)$ is the probability that the chain first hits y among all recurrent states, starting from x. Compute U. You may use

$$\begin{bmatrix} 1 & -1/5 & 0 & 0 \\ 0 & 1 & -1/4 & 0 \\ 0 & 0 & 1 & -1/3 \\ -1/2 & 0 & 1 & 0 \end{bmatrix}^{-1} = \frac{1}{119}\begin{bmatrix} 120 & 24 & 6 & 2 \\ 5 & 20 & 30 & 10 \\ 20 & 4 & 120 & 40 \\ 60 & 12 & 3 & 120 \end{bmatrix}. \tag{356}$$

(iii) Compute $\rho_{1,5}$ and $\rho_{1,6}$. (Hint: Use U in part **(ii)**.)

(iv) For each $x \in \mathscr{T}$, let $W(x)$ denote the expected hitting time of any recurrent state starting from x. Compute $W : \mathscr{T} \to [0,\infty)$.

(v) Compute all stationary distributions of the Markov chain.

3.6. Existence and unieuqness of stationary distribution

Does a Markov chain always have a stationary distribution? If yes, when there is a unique stationary distribution? In Exercise 3.2.3 (only for finite state spaces) we have exploited the relation between stationary distribution and eigenvectors associated with eigenvalue 1 to study existence and uniqueness of stationary distribution. Namely, the all-one column vector is a right-eigenvector associated with eigenvalue 1 of a given transition matrix P. Hence its transpose P^T has an eigenvector v associated with eigenvalue 1. That is, $P^T v = v$. Then taking transpose, we get $v^T P = v^T$, so $\pi = v^T/D$, where D is the sum of all entries in v, is a stationary distribution of P. For the uniqueness, see part (iii) of Exercise 3.2.3.

However, the linear algebra approach do not provide us a useful intuition for the behavior of the Markov chain itself. In this section, we give an alternative and constructive argument to show that every Markov chain has a stationary distribution. This is valuable since we can write down what the stationary distribution is, whereas the linear algebra argument doesn't give this information.

For each $x, y \in \Omega$, let $V_n(x, y)$ denote the number of visits to y in the first n steps given that $X_0 = x$,

$$V_n(x, y) := \sum_{k=1}^{n} \mathbf{1}(X_k = y \,|\, X_0 = x). \tag{357}$$

We will see that with reasonable assumptions, a Markov chain will admit a unique stationary distribution given by

$$\pi(y) := \lim_{t\to\infty} \frac{1}{n} \mathbb{E}[V_n(x, y)], \tag{358}$$

which is the expected proportion of times spending at y starting at x and it turns out that it does not depend on x. The main result in this section, Theorem 3.6.4, states that when the Markov chain satisfies certain reasonable conditions, (381) is indeed a stationary distribution of the Markov chain. However, it is not a priori clear that the limit in (381) exists. Instead of taking the limiting frequency of visits to y, we will instead construct a measure by only counting visits to y during a single 'excursion' from x to x, which occurs during the time interval $[0, T_x]$. As we will see later, such a construction will give a valid probability distribution if and only if $\mathbb{E}[T_x \,|\, X_0 = x]$ is finite. For a further discussion, we introduce the following notion.

Definition 3.6.1. Let $(X_n)_{n\geq 0}$ be a Markov chain on a state space Ω. We say a recurrent state $x \in \Omega$ is *positive recurrent* if $\mathbb{E}[T_x \,|\, X_0 = x] < \infty$ and *null recurrent* if $\mathbb{E}[T_x \,|\, X_0 = x] = \infty$.

Proposition 3.6.2 (Poisitive recurrence is a class property)**.** *Let x, y be two communicating states. If one is positive recurrent, then so is the other .*

Example 3.6.3 (SRW on $\mathbb{Z}$ is null recurrent)**.** In Example 3.5.7, we have seen that SRW on $\mathbb{Z}$ is recurrent, which is equivalent to $\mathbb{P}(T_x < \infty \,|\, X_0 = x) = 1$. However, this does not necessarily mean that we have finite expected return time, $\mathbb{E}[T_x \,|\, X_0] = 0$. This is indeed the case for SRW on $\mathbb{Z}$, which turns out ot be null recurrent. For this, we remark the following asymptotic of 'persistence probability' of SRW on $\mathbb{Z}$:

$$\mathbb{P}(X_1 \geq 1, X_2 \geq 1, \ldots, X_n \geq 1 \,|\, X_0 = 0) \sim \sqrt{\frac{2}{\pi n}}. \tag{359}$$

(For reference, see, e.g., [LS19].) It follows that, by the tail-sum formula for the expectation of nonnegative RVs,

$$\mathbb{E}[T_0 \,|\, X_0 = 0] = \sum_{n=1}^{\infty} \mathbb{P}(T_0 \geq n) \sim \sum_{n=1}^{\infty} \sqrt{\frac{2}{\pi n}} = \infty. \tag{360}$$

Thus the origin is null recurrent. By symmetry, all other states are also null recurrent. ▲

The full theorem regarding the existence and uniqueness of the stationary distribution is the following:

Theorem 3.6.4 (Existence and uniqueness of stationary distribution)**.** *Assume P is irreducible. Then P has a stationary distribution if and only if all states are positive recurrent. In this case the stationary distribution π is unique and its values satisfy*

$$\pi(x) = \frac{1}{\mathbb{E}[T_x \,|\, X_0 = x]} \qquad \forall\ x \in \Omega. \tag{361}$$

Corollary 3.6.5. *Assume Ω is finite and P is irreducible. Then P has a unique stationary distribution π given by* (361).

PROOF. By Theorem 3.6.4, we only need to verify if all states in Ω are positive recurrent. This is done in Exercise 3.6.6. □

Exercise 3.6.6 (Finite irreducible implies positive recurrence)**.** Let $(X_n)_{n\geq 0}$ be an irreducible Markov chain on a finite state space Ω with transition matrix P. Let $X_0 = x \in \Omega$. For any $y \in \Omega$ and $t \geq 0$, define the *first hitting time* of y by

$$T_y = \min\{k \geq 1 \,|\, X_k = y\}. \tag{362}$$

We will show that, for all $x, y \in \Omega$,

$$\mathbb{E}[T_y \,|\, X_0 = x] < \infty. \tag{363}$$

That is, the expected hitting time of $(X_n)_{n\geq 0}$ to any state starting from any state is finite.

(i) Recall that by the irreducibility and Exercise 3.5.2, for any $z, y \in \Omega$, $P^r(z, y) > 0$ for some $r \geq 1$ (r may depend on z and y). Let

$$d = \max_{z,y\in\Omega} \min\{r \geq 1 \,|\, P^r(z, y) > 0\}. \tag{364}$$

Show that there exists some constant $\delta > 0$ such that

$$\min_{y,z\in\Omega} \mathbb{P}(X_n \text{ visits } y \text{ some time during } [0, d] \,|\, X_0 = z) \geq \delta > 0. \tag{365}$$

(ii) Show that for any $k \geq 1$, (You may use strong Markov property)

$$\mathbb{P}(T_y > kd) \leq (1 - \delta)\mathbb{P}(T_y > (k - 1)d). \tag{366}$$

Use induction to conclude that

$$\mathbb{P}(T_y > kd) \leq (1 - \delta)^k. \tag{367}$$

(iii) Noting that $\mathbb{P}(T_y \geq t)$ is a non-increasing function in t, show that

$$\mathbb{E}[T_y] = \sum_{t=1}^{\infty} \mathbb{P}(T_y \geq t) \leq d \sum_{k=0}^{\infty} \mathbb{P}(T_y \geq kd) \leq d \sum_{k=0}^{\infty} (1 - \delta)^k = d/\delta < \infty. \tag{368}$$

Conclude that all states in an finite irreducible Markov chain are positive recurrent.

3.7. Uniqueness of stationary distribution

In establishing Theorem 3.6.4, we first consider the uniqueness of stationary distribution and when it should hold. Before we state our main result, we first take a look at random walk on disconnected graphs and that it must have multiple stationary distribution.

Exercise 3.7.1 (Random walk on disconnected graphs)**.** Let $G = (V, E)$ be a graph with two disjoint components G_1 and G_2. Let $(X_n)_{n\geq 0}$ be a random walk on G and denote by P its transition matrix.

(i) Let P_1 and P_2 be the transition matrices for random walks on G_1 and G_2, respectively. Let $V_1 = \{1, 2, \cdots, n\}$ and $V_2 = \{n + 1, n + 2, \cdots, n + m\}$ be the set of nodes in G_1 and G_2. Then show that P is of the following block diagonal form

$$P = \begin{bmatrix} P_1 & O \\ O & P_2 \end{bmatrix}. \tag{369}$$

(ii) Let $\pi_1 = [\pi_1^{(1)}, \cdots, \pi_1^{(n)}]$ and $\pi_2 = [\pi_2^{(1)}, \cdots, \pi_2^{m}]$ be any stationary distributions for P_1 and P_2. Let

$$\bar{\pi}_1 = [\pi_1^{(1)}, \cdots \pi_1^{(n)}, 0 \cdots, 0] \tag{370}$$

$$\bar{\pi}_2 = [0, \cdots, 0, \pi_2^{(1)}, \cdots, \pi_2^{m}] \tag{371}$$

be distributions on $V = V_1 \cup V_2 = \{1, 2, \cdots n, n+1, \cdots, n+m\}$. Show that for any $\lambda \in [0,1]$, $\pi := \lambda\bar{\pi}_1 + (1-\lambda)\bar{\pi}_2$ is a stationary distribution for P.

(iii) Recall that random walk on arbitrary graph has a stationary distribution, where probability of each node is proportional to its degree. Conclude that the random walk $(X_n)_{n\geq 0}$ on G has infinitely many stationary distribution.

A graph G is said to be *connected* if there exists a 'path' between any given two nodes. A *path* in G is a sequence of distinct nodes $v_0, v_1, \cdots, v_k$ such that consecutive nodes are adjacent in G. In Exercise 3.7.1, we have seen that the random walk on a disconnected graph has infinitely many stationary distribution. How about random walk on connected graphs? We will see that they have a unique stationary distribution, which must be the typical one that we know.

Proposition 3.7.2 (RW on connected graphs)**.** *Let $G = (V, E)$ be a connected graph. Then the random walk on G has a unique stationary distribution π, which is given by*

$$\pi(i) = \frac{\deg_G(i)}{2|E|}. \tag{372}$$

PROOF. Let P denote the transition matrix of random walk on G. Let $V = \{1, 2, \cdots, N\}$ and let E_1 be the column eigenspace of P associated with eigenvalue 1. That is,

$$E_1 = \{\mathbf{v} \in \mathbb{R}^N \mid P\mathbf{v} = \mathbf{v}\}. \tag{373}$$

Note that $[1, 1, \cdots, 1]^T \in E_1$, so E_1 has dimension ≥ 1. According to Exercise 3.2.3, it suffice to show that E_1 has dimension 1. This is equivalent to say that E_1 is spanned by $[1, \cdots, 1]^T$.

The argument uses the 'maximum principle' for 'harmonic functions'. Let $\mathbf{x} \in E_1$ be arbitrary. We will view $\mathbf{x}$ as a function on the node set V. So we want to show that $\mathbf{x}$ is a constant function on V. Since V is finite, there is some node k where $\mathbf{x}$ attains its global maximum. Suppose node k has some neighbor j such that $\mathbf{x}(j) < \mathbf{x}(k)$. Then from the eigenvector condition $P\mathbf{x} = \mathbf{x}$,

$$\mathbf{x}(k) = \sum_{i=1}^{N} P(k,i)\mathbf{x}(i) \tag{374}$$

$$= P(k,j)\mathbf{x}(j) + \sum_{i \neq j} P(k,i)\mathbf{x}(i) \tag{375}$$

$$< P(k,j)\mathbf{x}(k) + \sum_{i \neq j} P(k,i)\mathbf{x}(i) \tag{376}$$

$$\leq P(k,j)\mathbf{x}(k) + \sum_{i \neq j} P(k,i)\mathbf{x}(k) = \mathbf{x}(k), \tag{377}$$

which is a contradiction. This implies that $\mathbf{x}$ also attains its global maximum. Apply the similar argument to the neighbors of k. Repeating the same argument, we see that $\mathbf{x}$ must be constant on a 'connected component' of G containing k. But since G is connected, it follows that $\mathbf{x}$ is constant on V. This shows the assertion. □

If we carefully examine the proof of above result, we find that we haven't really used the value of entries of the transition matrix P. Rather, we only used an abstract property following from the connectivity of G: We can reach any node from any other node. We extract this as a general property of Markov chains.

Definition 3.7.3 (Irreducibility)**.** Let P be the transition matrix of a Markov chain $(X_n)_{n\geq 0}$ on a finite state space Ω. We say the chain (or P) is *irreducible* if for any $i, j \in \Omega$,

$$\mathbb{P}(X_k = j \mid X_0 = i) > 0 \tag{378}$$

for some integer $k \geq 0$.

In words, a Markov chain is irreducible if every state is 'accessible' from any other state.

Exercise 3.7.4 (RW on connected graphs is irreducible)**.** Let $G = (V, E)$ be a connected graph. Let $(X_n)_{n\geq 0}$ be a random walk on G and denote by P its transition matrix. Let $v_0, v_1, \cdots, v_k$ such that consecutive nodes are adjacent in G. Show that

$$P^k(v_0, v_k) = \mathbb{P}(X_k = v_k \,|\, X_0 = v_0) \geq P(v_0, v_1)P(v_1, p_2)\cdots P(v_{k-1}, v_k) \tag{379}$$

$$= \frac{1}{\deg_G(v_0)} \frac{1}{\deg_G(v_1)} \cdots \frac{1}{\deg_G(v_k)} > 0. \tag{380}$$

Conclude that random walks on connected graphs are irreducible.

Now we state the general uniqueness theorem of stationary distribution.

Lemma 3.7.5 (Uniqueness of stationary distribution)**.** *Let $(X_n)_{n\geq 0}$ be an irreducible Markov chain. Then it has at most a unique stationary distribution.*

PROOF. Exercise. Mimic the proof of Proposition 3.7.2. □

3.7.1. Construction of stationary distribution (Optional*). The gist of the to construct of a stationary distribution using the visit counts $V_n(x, y)$ is given in the following proposition. It constructs a stationary distribution for general Markov chains with a positive recurrent state.

Lemma 3.7.6. *Consider a Markov chain $(X_n)_{n\geq 0}$ on Ω with transition matrix P. Fix a recurrent state $x \in \Omega$ and for each $y \in \Omega$, define*

$$\lambda_x(y) := \mathbb{E}\left[V_{T_x}(x, y)\right], \tag{381}$$

which is the expected number of visits to y during $[0, T_x]$ (a single excursion from x to x).

(i) *For each $x \in \Omega$, $\lambda_x(x) = 1$ and $\sum_{y\in\Omega} \lambda_x(y) = \mathbb{E}[T_x \,|\, X_0 = x]$.*
(ii) *For each $x \in \Omega$, $\lambda_x = \lambda_x P$.*
(iii) *If x is positive recurrent, then $\pi_x := \frac{1}{\mathbb{E}[T_x \,|\, X_0 = x]} \lambda_x$ is a stationary distribution for P.*

PROOF. To show **(i)**, note that $N_{T_x}(x, x) = 1$ so $\lambda_x(x) = 1$. Also,

$$\sum_{y\in\Omega} \lambda_x(y) = \sum_{y\in\Omega} \mathbb{E}\left[V_{T_x}(x, y)\right] = \sum_{y\in\Omega} \mathbb{E}\left[\sum_{n=1}^{T_x} \mathbf{1}(X_n = y) \,\middle|\, X_0 = x\right] \tag{382}$$

$$= \mathbb{E}\left[\sum_{y\in\Omega} \sum_{n=1}^{T_x} \mathbf{1}(X_n = y) \,\middle|\, X_0 = x\right] \tag{383}$$

$$= \mathbb{E}\left[\sum_{n=1}^{T_x} \underbrace{\sum_{y\in\Omega} \mathbf{1}(X_n = y)}_{=1} \,\middle|\, X_0 = x\right] = \mathbb{E}\left[\sum_{n=1}^{T_x} 1 \,\middle|\, X_0 = x\right] = \mathbb{E}[T_x \,|\, X_0 = x], \tag{384}$$

where the third equality uses Fubini's theorem for nonnegative summands.

To show **(ii)**, our goal is to verify for each $y \in \Omega$,

$$\sum_{z\in\Omega} \lambda_x(z)P(z, y) = \lambda_x(y). \tag{385}$$

Indeed, first observe that

$$\mathbb{E}\left[\sum_{n=1}^{T_x}\mathbf{1}(X_n=z)\,\middle|\,X_0=x\right]=\mathbb{E}\left[\sum_{n=1}^{\infty}\mathbf{1}(X_n=z,\,T_x\geq n)\,\middle|\,X_0=x\right] \tag{386}$$

$$=\sum_{n=1}^{\infty}\mathbb{P}\left(X_n=z,\,T_x\geq n\,\middle|\,X_0=x\right) \tag{387}$$

$$=\sum_{n=1}^{\infty}\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z\,\middle|\,X_0=x\right), \tag{388}$$

where for the first equality, we have converted a random number of summation into an infinite sum with additioanl indicator. Next, by time-homogeneity and Markov property, we observe that

$$\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z\,\middle|\,X_0=x\right)P(z,y) \tag{389}$$

$$=\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z\,\middle|\,X_0=x\right)\mathbb{P}(X_{n+1}=y\,|\,X_n=z) \tag{390}$$

$$=\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z\,\middle|\,X_0=x\right)\mathbb{P}(X_{n+1}=y\,|\,X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z) \tag{391}$$

$$=\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z,\,X_{n+1}=y\,\middle|\,X_0=x\right). \tag{392}$$

Hence, noting that $\lambda_x(x)=1$ from Lemma 3.7.6,

$$\sum_{z\in\Omega}\lambda_x(z)P(z,y)=P(x,y)+\sum_{z\in\Omega\setminus\{x\}}\mathbb{E}\left[\sum_{n=1}^{T_x}\mathbf{1}(X_n=z)\,\middle|\,X_0=x\right]P(z,y) \tag{393}$$

$$=P(x,y)+\sum_{z\in\Omega\setminus\{x\}}\sum_{n=1}^{\infty}\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z,\,X_{n+1}=y\,\middle|\,X_0=x\right) \tag{394}$$

$$=P(x,y)+\sum_{n=1}^{\infty}\sum_{z\in\Omega\setminus\{x\}}\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n=z,\,X_{n+1}=y\,\middle|\,X_0=x\right) \tag{395}$$

$$=P(x,y)+\sum_{n=1}^{\infty}\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n\neq x,\,X_{n+1}=y\,\middle|\,X_0=x\right) \tag{396}$$

$$=\sum_{n=1}^{\infty}\mathbb{P}\left(X_1\neq x,\ldots,X_{n-1}\neq x,\,X_n\neq y\,\middle|\,X_0=x\right) \tag{397}$$

$$=\mathbb{E}\left[\sum_{n=1}^{T_x}\mathbf{1}(X_n=z)\,\middle|\,X_0=x\right]=\lambda_x(y). \tag{398}$$

This shows **(ii)**.

Lastly, if x is positive recurrent, then one can normalize λ_x by the total sum $\mathbb{E}[T_x\,|\,X_0=x]<\infty$ (see **(i)**) to obtain a probability distribution π_x that is invariant under right-multiplication of P. This shows **(iii)**. □

The following result shows the (the more important) half of Theorem 3.6.4.

Theorem 3.7.7. *Consider an irreducible Markov chain $(X_n)_{n\geq 0}$ on Ω with transition matrix P. Suppose all states are positive recurrent. Then there is a unique stationary distribution π for P given by* (361).

PROOF. According to Lemma 3.7.6, for each $x\in\Omega$, we have stationary distribution π_x defined in Lemma 3.7.6 **(iii)**. But by Lemma 3.7.5, there is a unique stationary distribution for P, which we will denote as π. Then for each $x\in\Omega$, it holds that

$$\pi(x)=\pi_x(x)=\frac{\lambda_x(x)}{\mathbb{E}[T_x\,|\,X_0=x]}=\frac{1}{\mathbb{E}[T_x\,|\,X_0=x]}. \tag{399}$$

This shows the assertion. □

In the following proposition, we make a simple observation that any stationary distribution will take value zero on transient states.

Proposition 3.7.8 (Stationary distribution is supported on recurrent states)**.** *Let a transition matrix P on Ω has an stationary distribution π. Then for each $x \in \Omega$, $\pi(x) > 0$ implies that x is recurrent.*

PROOF. Suppose $\pi(x) > 0$. Note that $\pi = \pi P^n$ for all $n \geq 1$. Hence

$$\infty = \sum_{n=1}^{\infty} \pi(x) = \sum_{n=1}^{\infty} \sum_{z\in\Omega} \pi(z)P^n(z,x) = \sum_{z\in\Omega} \sum_{n=1}^{\infty} \pi(z)P^n(z,x) \tag{400}$$

$$= \sum_{z\in\Omega} \pi(z) \sum_{n=1}^{\infty} P^n(z,x) \tag{401}$$

$$= \sum_{z\in\Omega} \pi(z)\mathbb{E}[N_x \,|\, X_0 = z], \tag{402}$$

where we have used Fubini's theorem to switch the order of double summation of nonnegative terms and the last equality uses (296). This implies that there exists some $z \in \Omega$ such that $\pi(z) > 0$ and $\mathbb{E}[N_x \,|\, X_0 = z] = \infty$. But by Proposition 3.5.5 this implies that x is recurrent (as well as $\rho_{zx} > 0$), as desired. □

PROOF OF THEOREM 3.6.4. According to Theorem 3.7.7, it suffices to show that, assuming irreducibility and the existence of stationary distribution π, all states are positive recurrent. Indeed, since π is a probability distribution, $\pi(x) > 0$ for some $x \in \Omega$, so by Proposition 3.7.8, x is recurrent. But since P is irreducible, it follows that all states are recurrent. Next, we need the following lemma:

Lemma: For irreducible and recurrent Markov chains, any two invariant measures for P is a constant multiple of each other. That is, if $\lambda = \lambda P$ and $\mu = \mu P$, then $\lambda = c\mu$ for some $c > 0$.

For the proof of the above lemma, see [SV, Thm. 2.67]. Now, by Lemma 3.7.6, λ_x is an invariant measure for P for each $x \in \Omega$. In particular, we have $\lambda_x = c_x\pi$ for some $c_x > 0$. Then

$$\mathbb{E}[T_x \,|\, X_0 = x] = \sum_{y\in\Omega} \lambda_x(y) = \sum_{y\in\Omega} c_x\pi(y) = c_x < \infty, \tag{403}$$

so x is positive recurrent. Since this holds for all x, we conclude that all states are positive recurrent. □

PROOF OF PROPOSITION 3.6.2. Suppose two states $x, y \in \Omega$ communicate and x is positive recurrent. By restricting onto the communicating class that contains x and y, we may assume that the Markov chain is irreducible and recurrent. By Lemma 3.7.6, there is a statioanry distribution, say, π_x. Then by a similar argument as in the the last part of the proof of Theorem 3.6.4, we have $\lambda_y = c_x\pi_x$ and this implies that y is also positive recurrent. □

3.8. Limit theorems for Markov chains

In this section, we establish a Markov chain version of the strong law of large numbers. Namely, consider the sequence of RVs $f(X_0), f(X_1), \ldots$ where $(X_n)_{n\geq 0}$ is a Markov chain and $f : \Omega \to \mathbb{R}$ is a function. Can a version of law of large number hold for the random increments $f(X_0), f(X_1), \ldots$? Namely, does the average of such increments converge to 'the' expected value? These are certainly not i.i.d. since they are driven by the underlying Markov chain X_n. However, if the chain starts from a positive recurrent state, say x, then one can group those increments according to the excursions of the chain. By the strong Markov property such excursions are i.i.d., so we can use the usual laws of large numbers for i.i.d. increments. This is the basic idea behind the proof of the following result, which states that "*time-average of a function converges to a space-average given by the stationary distribution*".

Theorem 3.8.1 (SLLN for finite Markov chains)**.** *Let $(X_n)_{n\geq 0}$ be an irreducible Markov chain on finite state space Ω with transition matrix P with arbitrary initial distribution. Let $f:\Omega\to\mathbb{R}$ denote a function. Then almost surely,*

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k)=\mathbb{E}_{X\sim\pi}\left[f(X)\right]=\sum_{x\in\Omega}f(x)\pi(x), \tag{404}$$

where π denotes the unique stationary distribution for P.

PROOF. Since Ω is finite and P is irreducible, all states are positive recurrent (see Exercise 3.6.6) and there exists a unique stationary distribution π (see Theorem 3.6.4). Let $V_x(n)=\sum_{k=1}^{n}\mathbf{1}(X_k=x)$ denote the number of visits to x in the first n steps of the Markov chain. The key step in converting time-average into space-average is the following observation: for each $x\in\Omega$, $f(x)$ is added up by $V_x(n)$ times:

$$\sum_{k=1}^{n}f(X_k)=\sum_{k=1}^{n}f(X_k)\sum_{x\in\Omega}\mathbf{1}(X_k=x)=\sum_{x\in\Omega}\sum_{k=1}^{n}f(X_k)\mathbf{1}(X_k=x) \tag{405}$$

$$=\sum_{x\in\Omega}\sum_{k=1}^{n}f(x)\mathbf{1}(X_k=x) \tag{406}$$

$$=\sum_{x\in\Omega}f(x)\sum_{k=1}^{n}\mathbf{1}(X_k=x) \tag{407}$$

$$=\sum_{x\in\Omega}f(x)V_x(n). \tag{408}$$

Now dividing by n and taking the limit $n\to\infty$, we get

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k)=\lim_{n\to\infty}\sum_{x\in\Omega}f(x)\frac{V_x(n)}{n}=\sum_{x\in\Omega}f(x)\lim_{n\to\infty}\frac{V_x(n)}{n}=\sum_{x\in\Omega}f(x)\pi(x), \tag{409}$$

where the second equality of exchanging limit and summation uses the fact that Ω is finite (it would be non-trivial if Ω is infinite) and the last equality uses Exercise 3.8.2. □

Exercise 3.8.2 (Convergence of empirical distribution)**.** Let $(X_t)_{t\geq 0}$ be an irreducible Markov chain on Ω with a unique stationary distribution π. For each $x\in\Omega$, define the following quantities

$$T_x^{(k)}:=\text{time of the } k\text{th visit to } x \tag{410}$$

$$V_x(n):=\sum_{k=1}^{n}\mathbf{1}(X_k=x)=\text{number of visits to } x \text{ during } [0,n]. \tag{411}$$

Here X_0 follows an arbitrary initial distribution μ on Ω.

(i) Let $\tau_k=T_x^{(k)}-T_x^{(k-1)}$ for $k\geq 2$. Show that τ_k's are i.i.d. and $0<\mathbb{E}[\tau_k]<\infty$. (For the i.i.d. part, use strong Markov property. For the finite expectation part, see Exercise 3.6.6 for the finite case and Theorem 3.6.4 for the general case.)

(ii) Use Markov property and strong law of large numbers to show

$$\mathbb{P}\left(\lim_{n\to\infty}\frac{\tau_2+\cdots+\tau_n}{n}=\mathbb{E}[\tau_2]\right)=1. \tag{412}$$

Furthermore, noting that $T_x^{(n)}=T_x^{(n)}+\tau_2+\cdots+\tau_n$, conclude that

$$\mathbb{P}\left(\lim_{n\to\infty}\frac{T_n}{n}=\mathbb{E}[\tau_2]\right)=1. \tag{413}$$

(iii) Let $V_x(t)=V(t)$ be the number of visits to x up to time t. Using the fact that $\mathbb{E}[\tau_k]<\infty$, show that $V(t)\to\infty$ as $t\to\infty$ a.s. Also, noting that $T_x^{(V_x(n))}\leq n<T_x^{(V_x(n)+1)}$, use **(ii)** to deduce

$$\mathbb{P}\left(\lim_{n\to\infty}\frac{V_x(n)}{n}=\frac{1}{\mathbb{E}[\tau_2]}\right)=1. \tag{414}$$

Furthermore, use Theorem 3.6.4 to deduce that

$$\mathbb{P}\left(\lim_{t\to\infty}\frac{V_x(t)}{t}=\pi(x)\right)=1. \tag{415}$$

Example 3.8.3. Let $(X_n)_{n\geq 0}$ be a Markov chain on $\Omega=\{0,1\}$ with transition matrix

$$P=\begin{bmatrix}1/4 & 3/4\\ 2/3 & 1/3\end{bmatrix}. \tag{416}$$

Given a function $f:\Omega\to\mathbb{R}$, we would like to find the limit

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k). \tag{417}$$

To this end, first note that P is irreducible. Since Ω is finite, it has a unique stationary distribution. More explictly, one can solve $\pi=\pi P$ for a probability distribution π on Ω and see that π must be $[8/17,9/17]$. Then by Theorem 3.8.1, almost surely,

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k)=f(0)\pi(0)+f(1)\pi(1)=\frac{8}{17}f(0)+\frac{9}{17}f(1). \tag{418}$$

Note that this holds regardless of the initial distribution of the Markov chain. ▲

The Markov chain SLLN stated in Theorem 3.8.1 holds for finite irreducible chains. When the state space is countably infinite, the analogous result continues to hold if we need to augment irreduciblity with positive recurrence:

Theorem 3.8.4 (SLLN for countable Markov chains)**.** *Let $(X_n)_{n\geq 0}$ be an irreducible and positive recurrent Markov chain on state space Ω with transition matrix P unique statioanry distribution π. Let $f:\Omega\to\mathbb{R}$ denote a function that is integerable w.r.t. π, that is, $\sum_{x\in\Omega}|f(x)|\pi(x)<\infty$. Then almost surely, for arbitrary initial distribution,*

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k)=\mathbb{E}_{X\sim\pi}\left[f(X)\right]=\sum_{x\in\Omega}f(x)\pi(x). \tag{419}$$

PROOF. Omitted. See [SV, Thm 2.94]. □

Example 3.8.5 (Success run)**.** Consider the success run chain in Exercise 3.2.5: It is a Markov chain $(X_n)_{n\geq 0}$ on the state space $\Omega=\mathbb{Z}_{\geq 0}$ of nonnegative integers with the following transition matrix P:

$$P(x,x+1)=p,\quad P(x,0)=1-p\qquad\text{for all } x\in\mathbb{Z}_{\geq 0}, \tag{420}$$

where $p\in[0,1]$ is a fixed parameter. We have shown that it has a unique stationary distribution π with $\pi(k)=(1-p)p^k$ for $k=1,2,\ldots$. Hence for a function $f:\mathbb{Z}_{\geq 0}\to\mathbb{R}$, by Theorem 3.8.4, we have

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k)=\sum_{k=1}^{\infty}f(k)(1-p)p^k\qquad\text{a.s.} \tag{421}$$

We take the following two special examples:

(a) Suppose one gets a dollor at each success, but only when the success run has length at least two. What is the long-term rate rewards, relative to the number of trials? In this case we set

$$f(k)=\mathbf{1}(k\geq 3). \tag{422}$$

Then the long-term rate of reward is

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k)=\sum_{k=3}^{\infty}(1-p)p^k=p^3\qquad\text{a.s.} \tag{423}$$

(b) What is the average length of success run in the long run? In this case we take $f(k)=k$ for $k\geq 1$ and

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}f(X_k)=\sum_{k=1}^{\infty}k(1-p)p^k=\frac{p}{1-p}\qquad\text{a.s.} \tag{424}$$

▲

3.9. Convergence to the stationary distribution

Let $(X_n)_{n\geq 0}$ be an irreducible Markov chain on a finite state space Ω. By Theorem 3.6.4, we know that the chain has unique stationary distribution π. Hence if we denote by π_t the distribution of X_t, then we should expect that π_t 'converges' to to π as $t \to \infty$ in some sense. We will prove a precise version of this claim in this section.

We start with an example, where we show the convergence of a 2-state chain using a diagonalization of its transition matrix.

Exercise 3.9.1. Let $(X_n)_{n\geq 0}$ be a Markov chain on $\Omega = \{1,2\}$ with the following transition matrix

$$P = \begin{bmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{bmatrix}. \tag{425}$$

(i) Show that P admits the following diagonalization

$$P = \begin{bmatrix} 1 & -4/3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2/5 \end{bmatrix} \begin{bmatrix} 1 & -4/3 \\ 1 & 1 \end{bmatrix}^{-1}. \tag{426}$$

(ii) Show that P^t admits the following diagonalization

$$P^t = \begin{bmatrix} 1 & -4/3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & (-2/5)^t \end{bmatrix} \begin{bmatrix} 1 & -4/3 \\ 1 & 1 \end{bmatrix}^{-1}. \tag{427}$$

(iii) Let $\mathbf{r}_t$ denote the row vector of distribution of X_t. Show that

$$\mathbf{r}_t = \mathbf{r}_0 \begin{bmatrix} 1 & -4/3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & (-2/5)^t \end{bmatrix} \begin{bmatrix} 1 & -4/3 \\ 1 & 1 \end{bmatrix}^{-1}. \tag{428}$$

Also show that

$$\lim_{t\to\infty} \mathbf{r}_t = \mathbf{r}_0 \begin{bmatrix} 3/7 & 4/7 \\ 3/7 & 4/7 \end{bmatrix} = [3/7, 4/7]. \tag{429}$$

(iv) Show that $\pi = [3/7, 4/7]$ is the unique stationary distribution for P. Conclude that regardless of the initial distribution $\mathbf{r}_0$, the distribution of the Markov chain $(X_n)_{n\geq 0}$ converges to $[3/7, 4/7]$.

Next, we observe that an irreducible MC may not always converge to the stationary distribution. The key issue there is the notion of 'periodicity'.

Example 3.9.2. Let $(X_n)_{n\geq 0}$ be a 2-state MC on state space $\Omega = \{0,1\}$ with transition matrix

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{430}$$

Namely, the chain deterministically alternates between the two states. Note that it is irreducible and has a unique stationary distribution

$$\pi = [1/2, 1/2]. \tag{431}$$

Let π_t be the distribution of X_n, where the initial distribution is given by $\pi_0 = [1,0]$. Then we have

$$\pi_1 = [0,1] \tag{432}$$
$$\pi_2 = [1,0] \tag{433}$$
$$\pi_3 = [0,1] \tag{434}$$
$$\pi_4 = [1,0], \tag{435}$$

and so on. Hence the distributions π_n do not converge to the stationary distribution π. ▲

Example 3.9.3 (RW on torus)**.** Let $\mathbb{Z}_n$ be the set of integers modulo n. Let $G = (V, E)$ be a graph where $V = \mathbb{Z}_n \times \mathbb{Z}_n$ and two nodes $u = (u_1, u_2)$ and $v = (v_1, v_2)$ are adjacent if and only if

$$|u_1 - u_2| + |v_1 - v_2| = 1. \tag{436}$$

Such a graph G is called the $n \times n$ *torus* and we write $G = \mathbb{Z}_n \times \mathbb{Z}_n$. Intuitively, it is obtained from the $n \times n$ square grid by adding boundary edges to wrap around (see Figure 10 left).

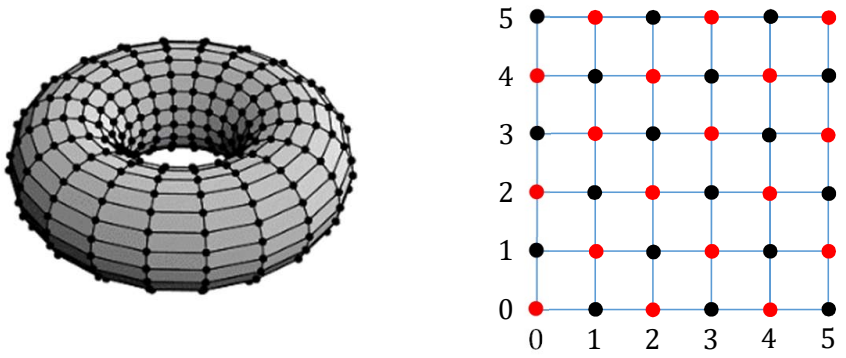


FIGURE 10. (Left) Torus graph (Figure excerpted from [LP17]). (Right) RW on torus $G = \mathbb{Z}_6 \times \mathbb{Z}_6$ has period 2.

Now let $(X_t)_{t\geq 0}$ be a random walk on G. Since G is connected, X_t is irreducible. Since all nodes in G have degree 4, the uniform distribution on $\mathbb{Z}_n \times \mathbb{Z}_n$, which we denote by π, is the unique stationary distribution of X_t. Let π_t denote the distribution of X_t.

For instance, consider $G = \mathbb{Z}_6 \times \mathbb{Z}_6$. As illustrated in Figure 10 below, observe that if X_0 is one of the red nodes (where sum of coordinates is even), then X_t is at a red node for any t = even and at a black node (where sum of coordinates is odd) at t = odd. Hence, π_t is supported only on the 'even' nodes for even times and on the 'odd' nodes for the odd times. Hence π_t does not converge in any sense to the uniform distribution π. ▲

The key issue in the 2-periodicity of RW on $G = \mathbb{Z}_6 \times \mathbb{Z}_6$ is that it takes even number of steps to return to any given node. Generalizing this observation, we introduce the following notion of periodicity.

Definition 3.9.4. Let P be the transition matrix of a Markov chain $(X_t)_{t\geq 0}$ on a finite state space Ω. For each state $x \in \Omega$, let $\mathcal{T}(x) = \{t \geq 1 \,|\, P^t(x, x) > 0\}$ be the set of times when it is possible for the chain to return to starting state x. We define the *period* of x by the greatest common divisor of $\mathcal{T}(x)$. We say the chain X_t is *aperiodic* if all states have period 1.

Example 3.9.5. Let $\mathcal{T}(x) = \{4, 6, 8, 10, \cdots\}$. Then the period of x is 2, even through it is not possible to go from x to x in 2 steps. If $\mathcal{T}(x) = \{4, 6, 8, 10, \cdots\} \cup \{3, 6, 9, 12, \cdots\}$, then the period of x is 1. This means the return times to x is irregular. For the RW on $G = \mathbb{Z}_6 \times \mathbb{Z}_6$ in Example 3.9.3, all nodes have period 2. ▲

Exercise 3.9.6 (Aperiodicity of RW on graphs)**.** Let $(X_t)_{t\geq 0}$ be a random walk on a connected graph $G = (V, E)$.

(i) Show that all nodes have the same period.
(ii) If G contains an odd cycle C (e.g., triangle), show that all nodes in C have period 1.
(iii) Show that X_t is aperiodic if and only if G contains an odd cycle.
(iv)* Show that X_t is periodic if and only if G is *bipartite*. (A graph G is bipartite if there exists a partition $V = A \cup B$ of nodes such that all edges are between A and B.)

Remark 3.9.7. If $(X_t)_{t\geq 0}$ is an irreducible Markov chain on a finite state space Ω, then all states $x \in \Omega$ must have the same period. The argument is similar to that for Exercise 3.9.6 (i).

Exercise 3.9.8 (Aperiodicity implies irreducibility for the product chain)**.** Let P be an irreducible transition matrix for a Markov chain on Ω. Let Q be a matrix on $\Omega \times \Omega$ defined by

$$Q((x,y),(z,w)) = P(x,z)\,P(y,w) \qquad \text{for } x,y,z,w \in \Omega. \tag{437}$$

(i) Consider a Markov chain $Z_t := (X_t, Y_t)$ on $\Omega \times \Omega$ that evolves by independently evolving its two coordinates according to P. Show that Q above is the transition matrix for Z_t.
(ii) Show that Q is irreducible if P is aperiodic.

We now state the convergence theorem for random walks on graphs.

Theorem 3.9.9 (Convergence of RW on graphs)**.** *Let $(X_t)_{t\geq 0}$ be a random walk on a connected graph $G = (V,E)$ with an odd cycle. Let π denote the unique stationary distribution π. Then for each $x, y \in V$,*

$$\lim_{t\to\infty} \mathbb{P}(X_t = y \,|\, X_0 = x) = \pi(y). \tag{438}$$

PROOF. Since G contains an odd cycle, random walk on G is aperiodic according to Exercise 3.9.6. Let $(Y_t)_{t\geq 0}$ be another RW on G. We evolve X_t and Y_t simultaneously independently until they meet, after which we evolve them in union. Namely, let τ be the 'meeting time' of these two walkers:

$$\tau = \min\{t \geq 0 \,|\, X_t = Y_t\}. \tag{439}$$

Then we let $X_t = Y_t$ for all $t \geq \tau$. Still, if we disguise one random walk, then the other behaves exactly as it should do. (This is a 'coupling' between the two random walks)

Now suppose $X_0 = x$ for some $x \in V$ and Y_0 is drawn from the stationary distribution π. In particular, the distribution of Y_t is π for all t. Let π_t denote the distribution of X_t. Then for any $y \in V$,

$$\begin{aligned}
|\pi_t(y) - \pi(y)| &= \left|\mathbb{P}(X_t = y) - \mathbb{P}(Y_t = y)\right| && (440)\\
&= \left|\mathbb{P}(X_t = y, Y_t = y) + \mathbb{P}(X_t = y, Y_t \neq y) - \mathbb{P}(X_t = y, Y_t = y) - \mathbb{P}(X_t \neq y, Y_t = y)\right| && (441)\\
&\leq \mathbb{P}(X_t = y,\, Y_t \neq y) + \mathbb{P}(X_t \neq y,\, Y_t = y) && (442)\\
&\leq \mathbb{P}(X_t \neq Y_t) && (443)\\
&= \mathbb{P}(\tau > t). && (444)
\end{aligned}$$

Hence if suffices to show that $\mathbb{P}(\tau > t) \to 0$ as $t \to \infty$, as this will yield $|\pi_t(y) - \pi(y)| \to 0$ as $t \to \infty$. This follows from the fact that $\mathbb{P}(\tau < \infty) = 1$, that is, the two independent random walks on G eventually meet with probability 1 (see Exercise 3.9.12). □

The following exercise shows that if a RW on G is irreducible and aperiodic, then it is possible to reach any node from any other node in a fixed number of steps.

Exercise 3.9.10. Let $(X_t)_{t\geq 0}$ be a RW on a connected graph $G = (V,E)$. Let P denote its transition matrix. Suppose G contains an odd cycle, so that the walk is irreducible and aperiodic. For each $x \in V$, let $\mathcal{T}(x)$ denote the set of all possible return times to the state x.

(i) For any $x \in V$, show that $a, b \in \mathcal{T}(x)$ implies $a + b \in \mathcal{T}(x)$.
(ii) For any $x \in V$, show that $\mathcal{T}(x)$ contains 2 and some odd integer b.
(iii) For each $x \in V$, let b_x be the smallest odd integer contained in $\mathcal{T}(x)$. Show that $m \in \mathcal{T}(x)$ whenever $m \geq b_x$.
(iv) Let $b^* = \max_{x\in V} b_x$. Show that $m \in \mathcal{T}(x)$ for all $x \in V$ whenever $m \geq b^*$.
(v) Let $r = |V| + b^*$. Show that $P^r(x,y) > 0$ for all $x, y \in V$.

With a little more work, one can also show a similar statement for general irreducible and aperiodic Markov chains.

Exercise 3.9.11. Let P be the transition matrix of a Markov chain $(X_t)_{t\geq 0}$ on a finite state space Ω. Show that (i) implies (ii):

(i) P is irreducible and aperiodic.

(ii) There exists an integer $r \geq 0$ such that every entry of P^r is positive.

Furthermore, show that (ii) implies (i) if X_t is a RW on some graph $G = (V, E)$.

(Hint for (i)⇒(ii): You may use the fact that if a subset $\mathcal{T}$ of positive integers $\mathbb{N}$ is closed under addition with $\gcd(\mathcal{T}) = 1$, then it must contain all but finitely many integers. Proof is similar to Exercise 3.9.10, but a bit more number-theoretic. See [LP17, Lem. 1.27].)

Exercise 3.9.12. Let (X_t) and (Y_t) be independent random walks on a connected graph $G = (V, E)$. Suppose that G contains an odd cycle. Let P be the transition matrix of the random walk on G.

(i) Let $t \geq 0$ be arbitrary. Use Exercise 3.9.10 to deduce that for some integer $r \geq 1$, we have

$$\mathbb{P}(X_{t+r} = Y_{t+r} \mid X_t = x, Y_t = y) = \sum_{z \in V} \mathbb{P}(X_{t+r} = z = Y_{t+r} \mid X_t = x, Y_t = y) \tag{445}$$

$$= \sum_{z \in V} \mathbb{P}(X_{t+r} = z \mid X_t = x) \mathbb{P}(Y_{t+r} = z \mid Y_t = y) \tag{446}$$

$$= \sum_{z \in V} P^r(x, z) P^r(y, z) > 0. \tag{447}$$

(ii) Let $r \geq 1$ be as in (i) and let

$$\delta = \min_{x,y \in V} \sum_{z \in V} P^r(x, z) P^r(y, z) > 0. \tag{448}$$

Use (i) and Markov property that in every r steps, X_t and Y_t meet with probability $\geq \delta > 0$.

(iii) Let τ be the first time that X_t and Y_t meet. From (ii) and Markov property, deduce that

$$\mathbb{P}(\tau \geq kr) = \mathbb{P}(\tau \geq r) \mathbb{P}(X_t \text{ and } Y_t \text{ never meet during } [r, kr) \mid X_{r-1} \neq Y_{r-1}) \tag{449}$$

$$\leq (1 - \delta) \mathbb{P}(X_t \text{ and } Y_t \text{ never meet during } [r, kr) \mid X_{r-1} \neq Y_{r-1}). \tag{450}$$

By an induction on k, conclude that

$$\mathbb{P}(\tau \geq kr) \leq (1 - \delta)^k \to 0 \quad \text{as } k \to \infty. \tag{451}$$

The general convergence theorem is stated below.

Theorem 3.9.13 (Convergence Thm)**.** *Let $(X_t)_{t \geq 0}$ be an irreducible aperiodic Markov chain on a finite state space Ω. Let π denote the unique stationary distribution of X_t. Then for each $x, y \in V$,*

$$\lim_{t \to \infty} \mathbb{P}(X_t = y \mid X_0 = x) = \pi(y). \tag{452}$$

PROOF. As always, there is a linear algebra proof to this result (see, e.g., [LP17, Thm. 4.9]) Mimic the argument for Theorem 3.9.9 for a coupling based proof. □

3.10. Markov chain Monte Carlo

So far, we were given a Markov chain $(X_t)_{t \geq 0}$ on a finite state space Ω and studied existence and uniqueness of its stationary distribution and convergence to it. In this section, we will consider the reverse problem. Namely, given a distribution π on a sample space Ω, can we construct a Markov chain $(X_t)_{t \geq 0}$ such that π is a stationary distribution? If in addition the chain is irreducible an aperiodic, then by the convergence theorem (Theorem 3.9.13), we know that the distribution π_t of X_t converges to π. Hence if we run the chain for long enough, the state of the chain is asymptotically distributed as π. In other words, we can sample a random element of Ω according to the prescribed distribution π by emulating it through a suitable Markov chain. This method of sampling is called *Markov chain Monte Carlo* (MCMC).

Example 3.10.1 (Uniform distribution on regular graphs)**.** Let $G = (V, E)$ be a connected *regular* graph, meaning that all nodes have the same degree. Let μ be the uniform distribution on the node set V. How can we sample a random node according to μ? If we have a list of all nodes, then we can label them from 1 to $N = |V|$, choose a random number between 1 and N, and find corresponding node. But often

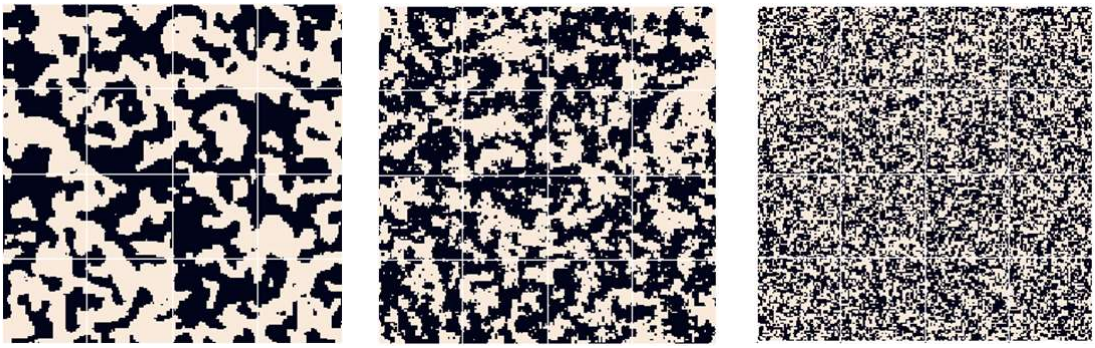


FIGURE 11. MCMC simulation of Ising model on 200 by 200 torus at temperature $T = 1$ (left), 2 (middle), and 5 (right).

times, we do not have the full list of nodes, especially when we want to sample a random node from a social network. Given only local information (set of neighbors for each given node), can we still sample a uniform random node from G?

One answer is given by random walk. Indeed, random walks on graphs are defined by only using local information of the underlying graph: Choose a random neighbor and move there. Moreover, since G is connected, there is a unique stationary distribution π for the walk, which is given by

$$\pi(x) = \frac{\deg_G(x)}{2|E|}. \tag{453}$$

Since G is regular, any two nodes have the same degree, so $\pi(x) = \pi(y)$ for all $x, y \in V$. This means π equals the uniform distribution μ on V. Hence the sampling algorithm we propose is as follows:

$(*)$ *Run a random walk on G for $t \gg 1$ steps, and take the random node that the walk sits on.*

However, there is a possible issue of convergence. Namely, if the graph G does not contain any odd cycle, then random walk on G is periodic (see Exercise 3.9.6), so we are not guaranteed to have convergence. We can overcome this by using a lazy random walk instead, which is introduced in Exercise 3.10.2. We know that the lazy RW on G is irreducible, aperiodic, and has the same set of stationary distribution as the ordinary RW on G. Hence we can apply the sampling algorithm $(*)$ above for lazy random walk on G to sample a uniform random node in G. ▲

Exercise 3.10.2 (Lazy RW on graphs)**.** Let $G = (V, E)$ be a graph. Let $(X_t)_{t\geq 0}$ be a Markov chain on the node set V with transition probabilities

$$\mathbb{P}(X_{t+1} = j \mid X_t = i) = \begin{cases} 1/2 & \text{if } j = i \\ 1/(2\deg_G(i)) & \text{if } j \text{ is adjacent to } i \\ 0 & \text{otherwise.} \end{cases} \tag{454}$$

This chain is called the *lazy random walk* on G. In words, the usual random walker on G now flips a fair coin to decide whether it stays at the same node or make a move to one of its neighbors.

(i) Show that for any connected graph G, the lazy random walk on G is irreducible and aperiodic.

(ii) Let P be the transition matrix for the usual random walk on G. Show that the following matrix

$$Q = \frac{1}{2}(P + I) \tag{455}$$

is the transition matrix for the lazy random walk on G.

(iii) For any distribution π on V, show that $\pi Q = \pi$ if and only if $\pi P = \pi$. Conclude that the usual and lazy random walks on G have the same set of stationary distribution.

Example 3.10.3 (Finding local minima)**.** Let $G = (V, E)$ be a connected graph and let $f : V \to [0, \infty)$ be a 'cost' function. The objective is to find a node $x^* \in V$ such that f takes global minimum at x^*. This problem has a lot of application in machine learning, for example. Note that if the domain V is very large, then an exhaustive search is too expensive to use.

Here is simple form of the popular algorithm of *stochastic gradient descent*, which lies at the heart of most of the important machine learning algorithms.

(i) Initialize the first guess $X_0 = x_0 \in V$.
(ii) Suppose $X_t = x \in V$ is chosen. Let

$$D_t = \{y \text{ a neighbor of } x \text{ or } x \text{ itself} \mid f(y) \le f(x)\}. \tag{456}$$

Define X_{t+1} to be a uniform random node from D_t.
(iii) The algorithm terminates if it finds a local minima.

In words, at each step we move to a random neighbor which could possibly decrease the current value of f. It is easy to see that one always converges to a local minima, which may not be a global minimum. In a very complex machine learning task (e.g., training a deep neural network), this is often good enough. Is this a Markov chain? Irreducible? Aperiodic? Stationary distribution? ▲

There is a neat solution to finding global minimum. The idea is to allow that we go uphill with a small probability.

Example 3.10.4 (Finding global minimum)**.** Let $G = (V, E)$ be a connected regular graph and let $f : V \to [0, \infty)$ be a cost function. Let

$$V^* = \left\{ x \in V \mid f(x) = \min_{y \in V} f(y) \right\} \tag{457}$$

be the set of all nodes where f attains global minimum.

Fix a parameter $\lambda \in (0, 1]$, and define a probability distribution π_λ on V by

$$\pi_\lambda(x) = \frac{\lambda^{f(x)}}{Z_\lambda}, \tag{458}$$

where $Z_\lambda = \sum_{x \in V} \lambda^{f(x)}$ is the normalizing constant. Since $\pi_\lambda(x)$ is decreasing in $f(x)$, it favors nodes x for which $f(x)$ is small.

Let $(X_t)_{t \ge 0}$ be Markov chain on V, whose transition is defined as follows. If $X_t = x$, then let y be a uniform random neighbor of x. If $f(y) \le f(x)$, then move to y; If $f(y) > f(x)$, then move to y with probability $\lambda^{f(y)-f(x)}$ and stay at x with probability $1 - \lambda^{f(y)-f(x)}$. We analyze this MC below:

(i) (Irreducibility) Since G is connected and we are allowing any move (either downhill or uphill) we can go from one node to any other in some number of steps. Hence the chain $(X_t)_{t \ge 0}$ is irreducible.

(ii) (Aperiodicity) By (i) and Remark 3.9.7, all nodes have the same period. Moreover, let $x \in V^*$ be an arbitrary node where f takes global minimum. Then all return times are possible, so x has period 1. Hence all nodes have period 1, so the chain is aperiodic.

(iii) (Stationarity) Here we show that π_λ is a stationary distribution of the chain. To do this, we first need to write down the transition matrix P. Namely, if we let $A_G(y, z)$ denote the indicator that y and z are adjacent, then

$$P(x, y) = \begin{cases} \frac{A_G(x,y)}{\deg_G(x)} \min(1, \lambda^{f(y)-f(x)}) & \text{if } x \ne y \\ 1 - \sum_{y \ne x} P(x, y) & \text{if } y = x. \end{cases} \tag{459}$$

To show $\pi_\lambda P = \pi_\lambda$, it suffices to show for any $y \in V$ that

$$\sum_{z \in V} \pi_\lambda(z) P(z, y) = \pi_\lambda(y). \tag{460}$$

Note that

$$\sum_{z\in V}\pi_\lambda(z)P(z,y)=\pi_\lambda(y)P(y,y)+\sum_{z\neq y}\pi_\lambda(z)P(z,y) \tag{461}$$

$$=\pi_\lambda(y)-\sum_{z\neq y}\pi_\lambda(y)P(y,z)+\sum_{z\neq y}\pi_\lambda(z)P(z,y). \tag{462}$$

Hence it suffices to show that

$$\pi_\lambda(y)P(y,z)=\pi_\lambda(z)P(z,y) \tag{463}$$

for any $z\neq y$. Indeed, considering the two cases $f(z)\leq f(y)$ and $f(z)>f(y)$, we have

$$\pi_\lambda(y)P(y,z)=\frac{\lambda^{f(y)}}{Z_\lambda}\frac{A_G(y,z)}{\deg_G(y)}\min(1,\lambda^{f(z)-f(y)})=\frac{1}{Z_\lambda}\frac{A_G(z,y)}{\deg_G(z)}\lambda^{\max(f(y),f(z))}, \tag{464}$$

$$\pi_\lambda(z)P(z,y)=\frac{\lambda^{f(z)}}{Z_\lambda}\frac{A_G(z,y)}{\deg_G(z)}\min(1,\lambda^{f(y)-f(z)})=\frac{1}{Z_\lambda}\frac{A_G(y,z)}{\deg_G(y)}\lambda^{\max(f(y),f(z))}. \tag{465}$$

Now since $A_G(z,y)=A_G(y,z)$ and we are assuming G is a regular graph, this yields (463), as desired. Hence π_λ is a stationary distribution for the chain $(X_t)_{t\geq 0}$.

(iv) (Convergence) By (i), (iii), Theorem 3.6.4, we see that π_λ is the unique stationary distribution for the chain X_t. Furthermore, by (i)-(iii) and Theorem 3.9.13, we conclude that the distribution of X_t converges to π_λ.

(v) (Global minimum) Let $f_*=\min_{x\in V}f(x)$ be the global minimum of f. Note that by definition of V^*, we have $f(x)=f_*$ for any $x\in V^*$. Then observe that

$$\lim_{\lambda\to 0}\pi_\lambda(x)=\lim_{\lambda\to 0}\frac{\lambda^{f(x)}}{\sum_{y\in V}\lambda^{f(y)}}=\lim_{\lambda\to 0}\frac{\lambda^{f(x)}/\lambda^{f*}}{\sum_{y\in V}\lambda^{f(y)}/\lambda^{f*}} \tag{466}$$

$$=\lim_{\lambda\to 0}\frac{\lambda^{f(x)-f_*}}{|V^*|+\sum_{y\in V\setminus V^*}\lambda^{f(y)-f_*}}=\frac{\mathbf{1}(x\in V^*)}{|V^*|}. \tag{467}$$

Thus for λ very small, π_λ is approximately the uniform distribution on the set of all nodes V^* where f attains global minimum. ▲

Exercise 3.10.5 (Detailed Balance equation)**.** Let P be a transition matrix of a Markov chain on state space Ω. Suppose π is a probability distribution on Ω that satisfies the following *detailed balance equation*

$$\pi(x)P(x,y)=\pi(y)P(y,x)\qquad\text{for all } x,y\in\Omega. \tag{468}$$

(i) Show that for all $x\in\Omega$,

$$\sum_{y\in\Omega}\pi(y)P(y,x)=\sum_{y\in\Omega}\pi(x)P(x,y)=\pi(x)\sum_{y\in\Omega}P(x,y)=\pi(x). \tag{469}$$

(ii) Conclude that π is a stationary distribution for P, that is, $\pi P=\pi$.

Exercise 3.10.6 (Metropolis-Hastings algorithm)**.** Let P be a transition matrix of a Markov chain on state space $\Omega=\{1,2,\cdots,m\}$. Let π be a probability distribution on Ω, which is not necessarily a stationary distribution for P. Our goal is to design a Markov chain on Ω that has π as a stationary distribution. Below we will derive the famous *Metropolis-Hastings algorithm* for MCMC sampling.

Fix a $m\times m$ matrix A of entries from $[0,1]$. Consider a Markov chain $(X_t)_{t\geq 0}$ on Ω that uses additional rejection step on top of the transition matrix P as follows:

(`Generation`) Suppose the current location $X_t=a$. Generate a candidate $b\in\Omega$ from the conditional distribution $P(a,\cdot)$.

(`Rejection`) Flip an independent coin with success probability $A(a,b)$. Upon success, accept the proposed move and set $X_{t+1}=b$; Otherwise, reject the move and set $X_{t+1}=a$.

Here, the (a,b) entry $A(a,b)$ is called the *acceptance probability* of the move $a\to b$.

(i) Let Q denote the transition matrix of the chain $(X_t)_{t\geq 0}$ defined above. Show that

$$Q(a,b) = \begin{cases} P(a,b)A(a,b) & \text{if } b \neq a \\ 1 - \sum_{c:c\neq a} P(a,c)A(a,c) & \text{if } b = a. \end{cases} \tag{470}$$

(ii) Show that $\pi(x)Q(x,y) = \pi(y)Q(y,x)\ \forall x,y \in \Omega,\ x \neq y$ implies $\pi Q = \pi$. Deduce that if

$$\pi(x)P(x,y)A(x,y) = \pi(y)P(y,x)A(y,x) \qquad \forall x,y \in \Omega,\ x \neq y, \tag{471}$$

then π is a stationary distribution for $(X_t)_{t\geq 0}$.

(iii) Since we also want the Markov chain to converge fast, we want to choose the acceptance probability $A(a,b) \in [0,1]$ as large as possible for each $a,b \in \Omega$. Show that the following choice (denoting $\alpha \wedge \beta := \min(\alpha,\beta)$)

$$A(x,y) = \frac{\pi(y)P(y,x)}{\pi(x)P(x,y)} \wedge 1 \qquad \forall x,y \in \Omega,\ x \neq y \tag{472}$$

satisfies the condition in (ii) and each $A(x,y)$ is maximized for all $x \neq y$.

(iv) Let $(Y_t)_{t\geq 0}$ be a random walk on the 5-wheel graph $G = (V,E)$ as shown in Figure 12. Show that $\pi = \left[\frac{5}{20}, \frac{3}{20}, \frac{3}{20}, \frac{3}{20}, \frac{3}{20}, \frac{3}{20}\right]$ is the unique stationary distribution of Y_t. Apply the Metropolis-Hastings algorithm derived in (i)-(iii) above to modify Y_t to obtain a new Markov chain X_t on V that converges to Uniform(V) in distribution.

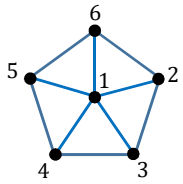


FIGURE 12. 5-wheel graph G

CHAPTER 4

Martingales

In order to properly develop our discussion on martingales in the following sections, we need to generalize the notion of conditional expectation $\mathbb{E}[X \mid Y]$ of a RV X given another RV Y. Recall that this was the a collection of 'best guesses' of X given $Y = y$ for all y. But what if we only know, say, $Y \geq 1$? Can we condition on this event as well?

More concretely, suppose Y takes values from $\{1,2,3\}$. Regarding Y, the following outcomes are possible:

$$\mathscr{E}_Y := \{\{Y=1\},\{Y=2\},\{Y=3\},\{Y=1,2\},\{Y=2,3\},\{Y=1,3\},\{Y=1,2,3\}\}. \tag{473}$$

For instance, the information $\{Y = 1,2\}$ could yield some nontrival implication on the value of X, so our best guess in this scenario should be

$$\mathbb{E}[X \mid \{Y=1,2\}] = \sum_x x\,\mathbb{P}(X=x|\{Y=1,2\}). \tag{474}$$

More generally, for each $A \in \mathscr{E}_Y$, the best guess of X given $A \in \mathbb{E}_Y$ is the following conditional expectation

$$\mathbb{E}[X \mid A] = \sum_x x\,\mathbb{P}(X=x|A). \tag{475}$$

Now, what if we don't know which event in the collection $\mathscr{E}_Y$ to occur? As we did before to define $\mathbb{E}[X \mid Y]$ from $\mathbb{E}[X \mid Y = y]$ by simply not specifying what value y that Y takes, we simply do not specify which event $A \in \mathscr{E}_A$ to occur. Namely,

$$\mathbb{E}[X \mid \mathscr{E}_Y] = \text{best guess on } X \text{ given the information in } \mathscr{E}_Y. \tag{476}$$

In general, this could be defined for any collection of events $\mathscr{E}$ in place of $\mathscr{E}_Y$. Mathematically, we understand $\mathbb{E}[X \mid \mathscr{E}]$ as[1]

$$\mathbb{E}[X \mid \mathscr{E}] = \text{the collection of } \mathbb{E}[X \mid A] \text{ for all } A \in \mathscr{E}. \tag{477}$$

4.1. Definition and examples of martingales

Let $(X_t)_{t\geq 0}$ be the sequence of observations of the price of a particular stock over time. Suppose that an investor has a strategy to adjust his portfolio $(M_t)_{t\geq 0}$ according to the observation $(X_t)_{t\geq 0}$. Namely,

$$M_t = \text{Net value of portfolio after observing } (X_k)_{0\leq k\leq t}. \tag{478}$$

We are interested in the long-term behavior of the 'portfolio process' $(M_t)_{t\geq 0}$. Martingales provide a very nice framework for this purpose.

Martingale is a class of stochastic processes, whose expected increment conditioned on the past is always zero. Recall that the simple symmetric random walk has this property, since each increment is i.i.d. and has mean zero. Martingales do not assume any kind of independence between increments, but it turns out that we can proceed quite far with just the unbiased conditional increment property.

[1] For more details, see [Dur10].

In order to define martingales properly, we need to introduce the notion of 'information up to time t'. Imagine we are observing the stock market starting from time t. We define

$$\mathscr{E}_t := \text{collection of all possible events we can observe at time } t \tag{479}$$

$$\mathscr{F}_t := \bigcup_{k=1}^{t} \mathscr{E}_k = \text{collection of all possible events we can observe up to time } t. \tag{480}$$

In words, $\mathscr{E}_t$ is the information available at time t and $\mathscr{F}_t$ contains all possible information that we can obtain by observing the market up to time t. We call $\mathscr{F}_t$ the *information* up to time t. As a collection of events, $\mathscr{F}_t$ needs to satisfy the following properties[2]:

(i) (*closed under complementation*) $A \in \mathscr{F}_t \Longrightarrow A^c \in \mathscr{F}_t$;
(ii) (*closed under countable union*) $A_1, A_2, A_3, \cdots \in \mathscr{F}_t \Longrightarrow \bigcup_{k=1}^{\infty} A_k \in \mathscr{F}_t$.

Note that as we gain more and more information, we have

$$\mathscr{F}_s \subseteq \mathscr{F}_t \qquad \forall\, t \geq s \geq 0. \tag{481}$$

In other words, $(\mathscr{F}_t)_{t\geq 0}$ is an increasing set of information, which we call a *filtration*. The roll of a filtration is to specify what kind of information is observable or not, as time passes by.

Example 4.1.1. Suppose $(\mathscr{F}_t)_{t\geq 0}$ is a filtration generated by observing the stock price $(X_t)_{t\geq 0}$ of company A in New York. Namely, $\mathscr{E}_t$ consists of the information on the values of the stock price X_t at day t. Given $\mathscr{F}_{10}$, we know the actual values of $X_0, X_1, \cdots, X_{10}$. For instance, X_8 is not random given $\mathscr{F}_{10}$, but X_{11} could still be random. On the other hand, if $(Y_t)_{t\geq 0}$ is the stock price of company B in Hong Kong, then we may have only partial information for $Y_0, \cdots, Y_{10}$ given $\mathscr{F}_t$. ▲

Now we define martingales.

Definition 4.1.2. Let $(\mathscr{F}_n)_{n\geq 0}$ be a filtration and $(M_n)_{n\geq 0}$ be discrete-time stochastic processes. We call $(M_n)_{n\geq 0}$ a *martingale* with respect to $(\mathscr{F}_n)_{n\geq 0}$ if the following conditions are satisfied: For all $n \geq 0$,

(i) (*finite expectation*) $\mathbb{E}[|M_n|] < \infty$.
(ii) (*measurability*[3]) $\{M_n = m\} \in \mathscr{F}_n$ for all $m \in \mathbb{R}$.
(iii) (*conditional increments*) $\mathbb{E}[M_{n+1} - M_n \,|\, A] = 0$ for all $A \in \mathscr{F}_n$.

If (iii) holds with "=" replaced by "≤" (resp., "≥"), then $(M_n)_{n\geq 0}$ is called a *supermartingale* (resp., *submartingale*) with respect to $(X_n)_{n\geq 0}$, respectively.

We will abbreviate the condition **(iii)** as

$$\mathbb{E}[M_{n+1} - M_n \,|\, \mathscr{F}_n] = 0. \tag{482}$$

In many cases, we work with filtrations generated by observing a single reference process. Let $(X_n)_{n\geq 1}$ denote a discrete-time stochastic process on Ω and for each n, let $\mathscr{F}_n = \sigma(X_1, \ldots, X_n)$ denote the collection of all events that can be determined by the values of $X_1, \ldots, X_n$. Then $(\mathscr{F}_n)_{n\geq 1}$ is called the *filtration generated by the process* $(X_n)_{n\geq 1}$. In this case, the conditions **(ii)**-**(iii)** in Definition 4.1.2 reduce to the following simpler conditions:

(ii)' (*measurability*) M_n is a function of $X_0, X_1, \ldots, X_n$.
(iii)' (*conditional increments*) $\mathbb{E}[M_{n+1} - M_n \,|\, X_0, X_1, \ldots, X_n] = 0$.

That is, in Definition 4.1.2 **(iii)**, we only need to consider events $A \in \mathscr{F}_n = \sigma(X_0, \ldots, X_n)$ of the form

$$A = \{X_0 = x_0, X_1 = x_1, \ldots, X_n = x_n\}, \tag{483}$$

which specifies the entire trejectory of the reference process $(X_n)_{n\geq 1}$. See Execise 4.1.3 for more details.

[2] We are requiring $\mathscr{F}_t$ to be a *σ-algebra*, but we avoid using this terminology.
[3] In this case, we say "M_n is measurable w.r.t. $\mathscr{F}_n$", but we avoid using this terminology.

Exercise 4.1.3 (Martingale w.r.t. the filtration generated by a stochastic process)**.** Let $(X_n)_{n\geq 1}$ be a discrete-time stochastic process on Ω. Let $(M_n)_{n\geq 1}$ denote a discrete-time stochastic process satisfying the following property:

(a) $\mathbb{E}[|M_n|] < \infty$ for $n \geq 1$;
(b) M_n is a function of $X_1, \ldots, X_n$ for $n \geq 1$;
(c) $\mathbb{E}[M_{n+1} - M_n \,|\, (X_1, \ldots, X_n)] = 0$ for $n \geq 1$.

Then show that $(M_n)_{n\geq 1}$ is a martingale (in the sense of Definition 4.1.2) with respect to the filtration $(\mathscr{F}_n)_{n\geq 1}$, where $\mathscr{F}_n = \sigma(X_0, \ldots, X_n)$.

Hint: The key step is to show that, for $A \in \mathscr{F}_n$, $\mathbb{E}[M_{n+1} - M_n \,|\, A] = \mathbb{E}[\mathbb{E}[M_{n+1} - M_n \,|\, A, X_1, \ldots, X_n]] = 0$. In order to apply property **(c)** here, justify the following identities

$$\mathbb{E}[M_{n+1} - M_n \,|\, A, X_1, \ldots, X_n] = \mathbb{E}[(M_{n+1} - M_n)\mathbf{1}((X_1, \ldots, X_n) \in A) \,|\, X_1, \ldots, X_n] \tag{484}$$

$$= \mathbf{1}((X_1, \ldots, X_n) \in A)\mathbb{E}[(M_{n+1} - M_n) \,|\, X_1, \ldots, X_n]. \tag{485}$$

If martingale is a fair gambling strategy, then one can think of supermartingale and submartingale as unfavorable and favorable gambling strategies, resepctively. For instance, expected winning in gambling on an unfavorable game should be non-increasing in time. This is an immediate consequence of the definition and iterated expectation.

Proposition 4.1.4. *Let $(M_t)_{t\geq 0}$ be a stochastic process and let $(\mathscr{F}_t)_{t\geq 0}$ be a filtration.*

(i) *If $(M_t)_{t\geq 0}$ is a supermartingale w.r.t. filtration $(\mathscr{F}_t)_{t\geq 0}$, then $\mathbb{E}[M_n] \leq \mathbb{E}[M_m]$ for all $n \geq m \geq 0$.*

(ii) *If $(M_t)_{t\geq 0}$ is a submartingale w.r.t. filtration $(\mathscr{F}_t)_{t\geq 0}$, then $\mathbb{E}[M_n] \geq \mathbb{E}[M_m]$ for all $n \geq m \geq 0$.*

(iii) *If $(M_t)_{t\geq 0}$ is a martingale w.r.t. filtration $(\mathscr{F}_t)_{t\geq 0}$, then $\mathbb{E}[M_n] = \mathbb{E}[M_m]$ for all $n \geq m \geq 0$.*

PROOF. (ii) and (iii) follows directly from (i), so we only show (i). Let $(M_t)_{t\geq 0}$ be a supermartingale w.r.t. $(X_t)_{\geq 0}$. Recall that for each $m \in \mathbb{R}$, $\{M_t = m\} \in \mathscr{F}_t$. Hence

$$\mathbb{E}[M_{t+1} - M_t \,|\, M_t = m] \leq 0 \tag{486}$$

since M_t is a supermartingale. Hence by conditioning on the values of M_t,

$$\mathbb{E}[M_{t+1} - M_t] = \mathbb{E}[\mathbb{E}[M_{t+1} - M_t \,|\, M_t]] \leq 0. \tag{487}$$

Then the assertion follows by an induction. □

In order to get familiar with martingales, it is helpful to envision them as a kind of simple symmetric random walk. In general, one can subtract off the mean of a given random walk to make it a martingale.

Example 4.1.5 (Random walks)**.** Let $(X_t)_{t\geq 1}$ be a sequence of i.i.d. increments with $\mathbb{E}[X_i] = \mu < \infty$. Let $S_t = S_0 + X_1 + \cdots + X_t$. Then $(S_t)_{t\geq 0}$ is called a *random walk*. Define a stochastic process $(M_t)_{t\geq 0}$ by

$$M_t = S_t - \mu t. \tag{488}$$

For each $t \geq 0$, let $\mathscr{F}_t$ be the information obtained by observing $S_0, S_1, \cdots, S_t$. Then $(M_t)_{t\geq 0}$ is a martingale with respect to the filtration $(\mathscr{F}_t)_{t\geq 0}$. Indeed, we have

$$\mathbb{E}[|M_t|] = \mathbb{E}[|S_t - \mu t|] \leq \mathbb{E}[|S_t| + |\mu t|] = \mathbb{E}[|S_t|] + \mu t < \infty, \tag{489}$$

and for any $m \in \mathbb{R}$,

$$\{M_t = m\} = \{S_t - \mu t = m\} = \{S_t = m + \mu t\} \in \mathscr{F}_t. \tag{490}$$

Furthermore, Since X_{t+1} is independent from $S_0, \cdots, S_t$, it is also independent from any $A \in \mathscr{F}_t$. Hence

$$\mathbb{E}[M_{t+1} - M_t \,|\, X_0, \ldots, X_t] = \mathbb{E}[X_{t+1} - \mu \,|\, X_0, \ldots, X_t] \tag{491}$$

$$= \mathbb{E}[X_{t+1} - \mu] = \mathbb{E}[X_{t+1}] - \mu = 0. \tag{492}$$

▲

Example 4.1.6 (Products of indep. RVs)**.** Let $(X_t)_{t\geq 0}$ be a sequence of independent RVs such that $X_t \geq 0$ and $\mathbb{E}[X_t] = 1$ for all $t \geq 0$. For each $t \geq 0$, let $\mathscr{F}_t$ be the information obtained by observing $M_0, X_0, \cdots, X_t$. Define

$$M_t = M_0 X_1 X_2 \cdots X_t, \tag{493}$$

where M_0 is a constant. Then $(M_t)_{t\geq 0}$ is a martingale with respect to $(\mathscr{F}_t)_{t\geq 0}$. Indeed, the assumption implies $\mathbb{E}[|M_t|] < \infty$ and that $\{M_t = m\} \in \mathscr{F}_t$ for all $m \in \mathbb{R}$ since M_t is determined by $M_0, X_1 \cdots, X_t$. Furthermore, since X_{t+1} is independent from $X_1, \cdots, X_t$, for each $A \in \mathscr{F}_t$,

$$\begin{aligned}
\mathbb{E}[M_{t+1} - M_t \,|\, X_0, \ldots, X_t] &= \mathbb{E}[M_t X_{t+1} - M_t \,|\, X_0, \ldots, X_t] && (494)\\
&= \mathbb{E}[(X_{t+1} - 1)(M_0 X_1 \cdots X_t) \,|\, X_0, \ldots, X_t] && (495)\\
&= \mathbb{E}[X_{t+1} - 1 \,|\, X_0, \ldots, X_t]\, \mathbb{E}[(M_0 X_1 \cdots X_t) \,|\, X_0, \ldots, X_t] && (496)\\
&= \mathbb{E}[X_{t+1} - 1]\, \mathbb{E}[(M_0 X_1 \cdots X_t) \,|\, X_0, \ldots, X_t] = 0. && (497)
\end{aligned}$$

This multiplicative model a reasonable for the stock market since the changes in stock prices are believed to be proportional to the current stock price. Moreover, it also guarantees that the price will stay positive, in comparison to additive models. ▲

Exercise 4.1.7 (Exponential martingale)**.** Let $(X_t)_{t\geq 0}$ be a sequence of i.i.d. RVs such that their moment generating function exists, namely, there exists $\theta > 0$ for which

$$\varphi(\theta) := \mathbb{E}[\exp(\theta X_k)] < \infty \qquad \forall k \geq 0. \tag{498}$$

Let $S_t = S_0 + X_1 + \cdots + X_t$. Define

$$M_t = \exp(\theta S_t)/\varphi(\theta)^t. \tag{499}$$

Show that $(M_t)_{t\geq 0}$ is a martingale with respect to filtration $(\mathscr{F}_t)_{t\geq 0}$, where $\mathscr{F}_t$ is the information obtained by observing $S_0, \cdots, S_t$.

The following lemma allows us to provide many examples of martingales from Markov chains.

Lemma 4.1.8. *Let $(X_t)_{t\geq 0}$ be a Markov chain on state space Ω with transition matrix P. For each $t \geq 0$, let f_t be a bounded function $\Omega \to \mathbb{R}$ such that*

$$f_t(x) = \sum_{y\in\Omega} P(x,y) f_{t+1}(y) \qquad \forall x \in \Omega. \tag{500}$$

Then $M_t = f_t(X_t)$ defines a martingale with respect to filtration $(\mathscr{F}_t)_{t\geq 0}$, where $\mathscr{F}_t = \sigma(X_0, \ldots, X_t)$.

PROOF. Since f_t is bounded, $\mathbb{E}[|f_t(M_t)|] < \infty$. By definition, M_t is given as a function $f_t(X_t)$ of the reference process $(X_n)_{n\geq 0}$. For the conditional increment property, note that for any $x_0, x_1, \ldots, x_{t-1}, x \in \Omega$,

$$\begin{aligned}
\mathbb{E}[M_{t+1} - M_t \,|\, X_0 = x_0, \ldots, X_t = x_{t-1}, X_t = x] &= \mathbb{E}[M_{t+1} - M_t \,|\, X_t = x] && (501)\\
&= \mathbb{E}\left[\left(\sum_{y\in\Omega} P(x,y) f_{t+1}(y)\right) - f_t(x)\right] = 0. && (502)
\end{aligned}$$

This shows the assertion. □

Example 4.1.9 (Simple random walk)**.** Let $(X_t)_{t\geq 1}$ be a sequence of i.i.d. RVs with

$$\mathbb{P}(X_k = 1) = p, \qquad \mathbb{P}(X_k = -1) = 1 - p. \tag{503}$$

Let $S_t = S_0 + X_1 + \cdots + X_t$. Note that $(S_t)_{t\geq 0}$ is a Markov chain on $\mathbb{Z}$. Define

$$M_t = \left(\frac{1-p}{p}\right)^{S_n}. \tag{504}$$

Then $(M_t)_{t\geq 0}$ is a martingale with respect to filtration $(\mathscr{F}_t)_{t\geq 0}$, where $\mathscr{F}_t$ is the information obtained by observing $S_0, \cdots, S_t$.

In order to see this, define a function $h_t(x) = ((1-p)/p)^x$. According to Lemma 4.1.8, it suffice to show that h is a harmonic function with respect to the Gambler's chain. (One can also check that M_t is a martingale by Example 4.1.6.) Namely,

$$\sum_{y\in\mathbb{Z}} P(x,y)h_t(y) = ph(x+1) + (1-p)h(x-1) \tag{505}$$

$$= p\left(\frac{1-p}{p}\right)^{x+1} + (1-p)\left(\frac{1-p}{p}\right)^{x-1} \tag{506}$$

$$= (1-p)\left(\frac{1-p}{p}\right)^{x} + p\left(\frac{1-p}{p}\right)^{x} = \left(\frac{1-p}{p}\right)^{x} = h_t(x). \tag{507}$$

Hence by Lemma 4.1.8, $(M_t)_{t\geq 0}$ is a martingale with respect to the filtration $(\mathscr{F}_t)_{t\geq 0}$. ▲

Example 4.1.10 (Simple symmetric random walk)**.** Let $(X_t)_{t\geq 1}$ be a sequence of i.i.d. RVs with

$$\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = 1/2. \tag{508}$$

Let $S_t = S_0 + X_1 + \cdots + X_t$. Note that $(S_t)_{t\geq 0}$ is a Markov chain on $\mathbb{Z}$. Define

$$M_t = S_t^2 - t. \tag{509}$$

Then $(M_t)_{t\geq 0}$ is a martingale with respect to $(S_t)_{t\geq 0}$.

One can check the martingale condition directly as follow:

$$\mathbb{E}[(S_{t+1}^2 - (t+1)) - (S_t^2 - t)\,|\,(X_0,\ldots,X_t)] = \mathbb{E}[(S_t + X_{t+1})^2 - S_t^2 - 1\,|\,(X_0,\ldots,X_t)] \tag{510}$$

$$= \mathbb{E}[2S_tX_{t+1} + \underbrace{X_{t+1}^2}_{=1} - 1\,|\,(X_0,\ldots,X_t)] \tag{511}$$

$$= 2S_t\mathbb{E}[X_{t+1}\,|\,(X_0,\ldots,X_t)] = 0, \tag{512}$$

where the last equality follows since $\mathbb{E}[X_{t+1}] = 0$ and X_{t+1} is independent from $X_0,\ldots,X_t$.

One can also check the martingale condition for $S_t^2 - t$ using Lemma 4.1.8. Namely, for each $t \geq 0$, define a function $f_t : \mathbb{Z} \to \mathbb{R}$ by $f_t(x) = x^2 - t$. By Lemma 4.1.8, it suffices to check if $f_t(x)$ is the average of $f_{t+1}(y)$ with respect to the transition matrix of S_t. Namely,

$$\sum_{y\in\mathbb{Z}} P(x,y)f_{t+1}(y) = \frac{1}{2}f_{t+1}(x+1) + \frac{1}{2}f_{t+1}(x-1) \tag{513}$$

$$= \frac{(x+1)^2 - (t+1)}{2} + \frac{(x-1)^2 - (t-1)}{2} = x^2 - t = f_t(x). \tag{514}$$

Hence by Lemma 4.1.8, $(M_t)_{t\geq 0}$ is a martingale with respect to filtration $(\mathscr{F}_t)_{t\geq 0}$, where $\mathscr{F}_t$ is the information obtained by observing $S_0,\cdots,S_t$. ▲

4.2. Gambling strategies and stopping times

In this section, we derive some properties of martingales in relation to stopping times viewed as gambling strategies. The following observation is natural if we think of supermartingale as a random walk with negative drift; or as betting on an unfavorable game.

Next, we prove one of the most famous result in martingale theory, which says

"You can't beat an unfavorable game." (515)

To formulate this, let $(M_t)_{t\geq 0}$ be a supermartingale w.r.t. $(X_t)_{t\geq 0}$. Think of $(X_t)_{t\geq 0}$ as the progression of stock market, and $(M_t)_{t\geq 0}$ is the price of the stock of company A. Say we have an investment strategy so that we can determine

$$H_{t+1} = \text{Amount of stock of company A that we hold during } [t, t+1), \tag{516}$$

based on the observation $X_0, \cdots, X_t$. Then the stock price will change from M_t to M_{t+1}, so

$$H_{t+1}(M_{t+1} - M_t) = \text{Net gain occured between time } t \text{ and } t+1. \tag{517}$$

Hence if we let W_0 denote the initial fortune, then

$$W_t := W_0 + \sum_{k=1}^{t} H_k(M_k - M_{k-1}) = \text{Total fortune at time } t. \tag{518}$$

The following theorem tells us that, if $(M_t)_{t\geq 0}$ is declining stock on average (supermartingale), then no matter what strategy $(H_t)_{t\geq 0}$ we use, we will always lose our fortune.

Theorem 4.2.1. *Let $(M_t)_{t\geq 0}$ be a supermartingale w.r.t. a filtration $(\mathscr{F}_t)_{t\geq 0}$. Suppose $(H_t)_{t\geq 0}$ is such that*

(i) *(Predictability) $\{H_{t+1} = h\} \in \mathscr{F}_t$ for all $h \in \mathbb{R}$.*

(ii) *(Boundedness) $0 \leq H_t \leq c_t$ for some constant $c_t \geq 0$ for all $t \geq 0$.*

Let $(W_t)_{t\geq 0}$ be as defined at (518). Then $(W_t)_{t\geq 0}$ is a supermartingale w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$.

PROOF. By (ii) and a triangle inequality,

$$|W_t| \leq |W_0| + \sum_{k=1}^{t} c_t(|M_k| + |M_{t-1}|). \tag{519}$$

Taking expectation shows $\mathbb{E}[|W_t|] < \infty$. Moreover, for each $w \in \mathbb{R}$,

$$\{W_t = w\} = \left\{ W_0 + \sum_{k=1}^{t} H_k(M_k - M_{k-1}) = w \right\} \in \mathscr{F}_t \tag{520}$$

since the event on the right hand side can be written as the union of suitable events involving the values of $H_0, \ldots, H_t$ and $M_0, \ldots, M_t$, which are all events belonging to $\mathscr{F}_t$.

Lastly, fix $A \in \mathscr{F}_t$. For each $h \in [0, c_t]$, since $\{H_{t+1} = h\} \in \mathscr{F}_t$, we also have $A \cap \{H_{t+1} = h\} \in \mathscr{F}_t$. Since M_t is a supermartingale, we have

$$\begin{aligned} \mathbb{E}[W_{t+1} - W_t \,|\, A \cap \{H_{t+1} = h\}] &= \mathbb{E}[H_{t+1}(M_{t+1} - M_t) \,|\, A \cap \{H_{t+1} = h\}] &(521)\\ &= h\mathbb{E}[M_{t+1} - M_t \,|\, A] \leq 0. &(522)\end{aligned}$$

Hence, by conditioning on the values of H_{t+1}, we get

$$\begin{aligned} \mathbb{E}[W_{t+1} - W_t \,|\, A] &= \mathbb{E}\left[\mathbb{E}[W_{t+1} - W_t \,|\, A, H_{t+1}] \,|\, A\right] &(523)\\ &= \mathbb{E}\left[\mathbb{E}[(W_{t+1} - W_t)\mathbf{1}(H_{t+1} \in A) \,|\, H_{t+1}] \,|\, A\right] &(524)\\ &= \mathbb{E}\left[\mathbf{1}(H_{t+1} \in A) \underbrace{\mathbb{E}[(W_{t+1} - W_t) \,|\, H_{t+1}]}_{=0} \,\middle|\, A \right] = 0. &(525)\end{aligned}$$

This shows the assertion. □

Next, we define stopping times with respect to a given filtration.

Definition 4.2.2. A random variable T taking valuse from $\{0, 1, 2, \ldots\} \cup \{\infty\}$ is a *stopping time* w.r.t. a filtration $(\mathscr{F}_t)_{t\geq 0}$ if

$$\{T = s\}, \{T \neq s\} \in \mathscr{F}_s \qquad \forall s \geq 0. \tag{526}$$

Note that the definition of stopping times w.r.t. a process $(X_n)_{n\geq 1}$ (see Definition 3.4.1) is a special case of the above definition when $\mathscr{F}_t = \sigma(X_1, \ldots, X_n)$ (see Exercise 4.1.3).

Example 4.2.3. Let $(N_1(t))_{t\geq 0}$ and $(N_2(t))_{t\geq 0}$ be the counting processes of two renewal processes that are not necessarily independent. For each $t \geq 0$, let $\mathscr{F}_t$ be the information obtained by observing $(N_i(t))_{0\leq s\leq t}$ for $i = 1, 2$. Let $T_k^{(i)}$ denote the kth arrival time for the ith process.

Clearly, for each $i \in \{1,2\}$ and $k \geq 0$, $T_k^{(i)}$ is a stopping time w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$. Namely, for each $t \geq 0$,

$$\{T_k^i = t\} = \{N^{(i)}(t) = k,\, N^{(i)}(t^-) = k-1\} \in \mathscr{F}_t. \tag{527}$$

Also, let

$$\tilde{T}_k := k\text{the arrival time for the merged process } N_1 + N_2. \tag{528}$$

This is also a stopping time w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$. Namely, for each $t \geq 0$,

$$\{\tilde{T}_k = t\} = \{N^{(1)}(t) + N^{(2)}(t) = k,\, N^{(1)}(t^-) + N^{(2)}(t^-) = k-1\} \in \mathscr{F}_t. \tag{529}$$

On the other hand, let $(N^{(3)}(t))_{t\geq 0}$ be the counting process of another renewal process, which is independent from the other processes. Let $T_k^{(3)}$ denote the kth arrival time of this process. Is this a stopping time w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$? No, since for each $t \geq 0$, the event

$$\{T_k^{(3)} = t\} = \{N^{(3)}(t) = k,\, N^{(3)}(t^-) = k-1\} \tag{530}$$

cannot be determined from observing the other two processes $N^{(1)}$ and $N^{(2)}$ up to time t. ▲

Example 4.2.4 (Constant betting up to a stopping time)**.** Let $(M_t)_{t\geq 0}$ be a supermartingale w.r.t. $(X_t)_{t\geq 0}$. Let T be a stopping time for $(X_t)_{t\geq 0}$. Consider the gambling strategy of betting \$1 up to time T. That is,

$$H_k = \mathbf{1}(k \leq T) = \begin{cases} 1 & \text{if } k \leq T \\ 0 & \text{if } k > T \end{cases}. \tag{531}$$

Since T is a stopping time, the above H_k defines a predictable sequence as in Theorem 4.2.1. Then the wealth at time t is

$$W_t = W_0 + \sum_{k=1}^{t} H_k(M_k - M_{k-1}) = W_0 + M_{T\wedge t} - M_0, \tag{532}$$

where $T \wedge t = \min(T, t)$. The above relation can be easily verified by considering whether $t > T$ or $t \leq T$. According to Theorem 4.2.1, we known that W_t is a supermartingale. It follows that $M_{T\wedge t}$ is also a supermaringale. ▲

Theorem 4.2.5 (Stopped martingale is a martingale)**.** *Let $(M_t)_{t\geq 0}$ be a supermartingale w.r.t. a filtration $(\mathscr{F}_t)_{t\geq 0}$. Let T be a stopping time for $(\mathscr{F}_t)_{t\geq 0}$. Then $M_{T\wedge t}$ is a supermartingale w.r.t. $(\mathscr{F}_t)_{t\geq 0}$. Furthermore,*

$$\mathbb{E}[M_{T\wedge t}] \leq \mathbb{E}[M_0] \qquad \forall\, t \geq 0. \tag{533}$$

In particular, if $(M_t)_{t\geq 0}$ is a martingale, then

$$\mathbb{E}[M_{T\wedge t}] = \mathbb{E}[M_0] \qquad \forall\, t \geq 0. \tag{534}$$

PROOF. Let W_t be as defined as in Example 4.2.4. Since $W_t = W_0 + M_{T\wedge t} - M_0$ and since W_t is a supermaringale by Theorem 4.2.1, and since W_0 and M_0 are constants, it follows that $M_{T\wedge t}$ is also a supermartingale. Then by Proposition 4.1.4, we have

$$\mathbb{E}[M_{T\wedge t}] \leq \mathbb{E}[W_t] \leq \mathbb{E}[W_0] = \mathbb{E}[M_0]. \tag{535}$$

The same argument holds for the submartingale case with the reversed inequalities. Then the martingale case follows. □

Exercise 4.2.6. Let $(M_t)_{t\geq 0}$ be a supermartingale w.r.t. a filtration $(\mathscr{F}_t)_{t\geq 0}$. We will directly show that $(M_{T\wedge t})_{t\geq 0}$ is a supermartingale w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$.

(i) For each fixed $A \in \mathscr{F}_t$, show that

$$A \cap \{T \le t\} = A \cap \left(\bigcup_{k=1}^{t} \{T = k\} \right) \in \mathscr{F}_t, \tag{536}$$

$$A \cap \{T \ge t+1\} = A \cap \left(\bigcap_{k=1}^{t} \{T \ne k\} \right) \in \mathscr{F}_t. \tag{537}$$

(ii) Use conditioning on whether $T \le t$ or not to show

$$\mathbb{E}[M_{T\wedge(t+1)} - M_{T\wedge t} \,|\, A] = \mathbb{E}[M_T - M_T \,|\, A \cap \{T \le t\}]\mathbb{P}(T \le t) \tag{538}$$

$$+ \mathbb{E}[M_{t+1} - M_t \,|\, A \cap \{T \ge t+1\}]\mathbb{P}(T \ge t+1) \tag{539}$$

$$\le 0. \tag{540}$$

Conclude that $(M_{T\wedge t})_{t\ge 0}$ is a supermartingale w.r.t. the filtration $(\mathscr{F}_t)_{t\ge 0}$.

4.3. Optional stopping time theorem and its applications

If a martingale is bounded up to a stopping time, then the expected value of the martingale at stopping time equals the initial expectation. This observation will be useful in many applications.

Theorem 4.3.1 (Optional stopping)**.** *Suppose $(M_t)_{\ge 0}$ is a martingale and T is a stopping time with respect to the same filtration $(\mathscr{F}_n)_{n\ge 1}$ such that $\mathbb{P}(T < \infty) = 1$. If any of the following conditions hold, then $\mathbb{E}[M_T] = \mathbb{E}[M_0]$.*

(i) *(Bounded stopping time) There is a constant $c > 0$ such that $\mathbb{P}(T \le c) = 1$;*
(ii) *(Bounded stopped martingale) There is a constant $c > 0$ such that $\mathbb{P}(M_{T\wedge n} \le c) = 1$ for all $n \ge 1$;*
(iii) *(Bounded conditional increments) There is a constant $c > 0$ such that $\mathbb{E}[|M_{T\wedge(n+1)} - M_{T\wedge n}| \,|\, \mathscr{F}_n] \le c$ for all $n \ge 1$.*

Example 4.3.2 (Gambler's ruin)**.** Let $(X_t)_{t\ge 1}$ be i.i.d. RVs with

$$\mathbb{P}(X = 1) = p, \qquad \mathbb{P}(X = -1) = 1 - p. \tag{541}$$

Let $S_t = S_0 + X_1 + \cdots + X_t$ with $S_0 = i$. Let $\mathscr{F}_t$ be the information obtained by observing $S_0, \cdots, S_t$. Let $h(x) = ((1-p)/p)^x$ as in Exercise 4.1.9. We have seen $M_t := h(S_t)$ is a martingale w.r.t. the filtration $(\mathscr{F}_t)_{t\ge 0}$.

Fix integers $a < S_0 < b$, and define

$$T = \min\{t \ge 0 \,|\, S_t = a \text{ or } S_t = b\}. \tag{542}$$

So T is the first time that the random walk hits a or b. Note that T is a stopping time, and since S_t is an irreducible Markov chain on $\mathbb{Z}$, its hitting time to any state is almost surely finite, so $\mathbb{P}(T < \infty) = 1$. Moreover, $S_T, S_{T\wedge t} \in [a, b]$, satisfying the bounded stopped martingale condition in the optional stopping (Theorem 4.3.1). Hence by that theroem, $\mathbb{E}[h(S_T)] = \mathbb{E}[h(S_0)] = \left(\frac{1-p}{p}\right)^i$, so we have

$$\left(\frac{1-p}{p}\right)^i = \mathbb{E}[h(S_T)] = \left(\frac{1-p}{p}\right)^a \mathbb{P}(S_T = a) + \left(\frac{1-p}{p}\right)^b (1 - \mathbb{P}(S_T = a)). \tag{543}$$

Assuming $p \ne 1/2$, solving for $\mathbb{P}(S_T = a)$, this gives

$$\mathbb{P}(S_T = a) = \frac{\left(\frac{1-p}{p}\right)^b - \left(\frac{1-p}{p}\right)^i}{\left(\frac{1-p}{p}\right)^b - \left(\frac{1-p}{p}\right)^a}. \tag{544}$$

Similarly,

$$\left(\frac{1-p}{p}\right)^i = \mathbb{E}[h(S_T)] = \left(\frac{1-p}{p}\right)^a (1-\mathbb{P}(S_T=b)) + \left(\frac{1-p}{p}\right)^b \mathbb{P}(S_T=b). \tag{545}$$

so solving for $\mathbb{P}(S_T=b)$,

$$\mathbb{P}(S_T=b) = \frac{\left(\frac{1-p}{p}\right)^a - \left(\frac{1-p}{p}\right)^i}{\left(\frac{1-p}{p}\right)^a - \left(\frac{1-p}{p}\right)^b}. \tag{546}$$

In particular, if $a \to -\infty$, then $\left(\frac{1-p}{p}\right)^a \to 0$, so

$$\lim_{a\to-\infty} \mathbb{P}(S_T=b) = \left(\frac{p}{1-p}\right)^{b-i}. \tag{547}$$

The above equals the probability that a simple symmetric random walk with up-step probability $p<1/2$ started at i ever reaches b. ▲

Example 4.3.3 (Duration of fair games)**.** Let $(X_t)_{t\geq 1}$ be i.i.d. RVs with

$$\mathbb{P}(X=1) = \mathbb{P}(X=-1) = 1/2. \tag{548}$$

Let $S_t = S_0 + X_1 + \cdots + X_t$ with $S_0 = 0$. Let $\mathscr{F}_t$ be the information obtained by observing $S_0, \cdots, S_t$. Note that $(S_t)_{t\geq 0}$ itself is a martingale w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$. Also, we have seen that $M_t := S_t^2 - t$ is a martingale w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$ in Exercise 4.1.10.

Fix integers $a<0<b$, and define

$$T = \min\{t\geq 0 \,|\, S_t = a \text{ or } S_t = b\}. \tag{549}$$

Again T is a stopping time w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$. As before, we also have $\mathbb{P}(T<\infty)$ and $S_T, S_{T\wedge t} \in [a,b]$. Hence by Theorem 4.3.1, we get

$$0 = \mathbb{E}[S_0] = \mathbb{E}[S_T] = a\mathbb{P}(S_T=a) + b\mathbb{P}(S_T=b). \tag{550}$$

Since $\mathbb{P}(S_T=b) = 1-\mathbb{P}(S_T=a)$, the first equation yield

$$\mathbb{P}(S_T=a) = \frac{b}{b-a}, \quad \mathbb{P}(S_T=b) = \frac{-a}{b-a}. \tag{551}$$

Next, we will again apply the optional stopping (Thoerem 4.3.1) for the martingale $M_t = S_t^2 - t$ with stopping time T. In order to do so, we check the bounded increment condition in Thoerem 4.3.1:

$$\begin{aligned} \left|M_{T\wedge(t+1)} - M_{T\wedge t}\right| &= \left|S^2_{T\wedge(t+1)} - S^2_{T\wedge t} - (T\wedge(t+1) - T\wedge t)\right| && (552)\\ &\leq S^2_{T\wedge(t+1)} + S^2_{T\wedge t} + 1 && (553)\\ &\leq \max(a^2,b^2)+1, && (554)\end{aligned}$$

where the last inequality above uses the fact that $S_{T\wedge n} \in [a,b]$ for all $n\geq 1$. Hence by optional stopping,

$$\begin{aligned} 0 = \mathbb{E}[M_0] = \mathbb{E}[M_T] = \mathbb{E}[S_T^2 - T] &= \mathbb{E}[S_T^2] - \mathbb{E}[T] && (555)\\ &= a^2\mathbb{P}(S_T=a) + b^2\mathbb{P}(S_T=b) - \mathbb{E}[T] && (556)\\ &= \frac{a^2 b}{b-a} - \frac{ab^2}{b-a} - \mathbb{E}[T] \qquad \text{(by (551))} && (557)\\ &= -ab - \mathbb{E}[T]. && (558)\end{aligned}$$

Therefore we deduce

$$\mathbb{E}[T] = -ab > 0. \tag{559}$$

Observe that for each fixed $a<0$, $\mathbb{E}[T] = |ab| \to \infty$ as $b\to\infty$. In other words, if one starts gambling on fair coin flips, each time winning or losing \$1, then the expected time $\mathbb{E}[T]$ that he reaches $-\$1$, or

ruins, is infinity. In terms of random walk, this shows that the first return time of a simple symmetric random walk has infinite expected value (recall null recurrence of SSRW on $\mathbb{Z}$ in Example 3.6.3) . In terms of $M/M/1$ queue, this shows that if the arrival and service rate is the same, then there is no stationary distribution for the queue size. ▲

Next, we recall the following computation of normal MGF.

Exercise 4.3.4 (MGF of normal RVs)**.** Let $X \sim N(\mu, \sigma^2)$ and $Z \sim N(0,1)$. Using the fact that $\mathbb{E}[e^{tZ}] = e^{t^2/2}$, show that

$$\mathbb{E}[e^{tY}] = \exp(\sigma^2 t^2/2 + t\mu). \tag{560}$$

The following example illustrates how martingales can be used in a risk management situation.

Example 4.3.5 (Cramér's estimate of ruin)**.** Let S_n denote the total assets of an insurance company at the end of year n. During year n, premiums totaling $c > 0$ dollars are received, while claims totaling Y_n dolors are paid. Hence

$$S_n = S_{n-1} + c - Y_n. \tag{561}$$

Let $X_n = c - Y_n$ be the net profit during year n. We may assume that X_n's are i.i.d. with distribution $N(\mu, \sigma^2)$. We are interested in estimating the probability of bankruptcy. Namely, let

$$B = \{S_n < 0 \text{ for some } n \geq 1\} = \{\text{Bankruptcy}\}. \tag{562}$$

We will show that

$$\mathbb{P}(B) \leq \exp\left(-\frac{2\mu S_0}{\sigma^2}\right). \tag{563}$$

Hence, the company will avoid bankruptcy by maximizing the mean profit per year μ and the initial asset S_0, while minimizing the uncertainty σ of profit per year.

To begin, notice that $S_n = S_0 + X_1 + \cdots + X_n$ is a random walk. Let $\mathscr{F}_t$ be the information obtained by observing $S_0, S_1, \cdots, S_t$. According to Exercise 4.3.4, we have

$$\varphi(\theta) := \mathbb{E}[e^{\theta X_k}] = \exp(\sigma^2\theta^2/2 + \theta\mu) < \infty. \tag{564}$$

Hence we can use the following exponential martingale (see Exercise 4.1.7)

$$M_n(\theta) := \frac{\exp(\theta S_n)}{\varphi(\theta)} \tag{565}$$

with respect to the filtration $(\mathscr{F}_t)_{t\geq 0}$. Moreover, note that

$$\varphi(-2\mu/\sigma^2) = \exp(2\mu^2/\sigma^2 - 2\mu^2/\sigma^2) = \exp(0) = 1. \tag{566}$$

Hence

$$M_n := M_n(-2\mu/\sigma^2) = \exp\left(-\frac{2\mu S_n}{\sigma^2}\right) \tag{567}$$

is a martingale w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$.

Let $T = \min\{k \geq 1 \,|\, S_k < 0\}$ be the first time of hitting negative asset, which is a stopping time w.r.t. the filtration $(\mathscr{F}_t)_{t\geq 0}$. Also note that

$$M_T = \exp\left(-\frac{2\mu S_T}{\sigma^2}\right) > 1 \quad \text{a.s.}, \tag{568}$$

since $S_T < 0$. Since S_t is a supercritical random walk ($\mu > 0$), $\mathbb{P}(T = \infty) > 0$. Hence we need to stop the martingale at truncated stopping time $T \wedge t$, using Theorem 4.3.1. Noting that $M_t \geq 0$ for all $t \geq 0$, this yields

$$\exp\left(-\frac{2\mu S_0}{\sigma^2}\right) = \mathbb{E}[M_0] = \mathbb{E}[M_{T\wedge t}] = \mathbb{E}\left[M_T \,|\, T \leq t\right]\mathbb{P}(T \leq t) + \mathbb{E}[M_t \,|\, T > t]\mathbb{P}(T > t) \tag{569}$$

$$\geq \mathbb{E}\left[M_T \,|\, T \leq t\right]\mathbb{P}(T \leq t) \geq \mathbb{P}(T \leq t). \tag{570}$$

Note that the last equality follows since S_T given $T \le t$ is zero almost surely and $M_t \ge 0$ for all $t \ge 0$. To finish, note that $\mathbb{P}(T \le t)$ is the probability of having bankruptcy by time t. Hence

$$\mathbb{P}(B) = \mathbb{P}(T < \infty) = \lim_{n\to\infty} \mathbb{P}(T \le n) \le \exp\left(-\frac{2\mu S_0}{\sigma^2}\right). \tag{571}$$

▲

PROOF OF THEOREM 4.3.1. (Optional*) Since $\mathbb{P}(T < \infty) = 1$, it holds that $\lim_{t\to\infty} T \wedge t = T$ almost surely. Hence by Theorem 4.2.5, one would have

$$\mathbb{E}[M_0] = \lim_{t\to\infty} \mathbb{E}[M_{T\wedge t}] \overset{?}{=} \mathbb{E}[\lim_{t\to\infty} M_{T\wedge t}] = \mathbb{E}[M_T] \tag{572}$$

provided the middle equality is verified.

To this end, fix $t > 0$, and by conditioning on whether $T > t$ or not, we can write

$$\mathbb{E}[M_T] = \mathbb{E}[M_T \,|\, T \le t]\mathbb{P}(T \le t) + \mathbb{E}[M_T \,|\, T > t]\mathbb{P}(T > t) \tag{573}$$

$$= \mathbb{E}[M_{T\wedge t} \,|\, T \le t]\mathbb{P}(T \le t) + \mathbb{E}[M_T \,|\, T > t]\mathbb{P}(T > t). \tag{574}$$

Similarly, we also write

$$\mathbb{E}[M_{T\wedge t}] = \mathbb{E}[M_{T\wedge t} \,|\, T \le t]\mathbb{P}(T \le t) + \mathbb{E}[M_{T\wedge t} \,|\, T > t]\mathbb{P}(T > t). \tag{575}$$

Subtracting these two equations, we get

$$|\mathbb{E}[M_T] - \mathbb{E}[M_{T\wedge t}]| \le |\mathbb{E}[M_T \,|\, T > t]|\mathbb{P}(T > t) + |\mathbb{E}[M_{T\wedge t} \,|\, T > t]|\mathbb{P}(T > t). \tag{576}$$

Now suppose **(i)** holds. Then $\mathbb{P}(T > t) = 0$ for all $t > c$. Hence (576) gives

$$\lim_{t\to\infty} |\mathbb{E}[M_T] - \mathbb{E}[M_{T\wedge t}]| = 0, \tag{577}$$

as desired. Simmilarly, suppose **(ii)** holds. Then $M_{T\wedge t} \le c$ for all $t \ge 0$, so (576) gives

$$|\mathbb{E}[M_T] - \mathbb{E}[M_{T\wedge t}]| \le 2c\,\mathbb{P}(T > t) \tag{578}$$

and the right hand side tends to zero as $t \to \infty$.

Lastly, suppose **(iii)** holds. Since $\mathbb{P}(T < \infty) = 1$, we have $T \wedge n \to T$ a.s. as $n \to \infty$, so $M_{T\wedge n} \to M_T$ a.s. as $n \to \infty$. In order to show $\lim_{n\to\infty} \mathbb{E}[M_{T\wedge n}] = \mathbb{E}[M_T]$, we will have to use the dominated convergence theorem (see [Dur10, Thm. 1.5.8]). For that, we only need to find a dominating and integrable RV M, that is, $M_{T\wedge n} \le M$ for all $n \ge 1$ (dominating) and $\mathbb{E}[|M|] < \infty$ (integrable). For this, we use triangle inequality to construct a dominating RV as follows:

$$M := |M_0| + \sum_{n=1}^{\infty} |M_n - M_{n-1}|\mathbf{1}(T \ge n). \tag{579}$$

Ineed, by triangle inequality,

$$M = M_0 + \sum_{n=1}^{\infty} (M_n - M_{n-1})\mathbf{1}(n \le T) \le M. \tag{580}$$

On the other hand, from **(iii)** and iterated expectation,

$$\mathbb{E}[|M|] = \mathbb{E}[M] = \mathbb{E}[|M_0|] + \sum_{n=1}^{\infty} \mathbb{E}\left[|M_n - M_{n-1}|\mathbf{1}(T \geq n)\right] \tag{581}$$

$$= \mathbb{E}[|M_0|] + \sum_{n=1}^{\infty} \mathbb{E}\left[\mathbb{E}\left[|M_n - M_{n-1}|\mathbf{1}(T > n-1)\,\middle|\,\mathscr{F}_{n-1}\right]\right] \tag{582}$$

$$= \mathbb{E}[|M_0|] + \sum_{n=1}^{\infty} \mathbb{E}\left[\mathbf{1}(T > n-1)\underbrace{\mathbb{E}\left[|M_{T\wedge n} - M_{T\wedge(n-1)}|\,\middle|\,\mathscr{F}_{n-1}\right]}_{\leq c}\right] \tag{583}$$

$$= \mathbb{E}[|M_0|] + \sum_{n=1}^{\infty} c\,\mathbb{E}[\mathbf{1}(T > n-1)] \tag{584}$$

$$= \mathbb{E}[|M_0|] + c\sum_{n=0}^{\infty} \mathbb{P}(T \geq n) = \mathbb{E}[|M_0|] + c\mathbb{E}[T] < \infty. \tag{585}$$

This is enough to conclude by dominiated convergence theorem. □

4.4. Martingale limit theorem

Recall the familiar fact in analysis that a non-decreasing sequence of reals bounded from above converges; a non-increasing sequence of reals bounded from below converges. Can we have a random variable version of this fact? Consider the two extreme cases:

(1) *A sequence of a.s. non-decreasing and bdd RVs:* $X_1 \leq X_2 \leq \cdots \leq c$. In this case indeed we have $\lim_{n\to\infty} X_n = X_\infty$ for some RV X_∞. But this setting is too restrictive for broad applications.

(2) *A sequence of RVs with non-decreasing and bdd expectations:* $\mathbb{E}[X_1] \leq \mathbb{E}[X_2] \leq \cdots \leq c$. In this case X_n does not always converge to a limiting RV. (Counterexample: $X_n = \text{Bernoulli}(1/2) - 1/n$.)

It turns out, the submartingale condition is in between the two extreme cases above and is flexible enough to have many applications. This is stated in the following the following "martingale limit theorem" is a martingale analgue of these facts:

Theorem 4.4.1 (Martingale convergence theorem)**.** *If $(M_n)_{n\geq 1}$ is a submartingale that is bounded from above, that is, there exists a constant $c > 0$ such that $\mathbb{P}(M_n < c) = 1$ for all $n \geq 1$, then there exists a random variable M_∞ that is almost surely finite, and $\lim_{n\to\infty} M_n = M_\infty$ with probability one.*

PROOF. See [Dur10, Thm. 4.2.11]. □

By taking the negative sign, the supermartingale counterpart of the above theorem also holds. More specifically, an important corollary of Theroem 4.4.1 is the following.

Corollary 4.4.2. *If $(M_n)_{n\geq 1}$ is a supermartingale and $M_n \geq 0$ for $n \geq 1$, then M_n converges almost surely to some random variable M_∞ such that $\mathbb{P}(0 \leq M_\infty < \infty) = 1$.*

The following corollary is immedaite from Theorem 4.4.1 with the observation that a martingale is both submartingale and supermatingale.

Corollary 4.4.3 (Limit theorem for bounded martingales)**.** *Let $(M_n)_{n\geq 0}$ be a martingale with respect to $(X_n)_{n\geq 0}$. Suppose that either all M_n are bounded from above or from below almost surely. Then there exists a random variable M_∞ that is almost surely finite, and $\lim_{n\to\infty} M_n = M_\infty$ with probability one.*

Example 4.4.4 (SSRW on $\mathbb{Z}$)**.** Let $(S_n)_{n\geq 0}$ with $S_0 = 0$ denote simple symmetric random walk on $\mathbb{Z}$. We will cook up both examples and non-examples of the martingale convergence theorem.

First, $(S_n)_{n\geq 0}$ is indeed a martingale. However, it is not bounded from above and also from below. That is, there is no constant $c \in \mathbb{R}$ such that either $S_n > c$ for all $n \geq 1$ or $S_n < c$ for all $n \geq 1$ almost surely. (Indeed, recall that S_n is recurrent, so it visits every integer infinitely often with probability 1.)

Thus Corollary 4.4.2 (as well as Theorem 4.4.1) cannot be applied. Indeed, since $S_{n+1} - S_n \in \{-1, 1\}$ for all $n \geq 1$, S_n cannot converge to any single random variable.

Second, by stopping the random walk, we can create a convergent martingale. Let $T = \inf\{n \geq 1 \mid S_n = 1\}$ denote the first hitting time of 1. Then T is a stopping time, so by Theroem 4.2.5, $M_n = S_{T \wedge n}$ is a martingale. The martingale satisfies $M_n \leq 1$ since S_n takes values from $\{\ldots, -2, -1, 0, 1\}$ until time T (recall that $S_0 = 0$) and it stays at 1 after time T. Hence by Corollary 4.4.2, $M_n \to M_\infty$ for some almost surely finite RV M_∞.

We can in fact identify the limiting RV M_∞. Recall that $T < \infty$ with probability 1 the SSRW S_n is recurrent (see Example 3.5.7 and Theorem 3.5.4). Thus $T \wedge n \to T$ a.s. as $n \to \infty$. Hence $\lim_{n\to\infty} M_n = \lim_{n\to\infty} S_{T\wedge n} = S_T = 1$. So $M_\infty = 1$ with probability 1. ▲

4.5. Branching processes

We introduce the Galton-Watson branching process $(X_n)_{n\geq 0}$ as follow, which is a simple population growth model, where X_n denotes the population at generation n. Let $X_0 = 1$, which means that initially there is one living specie. At each discrete time step, each individual in the previous generation gives birth to a (nonnegative) random number of offsprings and dies. Let

$$Z_{n,i} := \text{number of offspring of the } i\text{th individual at generation } n, \qquad 1 \leq i \leq X_n, \tag{586}$$

Then $(X_n)_{n\geq 1}$ satisfy the following recursion

$$X_{n+1} = \sum_{i=1}^{X_n} Z_{n,i} \qquad n \geq 0. \tag{587}$$

In order to analyze this process, we assume that the offspring numbers $Z_{n,i}$ are i.i.d. with the common distribution $[\beta_0, \beta_1, \ldots]$ on $\mathbb{Z}_{\geq 0}$:

$$\mathbb{P}(Z_{n,i} = k) = \beta_k, \qquad k \in \mathbb{Z}_{\geq 0}. \tag{588}$$

We denoted the mean offspring number as μ, that is,

$$\mu = \mathbb{E}[Z_{n,i}] = \sum_{k=0}^{\infty} k\beta_k. \tag{589}$$

Then we claim that

$$M_n := \mu^{-n} X_n, \qquad n \geq 1 \tag{590}$$

is a martingale with respect to the filtration generated by all offspring numbers until the current generation.

Proposition 4.5.1. *$M_n = \mu^{-n} X_n$ is a martingale with respect to the filteration $\mathscr{F}_n = \sigma(X_0, \ldots, X_n)$.*

PROOF. First we verify the conditional increment condition:

$$\mathbb{E}\left[M_{n+1} \,\middle|\, X_0, X_1, \ldots, X_n\right] = \mu^{-n-1}\mathbb{E}\left[Z_{n,1} + \cdots + Z_{n,X_n} \,\middle|\, X_0, X_1, \ldots, X_n\right] = \mu^{-n} X_n = M_n. \tag{591}$$

Since the offspring numbers $Z_{n,i}$ are nonnegative, by iterated expectation it follows that

$$\mathbb{E}[|M_{n+1}|] = \mathbb{E}[M_n] = \mathbb{E}\left[\mathbb{E}\left[M_{n+1} \,\middle|\, X_0, X_1, \ldots, X_n\right]\right] = \mathbb{E}[M_n] = \cdots = \mathbb{E}[M_0] = X_0 = 1. \tag{592}$$

Lastly, M_n is clearly measurable with respect to the offspring numbers $Z_{k,i}$ for all $1 \leq k \leq n$. Hence M_n is a martingale. □

As an application, by Proposition 4.1.4, we have

$$\mathbb{E}[X_n] = \mu^n \mathbb{E}[M_n] = \mu^n \mathbb{E}[M_0] = \mu^n. \tag{593}$$

Hence, if the mean offspring number μ is >1, then the expected population $\mathbb{E}[X_n]$ grows exponentially as μ^n (supercritical branching process); if $\mu<1$, then $\mathbb{E}[X_n]$ decays to zero exponentially as μ^n (subcritical branching process); if $\mu=1$, then $\mathbb{E}[X_n]=1$ for all $n\geq 1$ (critical branching process).

Since $M_n=\mu^{-n}X_n$ is a nonnegative martingale, it should converge to a limit $\lim_{n\to\infty}M_n=M_\infty$ by Corollary 4.4.2. This means that we can describe the generation n population X_n as

$$X_n\approx \mu^n M_\infty. \tag{594}$$

Therefore, not only in expectation as in (593), but also trajectory-wise, X_n grows/decays exponentailly fast depending on $\mu>1$ or $\mu<1$.

However, there is a subtle point here: What if $M_\infty=0$ with some positive probability? Then $\mu^n M_\infty$ does not necessarily diverge almost surely even if $\mu>1$. In fact, there are two cases depending on whether 'extinction event' occurs. Namely, define

$$\texttt{Extinction}:=\{X_n=0 \text{ for all sufficiently large } n\}=\text{'Extinction event'} \tag{595}$$

$$\pi:=\mathbb{P}(\texttt{Extinction})=\text{'extinction probability'}. \tag{596}$$

Then from (594), we can deduce

$$\text{Either extintion event occurs or } X_n \text{ grows exponentially in } n. \tag{597}$$

Below we identify the limit M_∞ starting from the subcritical case, $\mu<1$. Here, we will observe that the extinction event occurs. This should not be surprising since each individual gives birth to less then one child on average, so the population should die out.

Proposition 4.5.2. *If $\mu<1$, then $X_n=0$ for all sufficiently large n. (That is, $\mathbb{P}(\texttt{Ext})=0$.) Hence $M_n=\mu^{-n}X_n\to 0$ as $n\to\infty$ almost surely.*

PROOF. Since $M_n=\mu^{-n}X_n$ is a martingale, $\mathbb{E}[M_n]=\mathbb{E}[M_0]=1$ so we have $\mathbb{E}[X_n]=\mu^n$. Now recalling that X_n takes values from $\mathbb{Z}_{\geq 0}$, by using Markov's inequality,

$$\mathbb{P}(X_n>0)=\mathbb{P}(X_n\geq 1)\leq \mathbb{E}[X_n]=\mu^{-n}. \tag{598}$$

Thus, since $0<\mu<1$,

$$\sum_{n=1}^{\infty}\mathbb{P}(X_n>0)\leq\sum_{n=1}^{\infty}\mu^{-n}=\frac{1}{1-\mu^{-1}}<\infty. \tag{599}$$

Then by Borel-Cantelli lemma, $X_n>0$ only for finitely many n almost surely. In other words, $X_n=0$ for all sufficeintly large n almost surely. □

Proposition 4.5.3. *If $\mu=1$ and $\beta_1<1$, then $X_n=0$ for all sufficiently large n. (That is, $\mathbb{P}(\texttt{Ext})=0$.)*

PROOF. Since $\mu=1$, $X_n=\mu^{-n}X_n$ itself is a nonngeative martingale, so it should converge, that is, $X_n\to X_\infty$ for some a.s. finite RV X_∞. But since X_n takes values from nonnegative integers, we must have $X_n=Z_\infty$ for all large n. But since $\beta_1<1$ and $\mu=1$, X_n cannot stay at the same positive value after any large N. Hence $X_\infty=0$ almost surely. □

Now suppose $\mu>1$. For simple cases, note that $\beta_0=\mathbb{P}(Z_{n,i}=0)=1$ implies immedaite extinction $\beta_0=0$ implies no extinction. (Also note that, in general, $\pi\geq\beta_0$ so $\beta_0>0$ implies $\pi>0$.) In order to analyze the intermediate case $\beta_0\in(0,1)$, we will use a generating function argument. Recall that

$$g(s):=\sum_{k=1}^{\infty}s^k\beta_k=\text{generating function of the offspring distribution.} \tag{600}$$

Then by one-step analysis and using a 'renwal property of branching process', we can observe the following recursion:

$$\pi=\beta_0+\sum_{k=1}^{\infty}\beta_k\pi^k=g(\pi). \tag{601}$$

Namely, the sub-population trees descending from the generation 1 individuals follow exactly the same distribution as the whole population tree, so the probability that each such subtree goes extinct is again π, and these events are all independent. The above equation tells us that the extinction probability π is a fixed point of the generation function g of the offspring distribution.

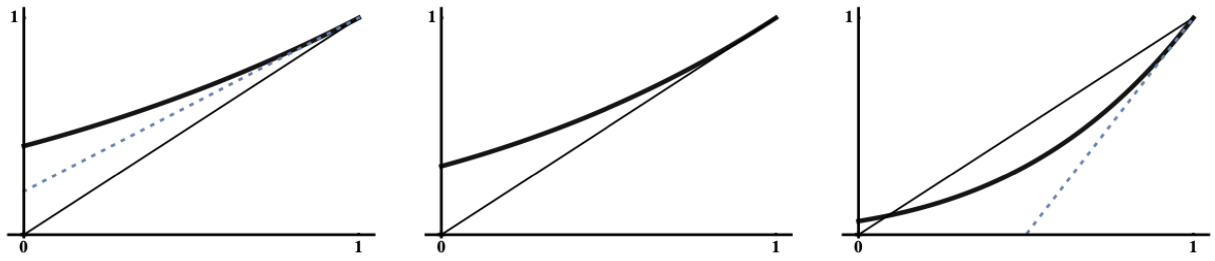


FIGURE 1. Generating functions $g(s)$ for cases when $\mu < 1$ (left), $\mu = 1$ (middle), $\mu > 1$ (right) in bold, and the lighter solid line is for the functino $y = x$. The dotted lines in the left and right plots are tengent lines to the generating functions at $s = 1^-$. The y-intercept is $g(0) = \beta_0$, and the x-coordinates of the intercepts between the curve and the line $y = x$ are the corresponding extinction probabilities π. Figure excerpted from [SV].

With a more refined analysis, we can show the following:

Proposition 4.5.4. *If $\mu > 1$ and $X_0 = 1$, then the extinction probability π is the only solution of $\pi = g(\pi)$ in $[0,1)$. In particular,*

$$\mathbb{P}(\text{The population survives forever}) = 1 - \pi > 0. \tag{602}$$

PROOF. See [Dur10, Thm 4.3.12] □

Finally, recall that, on the event that there is no extinction, X_n grows exponentially fast as $\mu^n M_\infty$, and on this even we also know $M_\infty > 0$ since otherwise $X_0 \to 0$, contradicting the no-extinction event.

Example 4.5.5. Consider a branching process with offspring distribution Poisson(λ). The generating function of the offspring distribution can be computed as

$$g(s) = \sum_{k=0}^{\infty} s^k \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=0}^{\infty} \frac{(s\lambda)^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(s\lambda)^k}{k!} = e^{-\lambda} e^{s\lambda} = e^{\lambda(s-1)}, \tag{603}$$

where we have used the Talyer expansion of the exponential function.

Now, suppose $\lambda \le 1$. Then the mean offspring number is $\lambda \le 1$, so the extinction probability π is 1. On the other hand, if $\lambda > 1$, then π is the unique solution of $\pi = e^{\lambda(\pi-1)}$ in $(0,1)$. This equation cannot be solved analytically, but with some computational aid, we can see $\pi \approx 0.2$ when $\lambda = 2$.

In fact, there is a good algorithm that computes the fixed point π with exponentially improving precision. Namely, consider generating a sequence $(\pi_n)_{n\ge 0}$ as $\pi_{n+1} = g(\pi_n) = e^{\lambda(\pi_n - 1)}$. For instance, set $\lambda = 2$ and $\pi_0 = 0.5$. Then $\pi_1 = 0.3678$, $\pi_2 = 0.2824$, $\pi_3 = 0.2380$, $\pi_4 = 0.2178$, $\pi_4 = 0.2092$, $\pi_5 = 0.2056$, and so on. ▲

4.6. Additional properties of martingales (Optional*)

In this section, we study some of the basic properties of martingales. We begin with the relation between martingale increments and convex functions. Recall that a function $\varphi : \mathbb{R} \to \mathbb{R}$ is *convex* if

$$\varphi(\lambda x + (1-\lambda)y) \le \lambda\varphi(x) + (1-\lambda)(\varphi(y)), \qquad \forall\ \lambda \in [0,1] \text{ and } x, y \in \mathbb{R}. \tag{604}$$

For instance, $\varphi(x) = x^2$ and $\exp(x)$ are convex functions.

Exercise 4.6.1 (Jensen's inequality)**.** Let X be any RV with $\mathbb{E}[X] < \infty$. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be any convex function, that is,

$$\varphi(\lambda x + (1-\lambda) y) \le \lambda\varphi(x) + (1-\lambda)(\varphi(y)), \qquad \forall\ \lambda \in [0,1] \text{ and } x, y \in \mathbb{R}. \tag{605}$$

Jensen's inequality states that

$$\varphi(\mathbb{E}[X]) \le \mathbb{E}[\varphi(X)]. \tag{606}$$

(i) Let $c := \mathbb{E}[X] < \infty$. Show that there exists a line $f(x) = ax + b$ such that $f(c) = \varphi(c)$ and $\varphi(x) \ge f(x)$ for all $x \in \mathbb{R}$.

(ii) Verify the following and prove Jensen's inequality:

$$\mathbb{E}[\varphi(X)] \ge \mathbb{E}[f(X)] = a\mathbb{E}[X] + b = f(c) = \varphi(c) = \varphi(\mathbb{E}[X]). \tag{607}$$

(iii) Let X be RV, A an event, φ be the convex function as before. Show the Jensen's inequality for the conditional expectation:

$$\varphi(\mathbb{E}[X|A]) \le \mathbb{E}[\varphi(X) \mid A]. \tag{608}$$

Proposition 4.6.2. *Let $(M_t)_{t\ge0}$ be a submartingale with respect to a filtration $(\mathscr{F}_t)_{t\ge0}$. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex function. Suppose that $\mathbb{E}[|\varphi(M_t)|] < \infty$ for all $t \ge 0$.*

(i) *If $(M_t)_{t\ge0}$ is a martingale, then $(\varphi(M_t))_{t\ge0}$ is a submartingale w.r.t. $(\mathscr{F}_t)_{t\ge0}$.*

(ii) *If $\varphi : \mathbb{R} \to \mathbb{R}$ is a non-decreasing function, then $(\varphi(M_t))_{t\ge0}$ is a submartingale w.r.t. $(\mathscr{F}_t)_{t\ge0}$.*

PROOF. We first show (i). Fix $A \in \mathscr{F}_t$. Since $(M_t)_{t\ge0}$ is a martingale, for each $A' \in \mathscr{F}_t$,

$$\mathbb{E}[M_{t+1} \mid A'] - \mathbb{E}[M_t \mid A'] = \mathbb{E}[M_{t+1} - M_t \mid A'] = 0. \tag{609}$$

Note that for each $m \in \mathbb{R}$, since $\{M_t = m\} \in \mathscr{F}_t$, by using Jensen's inequality,

$$\begin{aligned}
\mathbb{E}[\varphi(M_{t+1}) - \varphi(M_t) \mid A \cap \{M_t = m\}] &= \mathbb{E}[\varphi(M_{t+1}) \mid A \cap \{M_t = m\}] - \varphi(m) && (610)\\
&\ge \varphi(\mathbb{E}[M_{t+1} \mid A \cap \{M_t = m\}]) - \varphi(m) && (611)\\
&= \varphi(\mathbb{E}[M_t \mid A \cap \{M_t = m\}]) - \varphi(m) = \varphi(m) - \varphi(m) = 0. && (612)
\end{aligned}$$

Then by iterated expectation and using the fact that $A \cap \{M_t = m\} \in \mathscr{F}_t$,

$$\mathbb{E}[\varphi(M_{t+1}) - \varphi(M_t) \mid A] = \mathbb{E}\left[\mathbb{E}[\varphi(M_{t+1}) - \varphi(M_t) \mid A, M_t]\right] = 0. \tag{613}$$

So $(\phi(M_t))_{t\ge0}$ is a submartingale. This shows (i).

To show (ii), note that since $(M_t)_{t\ge0}$ is a submartingale, for any $A' \in \mathscr{F}_t$,

$$\mathbb{E}[M_{t+1} \mid A'] - \mathbb{E}[M_t \mid A'] = \mathbb{E}[M_{t+1} - M_t \mid A'] \ge 0. \tag{614}$$

By Jensen's inequality and since φ is non-decreasing,

$$\mathbb{E}[\varphi(M_{t+1}) \mid A'] \ge \varphi(\mathbb{E}[M_{t+1} \mid A']) \ge \varphi(\mathbb{E}[M_t \mid A']). \tag{615}$$

Then following the similar argument as before shows that $(\phi(M_t))_{t\ge0}$ is a submartingale. □

Example 4.6.3. Let $(M_t)_{t\ge0}$ be a martingale. Since $\varphi(x) = x^2$ is a convex function, $(M_t^2)_{t\ge0}$ is a submartingale. ▲

The following observation is a martingale analogue of the formula $\mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathrm{Var}(X)$.

Proposition 4.6.4. *If $(M_t)_{t\ge0}$ is a martingale with respect to filtration $(\mathscr{F}_t)_{t\ge0}$, then we have*

$$\mathbb{E}[M_{t+1}^2 - M_t^2 \mid \mathscr{F}_t] = \mathbb{E}[(M_{t+1} - M_t)^2 \mid \mathscr{F}_t] \ge 0. \tag{616}$$

PROOF. By the martingale condition, we have

$$\mathbb{E}[M_{t+1} - M_t \,|\, \mathscr{F}_t] = 0 \tag{617}$$

Hence by expanding the square, we get

$$\mathbb{E}[(M_{t+1} - M_t)^2 \,|\, \mathscr{F}_t] = \mathbb{E}[M_{t+1}^2 \,|\, \mathscr{F}_t] - 2M_t\mathbb{E}[M_{t+1} \,|\, \mathscr{F}_t] + \mathbb{E}[M_t^2 \,|\, \mathscr{F}_t] \tag{618}$$

$$= \mathbb{E}[M_{t+1}^2 \,|\, \mathscr{F}_t] - \mathbb{E}[M_t^2 \,|\, \mathscr{F}_t]. \tag{619}$$

□

Exercise 4.6.5 (Long range martingale condition)**.** Let $(M_t)_{t\geq 0}$ be a martingale with respect to a filtration $(\mathscr{F}_t)_{t\geq 0}$. For any $0 \leq k < n$, we will show that

$$\mathbb{E}[(M_n - M_k) \,|\, \mathscr{F}_k] = 0. \tag{620}$$

(i) Suppose (620) holds for fixed $0 \leq k < n$. For each $A \in \mathscr{F}_k$, to show that

$$\mathbb{E}[M_{n+1} - M_k \,|\, A] = \mathbb{E}[M_{n+1} - M_n \,|\, A] + \mathbb{E}[M_n - M_k \,|\, A] \tag{621}$$

$$= \mathbb{E}[M_{n+1} - M_n \,|\, A] = 0. \tag{622}$$

(ii) Conclude (620) for all $0 \leq k < n$ by induction.

Proposition 4.6.6 (Orthogonality of martingale increments)**.** *Let $(M_t)_{t\geq 0}$ be a martingale with respect to filtration $(\mathscr{F}_t)_{t\geq 0}$.*

(i) *For each $0 \leq j \leq k < n$, we have*

$$\mathbb{E}[(M_n - M_k)M_j] = 0. \tag{623}$$

(ii) *For each $0 \leq j \leq k < n$, we have*

$$\mathbb{E}[(M_n - M_k)(M_j - M_i)] = 0. \tag{624}$$

PROOF. To show (i), we use conditioning on the values of M_j. By noting that $\{M_j = m\} \in \mathscr{F}_j \subseteq \mathscr{F}_k$ for all $m \in \mathbb{R}$, we have

$$\mathbb{E}[(M_n - M_k)M_j \,|\, M_j = m] = m\mathbb{E}[M_n - M_k \,|\, M_j = m] = 0, \tag{625}$$

where we have used Exercise 4.6.5 for the last equality. Hence

$$\mathbb{E}[(M_n - M_k)M_j] = \mathbb{E}[\mathbb{E}[(M_n - M_k)M_j \,|\, M_j]] = 0. \tag{626}$$

Moreover, (ii) follows from (i) immediately since

$$\mathbb{E}[(M_n - M_k)(M_j - M_i)] = \mathbb{E}[(M_n - M_k)M_j] - \mathbb{E}[(M_n - M_k)M_i]. \tag{627}$$

This shows the assertion. □

Exercise 4.6.7 (Pythagorian theorem for martingales)**.** Let $(M_t)_{t\geq 0}$ be a martingale with respect to a filtration $(\mathscr{F}_t)_{t\geq 0}$. Use Proposition 4.6.6 to show that

$$\mathbb{E}[(M_t - M_0)^2] = \sum_{k=1}^{t} \mathbb{E}[(M_k - M_{k-1})^2]. \tag{628}$$

CHAPTER 5

Introduction to mathematical finance (Optional*)

We introduce basic notions of no arbitrage principle, binomial model, and connect to martingales.

5.1. Hedging and replication in the two-state world

We model the market as a probability space $(\Omega, \mathbb{P})$, where Ω consists of sample paths ω of the market, which describes a particular time evolution scenario. For each event $E \subseteq \Omega$, $\mathbb{P}(E)$ gives the probability that the event E occurs. A *portfolio* is a collection of assets that one has at a particular time. The value of a portfolio A at time t is denoted by V_t^A. If t denotes the current time, then V_t^A is a known quantity. However, at a future time $T \geq t$, V_T^A depends on how the market evolves during $[t, T]$, so it is a random variable. Also recall the definition of an arbitrage portfolio:

Definition 5.1.1. A portfolio A at current time t is said to be an *arbitrage portfolio* if its value V^A satisfies the followings:

(i) $V^A(t) \leq 0$.
(ii) There exists a future time $T \geq t$ such that $\mathbb{P}(V^A(T) \geq 0) = 1$ and $\mathbb{P}(V^A(T) > 0) > 0$.

Example 5.1.2 (A 1-step binomial model)**.** Suppose we have an asset with price $(S_t)_{t\geq 0}$. Consider a European call option at time $t = 0$ with strike $K = 110$ and maturity $t = 1$ (year). Suppose that $S_0 = 100$ and at time 1, S_1 takes one of the two values 120 and 90 according to a certain distribution. One can imagine flipping a coin with unknown probability, and according to whether it lands heads (H) or tail (T), the stock value S_1 takes values $S_1(H) = 120$ and $S_1(T) = 90$. Assume annually compounded interested rate $r = 4\%$. Can we determine its current value $c = C_{110}(0, 1)$? We will show $c = 14/3 \approx 4.48$ by using two arguments – hedging and replication.

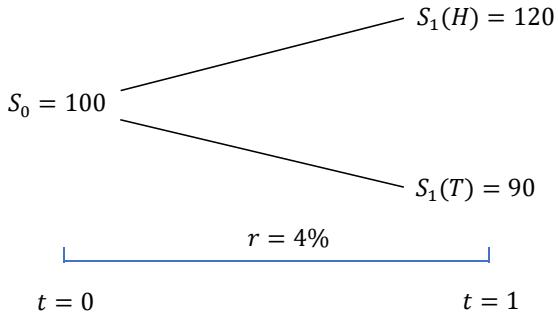


FIGURE 1. 1-step binomial model

Here we give a 'hedging argument' for option pricing. Consider the following portfolio at time $t = 0$:

Portfolio A: [x shares of the stock] + [y European call options with strike 110 and maturity 1].

The cost of entering into this portfolio (at time $t=0$) is $100x+cy$. Hence the profit of this portfolio takes the following two values

$$\begin{cases} V_1^A(H)-(100x+cy)(1.04)=[120x+y(120-110)^+]-[104x+(1.04)cy]=16x+(10-(1.04)c)y \\ V_1^A(T)-(100x+cy)(1.04)=[90x+y(90-110)^+]-[104x+(1.04)cy]=-14x-(1.04)cy. \end{cases} \tag{629}$$

In order for a perfect hedging, consider choosing the values of x and y such that the profit of this portfolio at maturity is the same for the two outcomes of the stock. Hence we must have

$$16x+(10-(1.04)c)y=-14x-(1.04)cy. \tag{630}$$

Solving this, we find

$$3x+y=0. \tag{631}$$

Hence if the above equation is satisfied, the profit of portfolio A is

$$V_1^A-(104x+(1.04)cy)=-14x-(1.04)c(-3x)=((3.12)c-14)\,x. \tag{632}$$

If $(3.12)c>14$, then portfolio A is an arbitrage portfolio; If $(3.12)c<14$, then the 'dual' of portfolio A, which consists of $-x$ shares of the stock and $-y$ European call options, is an arbitrage portfolio. Hence assuming no-arbitrage, the only possible value of c is $c=14/(3.12)$. ▲

5.2. The fundamental theorem of asset pricing

The observation we made in Example 5.1.2 can be generalized into the so-called 'fundamental theorem of asset pricing'. For its setup, consider a market where there are n different time evolution $\omega_1,\cdots,\omega_n$ between time $t=0$ and $t=1$, each occurs with a positive probability. Suppose there are assets $A^{(1)},\cdots,A^{(m)}$, whose price at time t is given by $S_t^{(i)}$ for $i=1,2,\cdots,m$ (see Figure 2). For each $1\le i\le m$ and $1\le j\le n$, define

$$\alpha_{i,j}=\begin{pmatrix}\text{profit at time } t=1 \text{ of buying one share of asset } A^{(i)} \\ \text{at time } t=0 \text{ when the market evolves via } \omega_j.\end{pmatrix} \tag{633}$$

Let $\mathbf{A}=(\alpha_{i,j})$ denote the $(m\times n)$ matrix of profits:

$$\mathbf{A}:=\begin{bmatrix}\alpha_{1,1} & \alpha_{1,2} & \cdots & \alpha_{1,n}\\ \alpha_{2,1} & \alpha_{2,2} & \cdots & \alpha_{2,n}\\ \vdots & \vdots & \cdots & \vdots \\ \alpha_{m,1} & \alpha_{m,2} & \cdots & \alpha_{m,n}\end{bmatrix}. \tag{634}$$

Consider the following portfolio at time $t=0$:

Portfolio A: [x_1 shares of asset $A^{(1)}$] $+\cdots+$ [x_m shares of asset $A^{(m)}$].

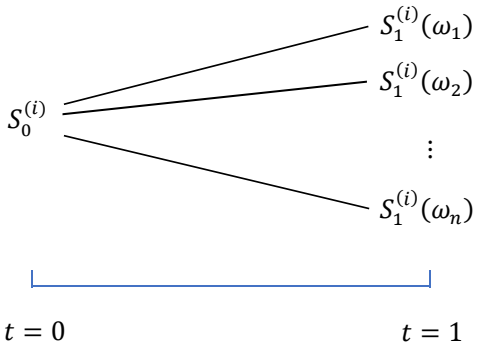


FIGURE 2. 1-step n-state model for asset $A^{(i)}$.

Theorem 5.2.1 (The fundamental theorem of asset pricing)**.** *Consider portfolio A and the profit matrix $\mathbf{A} = (\alpha_{i,j})$ as above. Then exactly one of the followings hold:*

(i) *There exists an investment allocation $(x_1, \cdots, x_m)$ such that portfolio A is an arbitrage portfolio, that is, the n-dimensional row vector*

$$[x_1, x_2, \cdots, x_m] \begin{bmatrix} \alpha_{1,1} & \alpha_{1,2} & \cdots & \alpha_{1,n} \\ \alpha_{2,1} & \alpha_{2,2} & \cdots & \alpha_{2,n} \\ \vdots & \vdots & \cdots & \vdots \\ \alpha_{m,1} & \alpha_{m,2} & \cdots & \alpha_{m,n} \end{bmatrix} \tag{635}$$

has nonzero coordinates and at least one strictly positive coordinate.

(ii) *There exists a strictly positive probability distribution $\mathbf{p}^* = (p_1^*, \cdots, p_n^*)$ under which the expected profit of each asset is zero:*

$$\begin{bmatrix} \alpha_{1,1} & \alpha_{1,2} & \cdots & \alpha_{1,n} \\ \alpha_{2,1} & \alpha_{2,2} & \cdots & \alpha_{2,n} \\ \vdots & \vdots & \cdots & \vdots \\ \alpha_{m,1} & \alpha_{m,2} & \cdots & \alpha_{m,n} \end{bmatrix} \begin{bmatrix} p_1^* \\ p_2^* \\ \vdots \\ p_n^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{636}$$

Remark 5.2.2. Theorem 5.2.1 (i) states that the portfolio A is an arbitrage portfolio for some $(x_1, \cdots, x_m)$. The probability distribution $\mathbf{p}^*$ in the above theorem is called the *risk-neutral probability distribution*. Hence Theorem 5.2.1 states that there is no way to make A into an arbitrage portfolio if and only if there exists a risk-neutral probability distribution under which the expected profit of each asset $A^{(i)}$ is zero.

Example 5.2.3 (Example 5.1.2 revisited)**.** Consider the situation described in Example 5.1.2. Let $A^{(1)}$ be the asset of price $(S_t)_{t\geq 0}$ and $A^{(2)}$ denote the European call with strike $K = 110$ on this asset with maturity T. Then the matrix $\mathbf{A}$ of profits is given by

$$\mathbf{A} = \begin{bmatrix} 16 & -14 \\ 10 - (1.04)c & -(1.04)c \end{bmatrix}, \tag{637}$$

where $c = C_{110}(0,1)$ denotes the price of this European option. Assuming no-arbitrage, the fundamental theorem implies that there exists risk-neutral probability distribution $\mathbf{p}^* = (p_1^*, p_2^*)$ such that $\mathbf{A}(\mathbf{p}^*)^T = \mathbf{0}^T$. Namely,

$$\begin{cases} 16p_1^* - 14p_2^* = 0 \\ (10 - (1.04)c)p_1^* - (1.04)cp_2^* = 0 \end{cases} \tag{638}$$

Since $p_1^* + p_2^* = 1$, the first equation implies $p_1^* = 7/15$ and $p_2^* = 8/15$. Then from the second equation, we get

$$(1.04)c = 10p_1^* = \frac{14}{3}. \tag{639}$$

This gives $c = 14/(3.12)$. ▲

Exercise 5.2.4. Rework Examples 5.1.2 and 5.2.3 with following parameters:

$$S_0 = 100, \quad S_1(H) = 130, \quad S_1(T) = 80, \quad r = 5\%, \quad K = 110. \tag{640}$$

PROOF OF THEOREM 5.2.1. Suppose (i) holds with $\mathbf{x} = (x_1, \cdots, x_n)$. We want to show that (ii) cannot hold. Fix a strictly positive probability distribution $\mathbf{p} = (p_1, \cdots, p_n)'$, where $'$ denotes the transpose so that $\mathbf{p}$ is an n-dimensional column vector. By (i), we have

$$\mathbf{x}(\mathbf{Ap}) = (\mathbf{xA})\mathbf{p} > 0. \tag{641}$$

It follows that $\mathbf{Ap}$ cannot be the zero vector in $\mathbb{R}^n$. Hence (636) cannot hold for $\mathbf{p}^* = \mathbf{p}$, as desired.

Next, suppose that (ii) holds for some strictly positive probability distribution $\mathbf{p}^* = (p_1^*, \cdots, p_n^*)'$. We use a linear algebra argument to show that (i) does not hold. For each m-dimensional row vector $\mathbf{x} =$

$(x_1,\cdots,x_m)\in\mathbb{R}^m$, one can correspond a n-dimensional column vector $\mathbf{xA}\in\mathbb{R}^n$. The condition (636) says that $\mathbf{Ap}^*=\mathbf{0}$, where T denotes the transpose. Hence for each $\mathbf{x}\in\mathbb{R}^m$, by using associativity of matrix multiplication,

$$(\mathbf{xA})\mathbf{p}^*=\mathbf{x}\left(\mathbf{Ap}^*\right)=\mathbf{x0}=0. \tag{642}$$

This shows that the image of the linear map $\mathbf{x}\mapsto\mathbf{xA}$, which is a linear subspace of $\mathbb{R}^n$, is orthogonal to the strictly positive vector $\mathbf{p}^*$. Hence this linear subspace intersects with the positive orthant $\{(y_1,\cdots,y_n)\,|\,y_1,\cdots,y_n\geq 0\}$ only at the origin. This shows that (i) does not hold, as desired. □

5.3. The binomial model

5.3.1. 1-step binomial model. Suppose we have an asset with price $(S_t)_{t\geq 0}$. Assume that a future time $t=1$, the stock price S_1 can take two values $S_1(H)=S_0u$ and $S_1(T)=S_0d$ according to an invisible coin flip, where $u,d>0$ are multiplicative factors for upward and downward moves for the stock price during the period $[0,1]$. Assume the aggregated interested rate during $[0,1]$ is $r>0$, so that the value of ZCB (Zero Cupon Bond) maturing at 1 is given by $Z(0,1)=1/(1+r)$.

Consider a general European option on this stock, whose value at time $t=0,1$ is denoted V_t. We would like to determine its initial value (price at $t=0$) V_0 in terms of its payoff V_1. In the previous section, we have seen that there are three ways to proceed: 1) hedging argument, 2) replication argument, and 3) risk-neutral probability distribution.

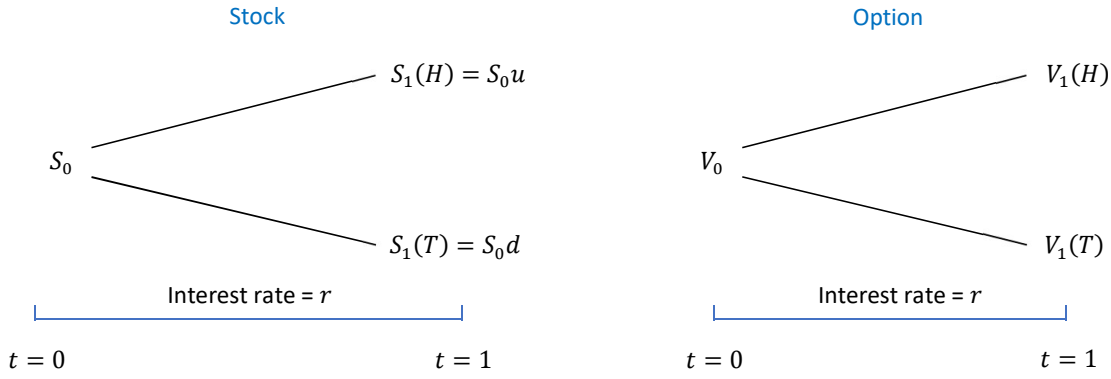


FIGURE 3. 1-step binomial model with general European option

Proposition 5.3.1. *In the above binomial model, the followings hold.*

(i) *There exists a risk-neutral probability distribution* $\mathbf{p}^*=(p_1^*,p_2^*)$ *if and only if*

$$0<d<1+r<u. \tag{643}$$

Furthermore, if the above condition holds, $\mathbf{p}^*$ *is uniquely given by*

$$p_1^*=\frac{(1+r)-d}{u-d},\qquad p_2^*=\frac{u-(1+r)}{u-d}. \tag{644}$$

(ii) *Suppose* (643) *holds. Then the initial value* V_0 *of the European option is given by*

$$V_0=\frac{1}{1+r}\mathbb{E}_{\mathbf{p}^*}[V_1]=\frac{1}{1+r}\left(\frac{(1+r)-d}{u-d}V_1(H)+\frac{u-(1+r)}{u-d}V_1(T)\right). \tag{645}$$

PROOF. To begin, we first need to compute the (2×2) profit matrix $\mathbf{A}=(\alpha_{i,j})$, whose rows and columns correspond to the two kinds of assets (stock and European option) and outcomes (coin flips), respectively. We find

$$\mathbf{A}=\begin{bmatrix} S_1(H)-S_0\cdot(1+r) & S_1(T)-S_0\cdot(1+r)\\ V_1(H)-V_0\cdot(1+r) & V_1(T)-V_0\cdot(1+r)\end{bmatrix}. \tag{646}$$

The risk-neutral probability $\mathbf{p}^* = (p_1^*, p_2^*)'$ satisfies $\mathbf{A}\mathbf{p}^* = \mathbf{0}$, so

$$\begin{cases} [S_1(H) - S_0 \cdot (1+r)]p_1^* + [S_1(T) - S_0 \cdot (1+r)]p_2^* = 0 \\ [V_1(H) - V_0 \cdot (1+r)]p_1^* + [V_1(T) - V_0 \cdot (1+r)]p_2^* = 0. \end{cases} \tag{647}$$

Using the fact that $p_1^* + p_2^* = 1$, the first equation gives

$$p_1^* = \frac{S_0(1+r) - S_1(T)}{S_1(H) - S_1(T)} = \frac{(1+r) - d}{u - d}, \qquad p_2^* = \frac{S_1(H) - S_0(1+r)}{S_1(H) - S_1(T)} = \frac{u - (1+r)}{u - d}. \tag{648}$$

Hence the desired expression for the risk-neutral probabilities p_1^* and p_2^* holds. Note that this gives a strictly positive probability distribution if and only if (643) holds. This shows (i). (Why does this condition make sense?)

Assuming (643), the second equation in (647) then gives

$$V_0 \cdot (1+r) = V_1(H)p_1^* + V_1(T)p_2^*. \tag{649}$$

The right hand side can be regarded as the expectation $\mathbb{E}_{\mathbf{p}^*}[V_1]$ of the value V_1 of the European option at time $t = 1$ under the risk-neutral probability distribution $\mathbf{p}^*$. Then (ii) follows from (i). □

Proposition 5.3.2. *In the 1-step binomial model as before, consider the following portfolios:*

Portfolio A: [Δ_0 shares of the stock] + [Short one European option].
Portfolio B: [Δ_0 shares of the stock] + [x cash],

Then the followings hold:

(i) *Portfolio A is perfectly hedged (i.e., constant payoff at time $t = 1$) if and only if*

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}. \tag{650}$$

(ii) *Portfolio B replicates long one European option if and only if we have (650) and*

$$x = \frac{1}{1+r} \frac{S_1(H)V_1(T) - S_1(T)V_1(H)}{S_1(H) - S_1(T)}. \tag{651}$$

Furthermore,

$$V_0 = x + \Delta_0 S_0 = \frac{1}{1+r}\left(V_1(H)\frac{(1+r) - u}{u - d} + V_1(T)\frac{u - (1+r)}{u - d}\right). \tag{652}$$

PROOF. To show (i), we equate the two payoffs of portfolio A at time $t = 1$ and obtain

$$\Delta_0 S_1(H) - V_1(H) = \Delta_0 S_1(T) - V_1(T). \tag{653}$$

Solving this for Δ_0 shows the assertion.

To show (ii), note that portfolio A replicates long one European option if and only if

$$\begin{cases} \Delta_0 S_1(H) + x \cdot (1+r) = V_1(H) \\ \Delta_0 S_1(T) + x \cdot (1+r) = V_1(T), \end{cases} \tag{654}$$

or in matrix form,

$$\begin{bmatrix} S_1(H) & 1+r \\ S_1(T) & 1+r \end{bmatrix} \begin{bmatrix} \Delta_0 \\ x \end{bmatrix} = \begin{bmatrix} V_1(H) \\ V_1(T) \end{bmatrix}. \tag{655}$$

This is equivalent to

$$\begin{bmatrix} \Delta_0 \\ x \end{bmatrix} = \frac{1}{(1+r)(S_1(H) - S_1(T))} \begin{bmatrix} 1+r & -1-r \\ -S_1(T) & S_1(H) \end{bmatrix} \begin{bmatrix} V_1(H) \\ V_1(T) \end{bmatrix}, \tag{656}$$

which is also equivalent to (650) and (651), as desired.

Lastly, suppose portfolio B replicates long one European option. Then by the monotinicty theorem, intitial value V_0 of the European option should equal to the value of portfolio A at time $t=0$. This shows

$$V_0 = x + \Delta_0 S_0. \tag{657}$$

Using (651), we also have

$$\begin{aligned} x + \Delta_0 S_0 &= \frac{1}{1+r}\frac{S_1(H)V_1(T) - S_1(T)V_1(H)}{S_1(H) - S_1(T)} + \frac{V_1(H)S_0 - V_1(T)S_0}{S_1(H) - S_1(T)} && (658)\\ &= \frac{1}{1+r}\left(V_1(H)\frac{S_0\cdot(1+r) - S_1(H)}{S_1(H) - S_1(T)} + V_1(T)\frac{S_1(H) - S_0\cdot(1+r)}{S_1(H) - S_1(T)}\right) && (659)\\ &= \frac{1}{1+r}\left(V_1(H)\frac{(1+r) - u}{u-d} + V_1(T)\frac{u - (1+r)}{u-d}\right). && (660) \end{aligned}$$

This shows (ii). (Remark: By using Proposition 5.3.1, one can avoid using the monotonicity theorem here.) □

Example 5.3.3 (Excerpted from [Dur99])**.** Suppose a stock is selling for \$60 today. A month from now it will be either \$80 or \$50, i.e., $u = 4/3$ and $d = 5/6$. Assume the interest rate $r = 1/18$ for this period. Then according to Proposition 5.3.2, the risk-neutral probability distribution $\mathbf{p}^* = (p_1^*, p_2^*)$ is given by

$$p_1^* = \frac{(1+\frac{1}{18}) - \frac{5}{6}}{\frac{4}{3} - \frac{5}{6}} = \frac{4}{9}, \qquad p_2^* = \frac{5}{9}. \tag{661}$$

Now consider a European call option on this stock with strike $K = 65$ maturing in a month. Then $V_1(H) = (80-65)^+ = 15$ and $V_1(T) = (50-65)^+ = 0$. By Proposition 5.3.2, the initial value V_0 of of this European option is

$$V_0 = \frac{1}{1+(1/18)}\mathbb{E}_{\mathbf{p}^*}[V_1] = \frac{18}{19}\cdot 15\cdot\frac{4}{9} = \frac{120}{19} = 6.3158. \tag{662}$$

Working in an investment bank, you were able to sell 10,000 calls to a customer for a slightly higher price each at \$6.5, receiving up-front payment of \$65,000. At maturity, the overall profit is given by

$$\begin{cases} (19/18)\cdot\$65,000 - 10,000\cdot(80-65)^+ = -\$81,389 & \text{if stock goes up} \\ (19/18)\cdot\$65,000 - 10,000\cdot(50-65)^+ = \$68,611 & \text{if stock goes down.} \end{cases} \tag{663}$$

Being worried about losing a huge amount if the stock goes up, you decided to hedge and lock the profit. According to Proposition 5.3.2, the hedge ratio Δ_0 is given by

$$\Delta_0 = \frac{(80-65)^+ - (50-65)^+}{80-50} = \frac{15}{30} = \frac{1}{2}. \tag{664}$$

Since you have shorted 10,000 calls, this means you need to buy 5,000 shares of the stock owing $5,000\cdot 60 - \$65,000 = \$235,000$ to the bank. This forms a portfolio of

$$[5,000 \text{ shares of stock}] + [10,000 \text{ short calls}]. \tag{665}$$

The overall profit at maturity is then

$$\begin{cases} 5,000\cdot\$80 - 10,000\cdot(80-65)^+ - (19/18)\cdot\$235,000 = \$1,944 & \text{if stock goes up} \\ 5,000\cdot\$50 - 10,000\cdot(50-65)^+ - (19/18)\cdot\$235,000 = \$1,944 & \text{if stock goes down.} \end{cases} \tag{666}$$

▲

Exercise 5.3.4. Rework Example 5.3.3 for the following parameters:

$$S_0 = 50, \quad S_1(T) = 70, \quad S_1(H) = 40, \quad r = 4\%, \quad K = 60. \tag{667}$$

5.3.2. The N-step binomial model. In this subsection, we consider the general N-step binomial model. Namely, staring at the current time $t=0$, we flip N coins at times $t=1,2,\cdots,N$ to determine the market evolution. More precisely, now the sample space of the outcomes is $\Omega=\{H,T\}^N$, which consists of sequences of length N strings of H's or T's. We assume constant interest rate for each periods $[k,k+1]$ for $k=0,1,\cdots,N-1$.

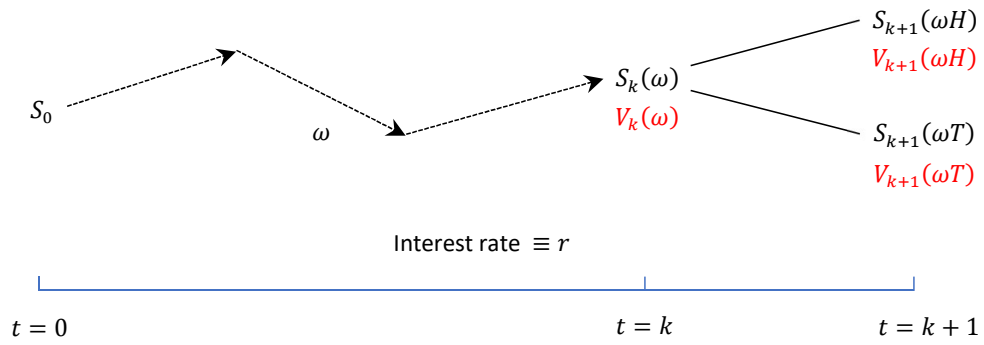


FIGURE 4. Illustration of the N-step binomial model. ω is a sample path of the market evolution in the first k steps. In the next period $[k,k+1]$, either up or down evolution occurs and the same path is extended to ωH or ωT accordingly. S_t and V_t denote the stock price and option payoff.

The following result for the general N-step binomial model is a direct analog of the 2-step case we have discussed in the previous subsection. To better understand its second part, recall the remark on a probabilistic interpretation on the 2-step value formula as in Proposition **??**.

Proposition 5.3.5. *Consider the N-step binomial model as above. Consider a European option on this stock wit value $(V_t)_{0\le t\le N}$.*

(i) *For each integer $0\le k<N$ and a sample path $\omega\in\{H,T\}^k$ for the first k steps, define the risk-neutral probability distribution $\mathbf{p}^*(\omega)=(p_1^*(\omega),p_2^*(\omega))$ by*

$$p_1^*(\omega)=\frac{(1+r)S_k(\omega)-S_{k+1}(\omega T)}{S_{k+1}(\omega H)-S_{k+1}(\omega T)},\qquad p_2^*(\omega)=\frac{S_{k+1}(\omega H)-(1+r)S_k(\omega)}{S_{k+1}(\omega H)-S_{k+1}(\omega T)}. \tag{668}$$

If $0<p_1^*(\omega)<1$, then

$$V_k(\omega)=\frac{1}{(1+r)}\mathbb{E}_{\mathbf{p}^*(\omega)}[V_{k+1}\,|\,\text{first } k \text{ coin flips}=\omega] \tag{669}$$

$$=\frac{1}{(1+r)}\left(V_{k+1}(\omega H)p_1^*(\omega)+V_{k+1}(\omega T)p_2^*(\omega)\right). \tag{670}$$

(ii) *Consider N consecutive coin flips such that given any sequence $x_1x_2\cdots x_k$ of the first k flips, the $(k+1)$st coin lands on heads with probability $p_1^*(x_1x_2\cdots x_k)$. Let $\mathbb{P}^*$ denote the induced probability measure (risk-neutral probability measure) on the sample space $\Omega=\{H,T\}^N$. Then*

$$V_0=\frac{1}{(1+r)^N}\mathbb{E}_{\mathbb{P}^*}[V_N]. \tag{671}$$

PROOF. The argument for (i) is exactly the same as in the proof of Proposition **??**. For (ii) we use an induction on the number of steps. The base case is verified by (i). For the induction step, we first use (i) to write

$$V_0=\frac{1}{1+r}(V_1(H)p_1^*(\emptyset)+V_1(T)p_2^*(\emptyset)). \tag{672}$$

Let $X_1, X_2, \cdots, X_N$ be a sequence of N (random) coin flips given by the risk-neutral probability measure $\mathbb{P}^*$. Denote the expectation under $\mathbb{P}^*$ by $\mathbb{E}^*$. By the induction hypothesis, we have

$$V_1(H) = \frac{1}{(1+r)^{N-1}}\mathbb{E}^*[V_N \,|\, X_1 = H], \qquad V_1(T) = \frac{1}{(1+r)^{N-1}}\mathbb{E}^*[V_N \,|\, X_1 = T]. \tag{673}$$

Hence we have

$$V_0 = \frac{1}{(1+r)^N}\left[\mathbb{E}^*[V_N \,|\, X_1 = H]\, p_1^*(\emptyset) + \mathbb{E}^*[V_N \,|\, X_1 = T]\, p_2^*(\emptyset)\right] \tag{674}$$

$$= \frac{1}{(1+r)^N}\mathbb{E}^*[\mathbb{E}^*[V_N \,|\, X_1]] = \frac{1}{(1+r)^N}\mathbb{E}^*[V_N], \tag{675}$$

where we have used iterated expectation for the last equality. This shows the assertion. □

5.3.3. The N-step binomial model revisited. In this subsection, we revisit the N-step binomial model in the framework of martingales. First recall the model. Staring at the current time $t = 0$, we flip N coins at times $t = 1, 2, \cdots, N$ to determine the market evolution. The sample space of the outcomes is $\Omega = \{H, T\}^N$, which consists of sequences of length N strings of H's or T's. We assume constant interest rate for each periods $[k, k+1]$ for $k = 0, 1, \cdots, N-1$.

Let $\mathscr{F}_t$ denote the information that we can obtain by observing the market up to time t. For instance, $\mathscr{F}_t$ contains the information of the first t coin flips, stock prices $S_0, S_1, \cdots, S_t$, European option values $V_0, V_1, \cdots, V_t$, and so on. Then $(\mathscr{F}_t)_{t\geq 0}$ defines a natural filtration for the N-step binomial model. Below we reformulate Proposition 5.3.5.

Proposition 5.3.6. *Consider the N-step binomial model as above. Let $\mathbb{P}^*$ denote the risk-neutral probability measure defined in Proposition 5.3.5. Consider a European option on this stock wit value $(V_t)_{0\leq t\leq N}$.*

(i) *The process $\left((1+r)^{-t}V_t\right)_{0\leq t\leq N}$ forms a martingale with respect to the filtration $(\mathscr{F}_t)_{t\geq 0}$ under the risk-neutral probability measure $\mathbb{P}^*$. That is,*

$$\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-(t+1)}V_{t+1} - (1+r)^{-t}V_t \,\middle|\, \mathscr{F}_t\right] = 0, \tag{676}$$

which is also equivalent to

$$V_t = \frac{1}{1+r}\mathbb{E}_{\mathbb{P}^*}\left[V_{t+1} \,\middle|\, \mathscr{F}_t\right] = 0. \tag{677}$$

(ii) *We have*

$$V_0 = \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-N}V_N\right]. \tag{678}$$

PROOF. To show (i), we first note that, conditioning on the information $\mathscr{F}_t$ up to time t, we know all the coin flips, stock prices, and European option values up to time t. Hence we have

$$\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-(t+1)}V_{t+1} \,\middle|\, \mathscr{F}_t\right] = (1+r)^{-(t+1)}\mathbb{E}_{\mathbb{P}^*}\left[V_{t+1} \,\middle|\, \mathscr{F}_t\right] \tag{679}$$

$$= (1+r)^{-(t+1)} \cdot (1+r)V_t = (1+r)^{-t}V_t. \tag{680}$$

This shows that $(1+r)^{-t}V_t$ is a martingale with respect to the natural filtration $(\mathscr{F}_t)_{t\geq 0}$ under the risk-neutral probability measure $\mathbb{P}^*$.

Now (ii) follows from (i) and Exercise 4.6.5. Namely, for each $A \in \mathscr{F}_0$, by Exercise 4.6.5,

$$\mathbb{E}_{\mathbb{P}^*}[(1+r)^{-N}V_N - (1+r)^0 V_0 \,|\, \mathscr{F}_0] = 0. \tag{681}$$

This yields

$$V_0 = \mathbb{E}_{\mathbb{P}^*}[V_0 \,|\, \mathscr{F}_0] = \mathbb{E}_{\mathbb{P}^*}[(1+r)^{-N}V_N \,|\, \mathscr{F}_0], \tag{682}$$

as desired. □

Exercise 5.3.7 (Risk-neutral probabilities make the stock price a martingale)**.** Consider the N-step binomial model with stock price $(S_t)_{0\le t\le N}$. Let $\mathbb{P}^*$ denote the risk-neutral probability measure defined in Proposition 3.10 in Lecture note 2. 5.3.5.

(i) Show that the discounted stock price $(1+r)^{-t}S_t$ forms a martingale with respect to the filtration $(\mathscr{F}_t)_{t\ge 0}$ under the risk-neutral probability measure $\mathbb{P}^*$. That is,

$$\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-(t+1)}S_{t+1}-(1+r)^{-t}S_t\,\middle|\,\mathscr{F}_t\right]=0, \tag{683}$$

which is also equivalent to

$$S_t=\frac{1}{1+r}\mathbb{E}_{\mathbb{P}^*}\left[S_{t+1}\,\middle|\,\mathscr{F}_t\right]=0. \tag{684}$$

(Hint: Use the fundamental thm of asset pricing.)

(ii) Show that

$$S_0=\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-N}S_N\right]. \tag{685}$$

CHAPTER 6

Poisson Processes

In this chapter, we introduce a particular class of renewal processes called the "Poisson processes", which is a arrival process whose inter-arrival times are i.i.d. exponential RVs. Due to the nice properties of exponential distribution, Poisson processes enjoy what's called the 'memoryless property', which says that Poisson processes can be restared at any stopping times, not necessarily at arrival times. Recall that renewal processs has renewal property, which means they can be restarted at arrival times. Hence the memoryless property of Poisson processes are stronger versions of renewal properties only enjoyed by Poisson processes. This allows one to analyze Poisson processes in greater details than general renewal processes.

6.1. Recap of exponential and Poisson random variables

In this section, we study two important properties of exponential and Poisson random variables, which will be crucial when we study Poisson processes from the following section.

Example 6.1.1 (Exponential RV)**.** X is an *exponential* RV with rate λ (denoted by $X \sim \text{Exp}(\lambda)$) if it has PDF

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}(x \geq 0). \tag{686}$$

Integrating the PDF gives its CDF

$$\mathbb{P}(X \leq x) = (1 - e^{-\lambda x}) \mathbf{1}(x \geq 0). \tag{687}$$

The following complimentary CDF for exponential RVs will be useful:

$$\mathbb{P}(X \geq x) = e^{-\lambda x} \mathbf{1}(x \geq 0). \tag{688}$$

▲

Exercise 6.1.2. Let $X \sim \text{Exp}(\lambda)$. Show that $\mathbb{E}(X) = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$.

Minimum of independent exponential RVs is again an exponential RV.

Exercise 6.1.3. Let $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ and suppose they are independent. Define $Y = \min(X_1, X_2)$. Show that $Y \sim \text{Exp}(\lambda_1 + \lambda_2)$. (Hint: Compute $\mathbb{P}(Y \geq y)$.)

The following example is sometimes called the 'competing exponentials'.

Example 6.1.4. Let $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ be independent exponential RVs. We will show that

$$\mathbb{P}(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2} \tag{689}$$

using the iterated expectation. Using iterated expectation for probability (see Exercise 1.4.3),

$$\mathbb{P}(X_1 < X_2) = \mathbb{E}\left[\mathbb{P}(X_2 > X_1 \mid X_1)\right] \tag{690}$$

$$= \int_0^\infty \mathbb{P}(X_1 < X_2 \mid X_1 = x_1)\lambda_1 e^{-\lambda_1 x_1}\, dx_1 \tag{691}$$

$$= \int_0^\infty \mathbb{P}(X_2 > x_1)\lambda_1 e^{-\lambda_1 x_1}\, dx_1 \tag{692}$$

$$= \lambda_1 \int_0^\infty e^{-\lambda_2 x_1} e^{-\lambda_1 x_1}\, dx_1 \tag{693}$$

$$= \lambda_1 \int_0^\infty e^{-(\lambda_1+\lambda_2)x_1}\, dx_1 = \frac{\lambda_1}{\lambda_1+\lambda_2}. \tag{694}$$

▲

When we add up independent continuous RVs, we can compute the PDF of the resulting RV by using the convolution formula or moment generating functions. In the following exercise, we compute the PDF of the sum of independent exponentials.

Exercise 6.1.5 (Sum of i.i.d. Exp is Erlang)**.** Let $X_1, X_2, \cdots, X_n \sim \text{Exp}(\lambda)$ be independent exponential RVs.

(i) Show that $f_{X_1+X_2}(z) = \lambda^2 z e^{-\lambda z}\mathbf{1}(z \geq 0)$.

(ii) Show that $f_{X_1+X_2+X_3}(z) = 2^{-1}\lambda^3 z^2 e^{-\lambda z}\mathbf{1}(z \geq 0)$.

(iii) Let $S_n = X_1 + X_2 + \cdots + X_n$. Use induction to show that $S_n \sim \text{Erlang}(n, \lambda)$, that is,

$$f_{S_n}(z) = \frac{\lambda^n z^{n-1} e^{-\lambda z}}{(n-1)!}. \tag{695}$$

Hint: Let X, Y be independent continuous RVs. Then the PDF of $X+Y$ is given by the convolution formula: $f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$.

Remark: For parameters (r, λ), a continuous random variable X is said to have the *Gamma distribution* with parameters (r, λ) (and denote $X \sim \text{Gamma}(r, \lambda)$) if

$$f_X(x) = \frac{\lambda^r z^{r-1} e^{-\lambda z}}{\Gamma(r)}\mathbf{1}(x \geq 0), \tag{696}$$

where $\Gamma(z)$ is the Euler Gamma function defined by

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx. \tag{697}$$

By an induction and integration by parts, one can show that $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$. Hence $\text{Gamma}(r, \lambda) = \text{Erlang}(r, \lambda)$ whenever r is a positive integer. So we can also say that $S_n \sim \text{Gamma}(r, \lambda)$.

Exponential RVs will be the builing block of the Poisson processes, because of their 'memoryless property'.

Exercise 6.1.6 (Memoryless property of exponential RV)**.** A continuous positive RV X is say to have *memoryless property* if

$$\mathbb{P}(X \geq t_1 + t_2) = \mathbb{P}(X \geq t_1)\mathbb{P}(X \geq t_2) \quad \forall x_1, x_2 \geq 0. \tag{698}$$

(i) Show that (698) is equivalent to

$$\mathbb{P}(X \geq t_1 + t_2 \mid X \geq t_2) = \mathbb{P}(X \geq t_1) \quad \forall x_1, x_2 \geq 0. \tag{699}$$

(ii) Show that exponential RVs have memoryless property.

(iii) Suppose X is continuous, positive, and memoryless. Let $g(t) = \log \mathbb{P}(X \geq t)$. Show that g is continuous at 0 and

$$g(x+y) = g(x) + g(y) \quad \text{for all } x, y \geq 0. \tag{700}$$

Using Exercise 6.1.7, conclude that X must be an exponential RV.

Exercise 6.1.7. Let $g : \mathbb{R}_{\geq 0} \to \mathbb{R}$ be a function with the property that $g(x+y) = g(x) + g(y)$ for all $x, y \geq 0$. Further assume that g is continuous at 0. In this exercise, we will show that $g(x) = cx$ for some constant c.

(i) Show that $g(0) = g(0+0) = g(0) + g(0)$. Deduce that $g(0) = 0$.
(ii) Show that for all integers $n \geq 1$, $g(n) = ng(1)$.
(iii) Show that for all integers $n, m \geq 1$,

$$ng(1) = g(n \cdot 1) = g(m(n/m)) = mg(n/m). \tag{701}$$

Deduce that for all nonnegative rational numbers r, we have $g(r) = rg(1)$.
(iv) Show that g is continuous.
(v) Let x be nonnegative real number. Let r_k be a sequence of rational numbers such that $r_k \to x$ as $k \to \infty$. By using (iii) and (iv), show that

$$g(x) = g\left(\lim_{k\to\infty} r_k\right) = \lim_{k\to\infty} g(r_k) = g(1) \lim_{k\to\infty} r_k = x \cdot g(1). \tag{702}$$

Lastly, we introduce the Poisson RVs and record some of their nice properties.

Example 6.1.8 (Poisson RV)**.** A RV X is a *Poisson RV* with rate $\lambda > 0$ if

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{703}$$

for all nonnegative integers $k \geq 0$. We write $X \sim \text{Poisson}(\lambda)$.

Exercise 6.1.9. Let $X \sim \text{Poisson}(\lambda)$. Show that $\mathbb{E}(X) = \text{Var}(X) = \lambda$.

Exercise 6.1.10 (Sum of ind. Poisson RVs is Poisson)**.** Let $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ be independent Poisson RVs. Show that $X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

Example 6.1.11 (MGF of Poisson RV)**.** Let $X \sim \text{Poisson}(\lambda)$. Then using the Taylor expansion of the exponential function, we can compute the moment generating function (MGF) of X:

$$\mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} e^{kt} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(e^t\lambda)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t-1)}. \tag{704}$$

Proposition 6.1.12 (Splitting a Poisson RV)**.** *Suppose there are $N \sim \text{Poisson}(\lambda)$ balls in a box. Each ball gets a color $i \in \{1, \ldots, m\}$ with probability p_i independently, where $p_1 + \cdots + p_m = 1$. Let N_i denote the number of balls of color i. Then $N_i \sim \text{Poisson}(\lambda p_i)$ for $1 \leq i \leq m$ and $N_1, \ldots, N_m$ are independent.*

PROOF. We directly compute the joint distribution of $N_1, \ldots, N_m$. The key is to recognize that the joint distribution of $(N_1, \ldots, N_m)$ conditional on $N = n$ is multinomial:

$$\begin{aligned}
\mathbb{P}(N_1 = k_1, \ldots, N_m = k_m) &= \mathbb{P}(N = n)\,\mathbb{P}(N_1 = k_1, \ldots, N_m = k_m \mid N = n) && (705)\\
&= \frac{\lambda^{-n} e^{-\lambda}}{n!} \frac{n!}{k_1! \cdots k_m!} p_1^{k_1} \cdots p_m^{k_m} && (706)\\
&= \left(\frac{(p_1\lambda)^{k_1} e^{-p_1\lambda}}{k_1!}\right) \cdots \left(\frac{(p_m\lambda)^{k_m} e^{-p_m\lambda}}{k_m!}\right) && (707)\\
&= \mathbb{P}(\text{Poisson}(p_1\lambda) = k_1) \cdots \mathbb{P}(\text{Poisson}(p_m\lambda) = k_m), && (708)
\end{aligned}$$

where we have used the fact that $\lambda = \lambda(p_1 + \cdots + p_m)$. Hence the joint distribution of $(N_1, \ldots, N_m)$ factorizes into the product of Poisson distributions with parameters $p_k\lambda$ for $1 \leq k \leq m$. This is enough to conclude the independence of $N_1, \ldots, N_m$ as well as their claimed Poisson distributions. □

6.2. Poisson processes as an arrival process

Recall the definition of arrival and renewal processes in Section 2.1. Below we define Poisson process.

Definition 6.2.1 (Poisson process)**.** An arrival process $(T_k)_{k\geq 1}$ is a *Poisson process of rate* λ if its inter-arrival times are i.i.d. Exp(λ) RVs. In this case, we denote $(T_k)_{k\geq 1} \sim \text{PP}(\lambda)$ and equivalently $(N(t))_{t\geq 0} \sim \text{PP}(\lambda)$ for its associated counting process $(N(t))_{t\geq 0}$.

The choice of exponential inter-arrival times is special due to the memoryless property of exponential RVs (Exercise 6.1.6).

Exercise 6.2.2. Let $(T_k)_{k\geq 1}$ be a Poisson process with rate λ. Show that $\mathbb{E}[T_k] = k/\lambda$ and $\text{Var}(T_k) = k/\lambda^2$. Furthermore, show that $T_k \sim \text{Erlang}(k,\lambda)$, that is,

$$f_{T_k}(z) = \frac{\lambda^k z^{k-1} e^{-\lambda z}}{(k-1)!}. \tag{709}$$

The following exercise explains what is 'Poisson' about the Poisson process.

Exercise 6.2.3. Let $(T_k)_{k\geq 1}$ be a Poisson process with rate λ and let $(N(t))_{t\geq 0}$ be the associated counting process. We will show that $N(t) \sim \text{Poisson}(\lambda t)$.

(i) Using the relation $\{T_n \leq t\} = \{N(t) \geq n\}$ and Exercise 6.2.2, show that

$$\mathbb{P}(N(t) \geq n) = \mathbb{P}(T_n \leq t) = \int_0^t \frac{\lambda^n z^{n-1} e^{-\lambda z}}{(n-1)!}\, dz. \tag{710}$$

(ii) Let $G(t) = \sum_{m=n}^{\infty} (\lambda t)^m e^{-\lambda t}/m! = \mathbb{P}(\text{Poisson}(\lambda t) \geq n)$. Show that

$$\frac{d}{dt} G(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!} = \frac{d}{dt}\mathbb{P}(T_n \leq t). \tag{711}$$

Conclude that $G(t) = \mathbb{P}(T_n \leq t)$.

(iii) From (i) and (ii), conclude that $N(t) \sim \text{Poisson}(\lambda t)$.

6.3. Momoryless property and stopping times

Let $(N(t))_{t\geq 0}\text{PP}(\lambda)$. For any fixed $u \geq 0$, $(N(t+u) - N(t))_{t\geq 0}$ is a counting process itself, which counts the number of arrivals during the interval $[u, u+t]$. We call this the counting process $(N(t))_{t\geq 0}$ *restarted at time* u. One of the remarkable properties of a Poisson process is that their counting process restarted at any given time u is again the counting process of a Poisson process, which is independent of what have happened up to time u. This is called the memoryless property of Poisson processes.

Memoryless property of Poisson processes not only applies for a deterministic restart times, but also some class of random times called 'stopping times'. We introduce this notion in a greater generality.

Definition 6.3.1 (Stopping time)**.** Let $\mathscr{F}_t$ be the collection of all events that one could observe by time t[1]. The sequence $(\mathscr{F}_t)_{t\geq 0}$ is called a *filtration*[2]. A random variable $T \geq 0$ is a *stopping time* w.r.t. a given filtration $(\mathscr{F}_t)_{t\geq 0}$ if for each $u \geq 0$,

$$\{T \leq u\} \in \mathscr{F}_u \quad \text{or} \quad \{T > u\} \in \mathscr{F}_u. \tag{712}$$

The condition above is interpreted as follows: By using all information we gathered up to time u, we can determine whether $T \leq u$ or $T > u$, not using any future information that is yet to be observed.

Theorem 6.3.2 (Memoryless property of PP)**.** *Let* $(N(t))_{t\geq 0} \sim \text{PP}(\lambda)$ *and let* $T \geq 0$ *be a stopping time w.r.t. a filtration* $(\mathscr{F}_t)_{t\geq 0}$ *such that* $\mathscr{F}_t$ *contains all information of* $(N(u))_{0\leq u\leq t}$*. Then the following holds.*

[1] Proper name: σ-field generated by information up to time t

[2] Since one can only gain information, we have $\mathscr{F}_a \subseteq \mathscr{F}_b$ for each $0 \leq a \leq b$.

(i) $(N(T+t)-N(T))_{t\geq 0} \sim \mathrm{PP}(\lambda)$.
(ii) $(N(t))_{0\leq t\leq T}$ *and* $(N(T+t)-N(T))_{t\geq 0}$ *are independent.*

The following is a very useful observation, which allows us to restart a Poisson process at any stopping time and still get an independent Poisson process.

Lemma 6.3.3. *Let $(X(t))_{t\geq 0}$ be any stochastic processes and let T be a stopping time for $(X(t))_{t\geq 0}$. Suppose the followings hold:*

(i) *(Independent increments) For any $u \geq 0$, the processes $(X(t))_{0\leq t\leq u}$ and $(X(t)-X(u))_{t>u}$ are independent.*
(ii) *(Stationary increments) The distribution of the process $(X(t)-X(u))_{t>u}$ does not depend on u (stationary increment).*

Then $(X(t)-X(T))_{t>T}$ has the same distribution as $(X(t)-X(0))_{t>0}$ and is independent of $(X(t))_{t\leq T}$.

According to Lemma 6.3.3, Theorem 6.3.2 follows once we show that the counting process $(N(t))_{t\geq 0}$ of $\mathrm{PP}(\lambda)$ satisfies the independent and stationary increment properties stated in Lemma 6.3.3. This is verified by the following proposition.

Proposition 6.3.4 (PP has independent and stationary increments)**.** *Let $(T_k)_{k\geq 1}$ be a Poisson process of rate λ and let $(N(t))_{t\geq 0}$ be the associated counting process.*

(i) *For any $t \geq 0$, let $Z(t) = \inf\{s > 0 : N(t+s) > N(t)\}$ be the waiting time for the first arrival after time t. Then $Z(t) \sim \mathrm{Exp}(\lambda)$ and it is independent of the process up to time t.*
(ii) *For any $s \geq 0$, $(N(t+s)-N(t))_{s\geq 0}$ is the counting process of a Poisson process of rate λ, which is independent of the process $(N(u))_{t\leq u}$.*

Exercise 6.3.5 (Sum of independent Poisson RV's is Poisson)**.** Let $(T_k)_{k\geq 1}$ be a Poisson process with rate λ and let $(N(t))_{t\geq 0}$ be the associated counting process. Fix $t, s \geq 0$.

(i) Use memoryless property to show that $N(t)$ and $N(t+s)-N(t)$ are independent Poisson RVs of rates λt and λs.
(ii) Note that the total number of arrivals during $[0, t+s]$ can be divided into the number of arrivals during $[0,t]$ and $[t,t+s]$. Conclude that if $X \sim \mathrm{Poisson}(\lambda t)$ and $Y \sim \mathrm{Poisson}(\lambda s)$ and if they are independent, then $X+Y \in \mathrm{Poisson}(\lambda(t+s))$.

Exercise 6.3.6 (Poisson process restarted at stopping times)**.** Let $(T_k)_{k\geq 1} \sim \mathrm{PP}(\lambda)$ and let $(N(t))_{t\geq 0}$ be its counting process.

(i) Let T be any stopping time for $(N(t))_{t\geq 0}$. Use the memoryless property of Poisson processes and Lemma 6.3.3 to show that $(N(t)-N(T))_{t\geq T}$ is the counting process of a $\mathrm{PP}(\lambda)$, which is independent of $(N(t))_{0\leq t<T}$.
(ii) Let T be the first time that we see three arrivals during a unit time. That is,

$$T = \inf\{t \geq 0 \,|\, N(t) - N(t-1) = 3\}. \tag{713}$$

Show that T is a stopping time. According to (i), $(N(t)-N(T))_{t\geq T}$ is the counting process of a $\mathrm{PP}(\lambda)$, which is independent of what have happened during $[0,T]$.
(iii) Let T be the first time t that there is no arrival during the interval $[t,t+1]$. Is T a stopping time? Is $(N(t)-N(T))_{t\geq T}$ the counting process of a $\mathrm{PP}(\lambda)$?

Proof of Lemma 6.3.3. (Optional*) Let E, E' be any events for real stochastic processes. For any stochastic process $W = (W(t))_{t\geq 0}$, we write $W \in E$ if the event E occurs for the process W. We want to show

$$\mathbb{P}\left((X(t)-X(T))_{t>T} \in E \,|\, (X(t))_{0\leq t\leq T}\right) = \mathbb{P}\left((X(t)-X(T))_{t>T} \in E\right). \tag{714}$$

We show the above equation under the conditioning on $\{T = u\}$, and then undo the conditioning by iterated expectation.

Since T is a stopping time for $(X(t))_{t\geq 0}$, for any $u \geq 0$, the event

$$\{(X(t))_{0\leq t\leq T} \in E', T = u\} = \{(X(t))_{0\leq t\leq u} \in E', T = u\} \tag{715}$$

is independent of the process $(X(t) - X(u))_{t>u}$ on the time interval (u,∞). Hence by the independent increments assumption (i), the above event is independent from the event $\{(X(t) - X(u))_{t>u} \in E\}$. This gives that, for any $u \geq 0$,

$$\begin{aligned}
&\mathbb{P}\left((X(t) - X(T))_{t>T} \in E \,\middle|\, (X(t))_{0\leq t\leq T} \in E', T = u\right) && (716)\\
&\quad = \mathbb{P}\left((X(t) - X(u))_{t>u} \in E \,\middle|\, (X(t))_{0\leq t\leq u} \in E', T = u\right) && (717)\\
&\quad = \mathbb{P}\big((X(t) - X(u))_{t>u} \in E\big) && (718)\\
&\quad = \mathbb{P}\big((X(t) - X(0))_{t>0} \in E\big), && (719)
\end{aligned}$$

where for the last equality, we have used the stationary increments assumption (ii).

Hence by iterated expectation,

$$\begin{aligned}
&\mathbb{P}\left((X(t) - X(T))_{t>T} \in E \,\middle|\, (X(t))_{0\leq t\leq T} \in E'\right) && (720)\\
&\quad = \mathbb{E}\left[\mathbb{P}\left((X(t) - X(T))_{t>T} \in E \,\middle|\, (X(t))_{0\leq t\leq T} \in E', T\right) \,\middle|\, T\right] && (721)\\
&\quad = \mathbb{E}\left[\mathbb{P}\left((X(t) - X(0))_{t>0} \in E\right) \,\middle|\, T\right] && (722)\\
&\quad = \mathbb{P}\left((X(t) - X(0))_{t>0} \in E\right). && (723)
\end{aligned}$$

To finish the proof, let E' be the entire sample space. Then the above yields

$$\mathbb{P}\left((X(t) - X(T))_{t>T} \in E\right) = \mathbb{P}\left((X(t) - X(0))_{t>0} \in E\right). \tag{724}$$

This shows that $(X(t)-X(T))_{t>T}$ has the same distribution as $(X(t)-X(0))_{t>0}$. Furthermore, we conclude

$$\begin{aligned}
\mathbb{P}\left((X(t) - X(T))_{t>T} \in E \,|\, (X(t))_{0\leq t\leq T} \in E'\right) &= \mathbb{P}\left((X(t) - X(0))_{t>0} \in E\right) && (725)\\
&= \mathbb{P}\left((X(t) - X(T))_{t>T} \in E\right). && (726)
\end{aligned}$$

Since E, E' were arbitrary, this also shows that $(X(t) - X(T))_{t>T}$ and $(X(t))_{0\leq t\leq T}$ are independent. □

PROOF OF PROPOSITION 6.3.4 . We first show **(ii)**. Note that

$$T_{N(t)} \leq t < T_{N(t)+1}. \tag{727}$$

Hence we may write

$$Z(t) = T_{N(t)+1} - t = \tau_{N(t)+1} - (t - T_{N(t)}). \tag{728}$$

Namely, $Z(t)$ is the remaining portion of the $N(t)+1$st inter-arrival time $\tau_{N(t)+1}$ after we waste the first $t - T_{N(t)}$ of it. (See Figure 3).

Now consider restarting the arrival process $(T_k)_{k\geq 0}$ at time t. The first inter-arrival time is $T_{N(t)+1}-t = Z(t)$, which is Exp(λ) and independent from the past by (i). The second inter-arrival time is $T_{N(t)+2} - T_{N(t)+1}$, which is Exp(λ) and is independent of everything else by assumption. Likewise, the following inter-arrival times for this restarted arrival process are i.i.d. Exp(λ) variables. This shows (ii).

Next, we show (i). Let E by any event for the counting process $(N(s))_{0\leq s\leq t}$ up to time t. In order to show the remaining waiting time $Z(t)$ and the past process up to time t are independent and $Z(t) \sim$ Exp(λ), we want to show that

$$\mathbb{P}(Z(t) \geq x \,|\, (N(s))_{0\leq s\leq t} \in E) = \mathbb{P}(Z(t) \geq x) = e^{-\lambda x}. \tag{729}$$

for any $x \geq 0$. To this end,

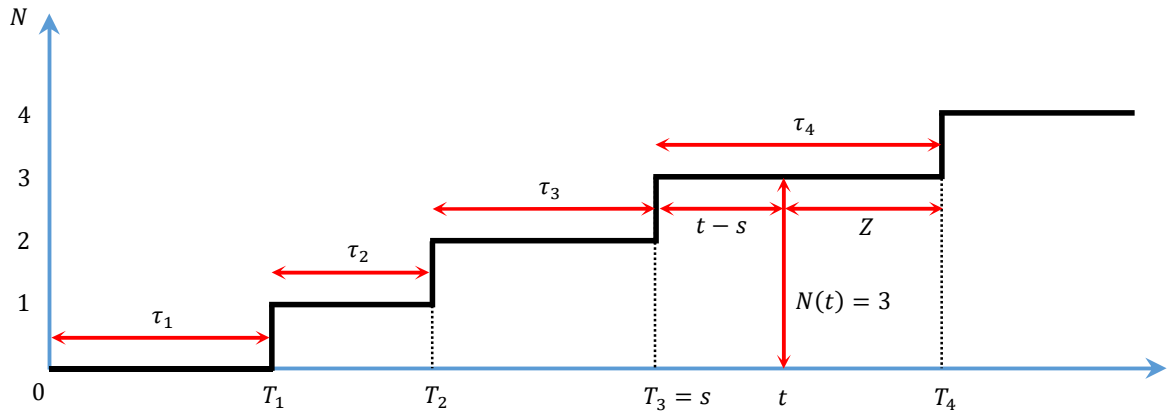


FIGURE 1. Assuming $N(t)=3$ and $T_3=s\le t$, we have $Z=\tau_4-(t-s)$. By memoryless property of exponential RV, Z follows Exp(λ) on this conditioning.

As can be seen from (6.3), $Z(t)$ depends on three random variables: $\tau_{N(t)+1}$, $N(t)$, and $T_{N(t)}$. To show , we argue by conditioning the last two RVs and use iterated expectation. Using 6.3, note that

$$\mathbb{P}\left(Z(t)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,N(t)=n,\,T_{N(t)}=s\right) \tag{730}$$

$$=\mathbb{P}\left(\tau_{n+1}-(t-s)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,N(t)=n,\,T_n=s\right) \tag{731}$$

$$=\mathbb{P}\left(\tau_{n+1}-(t-s)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,T_{n+1}>t,\,T_n=s\right) \tag{732}$$

$$=\mathbb{P}\left(\tau_{n+1}-(t-s)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,\tau_{n+1}>t-s,\,T_n=s\right). \tag{733}$$

Conditioned on $N(t)=n$, the event that $(N(s))_{0\le s\le t}\in E$ is determined by the arrival times $T_1,\cdots,T_n$ and the fact that $T_{n+1}\ge t$. Hence we can rewrite

$$\{(N(s))_{0\le s\le t}\in E,\,\tau_{n+1}>t-s,\,T_n=s\}=\left\{(\tau_1,\cdots,\tau_n)\in E',\,\tau_{n+1}>t-s\right\} \tag{734}$$

for some event E' to be satisfied by the first n inter-arrival times. Since inter-arrival times are independent, this gives

$$\mathbb{P}\left(Z(t)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,N(t)=n,\,T_{N(t)}=s\right) \tag{735}$$

$$=\mathbb{P}\left(\tau_{n+1}-(t-u)\ge x\,\middle|\,\tau_{n+1}\ge t-s\right) \tag{736}$$

$$=\mathbb{P}\left(\tau_{n+1}\ge x\right)=e^{-\lambda x}, \tag{737}$$

where we have used the memoryless property of exponential variables. Hence by iterated expectation,

$$\mathbb{P}\left(Z(t)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,N(t)=n\right)=\mathbb{E}_{T_{N(t)}}\left[\mathbb{P}\left(Z(t)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,N(t)=n,\,T_{N(t)}\right)\right] \tag{738}$$

$$=\mathbb{E}_{T_{N(t)}}[e^{-\lambda x}]=e^{-\lambda x}. \tag{739}$$

By using iterated expectation once more,

$$\mathbb{P}\left(Z(t)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\right)=\mathbb{E}_{N(t)}\left[\mathbb{P}\left(Z(t)\ge x\,\middle|\,(N(s))_{0\le s\le t}\in E,\,N(t)\right)\right] \tag{740}$$

$$=\mathbb{E}_{N(t)}[e^{-\lambda x}]=e^{-\lambda x}. \tag{741}$$

By taking E to be the entire sample space, this also gives

$$\mathbb{P}\left(Z(t)\ge x\right)=e^{-\lambda x}. \tag{742}$$

This shows (729). □

6.4. Merging and splitting of Poisson process

If customers arrive at a bank according to Poisson(λ) and if each one is male or female independently with probability q and $1-q$, then the 'thinned out' process of only male customers is a Poisson($q\lambda$); the process of female customers is a Poisson($(1-q)\lambda$).

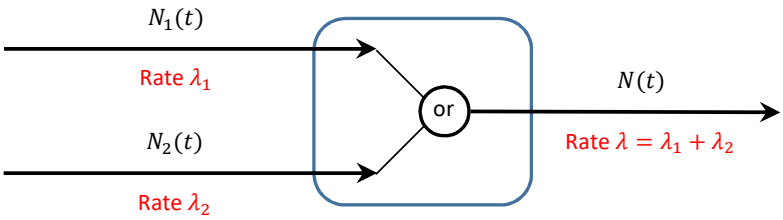


FIGURE 2. Merging two independent Poisson processes of rates λ_1 and λ_2 gives a new Poisson process of rate $\lambda_1 + \lambda_2$.

The reverse operation of splitting a given PP into two complementary PPs is call the 'merging'. Namely, imagine customers arrive at a register through two doors A and B independently according to PPs of rates λ_A and λ_B, respectively. Then the combined arrival process of entire customers is again a PP of the added rate.

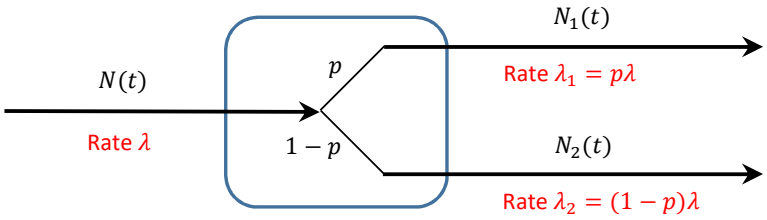


FIGURE 3. Splitting of Poisson process $N(t)$ of rate λ according to an independent Bernoulli process of parameter p.

Exercise 6.4.1 (Excerpted from [BT02])**.** Transmitters A and B independently send messages to a single receiver according to Poisson processes with rates $\lambda_A = 3$ and $\lambda_B = 4$ (messages per min). Each message (regardless of the source) contains a random number of words with PMF

$$\mathbb{P}(1 \text{ word}) = 2/6, \quad \mathbb{P}(2 \text{ words}) = 3/6, \quad \mathbb{P}(3 \text{ words}) = 1/6, \tag{743}$$

which is independent of everything else.

(i) Find $\mathbb{P}$(total nine messages are recieved during $[0, t]$).

(ii) Let $M(t)$ be the total number of words received during $[0, t]$. Find $\mathbb{E}[M(t)]$.

(iii) Let T be the first time that the receiver receives exactly three messages consisting of three words from transmitter A. Find distribution of T.

(iv) Compute $\mathbb{P}$(exactly seven messages out of the first ten messages are from A).

Exercise 6.4.2 (Order statistics of i.i.d. Exp RVs)**.** One hundred light bulbs are simultaneously put on a life test. Suppose the lifetimes of the individual light bulbs are independent Exp(λ) RVs. Let T_k be the kth time that some light bulb fails. We will find the distribution of T_k using Poisson processes.

(i) Think of T_1 as the first arrival time among 100 independent PPs of rate λ. Show that $T_1 \sim \text{Exp}(100\lambda)$.

(ii) After time T_1, there are 99 remaining light bulbs. Using memoryless property, argue that $T_2 - T_1$ is the first arrival time of 99 independent PPs of rate λ. Show that $T_2 - T_1 \sim \text{Exp}(99\lambda)$ and that $T_2 - T_1$ is independent of T_1.

(iii) As in the coupon collector problem, we break up

$$T_k = \tau_1 + \tau_2 + \cdots + \tau_k, \tag{744}$$

where $\tau_i = T_i - T_{i-1}$ with $\tau_1 = T_1$. Note that τ_i is the waiting time between $i-1$st and ith failures. Using the ideas in (i) and (ii), show that τ_i's are independent and $\tau_i \sim \text{Exp}((101-i)\lambda)$. Deduce that

$$\mathbb{E}[T_k] = \frac{1}{\lambda}\left(\frac{1}{100} + \frac{1}{99} + \cdots + \frac{1}{(100-k+1)}\right), \tag{745}$$

$$\text{Var}[T_k] = \frac{1}{\lambda^2}\left(\frac{1}{100^2} + \frac{1}{99^2} + \cdots + \frac{1}{(100-k+1)^2}\right). \tag{746}$$

(iv) Let $X_1, X_2, \cdots, X_{100}$ be i.i.d. Exp(λ) variables. Let $X_{(1)} < X_{(2)} < \cdots < X_{(100)}$ be their order statistics, that is, $X_{(k)}$ is the kth smallest among the X_i's. Show that $X_{(k)}$ has the same distribution as T_k, the kth time some light bulb fails. (So we know what it is from the previous parts.)

In the next two exercises, we rigorously justify merging and splitting of Poisson processes.

Exercise 6.4.3 (Merging of independent PPs)**.** Let $(N_1(t))_{t\geq 0}$ and $(N_2(t))_{t\geq 0}$ be the counting processes of two independent PPs of rates λ_1 and λ_2, respectively. Define a new counting process $(N(t))_{t\geq 0}$ by

$$N(t) = N_1(t) + N_2(t). \tag{747}$$

In this exercise, we show that $(N(t))_{t\geq 0} \sim \text{PP}(p\lambda)$.

(i) Let $\tau_k^{(1)}, \tau_k^{(2)}$, and τ_k be the kth inter-arrival times of the counting processes $(N_1(t))_{t\geq 0}$, $(N_2(t))_{t\geq 0}$, and $(N(t))_{t\geq 0}$. Show that $\tau_1 = \min(\tau_1^{(1)}, \tau_1^{(2)})$. Conclude that $\tau_1 \sim \text{Exp}(\lambda_1 + \lambda_2)$.

(ii) Let T_k be the kth arrival time for the joint process $(N(t))_{t\geq 0}$. Use memoryless property of PP and Exercise 6.3.6 to deduce that N_1 and N_2 restarted at time T_k are independent PPs of rates λ_1 and λ_2, which are also independent from the past (before time T_k).

(iii) From (ii), show that

$$\tau_{k+1} = \min(\tilde{\tau}_1, \tilde{\tau}_2), \tag{748}$$

where $\tilde{\tau}_1$ is the waiting time for the first arrival after time T_k for N_1, and similarly for $\tilde{\tau}_2$. Deduce that $\tau_{k+1} \sim \text{Exp}(\lambda_1 + \lambda_2)$ and it is independent of $\tau_1, \cdots, \tau_k$. Conclude that $(N(t))_{t\geq 0} \sim \text{PP}(\lambda_1 + \lambda_2)$.

Exercise 6.4.4 (Splitting of PP)**.** Let $(N(t))_{t\geq 0}$ be the counting process of a Poisson(λ), and let $(X_k)_{k\geq 0}$ be a sequence of i.i.d. Bernoulli(p) RVs. We define two counting processes $(N_1(t))_{t\geq 0}$ and $(N_2(t))_{t\geq 0}$ by

$$N_1(t) = \sum_{k=1}^{\infty} \mathbf{1}(T_k \leq t)\mathbf{1}(X_k = 1) = \#(\text{arrivals with coin landing on heads up to time } t), \tag{749}$$

$$N_2(t) = \sum_{k=1}^{\infty} \mathbf{1}(T_k \leq t)\mathbf{1}(X_k = 0) = \#(\text{arrivals with coin landing on tails up to time } t). \tag{750}$$

In this exercise, we show that $(N_1(t))_{t\geq 0} \sim \text{Poisson}(p\lambda)$ and $(N_2(t))_{t\geq 0} \sim \text{Poisson}((1-p)\lambda)$.

(i) Let τ_k and $\tau_k^{(1)}$ be the kth inter-arrival times of the counting processes $(N(t))_{t\geq 0}$ and $(N_1(t))_{t\geq 0}$. Let Y_k be the location of kth 1 in $(X_t)_{t\geq 0}$. Show that

$$\tau_1^{(1)} = \sum_{i=1}^{Y_1} \tau_i. \tag{751}$$

(ii) Show that

$$\tau_2^{(1)} = \sum_{k=Y_1+1}^{Y_2} \tau_i. \tag{752}$$

(iii) Show that in general,

$$\tau_k^{(1)} = \sum_{k=Y_{k-1}+1}^{Y_k} \tau_i. \tag{753}$$

(iv) Recall that $Y_k - Y_{k-1}$'s are i.i.d. Geom(p) RVs. Use Example 6.4.5 and (iii) to deduce that $\tau_k^{(1)}$'s are i.i.d. Exp$(p\lambda)$ RVs. Conclude that $(N_1(t)) \sim$ Poisson$(p\lambda)$. (The same argument shows $(N_2(t))_{t\geq 0} \sim$ Poisson$((1-p)\lambda)$.)

Example 6.4.5 (Sum of geometric number of Exp. is Exp.)**.** Let $X_i \sim \text{Exp}(\lambda)$ for $i \geq 0$ and let $N \sim \text{Geom}(p)$. Let $Y = \sum_{k=1}^N X_k$. Suppose all RVs are independent. Then $Y \sim \text{Exp}(p\lambda)$.

To see this, recall that their moment generating functions are

$$M_{X_1}(t) = \frac{\lambda}{\lambda - t}, \qquad M_N(t) = \frac{pe^t}{1-(1-p)e^t}. \tag{754}$$

Hence (see Remark 6.4.6)

$$M_Y(t) = M_N(\log M_{X_1}(t)) = \frac{p\frac{\lambda}{\lambda-t}}{1-(1-p)\frac{\lambda}{\lambda-t}} = \frac{p\lambda}{(\lambda-t)-\lambda(1-p)} = \frac{p\lambda}{p\lambda - t}. \tag{755}$$

Notice that this is the MGF of an Exp$(p\lambda)$ variable. Thus by uniqueness, we conclude that $Y \sim \text{Exp}(p\lambda)$. ▲

Remark 6.4.6. Let $Y = X_1 + X_2 + \cdots + X_N$, where X_i's are i.i.d. RVs and N is an independent RV taking values from positive integers. By iterated expectation, we have

$$\begin{aligned} M_Y(t) = \mathbb{E}[e^{tY}] &= \mathbb{E}[e^{tX_1}e^{tX_2}\cdots e^{tX_N}] && (756)\\ &= \mathbb{E}_N[\mathbb{E}[e^{tX_1}e^{tX_2}\cdots e^{tX_N}]\,|\,N] && (757)\\ &= \mathbb{E}_N[\mathbb{E}[e^{tX_1}]^N\,|\,N] && (758)\\ &= \mathbb{E}_N[M_{X_1}(t)^N] && (759)\\ &= \mathbb{E}_N[e^{N\log M_{X_1}(t)}] && (760)\\ &= M_N(\log M_{X_1}(t)). && (761) \end{aligned}$$

6.5. Poisson process and Poisson point process

In Section 6.2, we have defined an arrival process $(T_k)_{k\geq 1}$ to be a Poisson process of rate λ if its inter-arrival times are i.i.d. Exp(λ) variables (Definition 6.2.1). In this section, we provide equivalent definitions of Poisson processes in terms of the associated counting process (see (153)). This new perspective has many complementary advantages. Most importantly, this allows us to define the time-inhomogeneous Poisson processes, where the rate of arrival changes in time.

An arrival process gives rise to the notion of a 'size' of each subset of $A \subset \mathbb{R}$ as the numbe of arrival times that occur in A. This is called the associated 'counting measure' as defined below.

Definition 6.5.1. Let $(T_n)_{n\geq 0}$, $T_0 = 0$ denote the sequence of arrival times of an arrival process. The associated *counting measure* of a renewal process is the function $N(A)$ of subsets $A \subseteq \mathbb{R}$ that gives the number of arrival times that lie in the set A:

$$N(A) := \sum_{n=1}^{\infty} \mathbf{1}(T_n \in A) = \#\text{ of arrival times in } A \tag{762}$$

With the above definition, we can think of the associaved counting process $(N(t))_{t\geq 0}$ as $N(t) = N([0,t])$. Then, taking $A = (a,b]$ for $0 \leq a < b$, we can write $N(A) = N((a,b])$ as

$$N((a,b]) = N(b) - N(a) = \text{ (\# of arrivals during } [0,b]) \text{ -(\# of arrivals during } [0,a]) \,. \tag{763}$$

Definition 6.5.2 (Poisson point process)**.** A counting measure N on $\mathbb{R}$ is called a *Poisson point process* of intensity λ (denoted $N \sim \text{PPP}(\lambda)$) if it has the following properties:

(i) For each $0 \le a < b$, $N((a,b]) \sim \text{Poisson}(\lambda(b-a))$.
(ii) For disjoint intervals $(a_1,b_1],\ldots,,(a_n,b_n]$, the random variables $N((a_1,b_1]),\ldots,N((a_n,b_n])$ are independent.

The next theorem describes the properties of the counting measure of the Poisson process.

Proposition 6.5.3 (PP implies PPP)**.** *Let $(N(t))_{t\ge 0} \sim \text{PP}(\lambda)$ be the counting process of a Poisson process of rate λ. As a counting measure, $N((a,b]) := N(b) - N(a)$, N is a Poisson point process with intensity λ.*

PROOF. Follows from the memoryless process of Poisson process. Namely, restarting the PP at time $t = a$, we get $N((a-b)) = N(b) - N(a) \sim \text{PP}(\lambda(b-a))$, since $(N(a+t) - N(a))_{t\ge 0}$ is a PP(λ) by Proposition 6.3.4. Also, fix disjoint intervals $(a_1,b_1],\ldots,,(a_n,b_n]$. Without loss of generality, we may assume that $a_1 \le b_1 \le a_2 \le b_2 \le a_3 \le \cdots \le a_n \le b_n$. Then restarting the process at $t = b_1$ and using the memoryless property of PP, we see that $N((a_1,b_1])$ is independent of $N((a_k,b_k])$ for $2 \le k \le n$. Similarly, restarting the process at time $t = b_2$, we see that $N((a_2,b_2])$ is independent of $N((a_k,b_k])$ for $3 \le k \le n$. Proceeding in the same way, we see the RVs $N((a_1,b_1]),\ldots,N((a_n,b_n])$ are independent. □

Exercise 6.5.4 (PPP implies PP)**.** Let $N \sim \text{PPP}(\lambda)$. Consider the induced counting process $N(t) := N([0,t])$ for $t \ge 0$. Let $T_k = \inf\{u \ge 0 \,|\, N(u) = k\}$ be the kth 'arrival time' and let $\tau_k = T_k - T_{k-1}$ be the kth 'inter-arrival time'.

(i) Use the fact that T_k is a stopping time for $(N(t))_{t\ge 0}$ and Lemma 6.3.3 to deduce that $(N(T_k + t) - N(T_k))_{t\ge 0}$ is the counting process of a PP(λ) that is independent of $(N(t))_{t\le T_k}$.
(ii) Let $Z(t) = \inf\{u \ge 0 \,|\, N(t+u) > N(t)\}$ be the waiting time for the first arrival after time t. Show that $Z(t) \sim \text{Exp}(\lambda)$ for all $t \ge 0$.
(iii) Use **(ii)** and conditioning on T_{k-1} to show that $\tau_k \sim \text{Exp}(\lambda)$ for all $k \ge 1$.

Recall that in Exercise 6.4.4, we have established splitting of PP by directly analyzing the super-inter-arrival times obtained by merging a geometric number of Exp(λ) inter-arrival times. In the following exercise, we give steps to obtaining a similar statement via using PPP instead.

Exercise 6.5.5 (Splitting of PP via PPP)**.** Let $N \sim \text{PPP}(\lambda)$ and let $T_1 < T_2 < \ldots$ denote the associated arrival times (or particle locations), where for any interval I, the counting measure $N(I)$ of I equals the number of T_k's such that $T_k \in I$. Now suppose each particle at location T_k for $k \ge 1$ is either red with probability p and blue with probability $1-p$, independently of everything else. Define two induced counting measures N_1 and N_2 by

$$N_1(I) = \#\text{ of red points in } I, \tag{764}$$
$$N_2(I) = \#\text{ of blue points in } I. \tag{765}$$

(i) Show that $N_1 \sim \text{PPP}(p\lambda)$ and $N_2 \sim \text{PPP}((1-p)\lambda)$. (Use Proposition 6.4.4.)
(ii) For each interval I, show that $N_1(I)$ and $N_2(I)$ are independent. (Use Proposition 6.4.4.)
(iii) Show that N_1 and N_2 are independent. Namely, for intervals I_1 and I_2 (not necessarily disjoint), show that $N(I_1)$ and $N(I_2)$ are independent. (**Hint**: Decompose $N_1(I_1) = N_1(I_1 \setminus I_2) + N_1(I_1 \cap I_2)$ and similarly for $N_2(I_2)$. $N_1(I_1 \setminus I_2)$ and $N_2(I_2 \setminus I_1)$ are independent by the independence property of N and since $(I_1 \setminus I_2)$ and $(I_2 \setminus I_1)$ are disjoint. Use **(ii)** to conclude.)
(iv) By using Exercise 6.5.4, conclude that splitting a PP(λ) with independent Bernoulli(p) coin flips gives two independent PP($p\lambda$) and PP($(1-p)\lambda$).

6.6. Poisson process conditioned on the number of arrivals

Suppose we know there are total 10 arrivals during $[0,1]$ of a PP(λ). What can we tell about the arrival times $T_1 < \cdots < T_{10}$? Clearly they all lie in the interval $[0,t]$. It turns out that, if we sample n points $U_1,\ldots,U_n$ independently and uniformly at random from $[0,1]$, then the leftmost value is T_1, the one to its right is T_2, and so on, in distribution. Note that in this description, there is no involvement of the rate parameter λ. In other words, the role of the rate parameter λ in PP(λ) is only to modulate the number of arrivals in a given interval, and once we know the number, the specific arrival times within that interval are generic and do not depend on λ.

To make a more precise statement of the above sketch, recall the following notion of order statistics. Let $U_1,\ldots,U_n$ denote n RVs. Let $U_{n;k}$ denote the kth smallest among them for $1 \le k \le n$. Then $U_{n;1} \le \cdots \le U_{n;n}$, which are called the *order statistic* of $U_1,\cdots,U_n$. o

Proposition 6.6.1. *Let $(T_n)_{n\ge1}$ denote the arrival times of $(N(t))_{t\ge0} \sim$ PP(λ).*

(i) *The joint distribution of $(T_1,\ldots,T_n)$ conditional on $N(t)=n$ is the same as the joint distribution of the order statistics of n i.i.d.* Uniform($[0,t]$) *RVs.*

(ii) *The joint PDF of $(T_1,\ldots,T_n)$ is given as*

$$f_{(T_1,\ldots,T_n)|N(t)=n}(t_1,\ldots,t_n) = \frac{n!}{t^n}\mathbf{1}(0 \le t_1 < \cdots < t_n \le t). \tag{766}$$

PROOF. We first show **(ii)**. For this we give a 'physics' style proof here. First note that $0 < T_1 < \cdots < T_n < t$ with probability 1: the ordering follows from definition of arrival times and $T_n < t$ since we are conditioning on $N(t) = n$. Hence we may need to verify the formula (766) for $0 < t_1 < \cdots < t_n < t$. Fix such t_i's. Consider an ε-box at point $(t_1,\ldots,t_n)$: $(t_1,t_1+\varepsilon] \times \cdots \times (t_n,t_n+\varepsilon]$. We will assume that $0 < \varepsilon < \min(t_1, t_2-t_1, \ldots, t_n - t_{n-1}, t-t_n)$ so that the intervals $(t_1,t_1+\varepsilon],(t_1+\varepsilon,t_2],\ldots,(t_n,t_n+\varepsilon],(t_n+\varepsilon,t]$ are disjoint. By the mean value theorem for integrals,

$$\lim_{\varepsilon\to0}\frac{1}{\varepsilon^t}\mathbb{P}\left(T_1 \in (t_1,t_1+\varepsilon),\ldots,T_n\in(t_n,t_n+\varepsilon]\,\middle|\,N(t)=n\right) = f_{(T_1,\ldots,T_n)|N(t)=n}(t_1,\ldots,t_n). \tag{767}$$

Observe that

$$\mathbb{P}\left(T_1 \in (t_1,t_1+\varepsilon),\ldots,T_n\in(t_n,t_n+\varepsilon]\,\middle|\,N(t)=n\right) \tag{768}$$

$$= \mathbb{P}\left(N(t_1,t_1+\varepsilon]=1, N(t_1+\varepsilon,t_2]=0,\ldots,N(t_n,t_n+\varepsilon]=1, N(t_n+\varepsilon,t]=0\,\middle|\,N(t)=n\right) \tag{769}$$

$$= \frac{\mathbb{P}\left(N(t_1,t_1+\varepsilon]=1, N(t_1+\varepsilon,t_2]=0,\ldots,N(t_n,t_n+\varepsilon]=1, N(t_n+\varepsilon,t]=0\right)}{\mathbb{P}(N(t)=n)} \tag{770}$$

$$= \frac{n!}{(\lambda t)^n e^{-\lambda t}}\underbrace{\left((\lambda\varepsilon)e^{-\lambda\varepsilon}\right)\left(e^{-\lambda(t_2-t_1-\varepsilon)}\right)\left((\lambda\varepsilon)e^{-\lambda\varepsilon}\right)\left(e^{-\lambda(t_3-t_2-\varepsilon)}\right)\cdots\left((\lambda\varepsilon)e^{-\lambda\varepsilon}\right)\left(e^{-\lambda(t-t_n-\varepsilon)}\right)}_{t \text{ repetitions}} \tag{771}$$

$$= \frac{n!}{(\lambda t)^n e^{-\lambda t}}\lambda^t\varepsilon^t e^{-\lambda t} \tag{772}$$

$$= \frac{n!}{t^n}\varepsilon^t. \tag{773}$$

Hence dividing both sides by ε^t and taking the limit $\varepsilon\to0$, we obtain the desired result.

Next, we deduce **(i)**. Let $U_1,\ldots,U_n \overset{i.i.d.}{\sim}$ Uniform($[0,t]$). Then the joint PDF of $(U_1,\ldots,U_n)$ is constant over the cube $[0,t]^n$ with value $1/t^n$. But for each values $0 < t_1 < \cdots < t_n < t$ for the order statistic $U_{n;1},\ldots,U_{n;n}$, there are $n!$ ways that these values will be realized by $U_1,\ldots,U_n$, and each case will have the same probability by symmetry. Hence the joint PDF of $U_{n;1},\ldots,U_{n;n}$ is constant on the region $\{(t_1,\ldots,n_n) : 0 < t_1 < \ldots,< t_n < t\}$ with constant value $n!/t^n$. This is exactly the same joint PDF as in **(ii)**. □

Example 6.6.2. Let $N \sim \text{PPP}(\lambda)$. Let $0 < r < t$ and we compute $\mathbb{P}(T_2 < r \,|\, N(t) = 2)$, the probability that the second arrival is by time r given that there are total two arrivals up to time t. By using Proposition 6.6.1, we have

$$f_{(T_1,T_2)|N(t)=2}(t_1, t_2) = \frac{2}{t^2}\mathbf{1}(0 < t_1 < t_2 < t). \tag{774}$$

Hence we get

$$\mathbb{P}(T_2 < r \,|\, N(t) = 2) = \int_0^r \int_0^{t_2} \frac{2}{t^2}\, dt_1\, dt_2 = \frac{2}{t^2}\frac{r^2}{2} = \frac{r^2}{t^2}. \tag{775}$$

To go one step further, by differentiating the above by r, we can obtain the conditional PDF of T_2 given $N(t) = 2$:

$$f_{T_2|N(t)=2}(x) = \frac{2x}{t^2}\mathbf{1}(0 < x < t). \tag{776}$$

▲

6.7. Nonhomogeneous Poisson process

In this section, we introduce Poisson process with time-varying rate $\lambda(t)$.

Example 6.7.1. Consider an counting process $(N(t))_{t\geq 0}$, which follows $\text{PP}(\lambda_1)$ on interval $[1,2)$, $\text{PP}(\lambda_2)$ on interval $[2,3)$, and $\text{PP}(\lambda_3)$ on interval $[3,4]$. Further assume that the increments $N(2)-N(1)$, $N(3)-N(2)$, and $N(4)-N(3)$ are independent. Then what is the distribution of the total number of arrivals during $[1,4]$? Since we can add independent Poisson RVs and get a Poisson RV with added rates, we get

$$N(4) - N(1) = [N(4) - N(3)] + [N(3) - N(2)] + [N(2) - N(1)] \sim \text{Poisson}(\lambda_1 + \lambda_2 + \lambda_3). \tag{777}$$

Note that the combined rate $\lambda_1 + \lambda_2 + \lambda_3$ can be seen as the integral of the step function $f(t) = \lambda_1\mathbf{1}(t \in [1,2)) + \lambda_2\mathbf{1}(t \in [2,3)) + \lambda_3\mathbf{1}(t \in [3,4])$. ▲

In general, suppose we have a concatenation of Poisson processes on disjoint intervals of very small lengths. Then $N(t) - N(s)$ can be seen as the sum of independent increments over the interval $[t, s]$, and by additivity of independent Poisson increments, it follows a Poisson distribution with rate given by the 'Riemann sum'. As the lengths of the intervals go to zero, this Riemann sum of rates tend to the integral of the rate function $\lambda(r)$ over the interval $[s, t]$. This suggests the following definition of nonhomogeneous Poisson processes.

Definition 6.7.2. An arrival process $(T_k)_{k\geq 1}$ is said to be a Poisson process with rate $\lambda(t)$ if its counting process $(N_t)_{t\geq 0}$ satisfies the following properties:

(i) $N(0) = 0$.
(ii) $(N(t))_{t\geq 0}$ has independent increments.
(iii) For any $0 \leq s < t$, $N(t) - N(s) \sim \text{Poisson}(\mu)$ where

$$\mu = \int_s^t \lambda(r)\, dr. \tag{778}$$

Example 6.7.3. A store opens at 8 AM. From 8 until 10 AM, customers arrive at a Poisson rate of four per hour. Between 10 AM and 12 PM, they arrive at a Poisson rate of eight per hour. From 12 PM to 2 PM the arrival rate increases steadily from eight per hour at 12 PM to ten per hour at 2 PM; and from 2PM to 5 PM, the arrival rate drops steadily from ten per hour at 2 PM to four per hour at 5 PM. Let us determine the probability distribution of the number of customers that enter the store on a given day (between 8 AM and 5 PM).

Let $\lambda(t)$ be the rate function in the statement and let $N(t)$ be a nonhomogeneous Poisson process with this rate function. From the description above, we can compute $m = \int_8^{17} \lambda(s)\, ds = 63$. Hence

$$N(17) - N(8) \sim \text{Poisson}(63). \tag{779}$$

▲

Exercise 6.7.4. Let $(T_k)_{k\geq 1} \sim \text{PP}(\lambda(t))$. Let $(\tau_k)_{k\geq 1}$ be the inter-arrival times.

(i) Let $Z(t)$ be the waiting time for the first arrival after time t. Show that

$$\mathbb{P}(Z(t) \geq x) = \exp\left(-\int_t^{t+x} \lambda(t)\, dt\right). \tag{780}$$

(ii) From (i), deduce that τ_1 has PDF

$$f_{\tau_1}(t) = \lambda(t) e^{-\int_0^t \lambda(r)\, dr}. \tag{781}$$

(iii) Denote $\mu(t) = \int_0^t \lambda(s)\, ds$. Use (i) and conditioning to show

$$\mathbb{P}(\tau_2 > x) = \mathbb{E}_{\tau_1}[\mathbb{P}(\tau_2 > x \,|\, \tau_1) \,|\, \tau_1] \tag{782}$$

$$= \int_0^\infty \mathbb{P}(\tau_2 > x \,|\, \tau_1 = t) f_{\tau_1}(t)\, dt \tag{783}$$

$$= \int_0^\infty e^{-(\mu(t+x)-\mu(t))} \lambda(t) e^{-\mu(t)}\, dt \tag{784}$$

$$= \int_0^\infty \lambda(t) e^{-\mu(t+x)}\, dt. \tag{785}$$

Conclude that τ_1 and τ_2 do not necessarily have the same distribution.

The following exercise shows that the nonhomogeneous Poisson process with rate $\lambda(t)$ can be obtained by a time change of the usual Poisson process of rate 1.

Exercise 6.7.5. Let $(N_0(t))_{t\geq 0}$ be the counting process of a Poisson process of rate 1. Let $\lambda(t)$ denote a non-negative function of t, and let

$$m(t) = \int_0^t \lambda(s)\, ds. \tag{786}$$

Define $N(t)$ by

$$N(t) = N_0(m(t)) = \#\text{ arrivals during } [0, m(t)]. \tag{787}$$

Show that $(N(t))_{t\geq 0}$ is the counting process of a Poisson process of rate $\lambda(t)$.

6.8. PP via asymptotic properties of the counting process (Optional*)

In this section, we give yet another definition of Poisson process in terms of the asymptotic properties of its counting process. For this, we need something called the 'small-o' notation. We say a function $f(t)$ is of order $o(t)$ or write $f(t) = o(t)$ if

$$\lim_{t\to 0} \frac{f(t)}{t} = 0. \tag{788}$$

Definition 6.8.1 (Def of PP-counting-asymptotic)**.** Suppose a counting process $(N(t))_{t\geq 0}$satisfies the following conditions (and denote $(N(t))_{t\geq 0} \sim \text{PP}'(\lambda)$):

(i) $N(0) = 0$;
(ii) $\mathbb{P}(N(t) = 0) = 1 - \lambda t + o(t)$;
(iii) $\mathbb{P}(N(t) = 1) = \lambda t + o(t)$;
(iv) $\mathbb{P}(N(t) \geq 2) = o(t)$;
(v) (Independent increment) For any $t, s \geq 0$, $N(t+s) - N(t)$ is independent of $(N(u))_{u\leq t}$;
(vi) (Stationary increment) For any $t, s \geq 0$, the distribution of $N(t+s) - N(t)$ is invariant under t.

It is easy to see that our usual definition of Poisson process in Definition 6.2.1 satisfies the properties **(i)**-**(vi)** above.

Proposition 6.8.2. *Let $(N(t))_{t\geq 0} \sim \text{PP}(\lambda)$ (see Definition 6.2.1). Then $(N(t))_{t\geq 0} \sim \text{PP}'(\lambda)$.*

PROOF. Using Taylor expansion of exponential function, note that

$$e^{-\lambda t} = 1 - \lambda t + o(t) \tag{789}$$

for all $t > 0$. Hence

$$\mathbb{P}(N(t) = n) = e^{-\lambda t}\frac{(\lambda t)^n}{n!} = (1 - \lambda t + o(t))\frac{(\lambda t)^n}{n!}. \tag{790}$$

So plugging in $n = 0$ and 1 gives **(ii)** and **(iii)**. For **(iv)**, we use **(ii)** and **(iii)** to get

$$\mathbb{P}(N(t) \geq 2) = 1 - \mathbb{P}(N(t) \leq 1) = 1 - (1 - \lambda t + o(t)) - (\lambda t + o(t)) = o(t). \tag{791}$$

Lastly, **(v)** and **(iv)** follows from the memoryless property of Poisson process (Proposition 6.3.4). □

Next, we consider the converse implication. We will break this into several exercises.

Exercise 6.8.3. Let $(N(t))_{t\geq 0}$ is the Poisson process with rate $\lambda > 0$ in the sense of Definition 6.8.1. In this exercise, we will show that $\mathbb{P}(N(t) = 0) = e^{-\lambda t}$.

(i) Use independent/stationary increment properties to show that

$$\mathbb{P}(N(t+h) = 0) = \mathbb{P}(N(t) = 0, N(t+h) - N(t) = 0) \tag{792}$$
$$= \mathbb{P}(N(t) = 0)\mathbb{P}(N(t+h) - N(t) = 0) \tag{793}$$
$$= \mathbb{P}(N(t) = 0)(1 - \lambda h + o(h)). \tag{794}$$

(ii) Denote $f_0(t) = \mathbb{P}(N(t) = 0)$. Use (i) to show that

$$\frac{f_0(t+h) - f_0(h)}{h} = \left(-\lambda + \frac{o(h)}{h}\right) f_0(t). \tag{795}$$

By taking limit as $h \to 0$, show that $f(t)$ satisfies the following differential equation

$$\frac{df_0(t)}{dt} = -\lambda f_0(t). \tag{796}$$

(iii) Conclude that $\mathbb{P}(N(t) = 0) = e^{-\lambda t}$.

Next, we generalize the ideas used in the previous exercise to compute the distribution of $N(t)$.

Exercise 6.8.4. Let $(N(t))_{t\geq 0}$ is the Poisson process with rate $\lambda > 0$ in the sense of Definition 6.8.1. Denote $f_n(t) = \mathbb{P}(N(t) = n)$ for each $n \geq 0$.

(i) Show that

$$\mathbb{P}(N(t) \leq n-2, N(t+h) = n) \leq \mathbb{P}(N(t+h) - N(t) \geq 2). \tag{797}$$

Conclude that

$$\mathbb{P}(N(t) \leq n-2, N(t+h) = n) = o(h). \tag{798}$$

(ii) Use (i) and independent/stationary increment properties to show that

$$f_n(t+h) = \mathbb{P}(N(t+h) = n) = \mathbb{P}(N(t) = n, N(t+h) - N(t) = 0) \tag{799}$$
$$+ \mathbb{P}(N(t) = n-1, N(t+h) - N(t) = 1) \tag{800}$$
$$+ \mathbb{P}(N(t) \leq n-2, N(t+h) = n) \tag{801}$$
$$= f_n(t)(1 - \lambda h + o(h)) + f_{n-1}(t)(\lambda h + o(h)) + o(h). \tag{802}$$

(iii) Use (ii) to show that the following differential equation holds:

$$\frac{df_n(t)}{dt} = -\lambda f_n(t) + \lambda f_{n-1}(t). \tag{803}$$

(iv) By multiplying the integrating factor $\mu(t) = e^{\lambda t}$ to (803), show that

$$(e^{\lambda t} f_n(t))' = \lambda e^{\lambda t} f_{n-1}(t). \tag{804}$$

Use the initial condition $f_n(0) = \mathbb{P}(N(0) = n) = 0$ to derive the recursive equation

$$f_n(t) = \lambda e^{-\lambda t} \int_0^t e^{\lambda s} f_{n-1}(s)\, ds. \tag{805}$$

(v) Use induction to conclude that $f_n(t) = (\lambda t)^n e^{-\lambda t}/n!$.

(vi) Conclude that for all $t, s \geq 0$ and $n \geq 0$,

$$N(t+s) - N(s) \sim \text{Poisson}(\lambda t). \tag{806}$$

CHAPTER 7

Renewal Processes: Examples and Queuing Theory

7.1. Age and residual life

Suppose buses arrive at a stop according to a renewal process with arrival times $(T_n)_{n\geq 1}$ and counting process $(N(t))_{t\geq 0}$. Assume that we get to the stop at time s. How much time has passed from the last bus? How long should we wait for the next bus? These quantities are called the *age* and *residual live*, respectively. Below we give precise definitions.

$$Z_s := T_{N(s)+1} - s = \text{Residual life (wating time until the next bus from time } s) \tag{807}$$
$$A_s := s - T_{N(s)} = \text{Age (time passed from the previous bus till time } s)$$

What can we say about the above random variables for fixed s or as $s \to \infty$?

We first address the above question for the speical case of Poisson processes.

Proposition 7.1.1. *Let $(N(t))_{t\geq 0} \sim \mathrm{PP}(\lambda)$. Then the age and residual life, A_s and Z_s in (807) satisfy the following.*

(i) *For each fixed $s > 0$, Z_s and A_s are independent with distributions $Z_s \sim \mathrm{Exp}(\lambda)$ and $A_s \sim \tau \wedge s$, where $\tau \sim \mathrm{Exp}(\lambda)$.*

(ii) *As $s \to \infty$, the random vector (A_s, Z_s) converges in distribution to the pair of indpendent $\mathrm{Exp}(\lambda)$ RVs.*

PROOF. We first show **(i)**. By the memoryless property of PP, once we restart the process at time s, the future process $(N(s+t) - N(s))_{t\geq 0}$ is another $\mathrm{PP}(\lambda)$ that is independent of the past process $(N(t))_{0\leq t\leq s}$. Since Z_s and A_s are determined by the future and the past processes, respectively, we see that they are independent. Also, Z_s is exactly the first inter-arrival time of the restarted $\mathrm{PP}(\lambda)$, so it has the distribution $\mathrm{Exp}(\lambda)$. For A_s, note that for $x \geq 0$,

$$\mathbb{P}(A_s \geq x) = \mathbb{P}(\text{no arrival during } [s-x, s])\mathbf{1}(0 \leq x \leq s) \tag{808}$$
$$= e^{-\lambda x}\mathbf{1}(0 \leq x \leq s). \tag{809}$$

Note that, if $\tau \sim \mathrm{Exp}(\lambda)$, then for $x \geq 0$,

$$\mathbb{P}(\tau \wedge s \geq x) = \mathbb{P}(\tau \geq s)\mathbf{1}(0 \leq x \leq s) = e^{-\lambda x}\mathbf{1}(0 \leq x \leq s) = \mathbb{P}(A_s \geq x). \tag{810}$$

It follows that A_s and $\tau \wedge s$ have the same distribution, as desired.

Next, **(ii)** follows from **(i)**. Indeed, as $s \to \infty$, the distribution of Z_s remains the same and the distribution of A_s converges to $\mathrm{Exp}(\lambda)$, and these RVs remain independent for all s. □

Therefore, if busses arrive according to $\mathrm{PP}(\lambda)$, then the expected wait time until the next bus is always $\mathbb{E}[Z_s] = 1/\lambda$ regardless of s. So, regardless of how many students are still waiting in the stop, you may get your ride in $1/\lambda$ time on average!

Next, we discuss the age and residual live for the general arrival processes. Notice that, in the proof of Proposition 7.1.1, the use of memoryless property of PPs was crucial, which is not available for the general arrival processes. Hence we can expect the ansers in the general case would not be as simple as in the Poisson case.

Fix $0 \leq x \leq s$ and $y \geq 0$. Note that

$$\mathbb{P}(A_s > x,\, Z_s > y) = \mathbb{P}(\text{no arrival during } [s-x, s+y]). \tag{811}$$

The right hand side is $e^{-\lambda(x+y)}$ for PP(λ), but is not easy to evaluate for general renewal processes. Hence, we ask a bit of easier question instead. *In the long run, what portion of times do we have $A_s > x$ and $Z_s > y$?* Namely, this is the portion of times that we missed the previous bus by more than x and should wait at least y until the next bus.

Proposition 7.1.2. *Let $(T_n)_{n\geq 1}$ denote the arrival times of a renewal process and suppose the inter-arrival times $(\tau_n)_{n\geq 1}$ have finite expectation. Let A_s and Z_s denote the age and the residual life as in (807). Then*

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t \mathbf{1}(A_s > x,\, Z_s > y)\, ds = \frac{\mathbb{E}\left[(\tau_1 - (x+y))^+\right]}{\mathbb{E}[\tau_1]} = \frac{1}{\mathbb{E}[\tau_1]}\int_{x+y}^{\infty} \mathbb{P}(\tau_1 \geq z)\, dz. \tag{812}$$

In particular, by taking $x = 0$ or $y = 0$, we have

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t \mathbf{1}(A_s > x)\, ds = \frac{1}{\mathbb{E}[\tau_1]}\int_{x}^{\infty} \mathbb{P}(\tau_1 \geq z)\, dz, \tag{813}$$

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t \mathbf{1}(Z_s > y)\, ds = \frac{1}{\mathbb{E}[\tau_1]}\int_{y}^{\infty} \mathbb{P}(\tau_1 \geq z)\, dz. \tag{814}$$

PROOF. We will frame this problem as a renewal reward process. Namely, the renewal process is given by the same arrival times $(T_n)_{n\geq 1}$ of the buses, each 'cycle' is the inter-arrival interval $(T_{n-1}, T_n]$, and the 'reward' from that interval is the amount of times that the event $\{A_s \geq x, Z_s \geq s\}$ holds during that interval, which is given by

$$R_n := \int_{T_{n-1}}^{T_n} \mathbf{1}(A_s > x,\, Z_s > y)\, ds. \tag{815}$$

By the renewal property, note that R_n's are i.i.d. with common expectation $\mathbb{E}[R_1]$, which is the numerator in the right hand side above. Hence by the renewal reward SLLN (Theorem 2.2.1), we can deduce

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t \mathbf{1}(A_s > x,\, Z_s > y)\, ds = \frac{\mathbb{E}[R_1]}{\mathbb{E}[\tau_1]}. \tag{816}$$

Furthermore, observe that

$$\begin{aligned}
\mathbf{1}(A_s > s,\, Z_s > y,\, s \in [0, T_1]) &= \mathbf{1}(\text{no arrival during } [s-x, s+y],\ s \in [0, T_1]) && (817)\\
&= \mathbf{1}([s-x, s+y] \subseteq [0, T_1]) && (818)\\
&= \mathbf{1}(x \leq s \leq T_1 - y)\mathbf{1}(T_1 \leq x+y). && (819)
\end{aligned}$$

Therefore, we get

$$R_1 = (T_1 - (x+y))\mathbf{1}(T \leq x+y) = (\tau_1 - (x+y))^+. \tag{820}$$

Using the tail-sum formula for the expectation of nonnegative random variables,

$$\mathbb{E}[R_1] = \mathbb{E}\left[(\tau_1 - (x+y))^+\right] = \int_0^{\infty} \mathbb{P}\left((\tau_1 - (x+y))^+ \geq z\right) dz = \int_{x+y}^{\infty} \mathbb{P}(\tau_1 \geq z)\, dz. \tag{821}$$

Hence the asserion follows. □

Next, if we impose an additional restriction on the inter-arrival times that they do not take values from an arithmatic progression, then the long-run averages in Proposition 7.1.2 gives their limiting distributions.

Theorem 7.1.3. *Let $(T_n)_{n\geq 1}$ denote the arrival times of a renewal process and suppose the inter-arrival times $(\tau_n)_{n\geq 1}$ have finite expectation. Let A_s and Z_s denote the age and the residual life as in (807). Further assume that, for every $a > 0$,*

$$\mathbb{P}(\tau \in \{a, 2a, 3a, \dots\}) < 1. \tag{822}$$

Then the following thold:

$$\lim_{t\to\infty} \mathbb{P}(A_t > x,\, Z_t > y) = \frac{1}{\mathbb{E}[\tau_1]} \int_{x+y}^{\infty} \mathbb{P}(\tau_1 \geq z)\, dz, \tag{823}$$

$$\lim_{t\to\infty} \mathbb{P}(A_t > x) = \frac{1}{\mathbb{E}[\tau_1]} \int_{x}^{\infty} \mathbb{P}(\tau_1 \geq z)\, dz, \tag{824}$$

$$\lim_{t\to\infty} \mathbb{P}(Z_s > y) = \frac{1}{\mathbb{E}[\tau_1]} \int_{y}^{\infty} \mathbb{P}(\tau_1 \geq z)\, dz. \tag{825}$$

PROOF. Omitted. □

7.2. Examples of Renewal Processes

Example 7.2.1. Let $(X_t)_{t\geq 0}$ be an irreducible Markov chain on a finite state space Ω. Let $X_0 = x \in \Omega$ and let T_k be the kth time that the chain returns to x. By the strong Markov property, the inter-arrival times $\tau_k = T_k - T_{k-1}$ are i.i.d. We also know that $\mathbb{E}[\tau_k] < \infty$ since hitting times of finite-state irreducible chains have finite mean (see Exercise 4.6 in Lecture note 1). Hence $(T_k)_{k\geq 0}$ is a renewal process. ▲

Example 7.2.2. Let $(\xi)_{k\geq 0}$ be a sequence of i.i.d. RVs with

$$\mathbb{P}(\xi_k = 1) = \mathbb{P}(\xi_k = -1) = 1/2. \tag{826}$$

Define a random walk $(S_n)_{n\geq 0}$ by $S_0 = 0$ and $S_n = \xi_1 + \cdots + \xi_n$. Let T_k denote the kth time that the walk visits the origin. Note that one can view S_n as a Markov chain on state space $\Omega = \mathbb{Z}$. Hence again by the strong Markov property, the inter-arrival times $\tau_k = T_k - T_{k-1}$ are i.i.d. However, later we will see that $\mathbb{E}[\tau_k] = \infty$. Hence $(T_k)_{k\geq 0}$ is not a renewal process as we defined above. However, we can still call it a 'renewal process with infinite expected inter-arrival times'. This will make more sense since, albeit having infinite mean, the inter-arrival times are finite almost surely. In other words, S_n always returns to the origin with probability 1.

How do we know that the walk S_n will eventually return back to the origin? That is, do we know that the inter-arrival times are finite almost surely? It is easy to see that S_n is an irreducible Markov chain. However, since the state space Ω for S_n is infinite, this does not guarantee that S_n returns to zero with probability 1.

In order to see this, recall the Gambler's ruin problem (Exercise 7.1 in Lecture note 1). Namely, let $(X_t)_{t\geq 0}$ be a Markov chain on state space $\Omega_N = \{0, 1, \cdots, N\}$ with transition probabilities

$$\mathbb{P}(X_{t+1} = k+1 \,|\, X_t = k) = p, \qquad \mathbb{P}(X_{t+1} = k-1 \,|\, X_t = k) = 1-p \quad \forall 0 \leq k < N. \tag{827}$$

Let $\rho = (1-p)/p$. We have seen that, for any $0 < i < N$,

$$\mathbb{P}(X_t \text{ hits } N \text{ before } 0 \,|\, X_0 = i) = \frac{1 + \rho + \cdots + \rho^{i-1}}{1 + \rho + \cdots + \rho^{N-1}}. \tag{828}$$

For our case, let $p = 1/2$ so that $\rho = 1$. We can view X_t as the random walk S_n during it lies in the interval $[0, N]$. Hence

$$\mathbb{P}(S_n \text{ hits } N \text{ before } 0 \,|\, S_0 = i) = i/N. \tag{829}$$

Now by taking the limit $N \to \infty$, which makes the right barrier at N to fade away, we deduce

$$\mathbb{P}(S_n \text{ never hits } 0 \,|\, S_0 = i) = \lim_{N\to\infty} \mathbb{P}(S_n \text{ hits } N \text{ before } 0 \,|\, S_0 = i) = 0. \tag{830}$$

This shows S_0 eventually returns to 0 with probability 1 starting from any initial state $S_0 = i > 0$. Since S_n is symmetric, the same conclusion holds for negative starting location. Hence the inter-arrival times are finite with probability 1. ▲

Example 7.2.3. Let $(X_t)_{t\geq 0}$ be an irreducible Markov chain on a finite state space Ω. Let $X_0 = x \in \Omega$ and let T_k be the kth time that the chain returns to x. By the strong Markov property, the inter-arrival times $\tau_k = T_k - T_{k-1}$ are i.i.d. We also know that $\mathbb{E}[\tau_k] < \infty$ since hitting times of finite-state irreducible chains have finite mean (see Exercise 4.6 in Lecture note 1). Hence $(T_k)_{k\geq 0}$ is a renewal process. ▲

Example 7.2.4. Let $(\xi)_{k\geq 0}$ be a sequence of i.i.d. RVs with

$$\mathbb{P}(\xi_k = 1) = \mathbb{P}(\xi_k = -1) = 1/2. \tag{831}$$

Define a random walk $(S_n)_{n\geq 0}$ by $S_0 = 0$ and $S_n = \xi_1 + \cdots + \xi_n$. Let T_k denote the kth time that the walk visits the origin. Note that one can view S_n as a Markov chain on state space $\Omega = \mathbb{Z}$. Hence again by the strong Markov property, the inter-arrival times $\tau_k = T_k - T_{k-1}$ are i.i.d. However, later we will see that $\mathbb{E}[\tau_k] = \infty$. Hence $(T_k)_{k\geq 0}$ is not a renewal process as we defined above. However, we can still call it a 'renewal process with infinite expected inter-arrival times'. This will make more sense since, albeit having infinite mean, the inter-arrival times are finite almost surely. In other words, S_n always returns to the origin with probability 1.

How do we know that the walk S_n will eventually return back to the origin? That is, do we know that the inter-arrival times are finite almost surely? It is easy to see that S_n is an irreducible Markov chain. However, since the state space Ω for S_n is infinite, this does not guarantee that S_n returns to zero with probability 1.

In order to see this, recall the Gambler's ruin problem (Exercise 7.1 in Lecture note 1). Namely, let $(X_t)_{t\geq 0}$ be a Markov chain on state space $\Omega_N = \{0, 1, \cdots, N\}$ with transition probabilities

$$\mathbb{P}(X_{t+1} = k+1 \,|\, X_t = k) = p, \qquad \mathbb{P}(X_{t+1} = k-1 \,|\, X_t = k) = 1-p \quad \forall 0 \leq k < N. \tag{832}$$

Let $\rho = (1-p)/p$. We have seen that, for any $0 < i < N$,

$$\mathbb{P}(X_t \text{ hits } N \text{ before } 0 \,|\, X_0 = i) = \frac{1 + \rho + \cdots + \rho^{i-1}}{1 + \rho + \cdots + \rho^{N-1}}. \tag{833}$$

For our case, let $p = 1/2$ so that $\rho = 1$. We can view X_t as the random walk S_n during it lies in the interval $[0, N]$. Hence

$$\mathbb{P}(S_n \text{ hits } N \text{ before } 0 \,|\, S_0 = i) = i/N. \tag{834}$$

Now by taking the limit $N \to \infty$, which makes the right barrier at N to fade away, we deduce

$$\mathbb{P}(S_n \text{ never hits } 0 \,|\, S_0 = i) = \lim_{N\to\infty} \mathbb{P}(S_n \text{ hits } N \text{ before } 0 \,|\, S_0 = i) = 0. \tag{835}$$

This shows S_0 eventually returns to 0 with probability 1 starting from any initial state $S_0 = i > 0$. Since S_n is symmetric, the same conclusion holds for negative starting location. Hence the inter-arrival times are finite with probability 1. ▲

Exercise 7.2.5 (Markov chain with continuous-time jumps)**.** Let $(X_k)_{k\geq 0}$ be an irreducible and aperiodic Markov chain on state space $\Omega = \{1, 2, \cdots, m\}$ with transition matrix $P = (p_{ij})$. Let π be the unique stationary distribution of the chain.

Suppose the chain spends an independent amount of time at each state $x \in \Omega$, whose distribution F_x may depend only on x. For each real $t \geq 0$, let $Y(t) \in \Omega$ denote the state of the chain at time t.

(i) Fix $x \in \Omega$, and let $T_k^{(x)}$ denote the kth time that the Markov process $(Y(t))_{t\geq 0}$ returns to x. Let $(\tau_k^{(x)})_{k\geq 1}$ and $(N^{(x)}(t))_{t\geq 0}$ be the associated inter-arrival times and the counting process, respectively. Then

$$N^{(x)}(t) = \text{number of visits to } x \text{ that } (Y(t))_{t\geq 0} \text{ makes up to time } t. \tag{836}$$

Show that $(T_k^{(x)})_{k\geq 2}$ is a renewal process. Then using renewal SLLN, show that

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{N^{(x)}(t)}{t} = \frac{1}{\mathbb{E}[\tau_1^{(x)}]} \right) = 1. \tag{837}$$

(ii) Let T_k denote the kth time that the Markov process $(Y(t))_{t\geq 0}$ jumps. Let $(\tau_k)_{k\geq 1}$ and $(N(t))_{t\geq 0}$ be the associated inter-arrival times and the counting process, respectively. Note that

$$N(t) = N^{(1)}(t) + N^{(2)}(t) + \cdots + N^{(m)}(t). \tag{838}$$

Use **(i)** to derive that

$$\mathbb{P}\left(\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{\mathbb{E}[\tilde{\tau}_1^{(1)}]} + \cdots + \frac{1}{\mathbb{E}[\tilde{\tau}_1^{(m)}]}\right) = 1. \tag{839}$$

(iii) Using the Markov chain SLLN, show that

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}(X_k = x) = \pi(x)\right) = 1. \tag{840}$$

From this, deduce

$$\mathbb{P}\left(\lim_{t\to\infty} \frac{N^{(x)}(t)}{N(t)} = \pi(x)\right) = 1. \tag{841}$$

(iv) From the previous parts, conclude that

$$\pi(x) = \frac{\frac{1}{\mathbb{E}[\tau_1^{(x)}]}}{\frac{1}{\mathbb{E}[\tau_1^{(1)}]} + \cdots + \frac{1}{\mathbb{E}[\tau_1^{(m)}]}}. \tag{842}$$

In particular, suppose the chain spends unit time at each state $x \in \Omega$. Then each $\tau_1^{(x)}$ equals the discrete return time of the Markov chain X_k to x, so we know $\pi(x) = 1/\mathbb{E}[\tau_1^{(x)}]$. Check that the above formula is consistent in this case.

Exercise 7.2.6 (Alternating renewal process)**.** Let $(\tau_k)_{t\geq 1}$ be a sequence of independent RVs where

$$\mathbb{E}[\tau_{2k-1}] = \mu_1, \qquad \mathbb{E}[\tau_{2k}] = \mu_2 \qquad \forall k \geq 1. \tag{843}$$

Define and arrival process $(T_k)_{k\geq 1}$ by $T_k = \tau_1 + \cdots + \tau_k$ for all $k \geq 1$.

(i) Is $(T_k)_{k\geq 1}$ a renewal process?

(ii) Let $(X_k)_{k\geq 0}$ be a Markov chain on state space $\Omega = \{1,2\}$ with transition matrix

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{844}$$

Show that the chain has $\pi = [1/2, 1/2]$ as the unique stationary distribution.

(iii) Suppose the chain spends time τ_k at state $X_k \in \Omega$ in between the $k-1$st and kth jump. For each real $t \geq 0$, let $Y(t) \in \Omega$ denote the state of the chain at time t. Let $N(t)$ be the number of jumps that $Y(t)$ makes up to time t. Define

$$R^{(1)}(t) = \sum_{k=1}^{N^{(1)}(t)} \tau_k \mathbf{1}(X_k = 1), \tag{845}$$

which is the total amount of time that $(Y(t))_{t\geq 0}$ spends at state 1. Use Exercise 7.2.5 to deduce

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{R^{(1)}(t)}{t} = \frac{\mu_1}{\mu_1 + \mu_2}\right) = 1. \tag{846}$$

Exercise 7.2.7 (Poisson janitor)**.** (Excerpted from [Dur99]) A light bulb has a random lifespan with distribution F and mean μ_F. A janitor comes at times according to PP(λ) and checks and replace the bulb if it is burnt out. Suppose all bulbs have independent lifespans with the same distribution F.

(i) Let T_k be the kth time that the janitor arrives and replaces the bulb. Show that $(T_k)_{k\geq 0}$ with $T_0 = 0$ is a renewal process.

(ii) Let $(\tau_k)_{k\geq 1}$ be the inter-arrival times of the renewal process defined in (i). Using the memoryless property of Poisson processes to show that

$$\mathbb{E}[\tau_k] = \mu_F + 1/\lambda \qquad \forall k \geq 1. \tag{847}$$

(iii) Let $N(t)$ be the number of bulbs replaced up to time t. Show that

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{N(t)}{t} = \frac{1}{\mu_F + 1/\lambda}\right) = 1. \tag{848}$$

(iv) Let $B(t)$ be the total duration that bulb is working up to time t, that is,

$$B(t) = \int_0^t \mathbf{1}(\text{Bulb is on at time } s)\, ds. \tag{849}$$

Use renewal reward process to show that

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{B(t)}{t} = \frac{\mu_F}{\mu_F + 1/\lambda}\right) = 1. \tag{850}$$

(v) Let $V(t)$ denote the total number of visits that the janitor has made by time t. Show that

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{N(t)}{V(t)} = \frac{1/\lambda}{\mu_F + 1/\lambda}\right) = 1. \tag{851}$$

That is, the fraction of times that the janitor replaces the bulb converges to $\frac{1/\lambda}{(\mu_F+1/\lambda)}$ almost surely, which is also the fraction of times that the bulb is off by (iv).

7.3. Little's Law

In this section, we will learn one of the cornerstones of queuing theory, which is called Little's Law. Roughly speaking, this reads

$$\text{(average size of the system)} = \text{(arrival rate)} \times \text{(average time spent in the system)}, \tag{852}$$

or $\ell = \lambda w$ for short. The power of Little's law lies in its applicability in a very general situation; one can even choose a portion of a gigantic queuing network and apply Little's law to analyze the local behavior of the system. We also remark that Little's law applies for deterministic queuing system: For stochastic ones satisfying certain conditions, it will hold with probability 1.

Consider a queuing system, where the kth customer arrives at time t_k, spends w_k units of time, and the exits the system. We assume no two customers arrive at the same time, that is, $t_1 < t_2 < \cdots$. Let $N(t)$ and $N^d(t)$ denote the number of arrivals and departures up to time t, respectively. Finally, let $L(t)$ denote the number of customers in the system at time t. To summarize:

$$t_k = \text{Time that the } k\text{th customer } \textit{enters} \text{ the system.} \tag{853}$$

$$w_k = \text{Time that the } k\text{th customer spends in the system until he } \textit{exits}. \tag{854}$$

$$N(t) = \sum_{k=1}^{\infty} \mathbf{1}(t_k \leq t) = \text{Number of arrivals up to time } t \tag{855}$$

$$N^d(t) = \sum_{k=1}^{\infty} \mathbf{1}(t_k + w_k \leq t) = \text{Number of departures up to time } t \tag{856}$$

$$L(t) = N(t) - N^d(t) = \text{Size of the system at time } t. \tag{857}$$

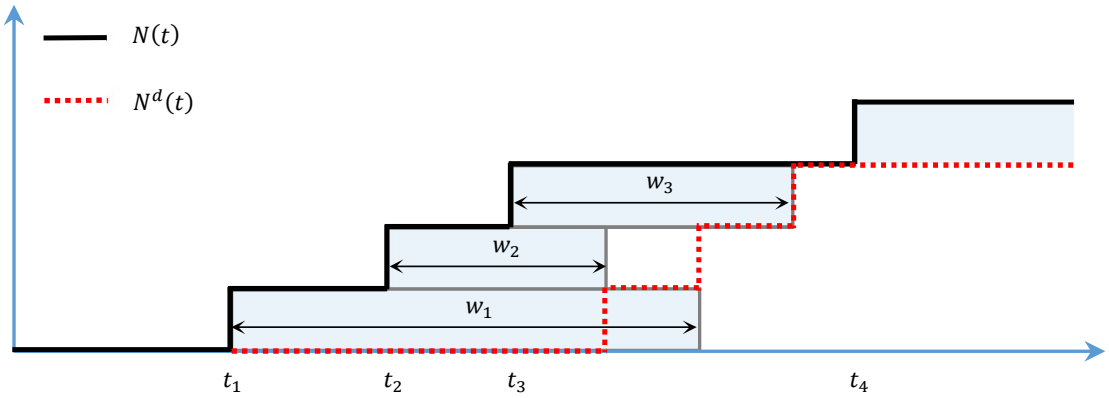


FIGURE 1. Arrival times, wait times, number of arrivals, and number of departures.

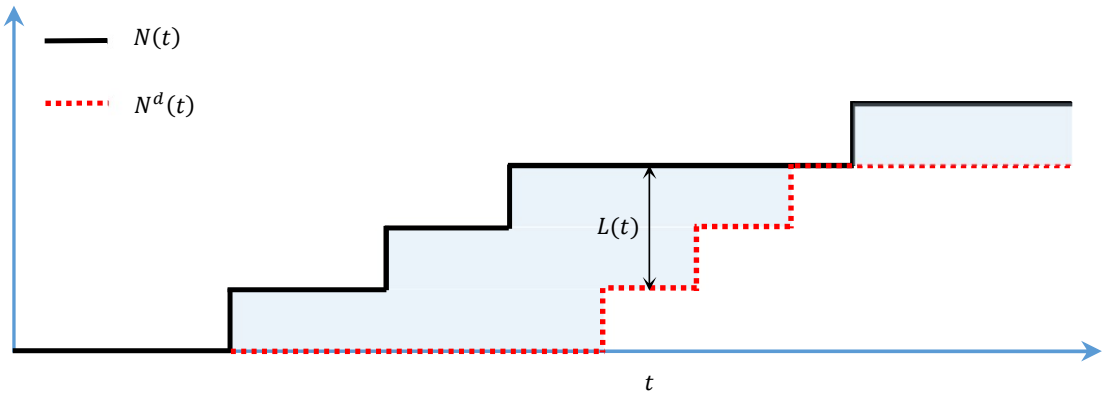


FIGURE 2. System size at time t

Now we introduce three key quantities which describe the average behavior of the system. Define the following quantities, whenever their limit exist:

$$\ell = \lim_{t\to\infty} \frac{1}{t}\int_0^t L(s)\,ds = \text{Average size of the queue} \tag{858}$$

$$\lambda = \lim_{t\to\infty} \frac{N(t)}{t} = \text{Arrival rate} \tag{859}$$

$$w = \lim_{t\to\infty} \frac{1}{N(t)} \sum_{k=1}^{N(t)} w_k = \text{Average wait time.} \tag{860}$$

Little's law gives a very simple relation between the above three average quantities.

Theorem 7.3.1 (Little's law)**.** *If both λ and w exist and are finite, then so does ℓ and*

$$\ell = \lambda w. \tag{861}$$

PROOF. Note that

$$\text{The } k\text{th customer is in the queue at time } t \iff t_k \le t < t_k + w_k. \tag{862}$$

Hence we may write

$$L(t) = \sum_{k=1}^{\infty} \mathbf{1}(t_k \le t < t_k + w_k). \tag{863}$$

Thus by Fubini's theorem,

$$\int_0^T L(t)\,dt = \int_0^T \sum_{k=1}^{\infty} \mathbf{1}(t_k \le t < t_k + w_k)\,dt \tag{864}$$

$$= \sum_{k=1}^{\infty} \int_0^T \mathbf{1}(t_k \le t < t_k + w_k)\,dt \tag{865}$$

$$= \sum_{k=1}^{N(T)} \min(T, t_k + w_k) - t_k. \tag{866}$$

This yields

$$\sum_{k=1}^{N^d(T)} w_k \le \int_0^T L(t)\,dt \le \sum_{k=1}^{N(T)} w_k. \tag{867}$$

Now observe that

$$\lim_{T\to\infty} \frac{1}{T} \sum_{k=1}^{N(T)} w_k = \lim_{T\to\infty} \frac{N(T)}{T} \left(\frac{1}{N(T)} \sum_{k=1}^{N(T)} w_k \right) = \lambda w. \tag{868}$$

On the other hand, using Exercise 7.3.2 we also have

$$\lim_{T\to\infty} \frac{1}{T} \sum_{k=1}^{N^d(T)} w_k = \lim_{T\to\infty} \frac{N^d(T)}{T} \left(\frac{1}{N^d(T)} \sum_{k=1}^{N^d(T)} w_k \right) = \lambda w. \tag{869}$$

Hence the assertion follows from (867). □

Exercise 7.3.2. Let $(t_k)_{k\ge 0}$ be a sequence of arrival times and let w_k be the time that the kth customer spends in the system. Also, let $N(t)$ and $N^d(t)$ denote the number of arrivals and departures up to time t. Suppose the following limit exist and finite:

$$\lambda := \lim_{t\to\infty} \frac{N(t)}{t}, \qquad w := \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} w_k. \tag{870}$$

We will show that

$$\lim_{T\to\infty} \frac{N^d(t)}{t} = \lambda. \tag{871}$$

(i) Write

$$\frac{w_n}{n} = \left(\frac{1}{n} \sum_{k=1}^{n} w_k \right) - \frac{n-1}{n} \left(\frac{1}{n-1} \sum_{k=1}^{n-1} w_k \right). \tag{872}$$

Deduce that $\lim_{n\to\infty} w_n/n = 0$. Hence for each $\delta > 0$, there exists $M(\delta) > 0$ such that

$$w_n \le \delta n \qquad \forall n \ge M(\delta). \tag{873}$$

(ii) Show that for each $\varepsilon > 0$, there exists $T_1(\varepsilon) > 0$ such that for all $t > T_1(\varepsilon)$,

$$(\lambda - \varepsilon)\varepsilon t \le N(\varepsilon t) \le (\lambda + \varepsilon)\varepsilon t, \tag{874}$$

$$(\lambda - \varepsilon)(1-\varepsilon)t \le N((1-\varepsilon)t) \le (\lambda + \varepsilon)(1-\varepsilon)t. \tag{875}$$

Also deduce that for all $t > T_1(\varepsilon)$,

$$\varepsilon t \le t_n < (1-\varepsilon)t \quad \Longrightarrow \quad (\lambda - \varepsilon)\varepsilon t \le n \le (\lambda + \varepsilon)(1-\varepsilon)t. \tag{876}$$

(iii) Using (i) and (ii), show that

$$t > \max\left(T_1(\varepsilon), \frac{1}{(\lambda-\varepsilon)\varepsilon} M\left(\frac{\varepsilon}{2(\lambda+\varepsilon)(1-\varepsilon)} \right) \right) \quad \text{and} \quad \varepsilon t \le t_n < (1-\varepsilon)t \quad \Longrightarrow \quad w_n \le (\varepsilon/2)t. \tag{877}$$

Deduce that for each t large enough, there are at least $(\lambda - \varepsilon - 2\varepsilon\lambda)t$ arrivals during $[\varepsilon t, (1-\varepsilon)t]$ and they all depart by time t.

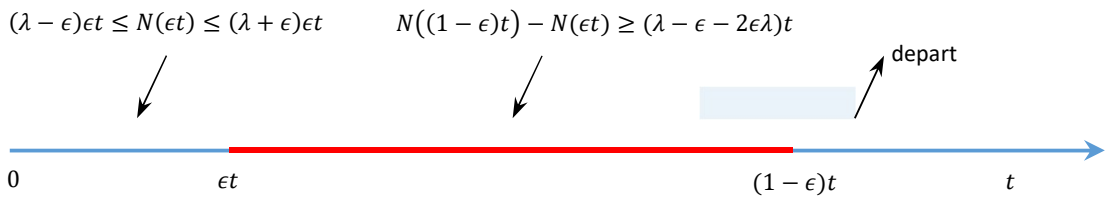


FIGURE 3. All customers arriving during $[\varepsilon t, (1-\varepsilon)t]$ departs by t

(iv) From (iii), show that for each $\varepsilon > 0$.

$$\liminf_{t\to\infty} \frac{N^d(t)}{t} \ge \lambda - \varepsilon - 2\varepsilon\lambda. \tag{878}$$

Finally, conclude (871).

Example 7.3.3 (Housing Market)**.** Suppose the local real estate in Westwood estimates that it takes 120 days on average to sell a house; This number does fluctuate with the economy and season, but it has been fairly stable over the past decade. We found out that at any given day last year, the number of houses for sale has ranged from 20 to 30, with average of 25. What can we say about the average number of transaction last year?

In order to apply Little's law, we view the housing market as a queuing system. Namely, we regard houses being put up for sale as an arrival to the system. The queue consists of unsold houses, and when houses are being sold, we regard them as exiting the queue. Now from the description above, we set the average wait time $w = 120$, and average queue length $\ell = 25$. Then by Little's law, we infer that the arrival rate is $\lambda = \ell / w = 25/120$ houses per day, or 75 houses per year. ▲

Exercise 7.3.4 (SLLN for weighted sum)**.** Let $(X_k)_{k\ge1}$ be a sequence of i.i.d. RVs with finite variance. Let $(w_k)_{k\ge1}$ be a sequence of real numbers such that

$$\bar{w} = \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} w_k < \infty, \qquad \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} w_k^2 < \infty. \tag{879}$$

Define $S_n = \sum_{k=1}^n X_k w_k$. In this exercise, we will show that, almost surely,

$$\lim_{n\to\infty} S_n/n = \mathbb{E}[X_1]\,\bar{w}. \tag{880}$$

(i) Write

$$\frac{S_n}{n} = \frac{1}{n}\sum_{k=1}^{n} \mathbb{E}[X_k] w_k + \frac{1}{n}\sum_{k=1}^{n} (X_k - \mathbb{E}X_k) w_k. \tag{881}$$

Show that it suffices to show the assertion assuming $\mathbb{E}[X_k] = 0$ for all $k \ge 1$. We may assume this for the following steps.

(ii) Use Chebyshef's inequality to show that

$$\mathbb{P}(S_n \ge t) \le t^{-2}\mathbb{E}[X_1^2]\left(\sum_{k=1}^{n} w_k^2\right). \tag{882}$$

(iii) Use (ii) to conclude that

$$\mathbb{P}\left(\frac{S_n}{n} \ge \frac{1}{\log n}\right) \le \frac{(\log n)^2}{n}\mathbb{E}[X_1^2]\left(\frac{1}{n}\sum_{k=1}^{n} w_k^2\right). \tag{883}$$

Deduce that

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\frac{S_{n^2}}{n^2} \ge \frac{1}{2\log n}\right) < \infty. \tag{884}$$

By Borel-Cantelli Lemma, this yields

$$\mathbb{P}\left(\lim_{n\to\infty}\frac{S_n^2}{n^2}=0\right)=1. \tag{885}$$

(iv) Use (iii) to show that, for any sequence (n_k) of integers such that $n_k\to\infty$ as $k\to\infty$, we can choose a further subsequence $(n_{k(r)})$ such that

$$\mathbb{P}\left(\lim_{n\to\infty}\frac{S_{n_k(r)}^2}{n_{k(r)}^2}=0\right)=1. \tag{886}$$

This implies that, almost surely as $n\to\infty$,

$$0=\liminf_{n\to\infty}\frac{S_n}{n}\le\limsup_{n\to\infty}\frac{S_n}{n}=0. \tag{887}$$

This implies $\lim_{n\to\infty}S_n/n=0$ almost surely. This shows (880).

Exercise 7.3.5 (Expected load in the server)**.** Consider a single-server queuing system, which is determined by the arrival times $(t_k)_{k\ge1}$ and total times spent in the system $(w_k)_{k\ge0}$ (a.k.a. sojourn times). Let $N(t)$ and $N^d(t)$ denote the number of arrivals and departures up to time t, respectively.

Each customer may wait for $\tilde{w}_k$ time in the queue, and then spends $\tilde{s}_k$ time the the server to get serviced, so that

$$w_k=\tilde{w}_k+\tilde{s}_k. \tag{888}$$

We may assume the following limits exist:

$$\lambda:=\lim_{t\to\infty}\frac{N(t)}{t},\qquad w:=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}w_k,\qquad \tilde{s}^2:=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\tilde{s}_k^2<\infty. \tag{889}$$

(i) Show that the kth customer is in the server if and only if

$$t_k+\tilde{w}_k\le t<t_k+\tilde{w}_k+\tilde{s}_k. \tag{890}$$

(ii) Let $R(t)$ denote the remaining service time of the current customer in the server. Use (i) to show that

$$R(t)=\sum_{k=1}^{\infty}(t_k+\tilde{w}_k+\tilde{s}_k-t)\mathbf{1}(t_k+\tilde{w}_k\le t<t_k+\tilde{w}_k+\tilde{s}_k). \tag{891}$$

(iii) Use Fubini's theorem to justify the following steps:

$$\int_0^T R(t)\,dt=\sum_{k=1}^{N(T)}\int_0^T(t_k+\tilde{w}_k+\tilde{s}_k-t)\mathbf{1}(t_k+\tilde{w}_k\le t<t_k+\tilde{w}_k+\tilde{s}_k)\,dt \tag{892}$$

$$=\sum_{k=1}^{N(T)}\int_0^{\min(\tilde{s}_k,T-t_k-\tilde{w}_k)}(\tilde{s}_k-t)\,dt. \tag{893}$$

(iv) From (iii), deduce

$$\sum_{k=1}^{N^d(T)}\frac{\tilde{s}_k^2}{2}\le\int_0^T R(t)\,dt\le\sum_{k=1}^{N(T)}\frac{\tilde{s}_k^2}{2}. \tag{894}$$

Finally, derive the following formula for the average load in the server:

$$r:=\lim_{T\to\infty}\frac{1}{T}\int_0^T R(t)\,dt=\lambda\tilde{s}^2/2. \tag{895}$$

Exercise 7.3.6 (Pollaczek–Khinchine formula)**.** Consider a $G/G/1$ queue, where arrivals are given by a renewal process of rate λ and service times are i.i.d. copies of a RV S with finite mean and variance. We use the following notations:

$$T_k = k\text{th arrival time} \tag{896}$$

$$\tilde{W}_k = \text{Time that the } k\text{th customer spends in the queue} \tag{897}$$

$$\tilde{S}_k = \text{Time that the } k\text{th customer spends in the server} \tag{898}$$

$$\tilde{W}(t) = \text{Remaining time until exit of the last customer in the queue at time } t. \tag{899}$$

Note that $\lim_{h\searrow 0} \tilde{W}(T_k - h)$ equals the waiting time $\tilde{W}_k$ of the kth customer in the queue. We will assume that the following limit exists almost surely:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \tilde{W}_k < \infty. \tag{900}$$

The goal of this exercise is to show the following Pollaczek–Khinchine formula for the mean waiting time: Almost surely,

$$\tilde{w} := \lim_{t\to\infty} \frac{1}{t} \int_0^t \tilde{W}(s)\, ds = \frac{\lambda \mathbb{E}[S^2]}{2(1-\lambda \mathbb{E}[S])}. \tag{901}$$

(i) Let $S(t)$ denote the sum of service times of all customers in the queue at time t. Show that

$$S(t) = \sum_{k=1}^{\infty} \tilde{S}_k \mathbf{1}(t_k \le t < t_k + \tilde{W}_k). \tag{902}$$

(ii) Let $N(t)$ and $N^d(t)$ denote the number of arrivals and departures up to time t. Use Fubini's theorem to show that

$$\int_0^T S(t)\, dt = \sum_{k=1}^{\infty} \int_0^T \tilde{S}_k \mathbf{1}(t_k \le t < t_k + \tilde{W}_k)\, dt \tag{903}$$

$$= \sum_{k=1}^{N(T)} \tilde{S}_k \left[\min(T, t_k + \tilde{W}_k) - t_k\right]. \tag{904}$$

Also, deduce that

$$\sum_{k=1}^{N^d(T)} \tilde{S}_k \tilde{W}_k \le \int_0^T S(t)\, dt \le \sum_{k=1}^{N(T)} \tilde{S}_k \tilde{W}_k. \tag{905}$$

(iii) From (ii) and Exercise 7.3.4, show that

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T S(t)\, dt = \lambda \mathbb{E}[S]\, \tilde{w}. \tag{906}$$

(iv) Let $R(t)$ denote the remaining service time of the current customer in the server. Then $\tilde{W}(t) = R(t) + S(t)$. Using (iii) and Exercise (7.3.5), conclude the PK formula (901).

CHAPTER 8

Continuous-time Markov Chains

In this chapter, we introduce the continuous-time Markov chains (CTMCs) and study various properties and examples. The continuous-time extension of MCs has a wide range of applications in simulating chemical or biological systems and also serves as a building block for Brownian motion.

8.1. Definition of CTMC

A continuous-time process $(X_t)_{t\geq 0}$ on state space Ω is said to have *Markov property* if for every $t_0 > 0$ and $x \in \Omega$, the distribution of the future process $(X_t)_{t\geq t_0}$ given the present state $X_{t_0} = x$ depends only on x and not on the past process $(X_t)_{0\leq t<t_0}$. Moreover, as in the discrete-time case, we require the transition probabilities do not depend on the time that the transition takes place. These two properties are stated below:

(i) (Continuous-time Markov property) The distribution of X_t given previous states $X_{s_0}, X_{s_1}, \cdots, X_{s_n}$, $s_0 < s_1 < \cdots < s_n < t$, depends only on X_{s_n}. That is, for every $0 \leq s_0 < \cdots < s_n < t$ and $x_0, \ldots, x_n, y \in \Omega$,

$$\mathbb{P}\left(X_t = y \,\middle|\, X_{s_n} = x_n, \ldots, X_{s_0} = x_0\right) = \mathbb{P}\left(X_t = y \,\middle|\, X_{s_n} = x_n\right). \tag{907}$$

(ii) (Time-homogeneity) There exists a *transition probability function* $(t, x, y) \mapsto \mathbf{P}_t(x, y)$ for $t \geq 0$, $x, y \in \Omega$ such that for each $0 \leq s \leq t$ and $x, y \in \Omega$,

$$\mathbf{P}_{t-s}(x, y) = \mathbb{P}\left(X_t = y \,\middle|\, X_s = x\right). \tag{908}$$

A continuous-time process $(X_t)_{t\geq 0}$ on state space Ω is called a *continuous-time Markov chain* (CTMC) if it satisfies the continuous-time Markov property. A time-homogeneous CTMC $(X_t)_{t\geq 0}$ on state space Ω and transition probability function $\mathbf{P}_t$ is a CTMC that satisfies time-homogeneity with the probability transition function $\mathbf{P}_t$.

The following proposition gives basic properties of transition probability function.

Proposition 8.1.1. *The transition probability function $\mathbf{P}_t$ of a time-homogeneous CTMC $(X_t)_{t\geq 0}$ on state space Ω satisfies the following properties:*

(i) $\mathbf{P}_t(x, y) \geq 0$, $\sum_{y\in\Omega} \mathbf{P}_t(x, y) = 1$ *for all* $t \geq 0$ *and* $x, y \in \Omega$.

(ii) $\mathbf{P}_0(x, y) = \mathbf{1}(x = y)$ *for* $x, y \in \Omega$.

(iii) *(Kolmogorov-Champman eg.)* $\mathbf{P}_{t+s}(x, z) = \sum_{y\in\Omega} \mathbf{P}_t(y, z)\mathbf{P}_s(x, y)$ *for all* $s, t \geq 0$ *and* $x, y \in \Omega$.

PROOF. Note that $\mathbf{P}_t(x, y) = \mathbb{P}\left(X_t = y \,\middle|\, X_0 = x\right)$ is a conditional probability so **(i)** and **(ii)** follow from properties of conditional probability. To see **(iii)**,

$$\mathbf{P}_{t+s}(x, z) = \mathbb{P}\left(X_{t+s} = z \,\middle|\, X_0 = x\right) \tag{909}$$

$$= \sum_{y\in\Omega} \mathbb{P}\left(X_{t+s} = z,\, X_s = y \,\middle|\, X_0 = x\right) \tag{910}$$

$$= \sum_{y\in\Omega} \mathbb{P}\left(X_{t+s} = z \,\middle|\, X_s = y,\, X_0 = x\right) \mathbb{P}\left(X_s = y \,\middle|\, X_0 = x\right) \tag{911}$$

$$= \sum_{y\in\Omega} \mathbb{P}\left(X_{t+s} = z \,\middle|\, X_s = y\right) P_s(x,y) \tag{912}$$

$$= \sum_{y\in\Omega} \mathbb{P}\left(X_t = z \,\middle|\, X_0 = y\right) P_s(x,y) \tag{913}$$

$$= \sum_{y\in\Omega} \mathbf{P}_t(y,z)\, P_s(x,y), \tag{914}$$

where the last two equalities follow from the continuous Markov property and time-homogeneity. □

Remark 8.1.2. Proposition 8.1.1 **(iii)** is called a 'semi-group' property or Kolmogorov-Champman equation.

Example 8.1.3 (Homogeneous Poisson Process)**.** Let $(N(t))_{t\geq 0} \sim \mathrm{PP}(\lambda)$. We will show that it is a time-homogeneous CTMC on state space $\mathbb{Z}_{\geq 0}$. Note that

$$\mathbb{P}\left(N(t) = y \,\middle|\, N(s_n) = x_n, \ldots, N(s_0) = x_0\right) \tag{915}$$

$$= \mathbb{P}\left(N((s_n,t]) = y - x_n \,\middle|\, N([s_0,s_1]) = x_1, \ldots, N((s_{n-1},s_n]) = x_n - x_{n-1}\right) \tag{916}$$

$$= \mathbb{P}\left(N((s_n,t]) = y - x_n\right) \tag{917}$$

$$= \mathbb{P}\left(\mathrm{Poisson}(\lambda(t-s_n)) = y - x_n\right) \tag{918}$$

$$= \mathbf{1}(y - x_n \geq 0)\frac{(\lambda(t-s_n)^{y-x_n})e^{-\lambda(t-s_n)}}{(y-x_n)!}, \tag{919}$$

where for the second equality, we used independent increment property of the Poisson counting process N over disjoint intervals. An identical computation shows

$$\mathbb{P}\left(N(t) = y \,\middle|\, N(s_n) = x_n\right) = \mathbf{1}(y - x_n \geq 0)\frac{(\lambda(t-s_n)^{y-x_n})e^{-\lambda(t-s_n)}}{(y-x_n)!}. \tag{920}$$

Hence we verify the continuous Markov property as well as time-homogeneity, where the last expression gives the transition probability function:

$$\mathbf{P}_t(x,y) = \mathbf{1}(y \geq x)\frac{(\lambda t)^{y-x}e^{-\lambda t}}{(y-x)!} = \mathbb{P}\left(\mathrm{Poisson}(\lambda t) = y - x\right). \tag{921}$$

The indicator $\mathbf{1}(y \geq x)$ above indicates that the counting process $N(t)$ can only increase. ▲

Example 8.1.4 (Discrete-time Markov chain run by a Poisson process)**.** Let $(Y_n)_{n\geq 1}$ denote a discrete-time Markov chain on a countable state space Ω with transition matrix P. Suppose we have a Poisson arrival process $(N(t))_{t\geq 0} \sim \mathrm{PP}(\lambda)$ independent of the chain $(Y_n)_{n\geq 1}$. We define a continuous-time process $(X_t)_{t\geq 0}$ on the same state space Ω by

$$X_t = Y_{N(t)}. \tag{922}$$

We call $(X_t)_{t\geq 0}$ the Markov chain $(Y_n)_{n\geq 1}$ run by the Poisson process $N \sim \mathrm{PP}(\lambda)$. For instance, if there has been ten arrivals up to time t ($N(t) = 10$), then $X_t = Y_{10}$. In other words, X_t is a version of Y_n where we make the jumps given by Y_n at arrival times T_k of the PP. Alternatively, the continuous-time chain X_t waits an independent $\mathrm{Exp}(\lambda)$ time between each jump (inter-arrival time of the PP) and makes the jumps modulated by Y_n.

We will show that $(X_t)_{t\geq 0}$ is a time-homogeneous CTMC with transition probability function

$$\mathbf{P}_t(x,y) = \sum_{k=0}^{\infty} \frac{(\lambda t)^k e^{-\lambda t}}{k!} P^k(x,y). \tag{923}$$

Namely, the chain X_t jumps a random amount of times during a period of length t, which has distribution $\mathrm{Poisson}(\lambda t)$. Once this number is given, say k, then one has to make discrete-time jumps from x to y in k steps according to the transition matrix Q, which occurs with probability $Q^k(x,y)$. Since k

is random, the right hand side of (923) takes the form of the conditional expectation of $Q^k(x,y)$ where $k \sim \text{Poisson}(\lambda t)$[1].

Fix times $0 \le s_0 < s_1 < \cdots < s_{n-1} < s_n (= s) < t$ and states $x_0, \ldots, x_{n-1}, x_n (= x), y \in \Omega$. Note that the number of arrivals during the disjoint intervals $[s_0, s_1]$, …, $(s_{n-1}, s_n]$, $(s_n, t]$ are independent and follows Poisson distributions with rate λ times the length of the corresponding intervals. We will partition the probability of interest according to the values of these arrival counts. Namely, fix integers $k_0, \ldots, k_{n+1} \ge 0$, and denote $t_i := k_0 + \cdots + k_i$ for $0 \le i \le n$. Then note that

$$\begin{aligned}
&\mathbb{P}\left(Y_{N(t)} = y,\, N((s,t]) = k_{n+1} \,\middle|\, N(s_0) = k_0,\, Y_{N(s_0)} = x_0,\, N([s_0,s_1]) = k_1, \ldots N((s_{n-1},s_n]) = k_n,\, Y_{N(s)} = x\right) \\
&\quad = \mathbb{P}\left(Y_{t_n + k_{n+1}} = y,\, N((s,t]) = k_{n+1} \,\middle|\, N(t_0) = k_0,\, Y_{t_0} = x_0,\, N([s_0,s_1]) = k_1, \ldots N((s_{n-1},s_n]) = k_n,\, Y_{t_n} = x\right) \\
&\quad = \mathbb{P}\left(Y_{t_n + k_{n+1}} = y,\, N((s,t]) = k_{n+1} \,\middle|\, N(t_0) = k_0,\, N([s_0,s_1]) = k_1, \ldots N((s_{n-1},s_n]) = k_n,\, Y_{t_n} = x\right) \\
&\quad = \mathbb{P}\left(Y_{t_n + k_{n+1}} = y,\, N((s,t]) = k_{n+1} \,\middle|\, Y_{t_n} = x\right) \\
&\quad = \mathbb{P}\left(Y_{t_n + k_{n+1}} = y, \,\middle|\, Y_{t_n} = x\right) \mathbb{P}\left(N((s,t]) = k_{n+1}\right) \\
&\quad = P^{k_{n+1}}(x,y)\, \mathbb{P}\left(N((s,t]) = k_{n+1}\right),
\end{aligned}$$

where the second equality uses the Markov property of Y_n, the following equality uses independence of N from Y_n as well as the independent increment property of N over disjoint intervals. Since the above holds for all $k_0, \ldots, k_n \ge 0$, by averaging out the counting variables $N(s_0)$, $N((s_0,s_1])$, …, $N((s_{n-1},s_n])$, we get

$$\mathbb{P}\left(Y_{N(t)} = y,\, N((s,t]) = k_{n+1} \,\middle|\, Y_{N(s_0)} = x_0, \ldots Y_{N(s)} = x\right) = P^{k_{n+1}}(x,y)\, \mathbb{P}\left(N((s,t]) = k_{n+1}\right),$$

Thus, by summing over the above expressions over $k_{n+1} \ge 0$ and using the definition $X_t = Y_{N(t)}$, we obtain

$$\mathbb{P}\left(X_t = y \,\middle|\, X_{s_0} = x_0, \ldots X_{s_n} = x\right) = \sum_{k=0}^{\infty} P^k(x,y)\, \mathbb{P}\left(N((s_n,t]) = k\right). \tag{924}$$

Since the right hand side above does not depend on $x_0, \ldots, x_{n-1}$, averaging out $X_{s_0}, \ldots, X_{s_n}$ gives

$$\mathbb{P}\left(X_t = y \,\middle|\, X_{s_n} = x\right) = \sum_{k=0}^{\infty} P^k(x,y)\, \mathbb{P}\left(N((s_n,t]) = k\right). \tag{925}$$

This verifies both the continuous-time Markov property and the time-homogenity. Noting that $N((s_n,t)) \sim \text{Poisson}(\lambda(t-s_n))$, we have also obtained the desired transition probability function in (923). ▲

Example 8.1.5. Let $(Y_n)_{n\ge1}$ be a discrete-time Markov chain on state space $\mathbb{Z}_{\ge0}$ with deterministic jumps $\mathbb{P}(Y_{n+1} = Y_n + 1 \,|\, Y_n) = 1$ for $n \ge 1$. Suppose we have an independent Poisson arrival process $(N(t))_{t\ge0} \sim \text{PP}(\lambda)$, and let $X_t = Y_{N(t)}$. Then in Example 8.1.4, we have seen that $(X_t)_{t\ge0}$ is a time-homogeneous CTMC on $\mathbb{Z}_{\ge1}$. In fact, in this case,

$$X_t = Y_{N(t)} = N(t). \tag{926}$$

Hence the Poisson counting process $N(t)$ as a CTMC is a special case of a MC run by PP. ▲

Example 8.1.6 (Countinuous-time SSRW)**.** Let $(Y_n)_{n\ge1}$ be a SSRW on $\mathbb{Z}$ with $Y_0 = 0$. That is, $\mathbb{P}(Y_{n+1} - Y_n = 1) = \mathbb{P}(Y_{n+1} - Y_n = -1) = 1/2$ for $n \ge 1$. Suppose $(N(t))_{t\ge0} \sim \text{PP}(1)$ is an independent Poisson counting process and let $X_t = Y_{N(t)}$. In Example 8.1.4, we have seen that $(X_t)_{t\ge0}$ is a time-homogeneous CTMC on $\mathbb{Z}$. Probabilistically, X_t operates as follow: We wait for an independent Exp(1) amount of time and the flip a fair coin independently of everything else.

[1]The formula (923) is also called the 'Poissonization' (Poisson average) of the quantity $P^k(x,y)$.

Let Q denote the transition matrix of the SSRW Y_n on $\mathbb{Z}$. From (923), we have

$$\mathbf{P}_t(x,x) = \sum_{k=0}^{\infty} \frac{t^k e^{-t}}{k!} P^k(x,x) \tag{927}$$

$$= \sum_{n=0}^{\infty} \frac{t^{2n} e^{-t}}{(2n)!} P^{2n}(x,x) \tag{928}$$

$$= \sum_{n=0}^{\infty} \frac{t^{2n} e^{-t}}{(2n)!} \binom{2n}{n} 2^{-2n} \tag{929}$$

$$= e^{-t} \sum_{n=0}^{\infty} \frac{(t/2)^{2n}}{(n!)^2}. \tag{930}$$

Note that for small t, we have $\mathbf{P}_t(x,x) = 1 + O(t^2)$, so $\mathbf{P}_t(x,x) \approx 1$ for small t. (Similar observation holds for general CTMC. Check!) ▲

8.2. Construction of CTMC

8.2.1. Three constructions of CTMC. In this subsection, we give three constructions of CTMC that satisfy the continuous-time Markov property as well as time-homogenity. Each construction has its own advantage. Throughout this subsection, Ω denotes the state space of the CTMC that we are going to construct.

Definition 8.2.1 (Construction I: Poisson clock on edges)**.** Suppose we have the following objects:

(i) (*Initial distribution*) A probability distribution μ on Ω;
(ii) (*Jump rate*) A matrix $q : \Omega \times \Omega \to [0,\infty)$ with $q(x,x) \equiv 0$.

We then generate a continuous-time process $(X_t)_{t\geq 0}$ as follow:

1. Sample $X_0 \in \Omega$ according to μ.
2. At each edge $(x,y) \in \Omega^2$, place a 'Poisson clock', which rings according to $\mathrm{PP}(q(x,y))$.
3. If at any time $X_t = x$, then X_t stays there until the first time some clock on (x,y) rings and jumps to y.
4. Go to 3.

Remark: If the jump rate $q(x,y)$ of the edge (x,y) is large, then it is more likely to make the jump $x \to y$ at x. At each time at x, there is competition between exponential variables of distribution $\mathrm{Exp}(q(x,y))$ for $y \neq x$. The jump rate matrix q do not need to have row sums equal to 1 (as for Markov transition matrices). This construction is also known as the graphical construction.

Definition 8.2.2 (Construction II: Jump chain and holding times)**.** Suppose we have the following objects:

(i) (*Initial distribution*) A probability distribution μ on Ω;
(ii) (*Routing matrix*) A transition matrix R for a (discrete-time) Markov chain on Ω;
(iii) (*Exponential holding parameter*) A function $\lambda : \Omega \to [0,\infty)$.

We then generate a continuous-time process $(X_t)_{t\geq 0}$ as follow:

1. Sample $X_0 \in \Omega$ according to μ.
2. If at any time $X_t = x$, then sample $\tau \in \mathrm{Exp}(\lambda(x))$ and wait at x until time $t+\tau$, sample $y \sim R(x,\cdot)$, and jump to y.
3. Go to 2.

Note that if at any time $X_t = x$ and $\lambda(x) = 0$, then X_t stays at x forever thereafter, that is, x is an absorbing state. This is because $\tau = \infty$ almost surely if $\tau \sim \mathrm{Exp}(0)$.

Remark: If $x \in \Omega$ is absorbing, then we can set $\lambda(x) = 0$ and $R(x,x) = 1$. Otherwise, we should have $\lambda(x) > 0$ and X_t eventually jumps to some $y \neq x$. Since jumping back to x itself does not give a proper jump, we can set $R(x,x) = 0$ without loss of generality.

Example 8.2.3. Suppose $\Omega = \mathbb{Z}$, $X_0 = 0$, $\lambda(x) \equiv 1$, and $R(x,y) = \mathbf{1}(y = x+1)$. What is $(X_t)_{t\geq 0}$?

Example 8.2.4. Suppose $\Omega = \{0,1\}$, $X_0 = 0$, $\lambda(x) \equiv 1$, and $R(x,y) = \mathbf{1}(y \neq x)$. What is $(X_t)_{t\geq 0}$?

Exercise 8.2.5. Show that Definitions 8.2.2 and 8.2.1 are equivalent under the relation

$$q(x,y) = \lambda(x)\, R(x,y) \qquad \text{for } x \neq y \text{ in } \Omega, \tag{931}$$

assuming the routing matrix R satisfies $R(x,x) \equiv 0$. You may argue with the following steps.

(i) (From Poisson clocks) $R(x,y)$ equals the probability of the Poisson clock on (x,y) rings the first, which is $\frac{q(x,y)}{\sum_{y\neq x} q(x,y)}$ by competing exponentials (see Exercise 6.1.4). On the other hand, the holding time τ_x at x is the minimum of exponential RVs with parameters $q(x,y)$ for $y \neq x$, so by Exercise 6.1.3, $\tau_x \sim \text{Exp}(\lambda(x))$ where $\lambda(x) = \sum_{y\neq x} q(x,y)$. Combine these results.

(ii) (From holding times) Define the jump rates $q(x,y)$, $x \neq y$ according to (931). Since $R(x,x) \equiv 0$, one can show $\lambda(x) = \sum_{y\neq x} q(x,y)$ and $R(x,y) = \frac{q(x,y)}{\sum_{y\neq x} q(x,y)}$. Use the argument in **(i)** to conclude consistency.

Definition 8.2.6 (Construction III: MC run by PP)**.** Suppose we have the following objects:

(i) (*Initial distribution*) A probability distribution μ on Ω;

(ii) (*PP*) Poisson arrival process $(T_n)_{n\geq 1} \sim \text{PP}(\alpha)$

(iii) (*Jump rate*) A matrix $q : \Omega \times \Omega \to [0,\infty)$ with $q(x,x) \equiv 0$; Further assume $\sum_{y\neq x} q(x,y) \leq \alpha$ for all x

We then generate a continuous-time process $(X_t)_{t\geq 0}$ as follow:

1. Sample $X_0 \in \Omega$ according to μ.

2. If at any time $X_t = x$, then X_t stays there until time $T_{N(t)+1}$. At time $T_{N(t)+1}$, X_t jumps to y with probability $q(x,y)/\alpha$, and stays at x with probability $1 - \sum_{y\neq x} q(x,y)/\alpha$.

3. Go to 2.

Proposition 8.2.7. *Consider a CTMC $(X_t)_{t\geq 0}$ as in Definition 8.2.6. Define a transition matrix P as*

$$P(x,y) = \begin{cases} \frac{q(x,y)}{\alpha} & \text{if } y \neq x \\ 1 - \frac{1}{\alpha}\sum_{z\in\Omega\setminus\{x\}} q(x,z) & \text{if } y = x. \end{cases} \tag{932}$$

Then $(X_t)_{t\geq 0}$ is the Markov chain with transition matrix P run by $\text{PP}(\alpha)$ as in Example 8.1.4. Namely, if $(Y_n)_{n\geq 1}$ is a Markov chain with transition matrix P and if $(N(t))_{t\geq 0} \sim \text{PP}(\alpha)$, $X_t = Y_{N(t)}$. Furthermore, the probability transition function $\mathbf{P}_t(x,y)$ is given by

$$\mathbf{P}_t(x,y) = \sum_{k=0}^{\infty} \frac{(\alpha t)^k e^{-\alpha t}}{k!} P^k(x,y). \tag{933}$$

PROOF. The first part of the assertion is clear from the construction. The transition probability function formula in (933) has been shown in the computation given in Example 8.1.4 to derive (923). □

Exercise 8.2.8. Show that Definitions 8.2.6 and 8.2.1 are equivalent assuming that there exists $\alpha > 0$ such that $\sum_{y\neq x} q(x,y) \leq \alpha$ for all x. You may argue with the following steps.

(i) Under the hypothesis, one can define the process $(X_t)_{t\geq 0}$ in two ways. Suppose $X_t = x$ and $X_{t^-} \neq x$. It suffices to show that the two definitions agree in distribution until the next jump.

(ii) (Holding time at x) (Poisson clocks) By merging all $\text{PP}(q(x,y))$ for $y \neq x$, we see that X_t at x gets a clock ring according to $\text{PP}(\lambda(x))$ with $\lambda(x) := \sum_{y\neq x} q(x,y)$. In particular, X_t waits at x for $\text{Exp}(\lambda(x))$ amount of time. (Poisson arrivals) On the other hand, by thinning of $\text{PP}(\alpha)$ with probability $\lambda(x)/\alpha$, the holding time at x follows $\text{Exp}(\lambda(x))$.

(iii) (Jump distribution): (From Poisson clocks) $R(x,y)$ equals the probability of the Poisson clock on (x,y) rings the first, which is $\frac{q(x,y)}{\sum_{y\neq x} q(x,y)}$ by competing exponentials (see Exercise 6.1.4). (From Poisson arrivals) Once we know that we are jumping to some $y \neq x$, its probability is $\frac{q(x,y)}{\sum_{y\neq x} q(x,y)}$.

Now, we will show that the equivalent continuous-time processes $(X_t)_{t\geq 0}$ in Definitions 8.2.2-8.2.6 are time-homogeneous CTMCs if we can ensure that there is no explosion of jumps, that is, no infinite number of jumps during a finite time interval.

Theorem 8.2.9. *The continuous-time process $(X_t)_{t\geq 0}$ in Definition 8.2.2 is a time-homogeneous CTMC if there is no explosion of jumps. A sufficient condition is the following:*

$$\text{There exists } \alpha > 0 \text{ such that } \sum_{y\neq x} q(x,y) \leq \alpha \text{ for all } x \in \Omega. \tag{934}$$

PROOF. Omitted. One can immitate the proof of renewal SLLN (Theorem 2.1.6). □

Example 8.2.10 (M/M/1 queue)**.** Suppose customers arrive at a single server according to a Poisson process with rate $\lambda > 0$. Customers gets serviced one by one according to the first-in-first-out ordering, and suppose there is no cap in the queue. Finally, assume that each customer in the top of the queue takes an independent $\text{Exp}(\mu)$ time to get serviced and exit the queue.

This is called the *M/M/1 queue* in queuing theory. Here the name of this model follows Kendall's notation, where the first 'M' stands for memorylessness of the arrival process, the second 'M' stands for memorylessness of the service times, and '1' means there is a single server. One can think of $M/M/c$ queue for c servers, in general.

The $M/M/1$ queue can be described nicely as a CTMC with Poisson clocks. The state space is $\Omega = \mathbb{Z}_{\geq 0}$. Since customers arrive according to $\text{PP}(\lambda)$ and existing customers depart according to $\text{PP}(\mu)$, the rightward transitions have jump rate λ and leftward transitions have jump rate μ. At 0, which represents that there is no custer currently in the queue, the only possible jump is to 1 and has jump rate λ. Hence we have the following jump rate diagram determining the $M/M/1$ queue:

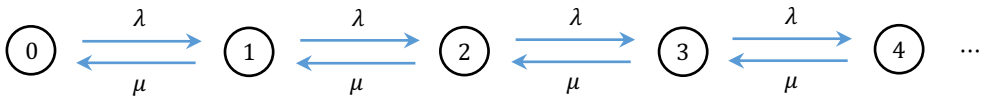


FIGURE 1. Jump rate diagram of the M/M/1 queue

As in the construction of CTMC using jump chains and holding times (Definition 8.2.1), we can also analyze the jump chain of the $M/M/1$ queue. According to Exercise 8.2.5, we can compute the exponential holding parameter λ and the routing matrix R as

$$\lambda(0) = \lambda,\ \lambda(x) = \lambda + \mu \quad \text{for } x \geq 1 \tag{935}$$

$$R(0,1) = 1, \qquad R(x, x+1) = \frac{\lambda}{\lambda+\mu} \quad \text{for } x \geq 1, \qquad R(x, x-1) = \frac{\mu}{\lambda+\mu} \quad \text{for } x \geq 1. \tag{936}$$

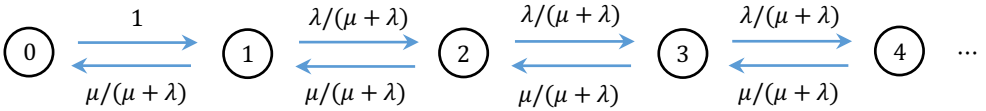


FIGURE 2. State space diagram of the jump chain of the M/M/1 queue

For instance, the above figure gives the space-state diagram for the jump chain with transition matrix R. ▲

Example 8.2.11 (Analyzing the jump chain of $M/M/1$ queue)**.** Consider the $M/M/1$ queue in Example 8.2.10. One of the main quantity of interest is the number of customers waiting in the queue at time t, which we denote by $Y(t)$. Then $(Y(t))_{t\geq 0}$ is a continuous-time stochastic process, which changes by 1 whenever a new customer arrives or the top customer in the queue leaves the system. In fact, this system can be modeled as a Markov chain, if we only think of the times when the queue state changes. Namely,

let $T_1 < T_2 < \cdots$ denote the times when the queue length changes. Let $X_k := Y(T_k)$. Then $(X_k)_{k\geq 0}$ forms a Markov chain. In fact, it is the Birth-Deach chain we have seen before.

(i) Let $(T_k^a)_{k\geq 0} \sim \text{PP}(\lambda)$ and $(T_k^d)_{k\geq 0} \sim \text{PP}(\mu)$. These are sequences of arrival and departure times, respectively. Let $\tilde{T}_i$ be the ith smallest time among all such arrival and departure times. In the next section, we will learn $(\tilde{T}_i)_{i\geq 0} \sim \text{PP}(\lambda+\mu)$. This is called 'merging' of two independent Poisson processes (see Exercise 6.4.3). Note that $\tilde{T}_i$ is the ith time that 'something happens' to the queue.

(ii) Define a Markov chain $(X_k)_{k\geq 0}$ on state space $\Omega = \{0,1,2,\cdots\}$ by

$$X_k = Y(\tilde{T}_k). \tag{937}$$

Namely, X_k is the number of customers in the queue at kth time that something happens to the queue.

(iii) What is the probability that $X_2 = 2$ given $X_1 = 1$? As soon as a new customer arrives at time T_1, she gets serviced and it takes an independent $\sigma_1 \sim \text{Exp}(\mu)$ time. Let τ_1 be the inter-arrival time between the first and second customer. Then by Example 6.1.4 (competing exponentials),

$$\mathbb{P}(X_2 = 2\, X_1 = 1) = \mathbb{P}(\text{New customer arrives before the previous customer exits}) \tag{938}$$

$$= \mathbb{P}(\tau_1 < \sigma_1) = \frac{\lambda}{\mu+\lambda}. \tag{939}$$

Similarly,

$$\mathbb{P}(X_2 = 0\, X_1 = 1) = \mathbb{P}(\text{New customer arrives after the previous customer exits}) \tag{940}$$

$$= \mathbb{P}(\tau_1 > \sigma_1) = \frac{\mu}{\mu+\lambda}. \tag{941}$$

(iii) In general, consider what has to happen for $X_{k+1} = n+1$ given $X_k = n \geq 1$:

$$\mathbb{P}(X_{k+1} = n+1 \,|\, X_k = n) = \mathbb{P}\begin{pmatrix}\text{remaining service time}\\ \text{after time } T_k\end{pmatrix} > \begin{pmatrix}\text{remaining time until first}\\ \text{arrival after time } T_k\end{pmatrix} \tag{942}$$

Note that the remaining service time after time T_k is still an $\text{Exp}(\mu)$ variable due to the memoryless property of exponential RVs. Moreover, by the memoryless property of Poisson processes, the arrival process restarted at time T_k is a Poisson(λ) process that is independent of the past. Hence (942) is the probability that an $\text{Exp}(\mu)$ RV is less than an independent $\text{Exp}(\lambda)$ RV. Thus we are back to the same computation as in (ii). Similar argument holds for the other possibility $\mathbb{P}(X_{k+1} = n-1 \,|\, X_k = n)$.

(iv) (Stationary distribution) From (i)-(iii), we conclude $(X_k)_{k\geq 0}$ is a Birth-Deach chain on state space $\Omega = \{0,1,2,\cdots\}$. By computing the transition matrix P (which is of infinite size!) and solving $\pi P = \pi$, one obtains that the stationary distribution for the M/M/1 queue is unique and it is a geometric distribution. Namely, if we write $\rho = \lambda/\mu$, then

$$\pi(n) = \rho^n(1-\rho) \qquad \forall\, n \geq 0. \tag{943}$$

Namely, π is the (shifted) geometric distribution with parameter ρ, which is well-defined if and only if $\rho < 1$, that is, $\mu > \lambda$. In words, the rate of service times should be larger than that of the arrival process in order for the queue to not to blow up.

Exercise 8.2.12 (RW perspective of M/M/1 queue)**.** Consider a $M/M/1$ queue with service rate μ and arrival rate λ. Denote $p = \lambda/(\mu+\lambda)$. Let $(X_k)_{k\geq 0}$ be the Markov chain where X_k is the number of customers in the queue at kth time that either departure or arrival occurs. (c.f. [Dur10, Example 6.2.4])

(i) Let $(\xi_k)_{k\geq 1}$ be a sequence of i.i.d. RVs such that

$$\mathbb{P}(\xi_k = 1) = p, \qquad \mathbb{P}(\xi_k = -1) = 1-p \tag{944}$$

Define a sequence of RVs $(Z_k)_{k\geq 0}$ by

$$Z_{k+1} = \max(0, Z_k + \xi_k). \tag{945}$$

Show that X_k and Z_k have the same distribution for all $k \geq 1$.

(ii) Define a simple random walk $(S_n)_{n\geq 0}$ by $S_n = \xi_1 + \cdots + \xi_n$ for $n \geq 1$ and $S_0 = 0$. Show that Z_k can also be written as

$$Z_k = S_k - \min_{0\leq i\leq k} S_i. \tag{946}$$

(iii) The simple random walk $(S_n)_{n\geq 0}$ is called *subcritical* if $p < 1/2$, *critical* if $p = 1/2$, and *supercritical* if $p > 1/2$. Below is a plot of $(Z_k)_{k\geq 0}$ when $p < 1/2$, in which case Z_k is more likely to decrease than to increase when $Z_k \geq 1$. Does it make sense that the $M/M/1$ queue has a unique stationary distribution for $p < 1/2$? Draw plots of Z_k for the critical and supercritical case. Convince yourself that the $M/M/1$ queue should not have a stationary distribution for $p > 1/2$. How about the critical case?



FIGURE 3. Plot of $Z_k = S_k - \min_{0\leq i\leq k} S_i$ for the subcritical case $p < 1/2$.

8.2.2. Completeness of the CTMC construction. We first start with some general discussion on what kind of information we need to specify. We first make the following assumption on the type of continuous-time processes we consider.

Assumption 8.2.13. Let $(X_t)_{t\geq 0}$ denote a continuous-time process on a state space Ω. We may assume that there is a sequence of jump times $0 < T_1 < T_2 < \ldots$ such that $X_t \in \Omega$ is constant on each interval $(0, T_1]$, $(T_1, T_2], \ldots$[2].

Since the process X_t can change its state only at jump times, we can record its states at jump times as a discrete-time stochastic process on Ω, as in the following definition.

Definition 8.2.14. $(X_t)_{t\geq 0}$ denote a continuous-time process on a state space Ω with jump times $(T_n)_{n\geq 1}$. Define the discrete-time process $(Z_n)_{n\geq 1}$, which is called the *jump chain* of $(X_t)_{t\geq 0}$, as $Z_n = X_{T_n}$.

Conversely, a sequence of jump times and a jump chain determines a continuous-time process.

Definition 8.2.15. Suppose we have a jump chain $(Z_n)_{n\geq 1}$ and jump times $(T_n)_{n\geq 1}$. Define the associated continuous-time process $(X_t)_{t\geq 0}$ by $X_t = Z_{N(t)}$, where $N(t) = \sum_{k=1}^{\infty} \mathbf{1}(T_k \leq t)$ is the associated counting process.

Proposition 8.2.16. *Suppose we have a jump chain $(Z_n)_{n\geq 1}$ and jump times $(T_n)_{n\geq 1}$. Let $(X_t)_{t\geq 0}$ denote the associated continuous-time process as in Definition (8.2.15). Then $(X_t)_{t\geq 0}$ has jump times $(T_n)_{n\geq 1}$ and jump chain $(Z_n)_{n\geq 1}$ in the sense of Definition 8.2.14.*

PROOF. Follows from definition. □

Proposition 8.2.17. *Suppose $(X_t)_{t\geq 0}$ is a continuous-time process on Ω with jump times $(T_n)_{n\geq 1}$. Further assume that $(X_t)_{t\geq 0}$ satisfies the continuous-time Markov property.*

(i) *The associated jump chain $(Z_n)_{n\geq 1}$ is a discrete-time Markov chain on Ω.*

[2]Not all CTMC satisfy this assumption, for instance, the Brownian motion, which changes it state at every single time instance almost surely. However, construction of such 'smooth CTMC' can be obtained by a suitable limiting procesure of 'discrete CTMC' that we consider here.

(ii) *Conditional on Z_n, the holding time $\tau_n := T_{n+1} - T_n$ at state $Z_n \in \Omega$ is independent from the past and has an exponential distribution.*

PROOF. Then the future states should only depend on the present state, so the jump chain $(Z_k)_{k\geq 1}$ should be a discrete-time Markov chain. Also, a continuous-time version of strong Markov property holds (see the proof of Proposition 3.4.4) the jump times are stopping times, since

$$\{T_n \leq s\} = \tag{947}$$

the amount of time $\tau_k = T_{k+1} - T_k$ spent at state $x := Z_k = X_{T_k}$ should be memoryless due to the continuous-time Markov property, so it has to be an exponential RV (see Exercise 6.1.6) with parameter λ. Finally, note that

$$\mathbb{P}(\tau_k \geq s \,|\, Z_n, Z_{n-1}, \ldots, Z_1) = \mathbb{P}\left(X_t \equiv Z_n \text{ for } t \in [T_n, T_n + s] \,|\, X_{T_n}, X_{T_{n-1}}, \ldots, X_{T_1}\right) \tag{948}$$

$$= \mathbb{P}\left(X_t \equiv Z_n \text{ for } t \in [T_n, T_n + s] \,|\, X_{T_n}\right) \tag{949}$$

$$= \mathbb{P}(\tau_k \geq s \,|\, Z_n), \tag{950}$$

where we have used a continuous-time version of strong Markov property and the fact that the jump times are stopping times. □

8.3. Evolution of the transition probability function

In this section, we give a differential equation point of view of CTMC, deriviing various quantities such as jump rates and exponential holding parameter from the transition probability function.

8.3.1. Differential Equations for TPF.

Proposition 8.3.1 (Rates from TPF)**.** *Consider a CTMC with bounded jump rates $q(x,y) \leq \alpha$ for all $x \neq y \in \Omega$. Let $(t,x,y) \mapsto \mathbf{P}_t(x,y)$ be its transition probability function. Then the following hold:*

(i) *(Jump rate)* $\quad q(x,y) = \lim_{h\to 0+} \dfrac{\mathbf{P}_h(x,y)}{h} = \dfrac{d}{dt}\mathbf{P}_t(x,y)\Big|_{t=0+}$ *for $x \neq y \in \Omega$;*

(ii) *(Holding parameter)* $\quad \lambda(x) = \lim_{h\to 0+} \dfrac{1-\mathbf{P}_h(x,x)}{h} = -\dfrac{d}{dt}\mathbf{P}_t(x,x)\Big|_{t=0+}$ *for $x \in \Omega$.*

PROOF. We prove the assertion only for the bounded jump rate case: $q(x,y) \leq \alpha$ for all $x \neq y \in \Omega$. In this case, we can construct the CTMC using Definition 8.2.6 (MC run by PP). Then we can use the formula (933). Suppose $x \neq y$ so that $P(x,y) = q(x,y)/\alpha$. Note that $P^0(x,y) = I(x,y) = \mathbf{1}(x=y) = 0$. Hence

$$\mathbf{P}_t(x,y) = \sum_{k=0}^{\infty} \frac{(\alpha t)^k e^{-\alpha t}}{k!} P^k(x,y) \tag{951}$$

$$= (\alpha t) e^{-\alpha t} \frac{q(x,y)}{\alpha} + \sum_{k=2}^{\infty} \frac{(\alpha t)^k e^{-\alpha t}}{k!} P^k(x,y). \tag{952}$$

Hene

$$\frac{\mathbf{P}_t(x,y)}{t} = e^{-\alpha t} q(x,y) + \alpha^2 t \sum_{k=2}^{\infty} \frac{(\alpha t)^{k-2} e^{-\alpha t}}{k!} P^k(x,y). \tag{953}$$

Note that $0 \leq \sum_{k=2}^{\infty} \frac{(\alpha t)^{k-2} e^{-\alpha t}}{k!} P^k(x,y) \leq \sum_{k=2}^{\infty} (\alpha t)^{k-2} = \frac{1}{1-\alpha t}$ for $t \in (0, 1/\alpha)$. Thus the last term above vanishes as $t \to 0+$. This shows **(i)**.

We show **(ii)** similarly. Since $P^0(x,x) = I(x,x) = \mathbf{1}(x=x) = 1$, we have

$$\mathbf{P}_t(x,x) = e^{-\alpha t} + \alpha t \sum_{k=1}^{\infty} \frac{(\alpha t)^{k-1} e^{-\alpha t}}{k!} P^k(x,x) \tag{954}$$

$$= e^{-\alpha t} + \alpha t e^{-\alpha t} Q(x,x) - \alpha^2 t^2 \sum_{k=2}^{\infty} \frac{(\alpha t)^{k-1} e^{-\alpha t}}{k!} P^k(x,x). \tag{955}$$

Divide both sides by t and take the limit $t \to 0+$. Then the second term in the last expression vanishes as before, and recall that $P(x,x) = 1 - \frac{1}{\alpha}\sum_{y\neq x} q(x,y) = 1 - \lambda(x)/\alpha$. So we get

$$\lim_{t\to 0+} \frac{\mathbf{P}_t(x,x)-1}{t} = \lim_{t\to 0+} \frac{e^{\alpha t}-1}{t} + \alpha\left(1 - \frac{\lambda(x)}{\alpha}\right) = -\lambda(x). \tag{956}$$

This shows the second part of the assertion. □

Definition 8.3.2 (Generator matrix)**.** Consider a CTMC with jump rates $\{q(x,y) \,|\, x \neq y \in \Omega\}$. Define the *generator matrix* $Q : \Omega^2 \to [0,\infty)$ by

$$Q(x,y) = \begin{cases} q(x,y) & \text{if } y \neq x \\ -\lambda(x) = -\sum_{y\neq x} q(x,y) & \text{if } y = x. \end{cases} \tag{957}$$

Example 8.3.3. Consider a CTMC $X_t = Y_{N(t)}$, which is a discrete-time Marvkov chain $(Y_n)_{n\geq 1}$ with transition matrix P run by a Poisson process $(N(t))_{t\geq 0} \sim \mathrm{PP}(\alpha)$. Then its generator matrix Q is given by

$$Q(x,y) = \begin{cases} q(x,y) & \text{if } y \neq x \\ -\lambda(x) & \text{if } y = x \end{cases} = \begin{cases} \alpha P(x,y) & \text{if } y \neq x \\ -\lambda(x) = \alpha(P(x,x)-1) & \text{if } y = x, \end{cases} \tag{958}$$

where the second equality follows from (932). In a matrix form, we have

$$Q = \alpha(P - I). \tag{959}$$

▲

Theorem 8.3.4 (Transition probability function and the generater matrix)**.** *Consider a CTMC with bounded jump rates $q(x,y) \leq \alpha$ for all $x \neq y \in \Omega$. Let $(t,x,y) \mapsto \mathbf{P}_t(x,y)$ be its transition probability function and Q denote its generator matrix. Then the following hold:*

(i) *(Forward Kolmogorov eq.)* $\dot{\mathbf{P}}_t = \mathbf{P}_t Q$ *for all* $t \geq 0$ *with initial condition* $\mathbf{P}_0 = I$*;*

(ii) *(Backward Kolmogorov eq.)* $\dot{\mathbf{P}}_t = Q\mathbf{P}_t$ *for all* $t \geq 0$ *with initial condition* $\mathbf{P}_0 = I$*;*

where $\dot{\mathbf{P}}_t := \frac{d}{dt}\mathbf{P}_t$ *denotes the entry-wise time drivative of* $\mathbf{P}_t$*.*

PROOF. We only give a proof that is rigorous for finite state space Ω. Recall that Kolmogorov-Champman equation in Proposition 8.1.1 **(iii)**. When Ω is finite, then $\mathbf{P}_t$ is a matrix-valued function in t so we can view it as the following matrix equation

$$\mathbf{P}_t\mathbf{P}_s = \mathbf{P}_{t+s}. \tag{960}$$

Now for the forward equation, note that

$$\frac{1}{h}(\mathbf{P}_{t+h} - \mathbf{P}_t) = \frac{1}{h}(\mathbf{P}_t\mathbf{P}_h - \mathbf{P}_t) = \mathbf{P}_t\frac{1}{h}(\mathbf{P}_h - \mathbf{P}_0). \tag{961}$$

Now taking the limit $h \to 0+$, we obtain the forward equation $\dot{\mathbf{P}}_t = \mathbf{P}_t Q$, where the identity $Q = \dot{\mathbf{P}}_0$ follows from Proposition 8.3.1 and the definition of the generator matrix Q. For the backward equation, note that

$$\frac{1}{h}(\mathbf{P}_{t+h} - \mathbf{P}_t) = \frac{1}{h}(\mathbf{P}_h\mathbf{P}_t - \mathbf{P}_t) = \frac{1}{h}(\mathbf{P}_h - \mathbf{P}_0)\mathbf{P}_t \tag{962}$$

and take the limit $h \to 0+$. □

Remark 8.3.5. Compare the forward and backward Kolmogorov-Champman equations with the following equations of discrete-time MC transition matrices:

$$P^{t+1} = PP^t = P^tP. \tag{963}$$

8.3.2. Kolmogorov's equation for finnite state space. Consider the following differential equation

$$\dot{x}(t) = \alpha x(t), \qquad x(0) = 1. \tag{964}$$

In order to solve it, we multiply an integrating factor $e^{-\alpha t} > 0$ and use product formula for derivative:

$$\dot{x}(t) = \alpha x(t) \quad \Leftrightarrow \quad \dot{x}(t) - \alpha x(t) = 0 \quad \Leftrightarrow \quad e^{-\alpha t}\dot{x}(t) - \alpha e^{-\alpha t}x(t) = 0 \tag{965}$$

$$\Leftrightarrow \quad \frac{d}{dt}e^{-\alpha t}x(t) = 0 \tag{966}$$

$$\Leftrightarrow \quad e^{-\alpha t}x(t) \equiv C, \tag{967}$$

where C is a constant. By using the initial condition $x(0) = 1$, we find $C = 1$. Hence $x(t) = e^{-\alpha t}$. Conversely, such solution satisfies the original boundary value problem (964). Hence it follows that

$$\dot{x}(t) = \alpha x(t), \qquad x(0) = 1 \quad \Longleftrightarrow \quad x(t) = e^{-\alpha t}. \tag{968}$$

We notice that Theorem 8.3.4 gives a matrix version of the analogous differential equation (964). So, is it true that the transition probability function $\mathbf{P}_t$ is also given by the 'exponential function' e^{tQ}, where Q is the generator matrix? It turns out to be true, as we will see shortly, but for that we first need to define exponential functions for matrices and discuss some basic properties. The idea is to adapt the Talyor expansion of the exponential function: $e^x = \sum_{k=0}^{\infty} \frac{t^k}{k!}$.

Definition 8.3.6 (Matrix exponential)**.** Let A be a matrix. Then we define

$$e^A := \sum_{k=1}^{\infty} \frac{A^k}{k!}. \tag{969}$$

Matrix exponentials have familiar properties such as $e^{A+B} = e^A e^B$. (Check!)

Remark 8.3.7. The matrix exponential e^A converges absolutely entrywise. To see this, we use a matrix norm $\|\cdot\|$ with the property $\|AB\| \le \|A\|\,\|B\|$. Then we have

$$\|e^A\| \le \sum_{k=1}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|} < \infty. \tag{970}$$

Examples of such matrix norms inlude the matrix Frobenius norm $\|A\|_F = \sqrt{\sum_{ij} A_{ij}^2}$ and the matrix 2-norm $\|A\|_2 = \sup_{x,\|x\|=1} \frac{\|Ax\|_2}{\|x\|_2}$ denotes the matrix 2-norm, where $\|\cdots\|_2$ denotes the usual Euclidean norm for vectors. This quantity measures the maximum amount that A stretches a given unit vector. Also, we have $\|A\|_2 = \sigma_1(A)$, the largest singular value of A.

Theorem 8.3.8. *Let Q be the generator matrix of a continuous-time Markov chain with finite state space Ω. Then both the Kolmogorov forward and backward equations in Theorem 8.3.4 have unique solution*

$$\mathbf{P}_t = e^{-tQ} = \sum_{k=0}^{\infty} \frac{t^k}{k!} Q^k. \tag{971}$$

PROOF. Convergent power series can be differentiated term by term. This applies also to a matrix-valued series. So we can verify that e^{-tQ} indeed solves both Komogorov's equation:

$$\frac{d}{dt}\sum_{k=0}^{\infty} \frac{t^k}{k!} Q^k = \sum_{k=0}^{\infty} \frac{d}{dt}\frac{t^k}{k!} Q^k = Q\left(\sum_{k=0}^{\infty} \frac{t^{k-1}}{(k-1!)} Q^{k-1}\right) = \left(\sum_{k=0}^{\infty} \frac{t^{k-1}}{(k-1!)} Q^{k-1}\right) Q \tag{972}$$

$$= Qe^{-tQ} = e^{-tQ}Q. \tag{973}$$

For the initial condition, note that $e^{0Q} = I$ by the definition. Hence $\mathbf{P}_t = e^{-tQ}$ satisfies both Komogorov's equation.

For the converse direction, we can mimic the integrating factor argument as follow:

$$\dot{\mathbf{P}}_t = Q\dot{\mathbf{P}}_t \quad \Leftrightarrow \quad \dot{\mathbf{P}}_t - Q\dot{\mathbf{P}}_t = O \quad \Leftrightarrow \quad e^{-Qt}\dot{\mathbf{P}}_t - e^{-tQ}Q\dot{\mathbf{P}}_t = O \tag{974}$$

$$\Leftrightarrow \quad e^{-Qt}\dot{\mathbf{P}}_t - Qe^{-Qt}\dot{\mathbf{P}}_t = O \tag{975}$$

$$\Leftrightarrow \quad \frac{d}{dt}e^{-Qt}\mathbf{P}_t = O \tag{976}$$

$$\Leftrightarrow \quad e^{-Qt}\mathbf{P}(t) \equiv C, \tag{977}$$

where the third equivalence uses the communtativity $Q\left(\sum_{k=0}^{\infty}\frac{t^k}{k!}Q^k\right) = \left(\sum_{k=0}^{\infty}\frac{t^k}{k!}Q^{k+1}\right) = \left(\sum_{k=0}^{\infty}\frac{t^k}{k!}Q^k\right)Q$ and the fourth equivalence uses the product rule for differentiation entrywise. Using the initial condition $\mathbf{P}_0 = I$, we find $\mathbf{P}_t = e^{Qt}$, as desired. □

Example 8.3.9 (MC run by PP – PTF)**.** Let $(Y_n)_{n\geq 1}$ denote a discrete-time Markov chain on a countable state space Ω with transition matrix P. Suppose we have a Poisson arrival process $(N(t))_{t\geq 0} \sim \mathrm{PP}(\lambda)$ independent of the chain $(Y_n)_{n\geq 1}$. We define a continuous-time process $(X_t)_{t\geq 0}$ on the same state space Ω by

$$X_t = Y_{N(t)}. \tag{978}$$

Recall that in Example 8.1.4, we derived using a conditinoing argument the transition probability function $\mathbf{P}_t$ of $(X_t)_{t\geq 0}$ as

$$\mathbf{P}_t(x,y) = \sum_{k=0}^{\infty}\frac{(\lambda t)^k e^{-\lambda t}}{k!}P^k(x,y). \tag{979}$$

We can use Theorem 8.3.8 to re-derive this. Note that the generator matrix Q of X_t is given by $Q = \alpha(P-I)$ (see Example 8.3.3).

8.4. Stationary distribution and limit theorems for CTMC

8.4.1. Stationary distribution for CTMC.

Definition 8.4.1. A probability measure π on the state space Ω is an invariant distribution (also stationary distribution) for the continuous-time Markov chain with transition probability function $\mathbf{P}_t$ if

$$\pi\mathbf{P}_t = \pi. \tag{980}$$

Example 8.4.2 (Continuation of Example 8.3.9)**.** Let $X_t = Y_{N(t)}$, where Y_n is a MC on Ω with transition matrix P and $N(t) \sim \mathrm{PP}(\alpha)$ is an independent Poisson process. Since the clock N_t ticks at a constant average rate, the switch between discrete and continuous time does not involve a time distortion. As a result X_t and Y_n have the same invariant distributions:

$$\pi\mathbf{P}_t = \pi \quad \Longleftrightarrow \quad \pi P = \pi. \tag{981}$$

This can be rigorously justified by using the formula (979).

Theorem 8.4.3. *The probability distribution π is invariant for a (non-explosive) continuous-time Markov chain if and only if*

$$\pi Q = 0. \tag{982}$$

By using the definition of Q in (957), this is equivalent to the following system of equations

$$\sum_{x:x\neq y}\pi(x)q(x,y) = \pi(y)\lambda(y) = \pi(y)\sum_{x:x\neq y}q(x,y) \qquad x,y\in\Omega. \tag{983}$$

(Incoming probability flow = outcoing probability flow)

PROOF. First assume $\pi \mathbf{P}_t \equiv \pi$. Then its derivative should be zero, so

$$0 = \frac{d}{dt}\pi \mathbf{P}_t = \pi \frac{d}{dt}\mathbf{P}_t = \pi Q \mathbf{P}_t \tag{984}$$

where we used Komogorov backward equation. Then since $\mathbf{P}_0 = I$, we have $\pi Q = 0$.

Conversely, suppose $\pi Q = 0$. Then

$$\frac{d}{dt}\pi \mathbf{P}_t = \pi \frac{d}{dt}\mathbf{P}_t = \pi Q \mathbf{P}_t \equiv 0, \tag{985}$$

so this implies that $\pi \mathbf{P}_t$ is constant. But by using $\mathbf{P}_0 = I$, we also get $\pi \mathbf{P}_t = \pi \mathbf{P}_0 = \pi$. □

For many classes of CTMCs, invariant distributions can be computed by solving a detailed balance equation, as in the discrete MC cases.

Definition 8.4.4. A probability distribution π on the state space Ω of A CTMC with transition probability function $\mathbf{P}_t$ is a *reversible distribution* for $(X_t)_{t\geq 0}$ if the detailed balance equation hold:

$$\pi(x)\mathbf{P}_t(x,y) = \pi(y)\mathbf{P}_t(y,x) \qquad \text{for all } t \geq 0 \text{ and } x, y \in \Omega. \tag{986}$$

As in the discrete-time case, reversible probability distributions are in fact stationary. Also, we can simply use the jump rates $q(x,y)$ in place of the transitino probability function $\mathbf{P}_t(x,y)$ in (986). These are justified in the following proposition.

Proposition 8.4.5. *Let $(X_t)_{t\geq 0}$ denote a CTMC on a state space Ω and transition probability function $\mathbf{P}_t$.*

(i) *If π is a reversible distribution for $(X_t)_{t\geq 0}$, then it is also an invariant distirbution.*

(ii) *A distribution π is a reversible distribution for $(X_t)_{t\geq 0}$ if and only if*

$$\pi(x)q(x,y) = \pi(y)q(y,x). \tag{987}$$

PROOF. For **(i)**, we mimic the proof of a similar statement for the discrete MC (Proposition 3.2.7). That is, note that

$$[\pi \mathbf{P}_t](x) = \sum_{y\in\Omega} \mu(y)\mathbf{P}_t(y,x) \overset{\text{reversibility}}{=} \sum_{y\in\Omega} \pi(x)\mathbf{P}_t(x,y) = \mu(x) \sum_{y\in\Omega} \mathbf{P}_t(x,y) = \pi(x). \tag{988}$$

That is, we have shown that $\pi \mathbf{P}_t = \pi$, so π is an invariant distribution for $(X_t)_{t\geq 0}$.

Next, we show **(ii)**. First suppose (986) holds. By differentiating both sides and plugging in $t = 0$ and using the Kolmogorov equations in Theorem 8.3.4, we get

$$\pi(x)Q(x,y) = \pi(y)Q(y,x) \qquad \text{for } x, y \in \Omega. \tag{989}$$

where Q is the generator matrix defined in terms of the jump rates q as in (957). If $x \neq y$, then $Q(x,y) = q(x,y)$ and $Q(y,x) = q(y,x)$, so (987) holds. If $x = y$, then (987) holds trivially.

Conversely, suppose (987) holds. Then by definition of the generator matrix Q in (957), we have (989). It follows that

$$\pi(x)Q^k(x,y) = \pi(y)Q^k(y,x) \qquad \text{for } k \in \mathbb{Z}_{\geq 0} \text{ and } x, y \in \Omega. \tag{990}$$

Indeed, the above is true for $k = 0$ since $Q = I$ and for $k = 1$ due to (989). Suppose (990) holds for some $k \geq 1$. Then

$$\pi(x)Q^{k+1}(x,y) = \pi(x) \sum_{z\in\Omega} Q^k(x,z)Q(z,y) \tag{991}$$

$$= \sum_{z\in\Omega} \pi(x)Q^k(x,z)Q(z,y) \tag{992}$$

$$= \sum_{z\in\Omega} Q^k(x,z)\pi(z)Q(z,y) \tag{993}$$

$$= \sum_{z\in\Omega} Q^k(x,z)Q(z,y)\pi(y) = \left(\sum_{z\in\Omega} Q^k(x,z)Q(z,y)\right)\pi(y) = Q^{k+1}(x,y)\pi(y). \tag{994}$$

This completes the induction for (990).

Now by using (990) and Theorem 8.3.8,

$$\pi(x)\mathbf{P}_t(x,y) = \pi(x)e^{-tQ(x,y)} = \sum_{k=0}^{\infty} \frac{t^k}{k!}\pi(x)Q^k(x,y) = \sum_{k=0}^{\infty} \frac{t^k}{k!}\pi(y)Q^k(y,x) \tag{995}$$

$$= \pi(y)\sum_{k=0}^{\infty} \frac{t^k}{k!}Q^k(y,x) = \pi(y)e^{-tQ(y,x)} = \pi(y)\mathbf{P}_t(y,x). \tag{996}$$

This shows the assertion. □

8.4.2. Limit theorems for CTMC. In this subsection, we study some basic limit theorems for CTMC and how to use stationary distributions for describing various asymptotic behavior of CTMC.

Definition 8.4.6. A CTMC $(X_t)_{t\geq 0}$ on Ω with transition probability function $\mathbf{P}_t$ is *irreducible* if for all $x, y \in \Omega$, $\mathbf{P}_t(x,y) > 0$ for some $t \geq 0$.

Proposition 8.4.7. *A CTMC $(X_t)_{t\geq 0}$ on Ω is irreducible if and only if its jump chain is irreducible.*

PROOF. Exericse. □

Theorem 8.4.8 (SLLN for CTMC)**.** *An irreducible continuous-time Markov chain with an invariant distribution satisfies the following limit theorems.*

(i) *(Limit distribution). For any initial distribution μ and any state y,*

$$\lim_{t\to\infty} \mathbb{P}_\mu(X_t = y) = \pi(y). \tag{997}$$

(ii) *(SLLN) Let $f : \Omega \to \mathbb{R}$ be a function such that $\mathbb{E}_{X\sim\pi}[|f(X)|] = \sum_x \pi(x)|f(x)| < \infty$. Then for any initial distribution μ, the limit below holds with probability one:*

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t f(X_s)\,ds = \sum_x \pi(x)f(x) = \mathbb{E}_{X\sim\pi}[f(X)]. \tag{998}$$

In particular, by taking $f(y) = \mathbf{1}(y \neq x)$,

$$\text{(Asymptotic fraction of time spent at } x) = \lim_{t\to\infty} \frac{1}{t}\int_0^t \mathbf{1}(X_s = x)\,ds = \pi(x). \tag{999}$$

PROOF. Omited. □

Think about why 'aperiodicity' is not needed for CTMC in order for the convergence to stationary distribution in Theorem 8.4.8 **(i)** to hold.

Theorem 8.4.9. *Suppose π is an invariant distribution for an irreducible continuous-time Markov chain. Then for all $x \in \Omega$,*

$$\pi(x) = \frac{1}{\lambda(x)\,\mathbb{E}[T_x \,|\, X_0 = x]}, \tag{1000}$$

where $T_x = \inf\{t > 0 \,|\, X_t = x\}$ denotes the first return time to x and $\lambda(x)$ denotes the exponential holding parameter.

PROOF. Let $T_x^{(k)}$ denote the k the return time to x, assuming $X_0 = x$. These are stopping times with respect to the CTMC $(X_t)_{t\geq 0}$. By strong Markov property, these are i.i.d. and hence define a renewal process. By Theorem 8.4.8 **(i)** and the renewal reward SLLN (Theorem 2.2.1),

$$\pi(x) = \lim_{T\to\infty} \frac{1}{T}\int_0^T \mathbf{1}(X_s = x)\,dx = \frac{\mathbb{E}[\text{time spent at } x]}{\mathbb{E}[\text{duraction of a cycle}]} = \frac{1/\lambda(x)}{\mathbb{E}[T_x \,|\, X_0 = x]}. \tag{1001}$$

This shows the assertion. □

Example 8.4.10. Consider a CTMC $(X_t)_{t\geq 0}$ on the state space $\Omega = \{0,1,2,3\}$ whose jump rates $q(x,y)$ are given by the following diagram in Figure 4. Recall that for each state x, the exponential holding parameter $\lambda(x)$ is given by $\lambda(x) = \sum_{y\neq x} q(x,y)$. Hence we have $\lambda(0) = 1$, $\lambda(1) = 2$, $\lambda(2) = 3$, and $\lambda(3) = 3$. Its generator matrix Q (see (957)) is given by

$$Q = \begin{bmatrix} -1 & 1 & & \\ 1 & -2 & 1 & \\ & 2 & -3 & 1 \\ & & 3 & -3 \end{bmatrix}. \tag{1002}$$

This CTMC is irreducible since the jump rate diagram gives a strongly connected directed graph. That is, for any two states x, y, there is a path from x to y with jumps of positive rates. Any such path has a positive probability to occur, so this yields that the CTMC can move from x to y with a positive probability. (Or derive the routing matrix and draw the state space diagram of the jump chain and see it is irreducible).

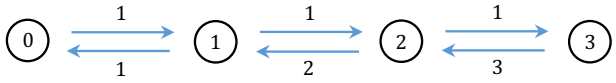


FIGURE 4. A jump rate diagram

By Theorem 8.4.8 **(i)** there is a unique stationary distribution π. By Theorem 8.4.3, π is given by $\pi Q = 0$. Solving this equation for $\pi = [\pi_0, \pi_1, \pi_2, \pi_3]$, we get

$$-\pi_0 + \pi_1 = 0 \tag{1003}$$
$$\pi_0 - 2\pi_1 + 2\pi_2 = 0 \tag{1004}$$
$$\pi_1 - 3\pi_2 + 3\pi_3 = 0 \tag{1005}$$
$$\pi_2 - 3\pi_3 = 0. \tag{1006}$$

From the first and last equations, we get $\pi_0 = \pi_1$ and $\pi_2 = 3\pi_3$. Plugging these into the middle two equations, we get

$$-\pi_1 + 2\pi_2 = 0 \tag{1007}$$
$$\pi_1 - 2\pi_2 = 0. \tag{1008}$$

Thus $\pi_1 = 2\pi_2$. So $\pi_2 = \pi_0/2$ and $\pi_3 = \pi_2/3 = \pi_0/6$. Since

$$1 = \pi_0 + \pi_1 + \pi_2 + \pi_3 = \pi_0(1 + 1 + (1/2) + (1/6)) = \pi_0 \frac{12+3+1}{6} = \pi_0 \frac{16}{6}. \tag{1009}$$

Hence $\pi_0 = 3/8$, so we conclude

$$\pi = \left[\frac{3}{8}, \frac{3}{8}, \frac{3}{16}, \frac{1}{16}\right]. \tag{1010}$$

Now by Theorem 8.4.8 **(ii)**, the asymototic fraction of times that X_t spends at each state x is $\pi(x)$. For example,

$$\text{(Asymptotic fraction of times spent not at 0)} = \pi(1) + \pi(2) + \pi(3) = \frac{5}{8}. \tag{1011}$$

Moreover, by Theorem 8.4.9, we can also find the expected return time to each state x:

$$\pi(x) = \frac{1}{\lambda(x)\,\mathbb{E}[T_x \,|\, X_0 = x]}. \tag{1012}$$

For example, $\mathbb{E}[T_0 \,|\, X_0 = 0] = 1/(\pi(0)\lambda(0)) = (8/3)$. So starting from 0, the expected time to return to 0 is $8/3$. ▲

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